CBSE - 2023

26. Name the factors on which electrical conductivity of a material depends. Obtain the relation between current density in a conductor and the conductivity of its material. *3*

Factors on which electrical conductivity depends	1½
Obtaining relation between current density and conductivity	1½

Electric conductivity $$\sigma = \frac{ne^2\tau}{m}$$ Factors on which conductivity depends (i) concentration of charge carrier (ii) relaxation time (iii) mass of charge carrier	 ½ ½ ½
In a conductor of resistance R, the potential difference V and Current I. $V = I R$ $El = I\rho\frac{l}{A}$ $j = \frac{I}{A} = \frac{E}{\rho}$ $j = \sigma E$	 ½ ½ ½

19. The terminal potential difference of a cell is 19 V when a current of 1·0 A flows in the circuit. It reduces to 17 V when the current supplied by the cell is 3·0 A. Find the emf and internal resistance of the cell. *2*

Finding the emf	1
Finding the internal resistance	1

19. The terminal potential difference of a cell is 19 V when a current of 1·0 A flows in the circuit. It reduces to 17 V when the current supplied by the cell is 3·0 A. Find the emf and internal resistance of the cell. *2*

$V = E - Ir$	½
$19 = E - 1.r$ ________ (i) $17 = E - 3.r$ ________ (ii)	½
$2r = 2 \quad \Rightarrow r = 1\,\Omega$	½
$E = 19+1 = 20V$	½

Finding the emf	1
Finding the internal resistance	1

2. A current flows through a series combination of two copper wires of equal length but their radii are in the ratio of 1 : 2. The ratio of drift velocities of free electrons in the wires will be :

(a) 8 (b) 4 (c) 2 (d) 1

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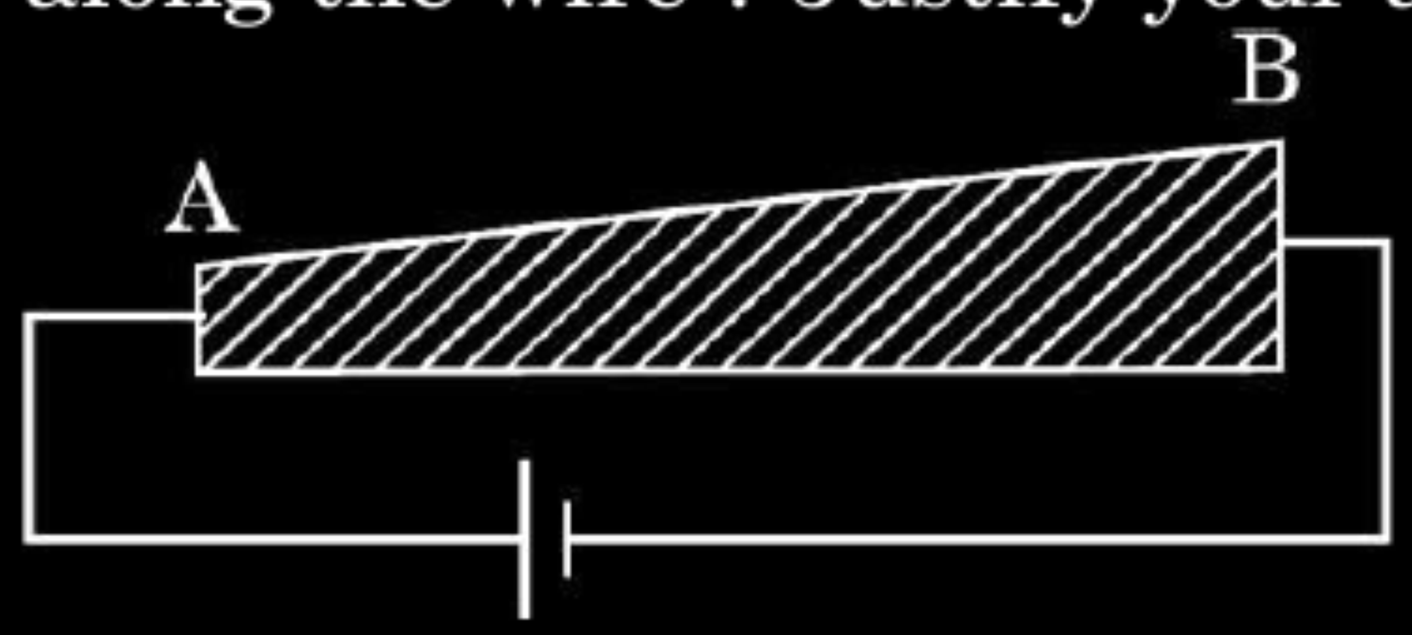
(a) 8 (b) 4 (c) 2 (d) 1

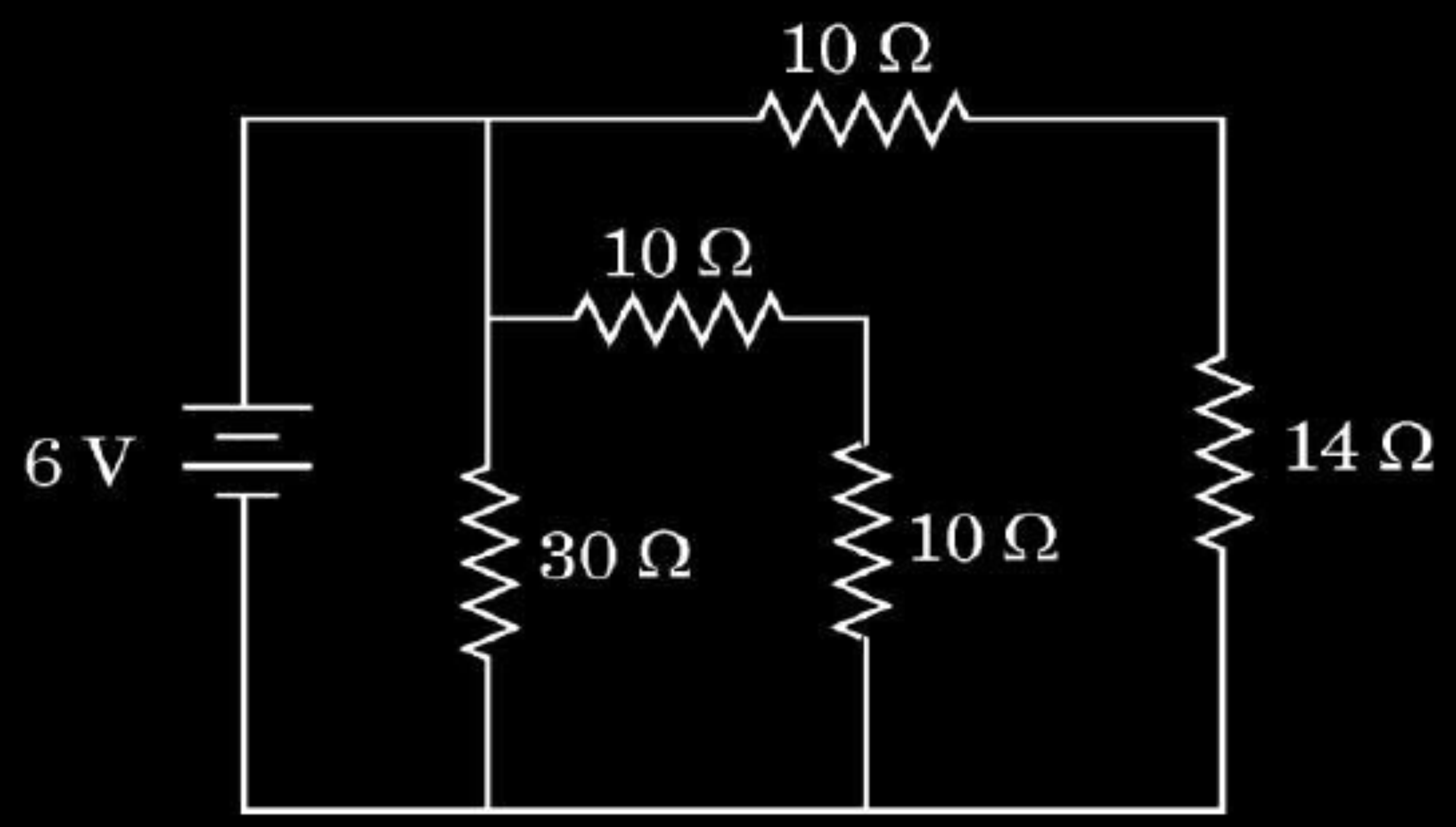
(b) 4

32. (a) (i) Define mobility of electrons. Give its SI units.

(ii) A steady current flows through a wire AB, as shown in the figure. What happens to the electric field and the drift velocity along the wire ? Justify your answer.



(iii) Consider the circuit shown in the figure. Find the effective resistance of the circuit and the current drawn from the battery. 5



(i) Definition & S.I. Unit	1+½
(ii)Change in Electric field and drift velocity along the wire	½+½
Justification	½+½
Effective resistance and current	1 ½

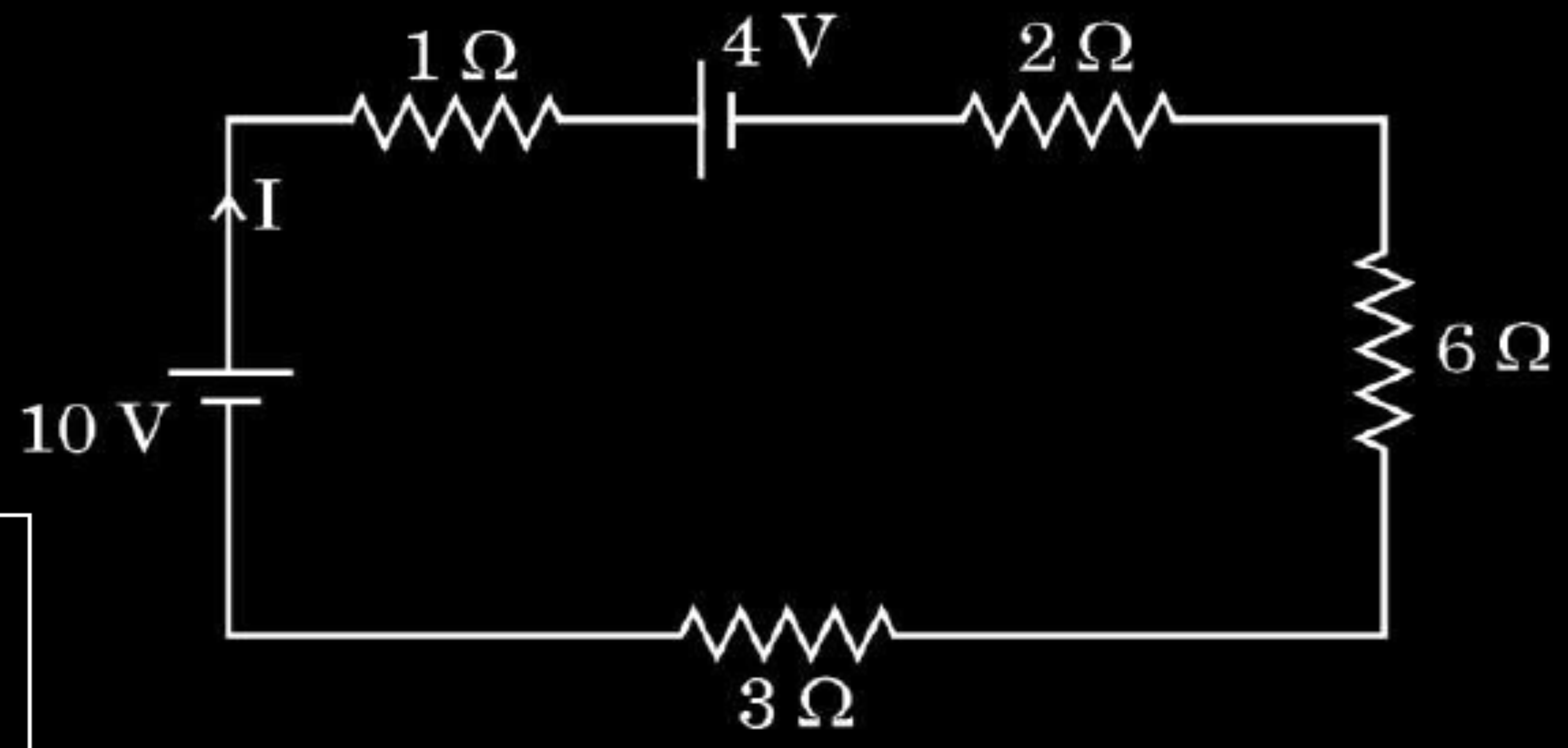
(i) Mobility: Mobility is defined as the magnitude of the drift velocity per unit electric field. S.I. Unit: $\frac{m^2}{V.s}$ or $\frac{C.s}{kg}$	1 ½
(ii) Both electric field and the drift velocity decreases. **Justification:** $v_d = \frac{I}{neA}$ As area increases across the wire, drift velocity decreases. $v_d = \frac{eE}{m}\tau$ As drift velocity decreases, electric field decreases (since e ,m and τ are constant).	½ + ½ ½ ½
(iii) From the diagram 10Ω and 14 Ω are in series R_1=10 Ω + 14 Ω = 24 Ω 10 Ω and 10 Ω are in series R_2=10 Ω+ 10 Ω = 20 Ω 24 Ω, 20 Ω and 30 Ω are in parallel. $\frac{1}{R} = \frac{1}{24} + \frac{1}{20} + \frac{1}{30} = \frac{5+6+4}{120} = \frac{15}{120}$ R= 8 Ω Electric current in the circuit $I = \frac{V}{R} = \frac{6}{8} = \frac{3}{4}A$ **Note: Full credit of 1 ½ marks is to be awarded for part (iii) even if a student does not attempt this part.**	½ 1

OR

(b) (i) Define electrical conductivity of a wire. Give its SI unit.

(ii) High current is to be drawn safely from (1) a low-voltage battery, and (2) a high-voltage battery. What can you say about the internal resistance of the two batteries ?

(iii) Calculate the total energy supplied by the batteries to the circuit shown in the figure, in one minute. *5*



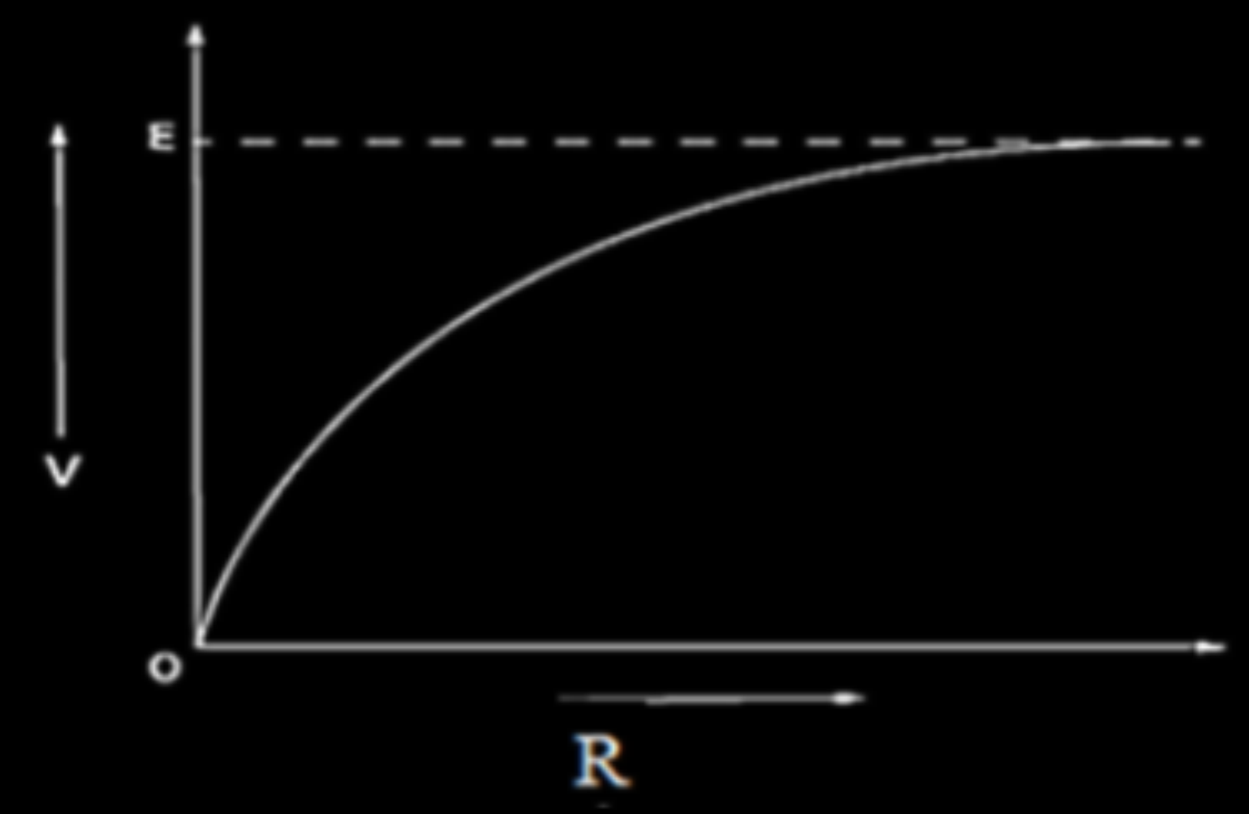
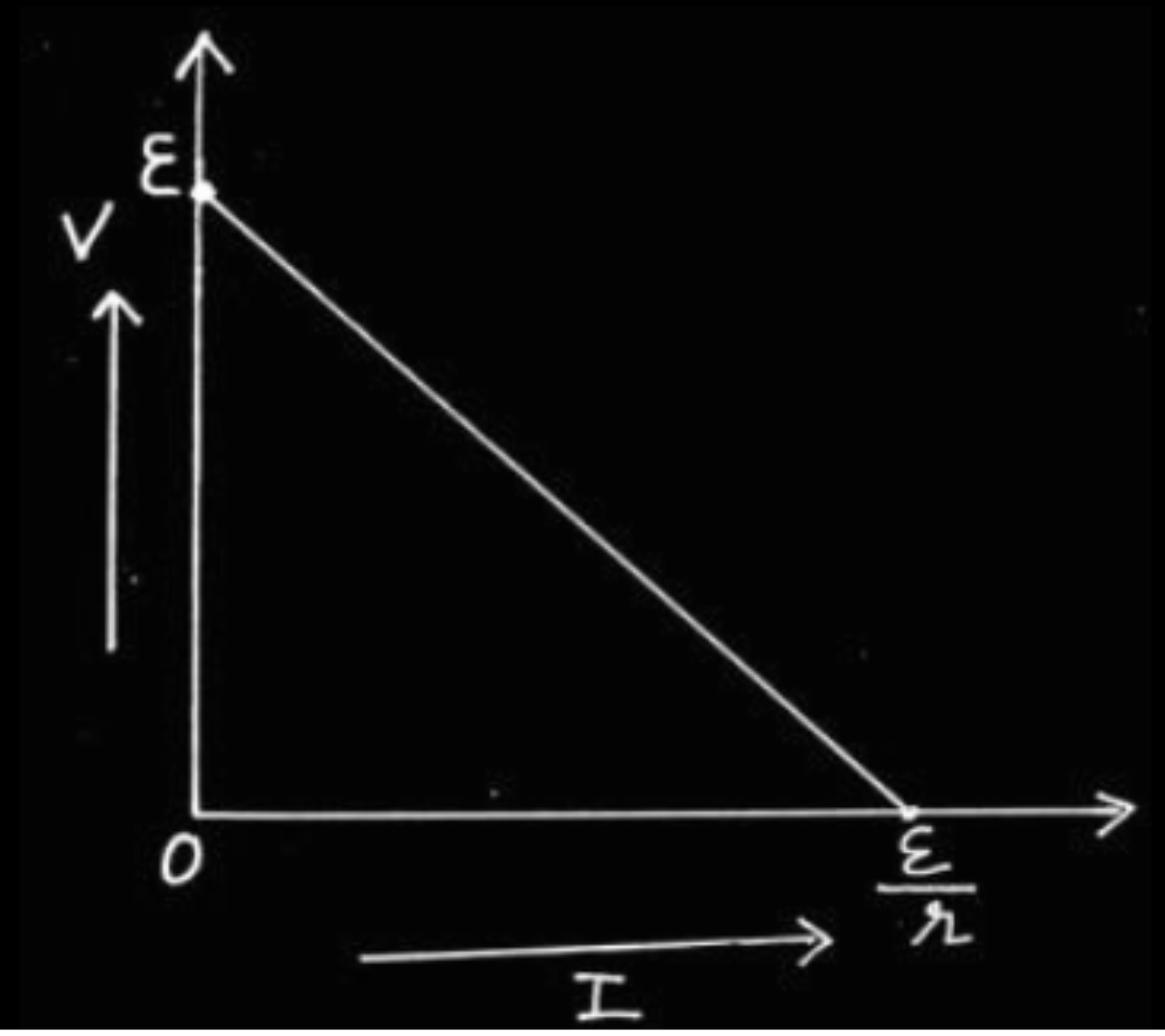
(i) Definition & S.I. Unit	1½
(ii) Explanation of internal resistance for low voltage and high voltage battery	1+1
(iii) Total energy	1½

(i) Definition : Electrical conductivity is defined as the measure of a material's ability to carry a current through it. **Alternatively**: It is the reciprocal of the resistivity. **Alternatively**: It is defined as the current density per unit electric field.	1
S.I. Unit: $(\text{ohm})^{-1}\text{-m}^{-1}$ or S-m^{-1}	½
(ii) **Low voltage Battery-**	
Internal resistance should be low.	1
High voltage Battery –	
Internal resistance should be high.	1
(iii) Applying Kirchhoff's loop rule	
$10 - I \times 1 - 4 - 2I - 6I - 3I = 0$	½
$12I = 6 \Rightarrow I = 0.5\text{ A}$	
Heat energy $H = I^2Rt$	½
$H = 0.25 \times 12 \times 60 = 180J$	½

22. A cell of emf E and internal resistance r is connected to a variable resistance R. Draw plots showing the variation of (a) terminal voltage V with R, and (b) V with current I, in the circuit. *2*

Plots of	
(a) Terminal voltage with R	1
(b) V with current I	1

22. A cell of emf E and internal resistance r is connected to a variable resistance R. Draw plots showing the variation of (a) terminal voltage V with R, and (b) V with current I, in the circuit. *2*

(a) 	**1**
(b) 	**1**

Plots of	
(a) Terminal voltage with R	1
(b) V with current I	1

17. *Assertion (A) :* When three electric bulbs of power 200 W, 100 W and 50 W are connected in series to a source, the power consumed by the 50 W bulb is maximum.

Reason (R) : In a series circuit, current is the same through each bulb, but the potential difference across each bulb is different.

17. *Assertion (A) :* When three electric bulbs of power 200 W, 100 W and 50 W are connected in series to a source, the power consumed by the 50 W bulb is maximum.

Reason (R) : In a series circuit, current is the same through each bulb, but the potential difference across each bulb is different.

(b) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of the Assertion (A).

25. Two identical cells, each of emf E and internal resistance r, are connected with a load resistance R, first in series and then in parallel. Obtain the condition under which the current through R is same in both cases. *2*

Calculation of current for cells in series and parallel	½+½
Obtaining the condition when current is the same in both the cases	1

In series,	
$\varepsilon_{eq} = \varepsilon + \varepsilon = 2\varepsilon$	
$I_s = \dfrac{2\varepsilon}{2r+R}$	½
In parallel,	
$\varepsilon_{eq} = \varepsilon$	
$r_{eq} = \dfrac{r}{2}$	
$I_p = \dfrac{\varepsilon}{\frac{r}{2}+R}$	½
$\because I_p = I_s$	
$\dfrac{2\varepsilon}{2r+R} = \dfrac{\varepsilon}{\frac{r}{2}+R}$	
$\dfrac{2}{2r+R} = \dfrac{1}{\frac{r}{2}+R}$	
$2\dfrac{r}{2} + 2R = 2r + R$	1
$r = R$	

20. Write two differences between the emf and terminal potential difference of a cell. What is the most important precaution that one should take while drawing current from a cell ? *2*

Two differences	½+ ½
Important precaution	1

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Two differences	
1. The potential difference across the electrodes in open circuit is e.m.f. (ε) and in closed circuit is terminal potential difference (V).	½
2. V depends on r and ε is independent of r.	½
Precaution-	
1. Some external resistance should be connected to cell in series.	
2. Short circuiting should be avoided.	1

Two differences	½+ ½
Important precaution	1

1. A battery of emf E and internal resistance r is connected to an external circuit. The potential drop within the battery is proportional to :

(a) current in the circuit

(b) total resistance of the circuit

(c) emf of the battery

(d) power dissipated in the circuit

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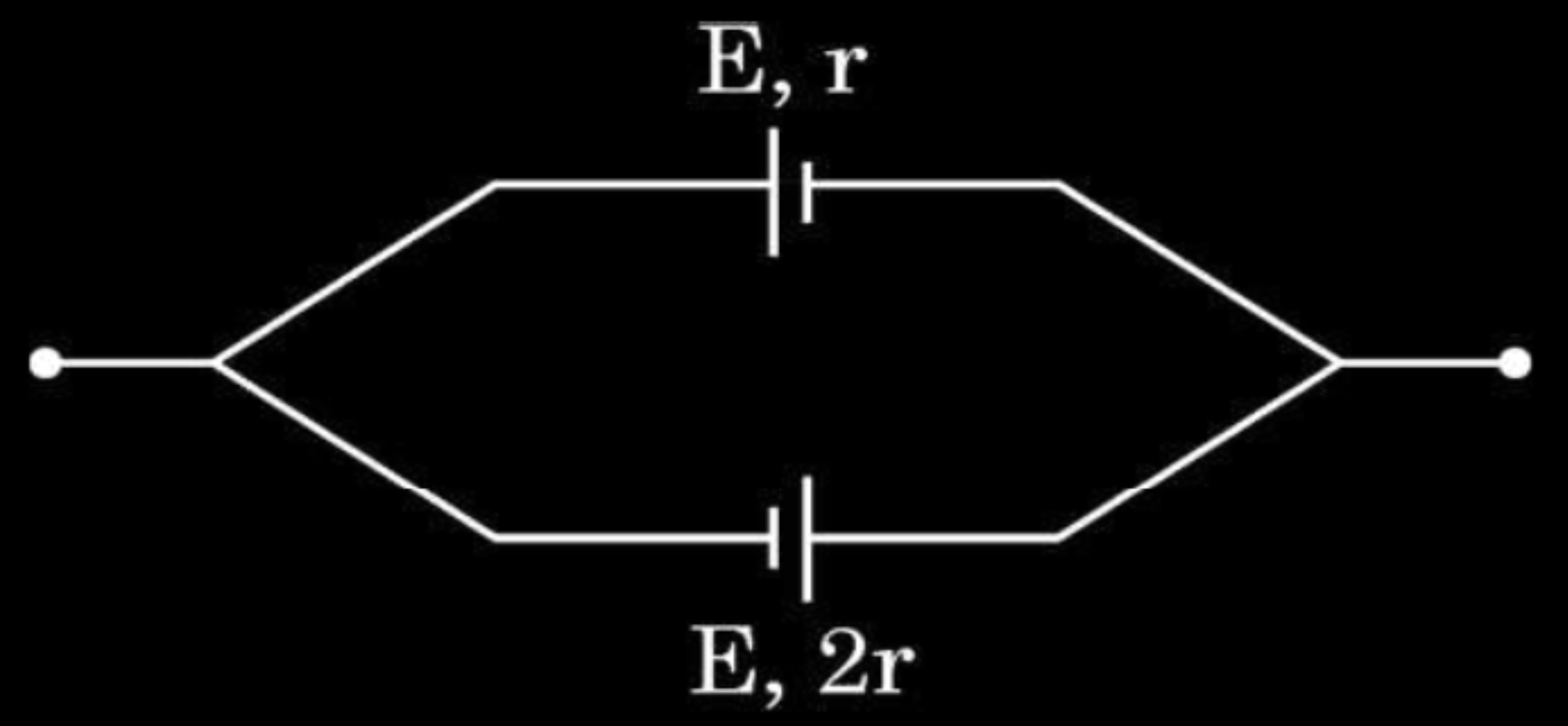
(a) Current in the circuit.

14. Two cells of emf E each and internal resistances r and 2r are connected in parallel as shown in the figure. The equivalent emf of the combination is :



(a) zero

(b) $\frac{E}{2}$

(c) $\frac{E}{3}$

(d) E

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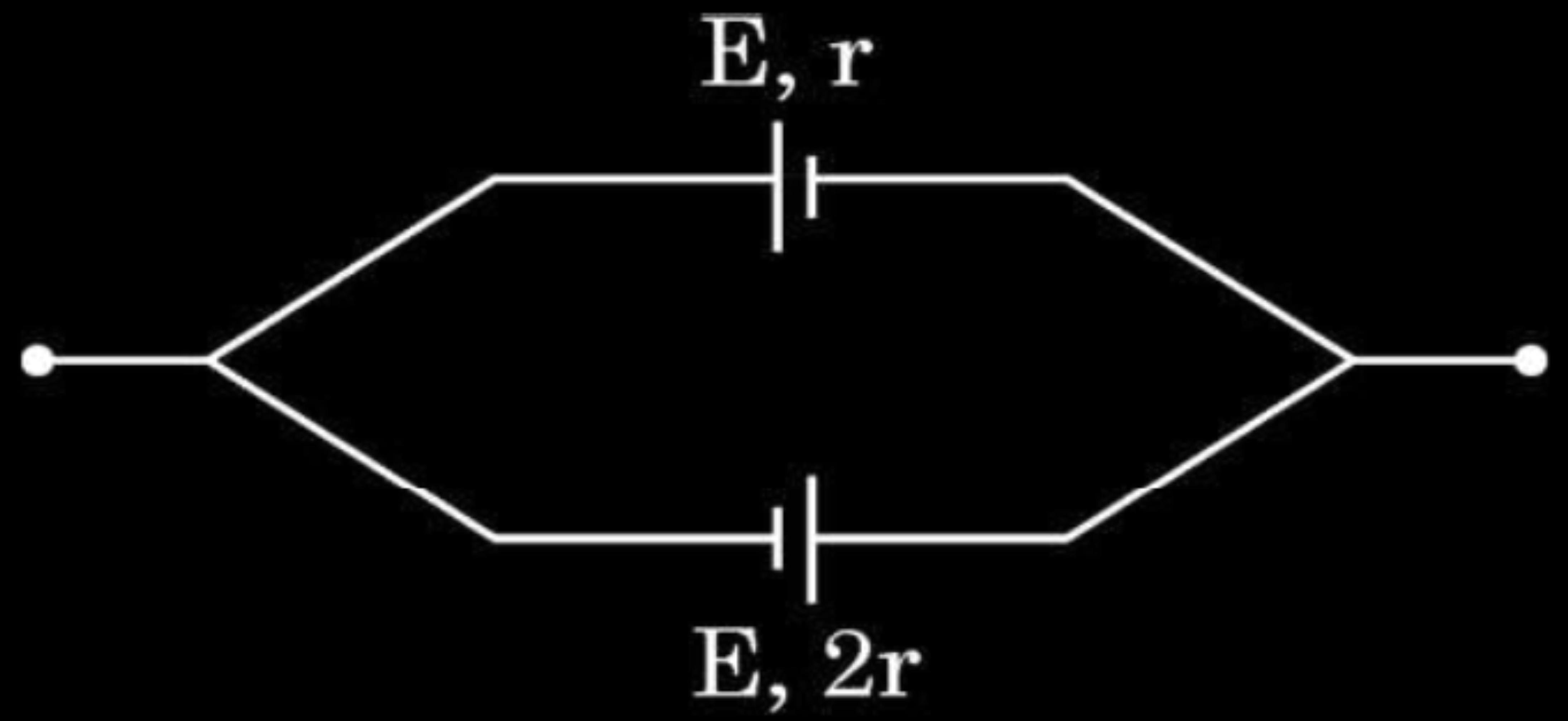

(a) zero

(b) $\frac{E}{2}$

(c) $\frac{E}{3}$

(d) E

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17. *Assertion (A) :* The internal resistance of a cell is constant.

Reason (R) : Ionic concentration of the electrolyte remains same during use of a cell.

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Reason (R) : Ionic concentration of the electrolyte remains same during use of a cell.

(d) Assertion (A) is false and Reason (R) is also false

8. The SI unit of mobility of charge carriers is :

(a) $\Omega\ s^{-1}$

(b) $m^2\ V^{-1}\ s^{-1}$

(c) $m\ s^{-1}\ V^{-1}$

(d) $\Omega\ m$

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(b) $m^2V^{-1}s^{-1}$

13. The emf and internal resistance of a cell are E and r respectively. It is connected across an external resistance R = 2r. The potential drop across the terminals of the cell will be :

(a) $\frac{E}{4}$

(b) $\frac{E}{2}$

(c) $\frac{2}{3}E$

(d) $\frac{E}{3}$

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(c) $\frac{2}{3}E$ (d) $\frac{E}{3}$

(c) $\frac{2}{3}E$

30. A potential difference 'V' is applied across a load resistor of resistance R. V and R can be varied. If the current that flows in the circuit is I, draw a plot showing the variation of power consumed by the resistor as a function of : *3*

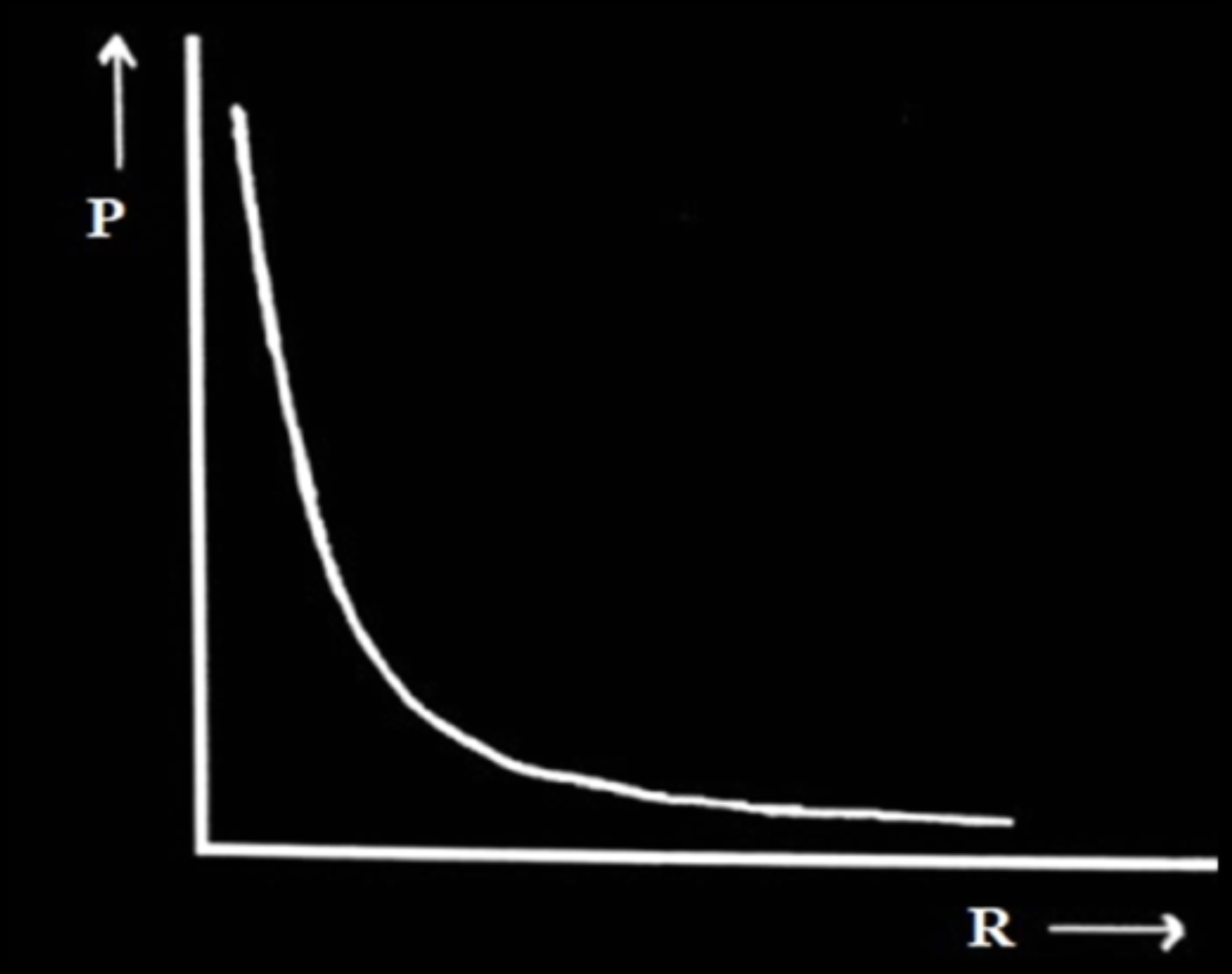
(a) R, keeping V constant

(b) I, keeping R constant

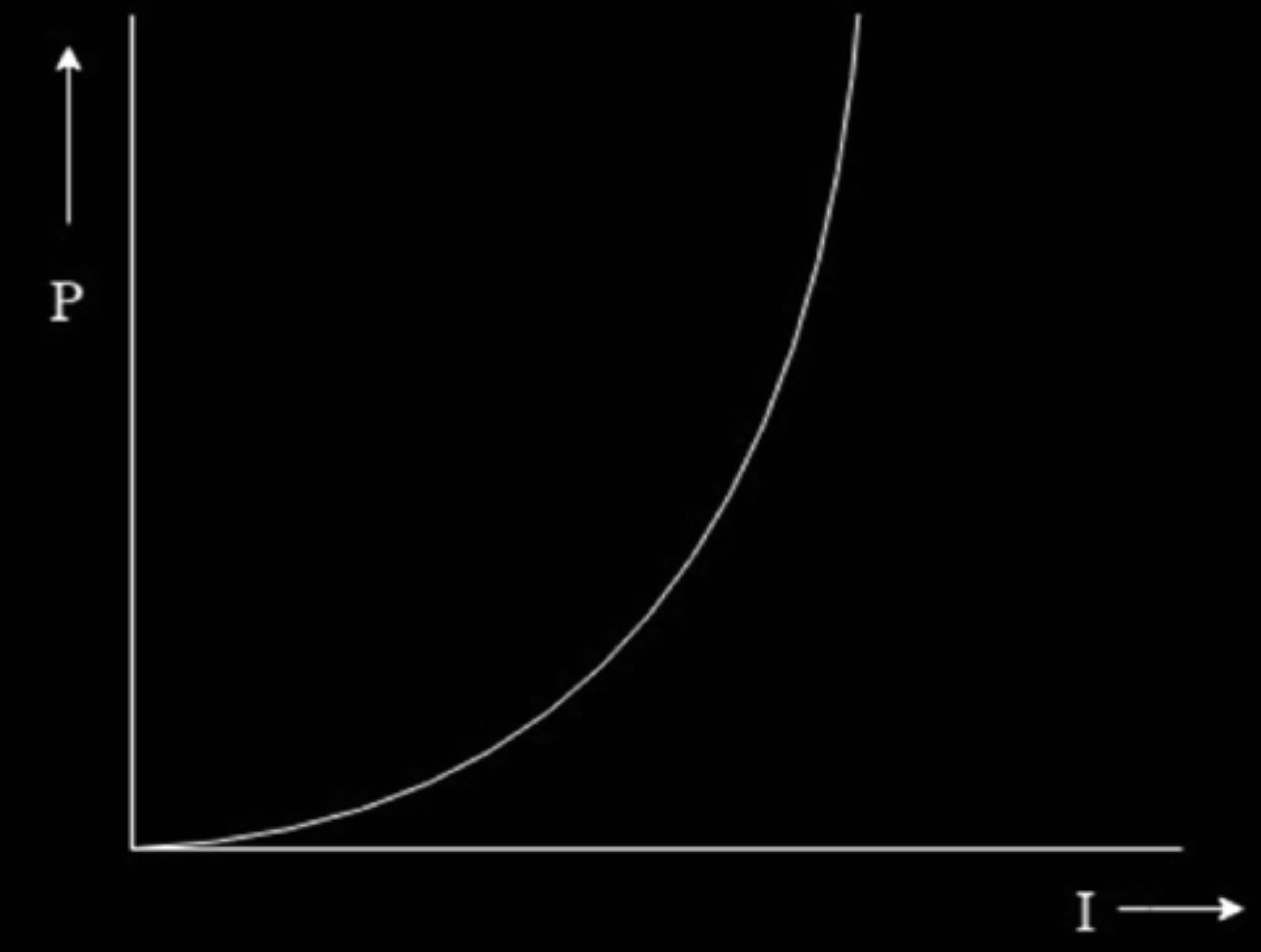
(c) V, keeping R constant

Graph showing variation of power consumed by the resistor as a function of:	
(a) R, keeping V constant	1
(b) I, keeping R constant	1
(c) V, keeping R constant	1

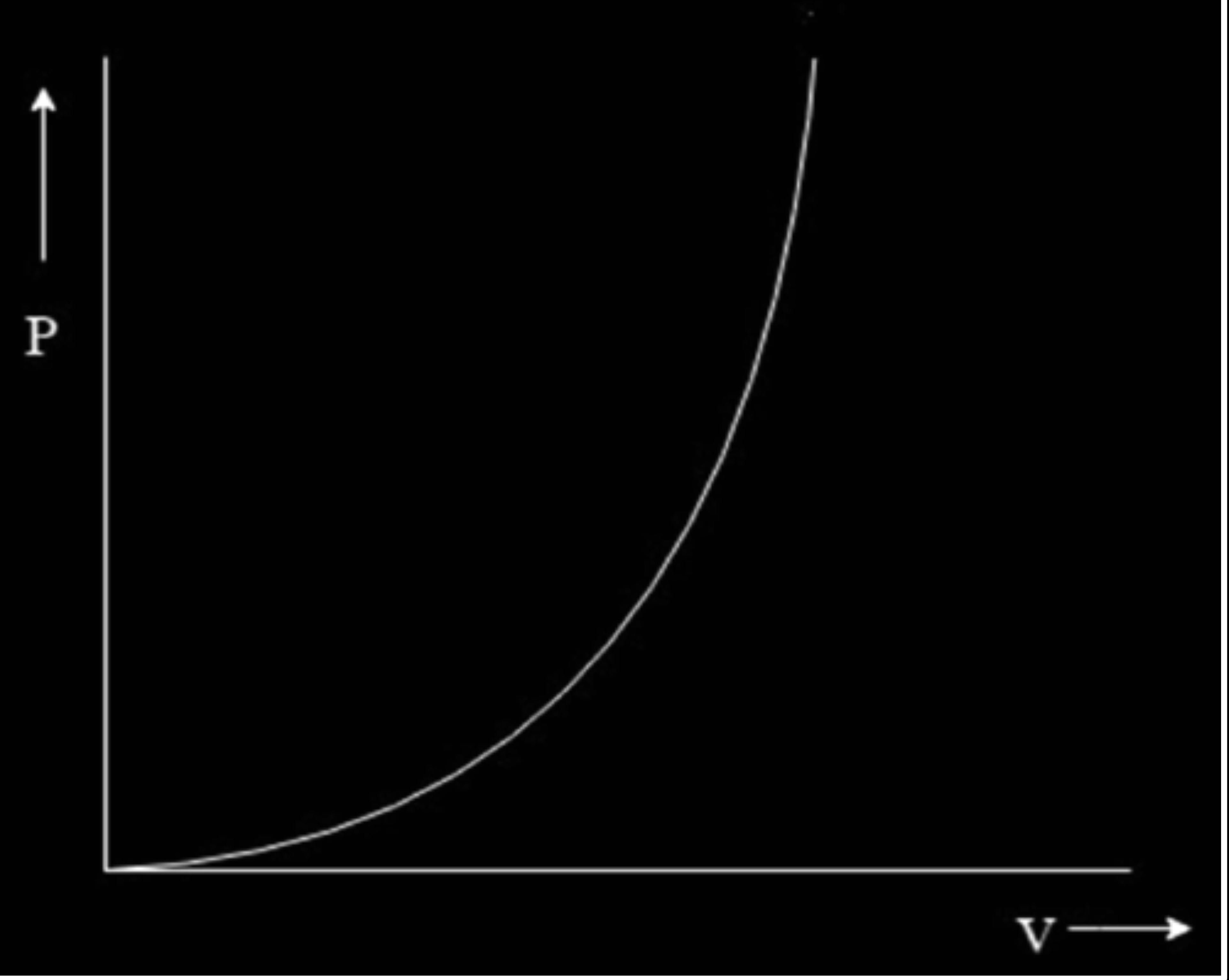
(a) $P = \frac{V^2}{R}$; keeping V constant



(b) $P = I^2 R$; keeping R constant



(c) $P = \frac{V^2}{R}$; keeping R constant



3. The current density due to drift of electrons in a conductor is given by (symbols have their usual meanings)

(a) $n\, e\, A\, v_d$

(b) $\dfrac{n\, A\, v_d}{e}$

(c) $\dfrac{n\, v_d}{e\, A}$

(d) $n\, e\, v_d$

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(c) $\frac{n\,v_d}{e\,A}$

(d) $n\,e\,v_d$

(d) nev_d

26. A potential difference V is applied across a conductor of length l and uniform cross-section area A. How will the (i) electric field E, (ii) drift velocity v_d, and (iii) current density j be affected when (a) V is doubled and (b) l is halved (keeping other factors constant) ? *3*

(a)	When V is doubled, effect on	
(i)	Electric field	½
(ii)	Drift velocity	½
(iii)	Current density	½
(b)	When l is halved, effect on	
(i)	Electric field	½
(ii)	Drift velocity	½
(iii)	Current density	½

(a) When V is doubled, (i) Electric field is doubled. (ii) Drift velocity is doubled. (iii) Current density is doubled (b) When l is halved, (i) Electric field is doubled. (ii) Drift velocity is doubled. (iii) Current density is doubled	 ½ ½ ½ ½ ½ ½	

Alternatively:

(i) $E=\dfrac{V}{l}$; $\quad \therefore E'=2E$

(ii) $v_d=\left(\dfrac{eE}{m}\right)\tau=\dfrac{e\tau}{m}\left(\dfrac{V}{l}\right)$

$v_d \,\alpha\, V$; $\quad v'_d=2v_d$

(iii) Current density $J=\dfrac{I}{A}=\dfrac{V}{RA}$

$\therefore$ J is also doubled.

When l is halved

(i) $E=\dfrac{V}{l}$; $\quad \therefore E'=2E$

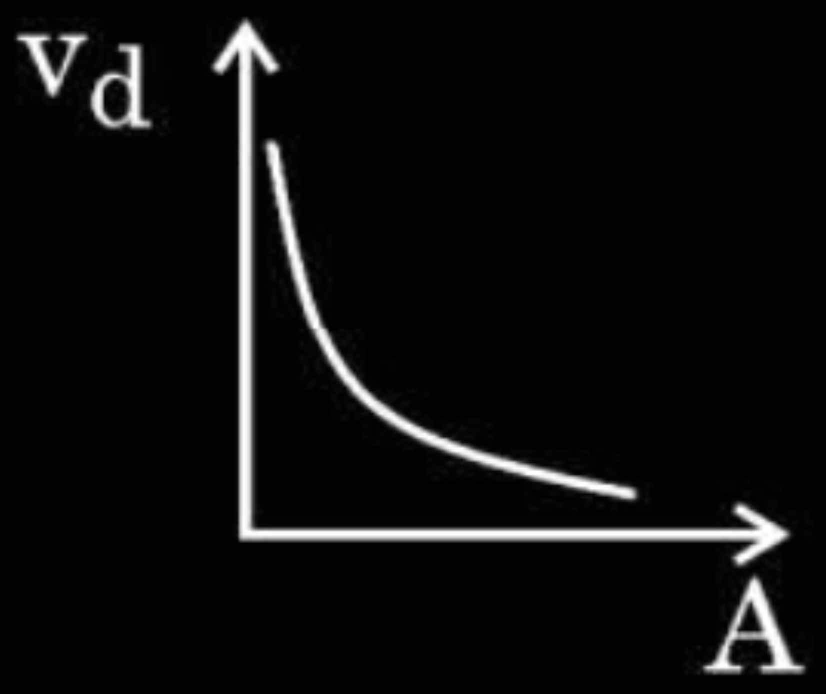
(ii) $v_d=\left(\dfrac{eE}{m}\right)\tau=\dfrac{e}{m}\left(\dfrac{V}{l}\right)\tau$

$\therefore v'_d=2v_d$

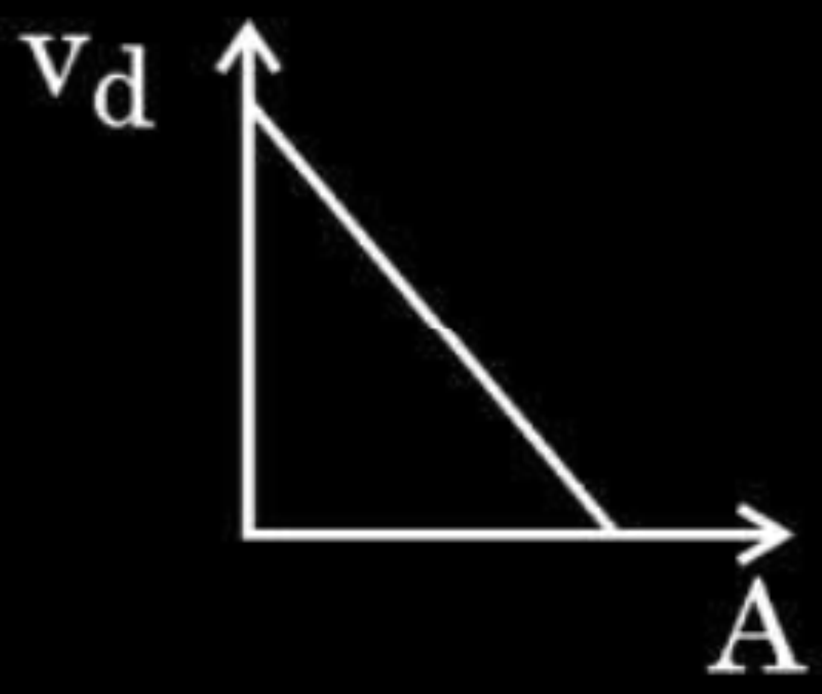
(iii) $J=\dfrac{I}{A}=\dfrac{V}{RA}=\dfrac{V}{A}\left(\dfrac{A}{\rho l}\right)=\dfrac{V}{\rho l}$

$\therefore$ J is also doubled.

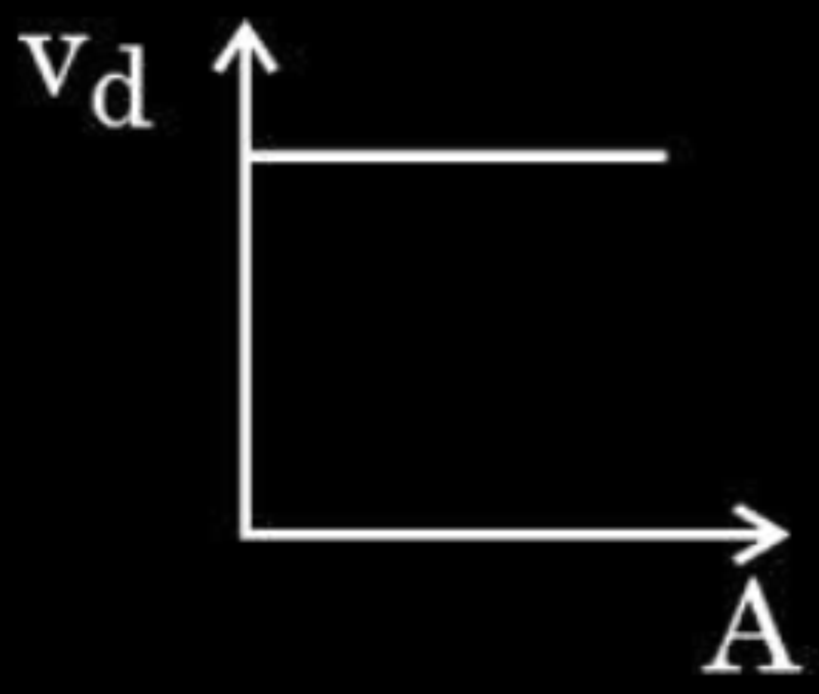
2. A steady current flows through a metallic wire whose area of cross-section (A) increases continuously from one end of the wire to the other. The magnitude of drift velocity (v_d) of the free electrons as a function of 'A' can be shown by :



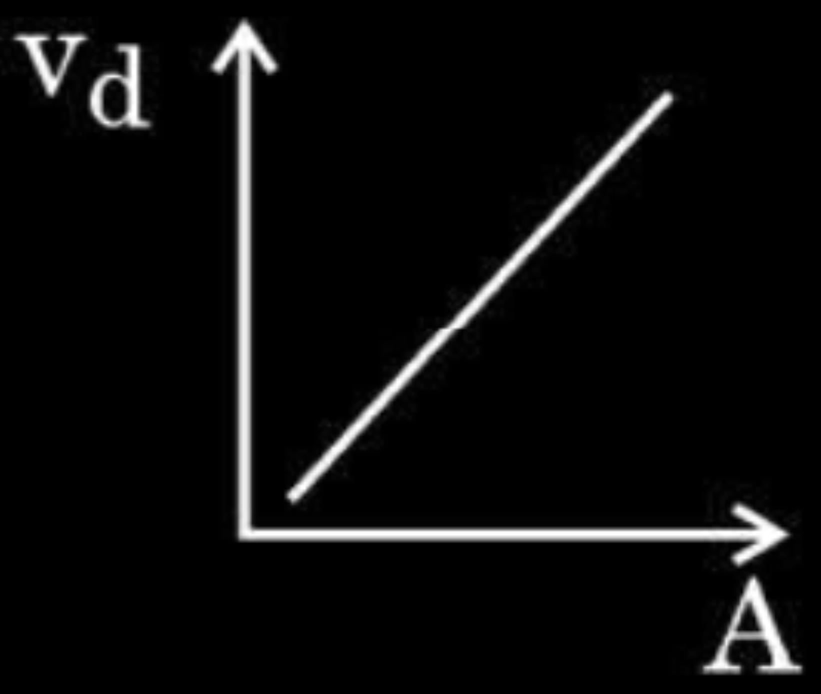
(a)



(b)



(c)



(d)

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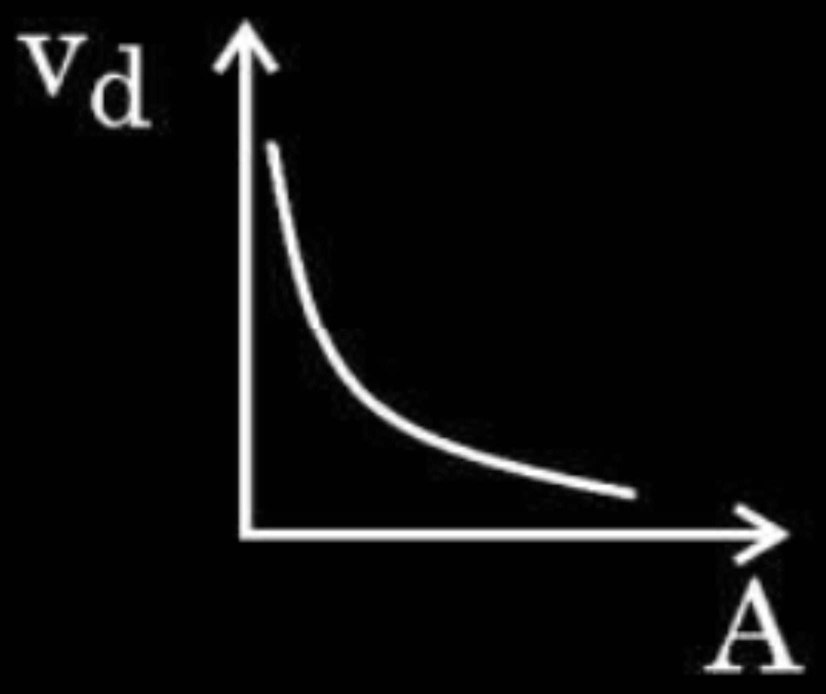
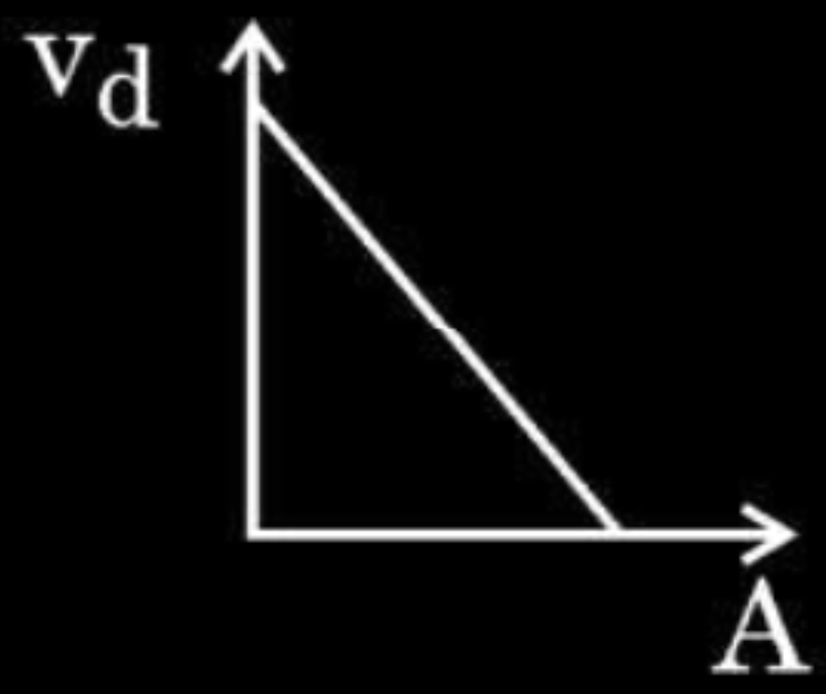


(a)



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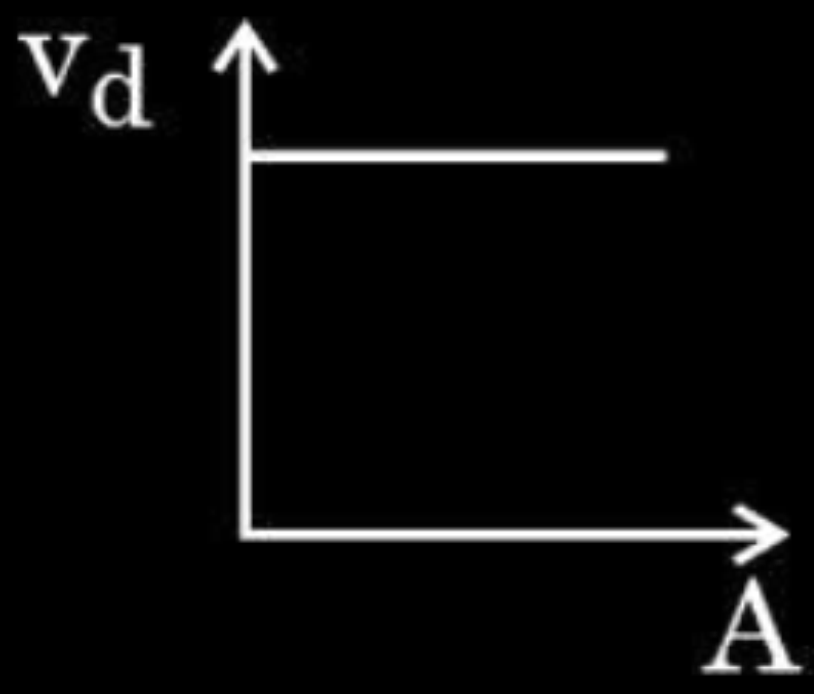

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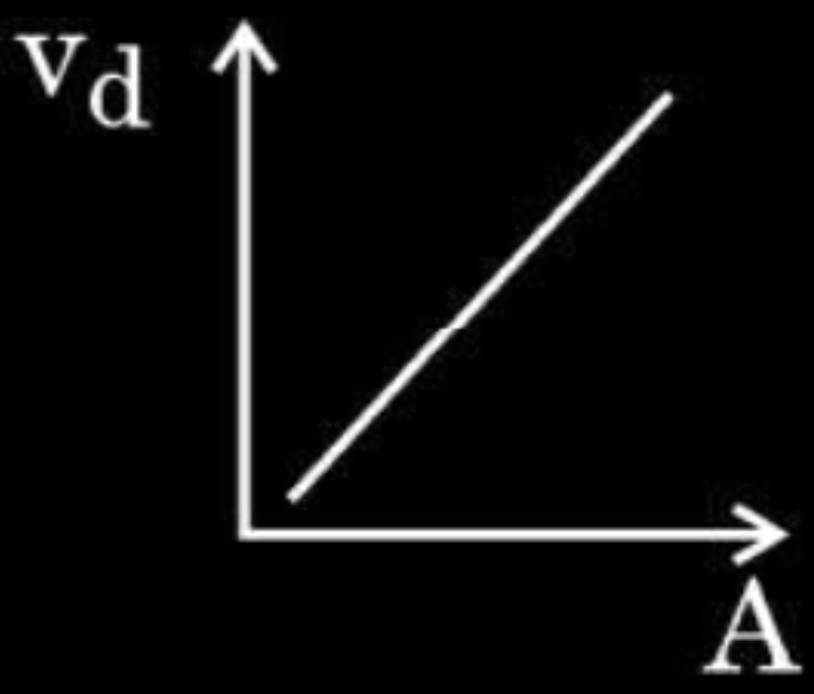
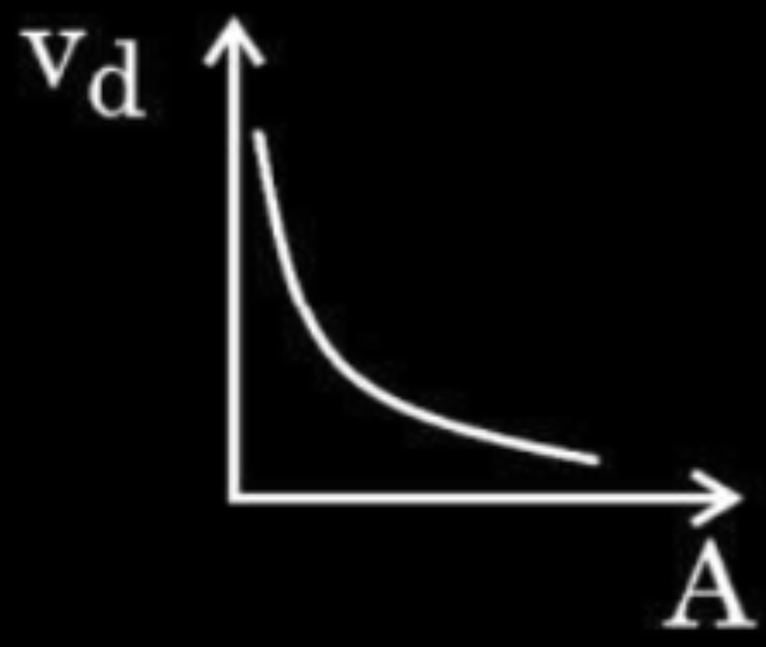
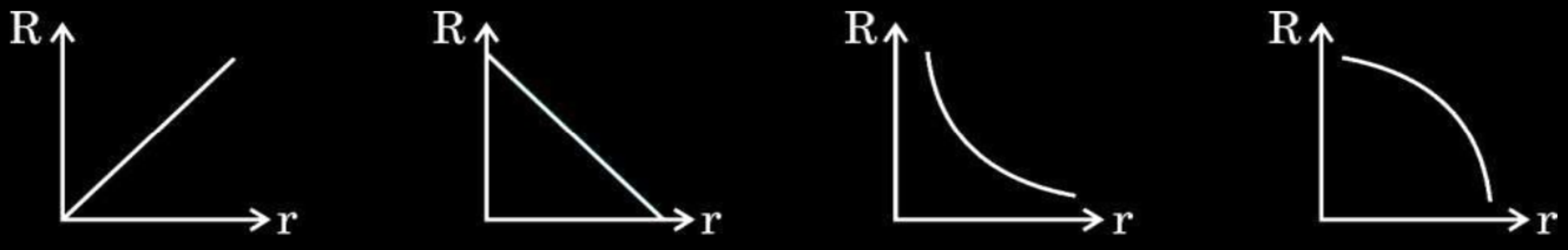


(d)



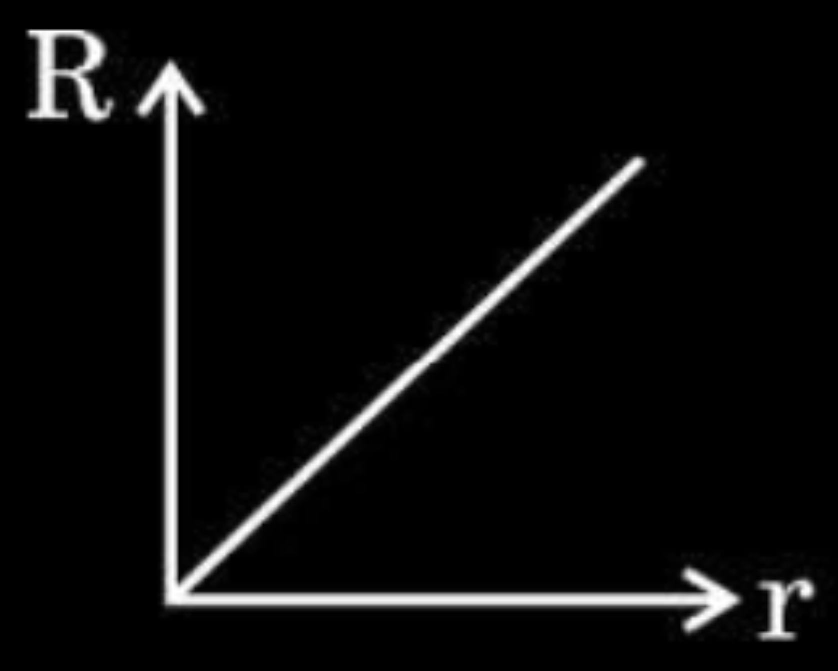
(a)

15. The correct graph showing the variation of the resistance (R) of a cylindrical metal wire as a function of its radius (r), keeping its length and temperature constant, is :

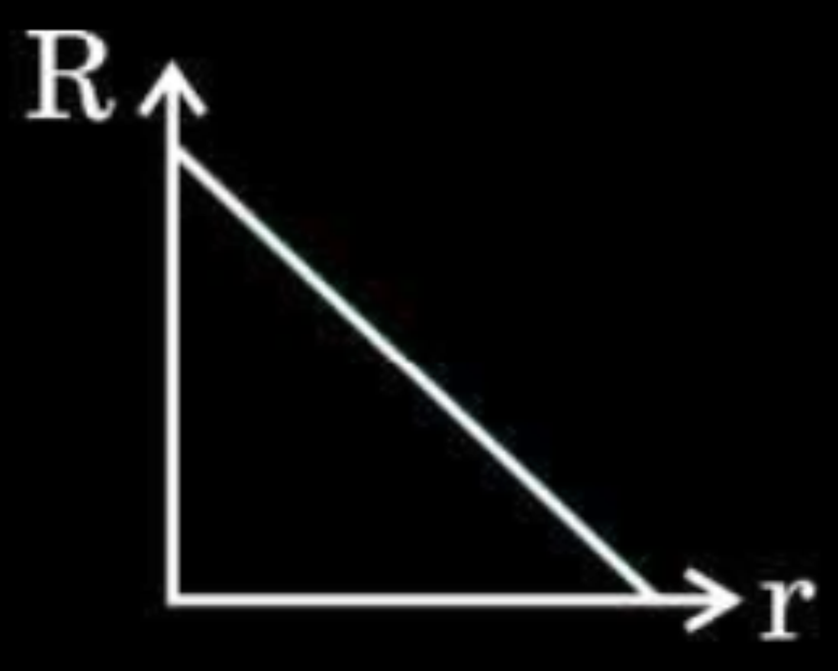


(a) (b) (c) (d)

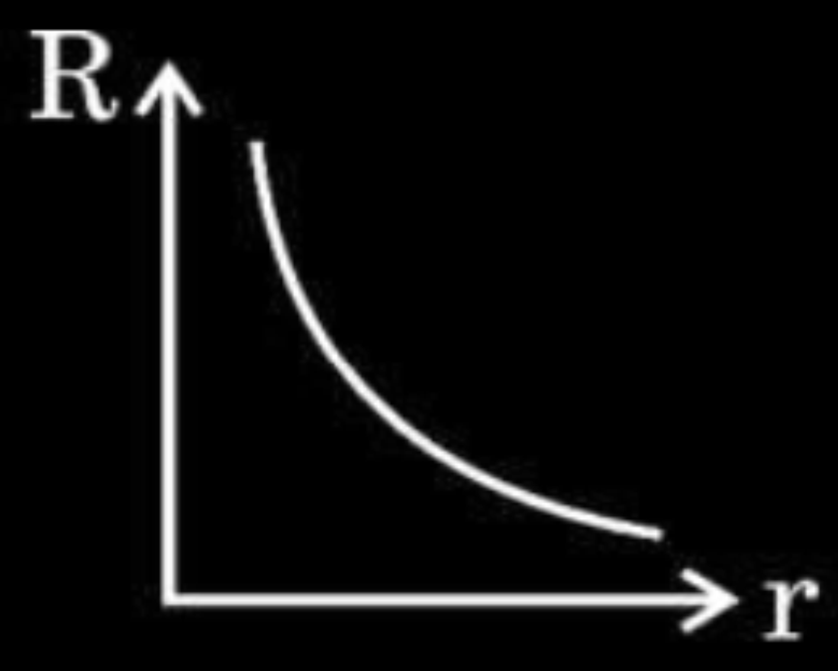
15. The correct graph showing the variation of the resistance (R) of a cylindrical metal wire as a function of its radius (r), keeping its length and temperature constant, is :



(a)



(b)



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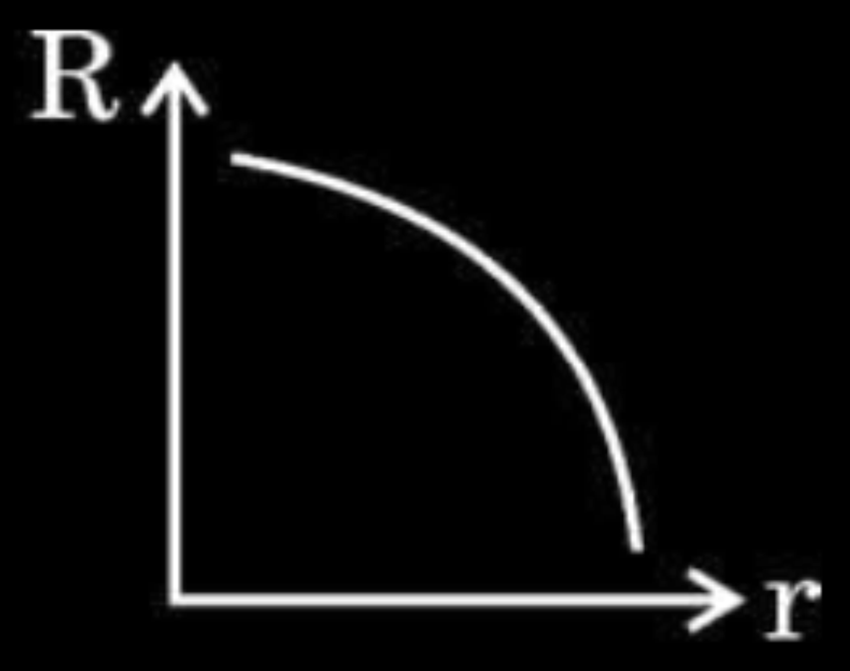

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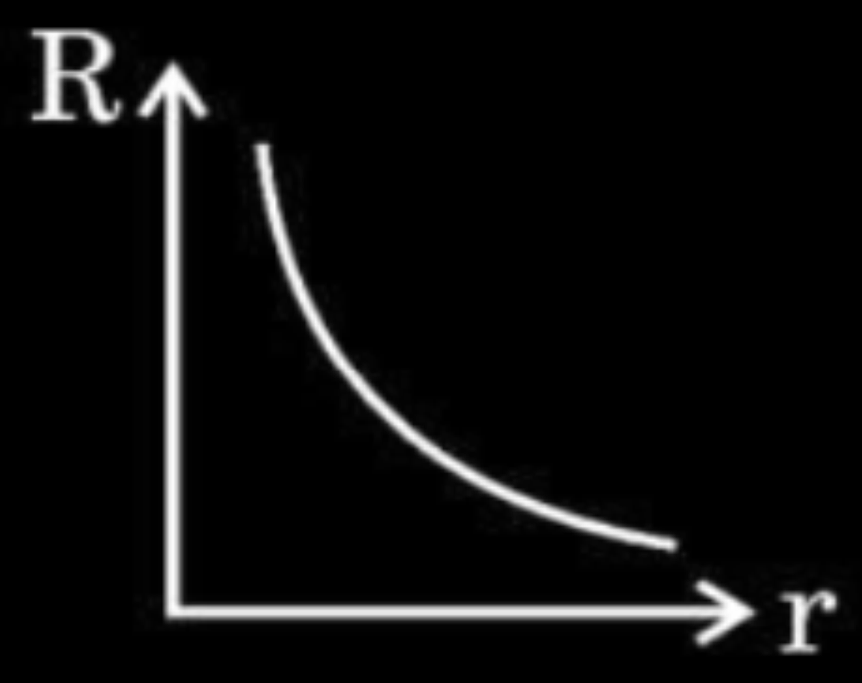
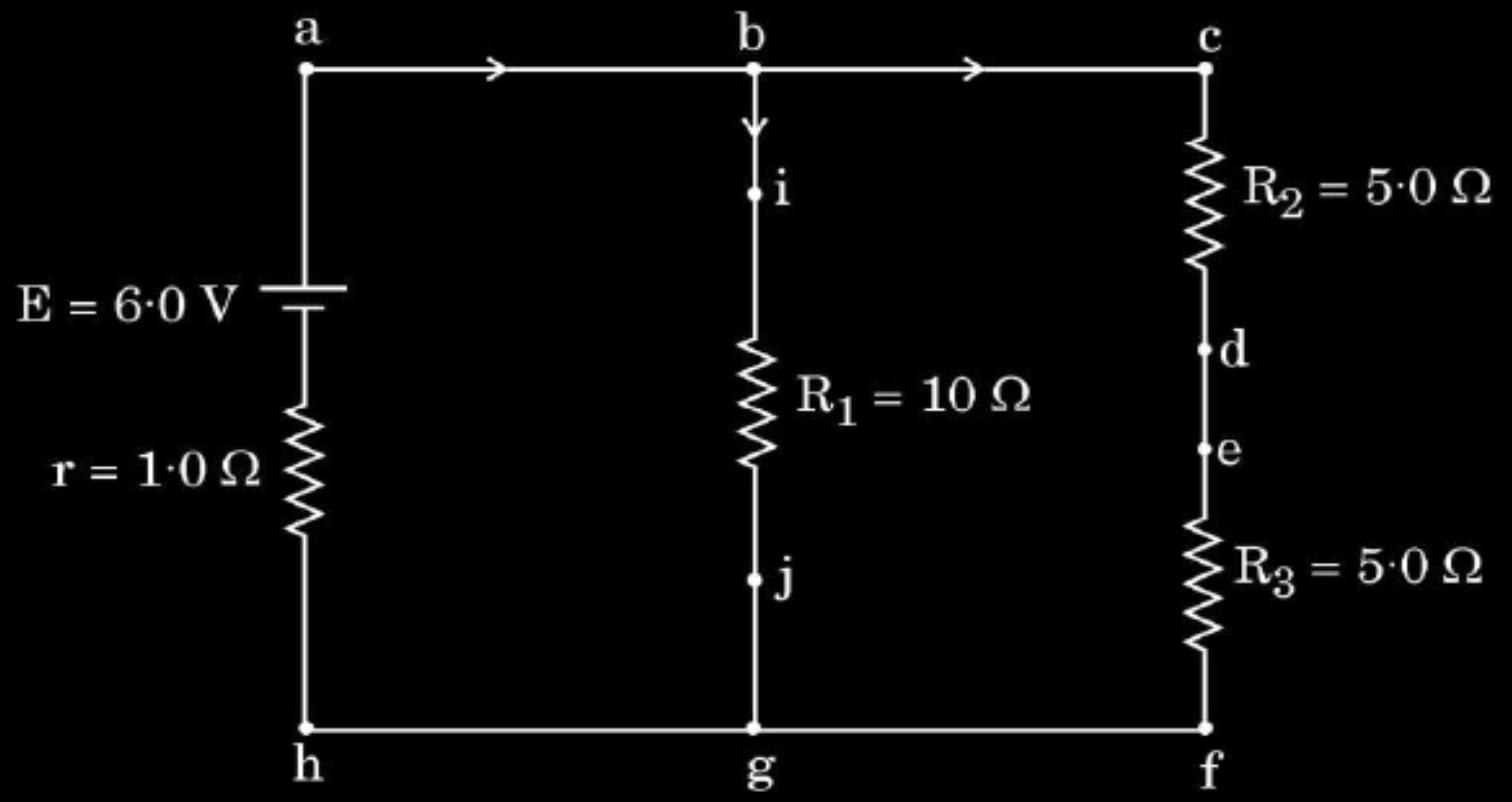


(c)

34. The following figure shows a circuit diagram. We can find the currents through and potential differences across different resistors using Kirchhoff's rules.



Answer the following questions based on the above :

(a) Which points are at the same potential in the circuit ? *1*

(b) What is the current through arm bg ? *1*

(c) Find the potential difference across resistance R_3. *2*

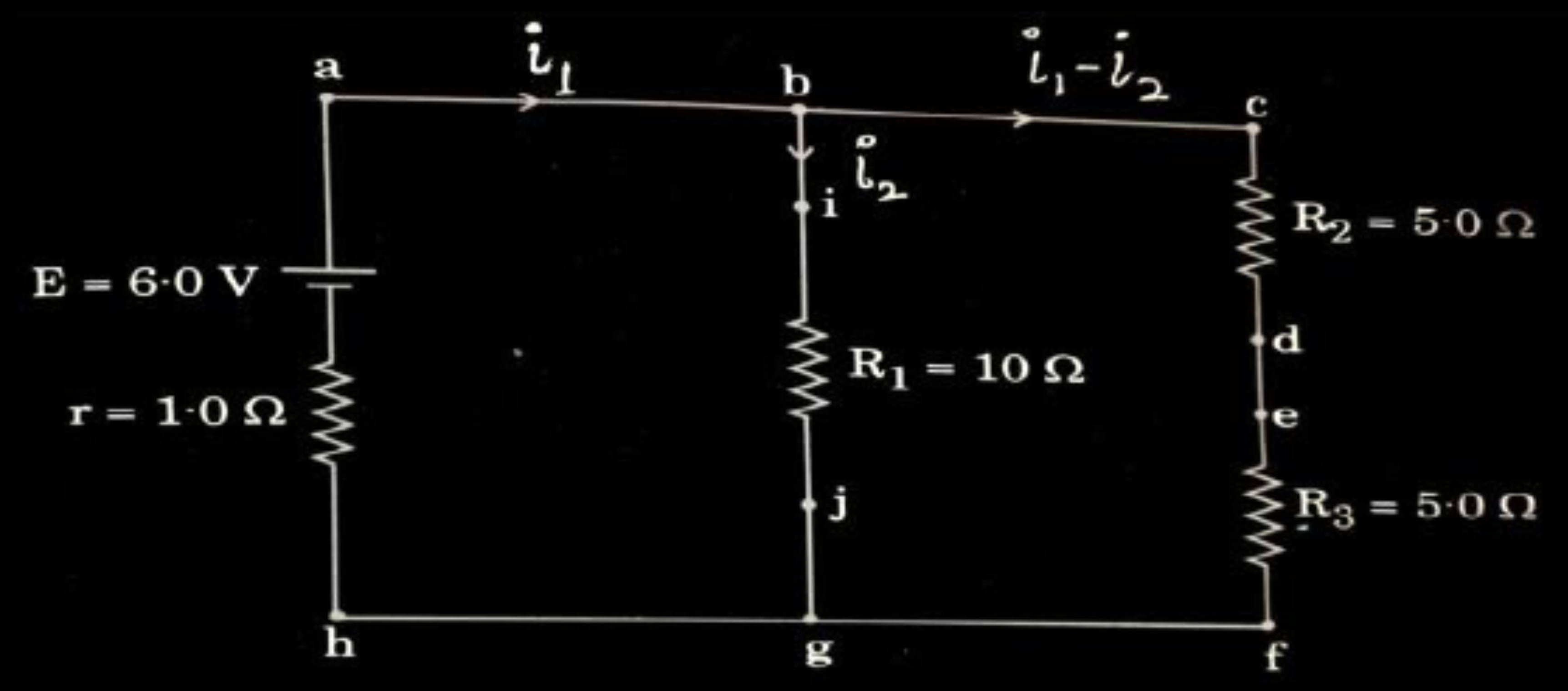
OR

(c) What is the power dissipated in resistance R_2 ? *2*

a) Points at same potential	1
b) Current through arm bg	1
c) Potential difference across R_3	
OR	
c) Power dissipated in R_2	2

a) Points (a, b, c) (d, e) (j, f, g ,h) are at same potential	1
Note: Give full credit if a student mentions any two points at same potential from the above.	

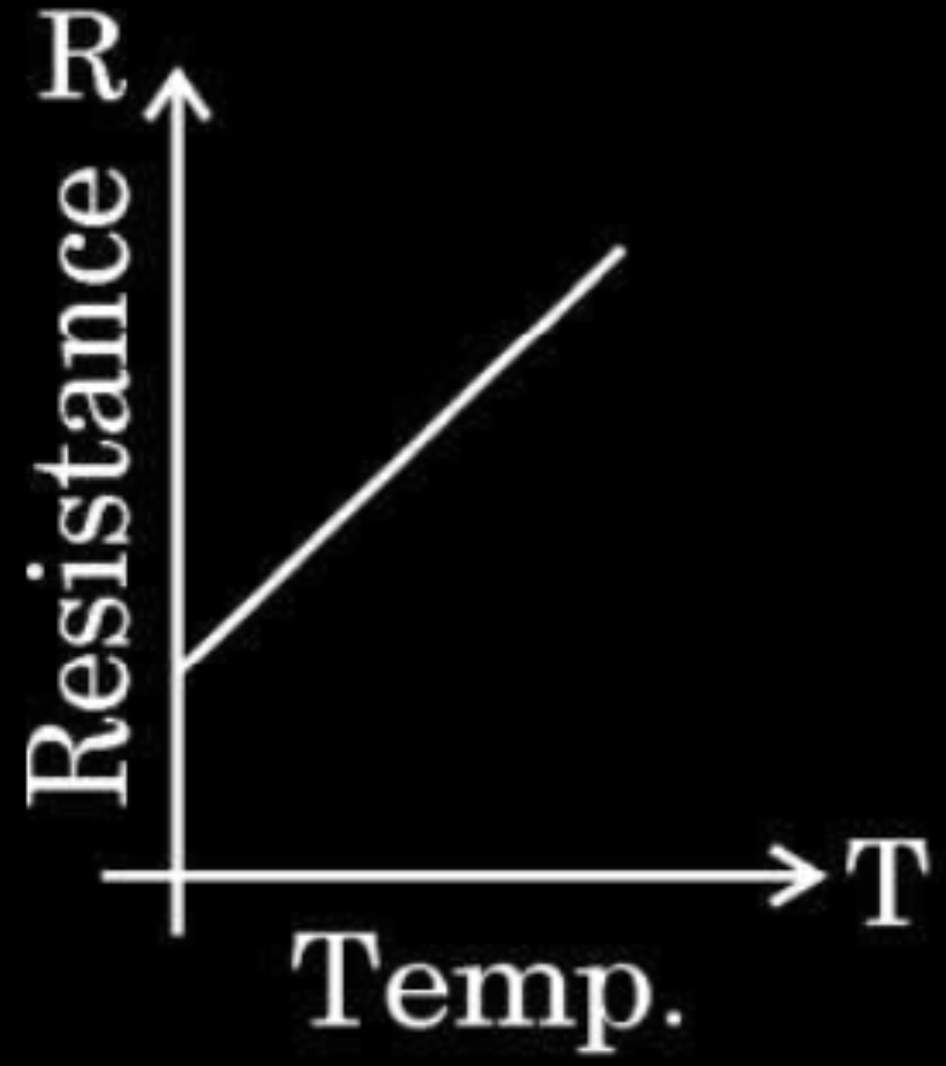
b)



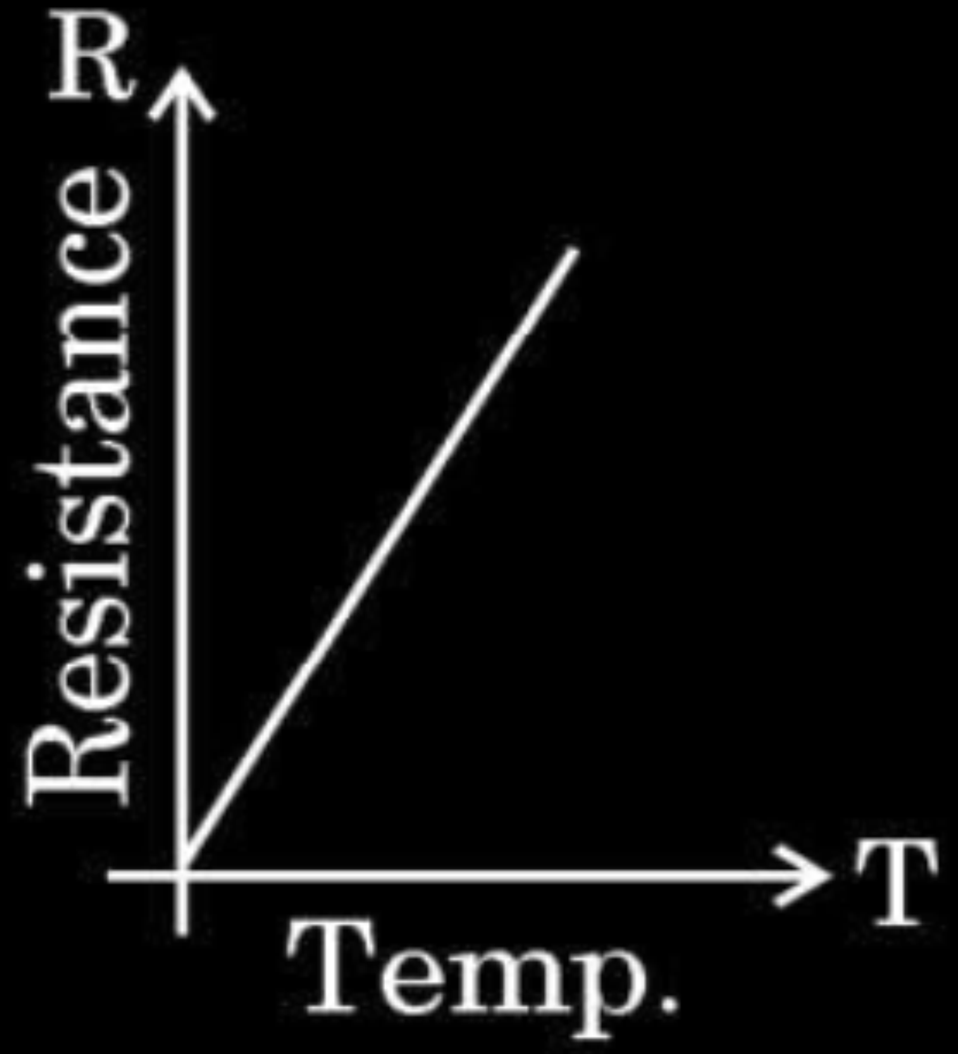
According to Kirchhoff's loop rule for closed loop abgha $-6 + 10\, I_2 + I_1 = 0$ $I_1 + 10\, I_2 = 6$ (i) for closed loop acfha $-6 + 10\,(I_1 - I_2) + I_1 = 0$ $11\, I_1 - 10\, I_2 = 6$ (ii)	½
Adding (i) and (ii) $12\, I_1 = 12$ $I_1 = 1$ A $I_2 = 0.5$ A = current through arm bg	½
Note: Award 1 mark if a student calculates the current by any other method.	

c) $V_{R3} = (I_1 - I_2) \times R_3$	1
$= 0.5 \times 5$ $= 2.5$ V	1
OR	
(c) $P = (I_1 - I_2)^2 \times R_2 = (0.5)^2 \times 5$	1
$= 1.25$ W	1

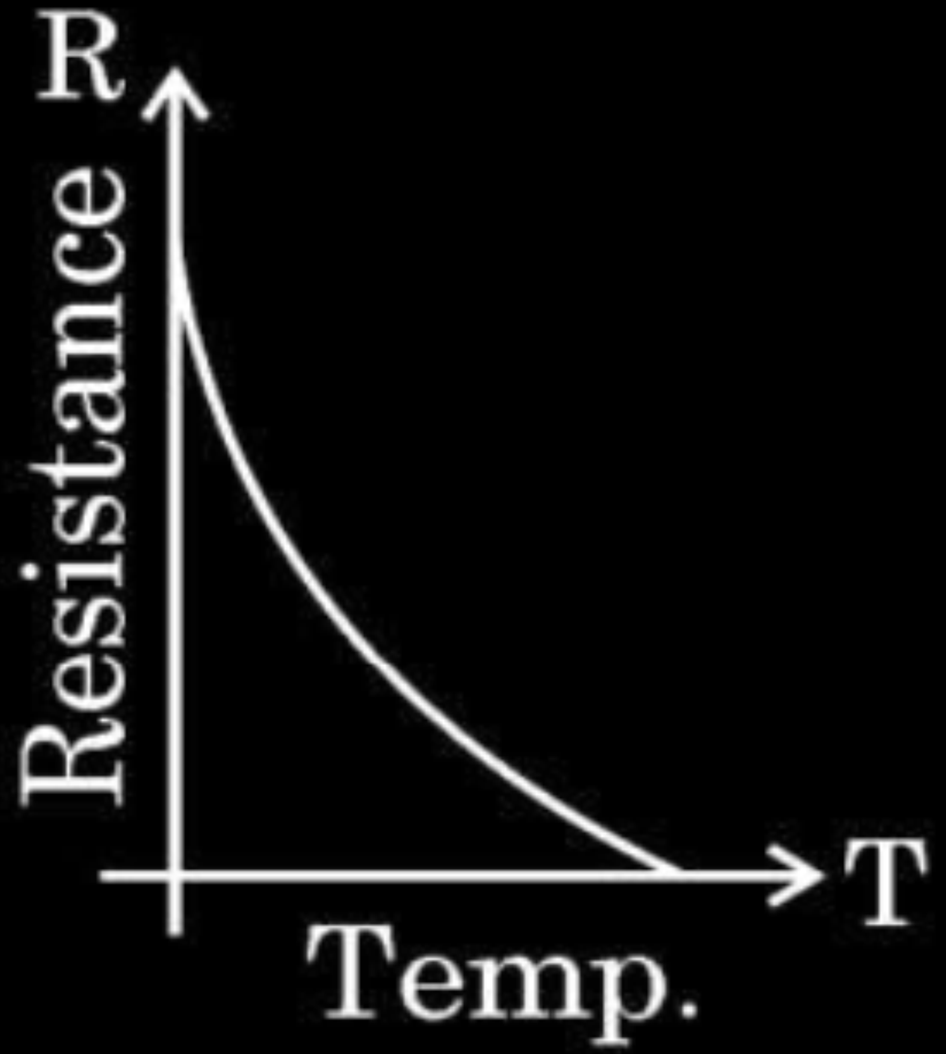
2. For a metallic conductor, the correct representation of variation of resistance R with temperature T is :



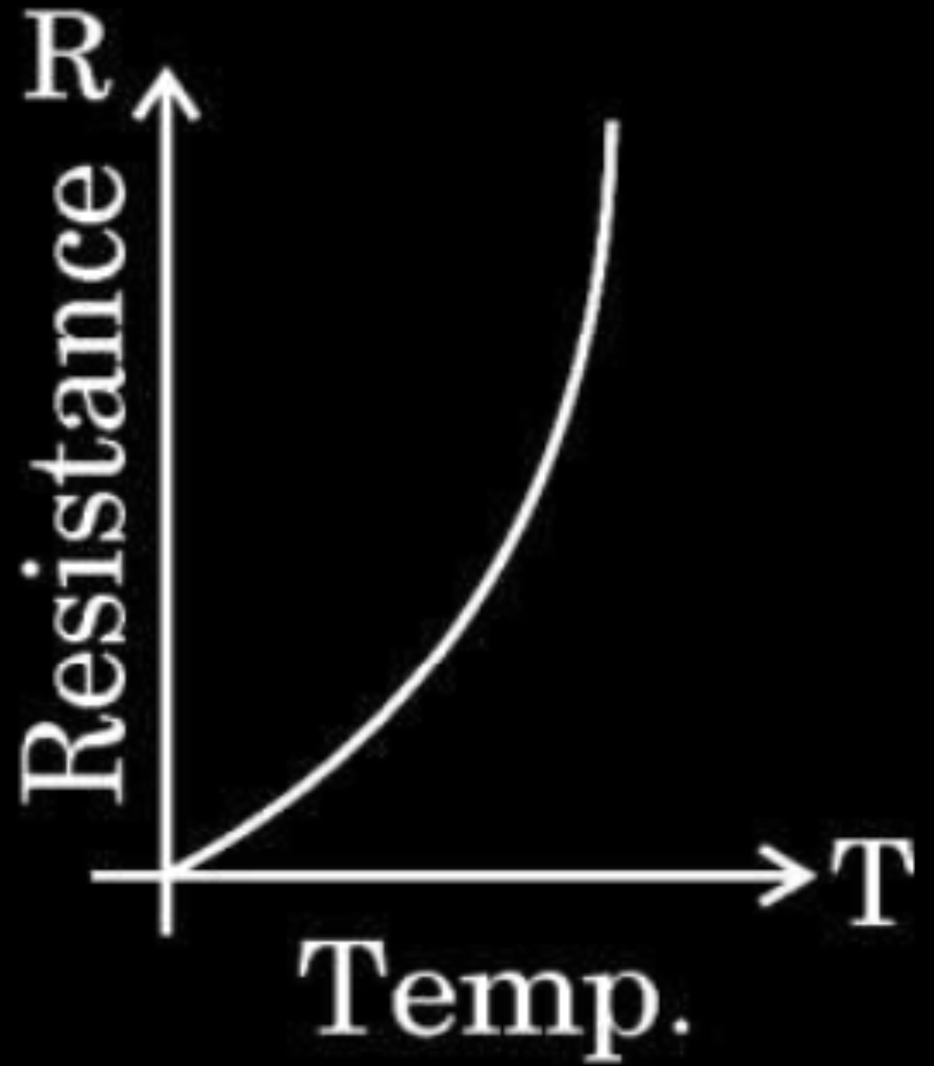
(a)



(b)

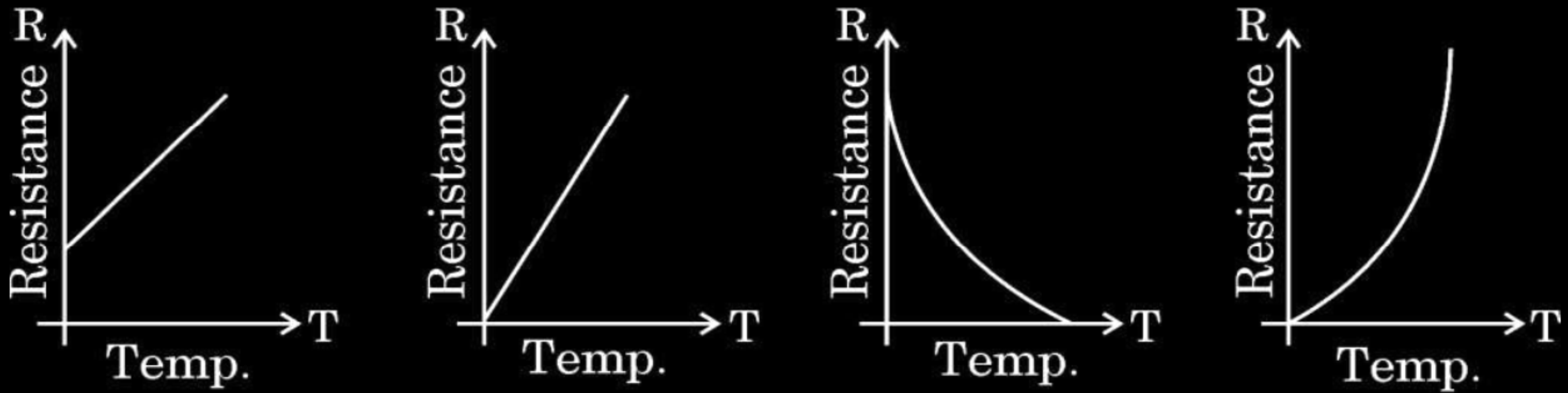


(c)

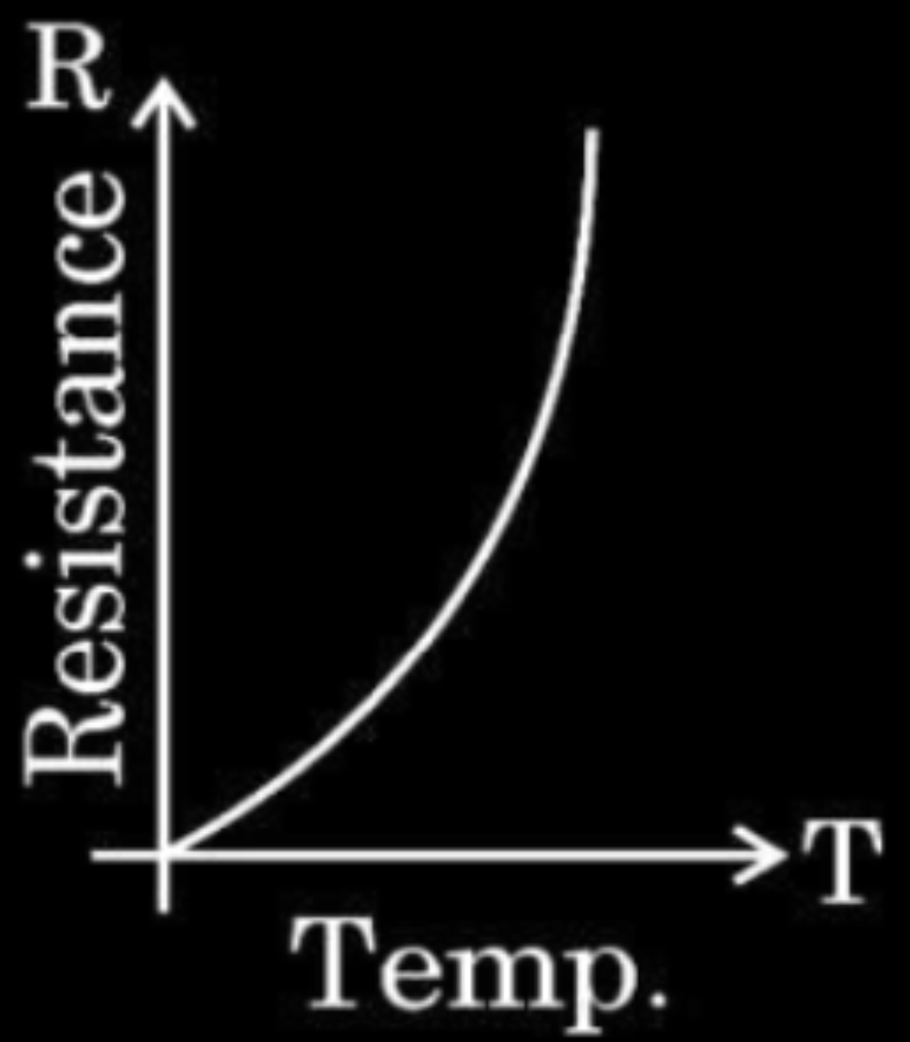


(d)

2. For a metallic conductor, the correct representation of variation of resistance R with temperature T is :



(a) (b) (c) (d)



(d)

11. The current in a device varies with time t as $I = 6\ t$, where I is in mA and t is in s. The amount of charge that passes through the device during $t = 0s$ to $t = 3s$ is **1**

(a) 10 mC (b) 18 mC

(c) 27 mC (d) 54 mC

11. The current in a device varies with time t as $I = 6\,t$, where I is in mA and t is in s. The amount of charge that passes through the device during $t = 0s$ to $t = 3s$ is 1

(a) 10 mC (b) 18 mC

(c) 27 mC (d) 54 mC

(c) 27 mC

13. A cell of emf E is connected across an external resistance R. When current 'I' is drawn from the cell, the potential difference across the electrodes of the cell drops to V. The internal resistance 'r' of the cell is **1**

(a) $\left(\frac{E-V}{E}\right)R$

(b) $\left(\frac{E-V}{R}\right)$

(c) $\frac{(E-V)\,R}{I}$

(d) $\left(\frac{E-V}{V}\right)R$

13. A cell of emf E is connected across an external resistance R. When current 'I' is drawn from the cell, the potential difference across the electrodes of the cell drops to V. The internal resistance 'r' of the cell is **1**

(a) $\left(\frac{E - V}{E}\right) R$

(b) $\left(\frac{E - V}{R}\right)$

(c) $\frac{(E - V)\, R}{I}$

(d) $\left(\frac{E - V}{V}\right) R$

(d) $\left(\frac{E - V}{V}\right) R$

29. A potential difference V is applied across a conductor of length l and cross-sectional area A. Briefly explain how the current density j in the conductor will be affected if

(a) the potential difference V is doubled,

(b) the conductor were gradually stretched to reduce its cross-sectional area to $\frac{A}{2}$ and then the same potential difference V is applied across it. **3**

(a) Effect of potential difference on current density	-	1 ½
(b) Effect of dimensions of conductor on current density	-	1 ½

(a) Current density j = $\dfrac{ne^2E\tau}{m}$ $= \dfrac{ne^2V\tau}{ml}$ Or $j \propto V$	1
If V is doubled , the current density will be doubled.	½
(b) As the conductor were gradually stretched to reduce its cross sectional area to half $(A/2)$, its length will become double $(2l)$ as volume of conductor remains the same.	½
$\because \; j \; \alpha \; \dfrac{1}{l}$	½
$\therefore$ If same potential difference is applied , the current density will become half.	½

7. A steady current of 8 mA flows through a wire. The number of electrons passing through a cross-section of the wire in 10 s is **1**

(a) 4.0×10^{16} (b) 5.0×10^{17}

(c) 1.6×10^{16} (d) 1.0×10^{17}

7. A steady current of 8 mA flows through a wire. The number of electrons passing through a cross-section of the wire in 10 s is **1**

(a) 4.0×10^{16} (b) 5.0×10^{17}

(c) 1.6×10^{16} (d) 1.0×10^{17}

(b) 5.0×10^{17}

9. A conductor of 10 Ω is connected across a 6 V ideal source. The power supplied by the source to the conductor is **1**

(a) 1.8 W (b) 2.4 W

(c) 3.6 W (d) 7.2 W

9. A conductor of 10 Ω is connected across a 6 V ideal source. The power supplied by the source to the conductor is **1**

(a) 1.8 W (b) 2.4 W

(c) 3.6 W (d) 7.2 W

(c) 3.6 W

29. Two cells of emf E_1 and E_2 and internal resistances r_1 and r_2 are connected in parallel, with their terminals of the same polarity connected together. Obtain an expression for the equivalent emf of the combination. **3**

Obtaining expression for equivalent emf of two cells in parallel	-	3

½

$$V \equiv V(B_1) - V(B_2) = \varepsilon_1 - I_1 r_1$$ ½

$$V \equiv V(B_1) - V(B_2) = \varepsilon_2 - I_2 r_2$$ ½

$$I = I_1 + I_2$$ ½

$$I = \frac{\varepsilon_1 - V}{r_1} + \frac{\varepsilon_2 - V}{r_2}$$

$$I = \left(\frac{\varepsilon_1}{r_1} + \frac{\varepsilon_2}{r_2}\right) - V\left(\frac{1}{r_1} + \frac{1}{r_2}\right)$$

$$V = \frac{\varepsilon_1 r_2 + \varepsilon_2 r_1}{r_1 + r_2} - I\,\frac{r_1 . r_2}{r_1 + r_2}$$ ½

$$\because \quad V = \varepsilon_{eq} - I r_{eq}$$

$$\varepsilon_{eq} = \frac{\varepsilon_1 r_2 + \varepsilon_2 r_1}{r_1 + r_2}$$ ½

3. A current of 0.8 A flows in a conductor of 40 Ω for 1 minute. The heat produced in the conductor will be **1**

(a) 1445 J (b) 1536 J

(c) 1569 J (d) 1640 J

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(a) 1445 J (b) 1536 J

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(b) 1536 J

14. Pieces of copper and of silicon are initially at room temperature. Both are heated to temperature T. The conductivity of **1**

(a) both increases.

(b) both decreases.

(c) copper increases and silicon decreases.

(d) copper decreases and silicon increases.

14. Pieces of copper and of silicon are initially at room temperature. Both are heated to temperature T. The conductivity of **1**

(a) both increases.

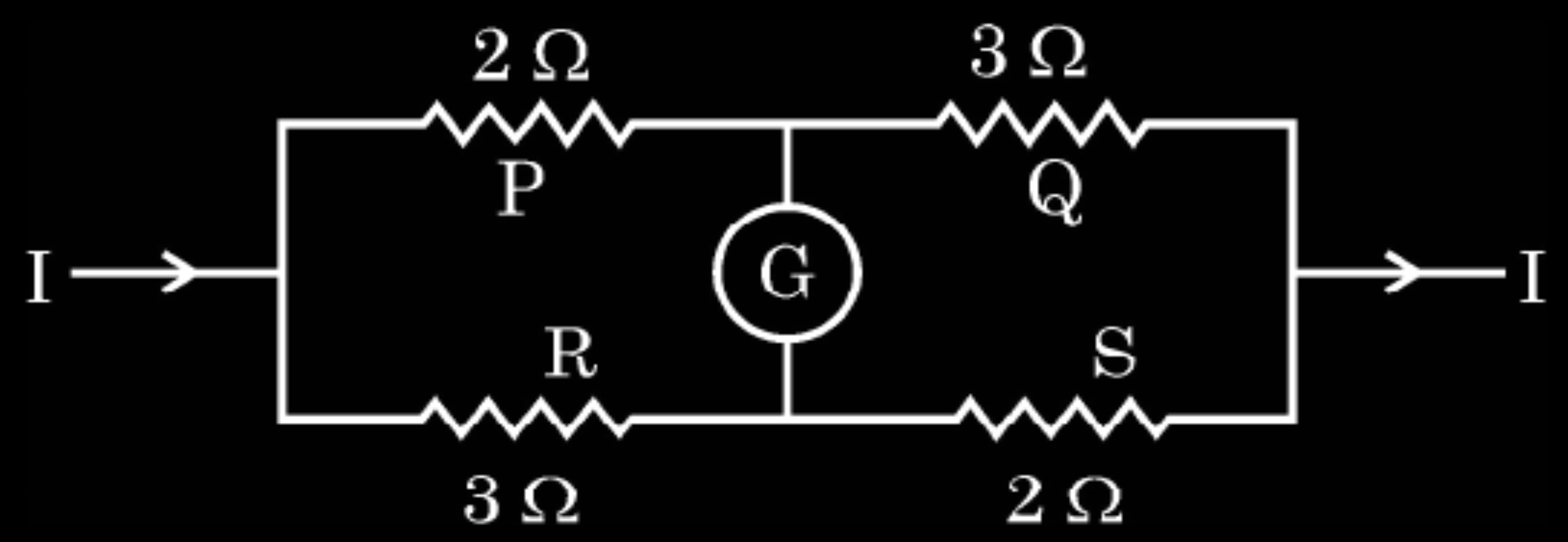
(b) both decreases.

(c) copper increases and silicon decreases.

(d) copper decreases and silicon increases.

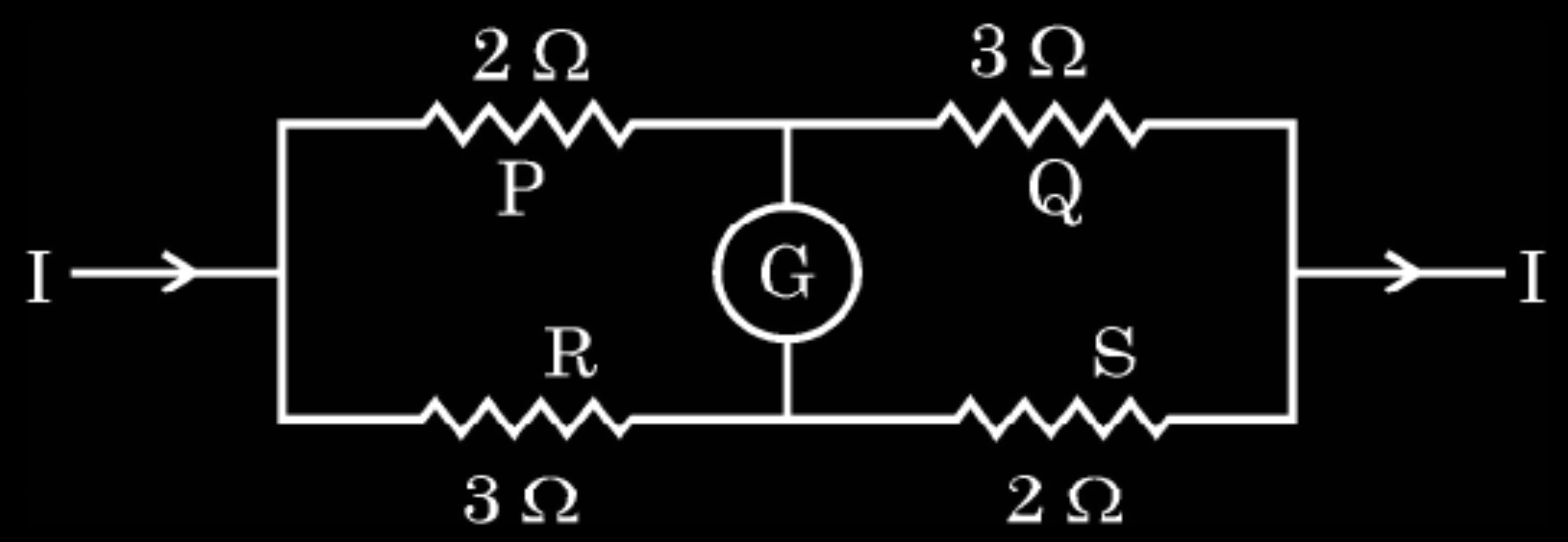
(d)
Copper decreases and silicon increases

17. **Assertion (A)** : The given figure does not show a balanced Wheatstone bridge.



Reason (R) : For a balanced bridge small current should flow through the galvanometer.

17. **Assertion (A)** : The given figure does not show a balanced Wheatstone bridge.



Reason (R) : For a balanced bridge small current should flow through the galvanometer.

(a)
Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion(A).

27. Define current density and relaxation time. Derive an expression for resistivity of a conductor in terms of number density of charge carriers in the conductor and relaxation time. **3**

Definition of		
Current density	-	½
Relaxation time	-	½
Derivation for resistivity of a conductor	-	2

Current density is defined as the current flowing per unit area of cross section of a conductor.	½
Alternatively Give full credit if a candidate writes j=I/A in place of definition	
Relaxation time is the average time interval between two successive collisions for drifting electrons in a conductor.	½
From I= $nAev_d$	½
but $v_d = \frac{eE}{m}\tau$	½
$\therefore \quad I = nAe.\frac{eE}{m}\tau$	
$j = \frac{I}{A} = \frac{ne^2E\tau}{m}$	
But $j = \frac{E}{\rho}$	½
$\therefore \quad \rho = \frac{m}{ne^2\tau}$	

24. Two conductors, made of the same material have equal lengths but different cross-sectional areas A_1 and A_2 ($A_1 > A_2$). They are connected in parallel across a cell. Show that the drift velocities of electrons in two conductors are equal. **2**

Showing the equal drift velocity in the two conductors (2)

24. Two conductors, made of the same material have equal lengths but different cross-sectional areas A_1 and A_2 ($A_1 > A_2$). They are connected in parallel across a cell. Show that the drift velocities of electrons in two conductors are equal. **2**

$I = neAVd$	½
$v_d = \frac{I}{neA} = \frac{VA}{neA\rho l}$	½
$v_d = \frac{V}{\rho lne}$	½
In parallel drift velocity is independent of area cross section	
So drift velocities are same in both the conductors	½

Showing the equal drift velocity in the two conductors (2)

33. (a) (i) Explain how free electrons in a metal at constant temperature attain an average velocity under the action of an electric field. Hence obtain an expression for it. **3**

(ii) Consider two conducting wires A and B of the same diameter but made of different materials joined in series across a battery. The number density of electrons in A is 1.5 times that in B. Find the ratio of drift velocity of electrons in wire A to that in wire B. **2**

(i) Explanation	(1)
Obtaining expression for average velocity	(2)
(ii) Finding ratio of drift velocities	(2)

(a) (i) Under the effect of external field, an electron experiences a force $\vec{F} = -e\vec{E}$ between collisions. Due to this force the electron is accelerated and attains a velocity. This velocity is different for different electrons, which averaged over all electrons gives average drift velocity. This drift velocity is constant for a given temperature.	1
Expression of average velocity:	
Under the action of an electric field electrons get accelerated with $a = -\frac{eE}{m}$	½
Velocity of an electron at any instant of time is $\vec{V}_i = \vec{v}_i - \frac{e\vec{E}}{m} t_i$	½
Average velocity of the electrons at time 't' is the drift velocity $\vec{v}_d = (\vec{V}_i)_{average}$ $(\vec{V}_i)_{average} = (\vec{v}_i)_{average} - \frac{e\vec{E}}{m}(t_i)_{average}$	½
But $(\vec{v}_i)_{average} = 0$ due to randomness $\vec{v}_d = 0 - \frac{e\vec{E}}{m}\tau$ $\vec{v}_d = -\frac{e\vec{E}}{m}\tau$	½

(ii) $v_d = \frac{I}{enA} = \left(\frac{4I}{e\pi D^2}\right) \times \frac{1}{n}$	½
$v_d \propto \frac{1}{n}$ for same diameter and current	½
$n_A = 1.5 n_B$ (Given)	
$\frac{v_{dA}}{v_{dB}} = \frac{n_B}{n_A} = \frac{2}{3}$	1

OR

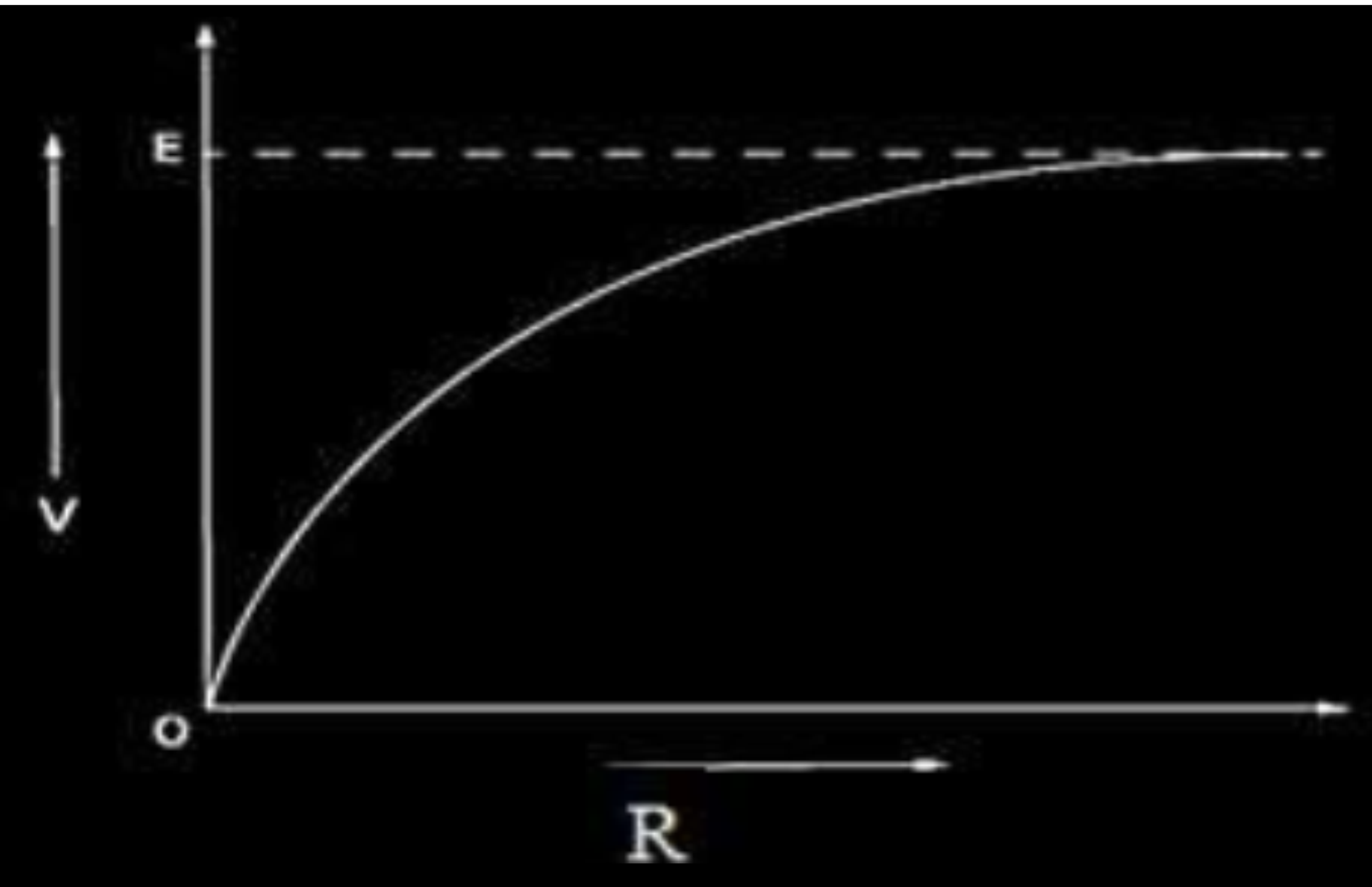
(b) (i) A cell emf of (E) and internal resistance (r) is connected across a variable load resistance (R). Draw plots showing the variation of terminal voltage V with (i) R and (ii) the current (I) in the load. **2**

(ii) Three cells, each of emf E but internal resistances 2r, 3r and 6r are connected in parallel across a resistor R.

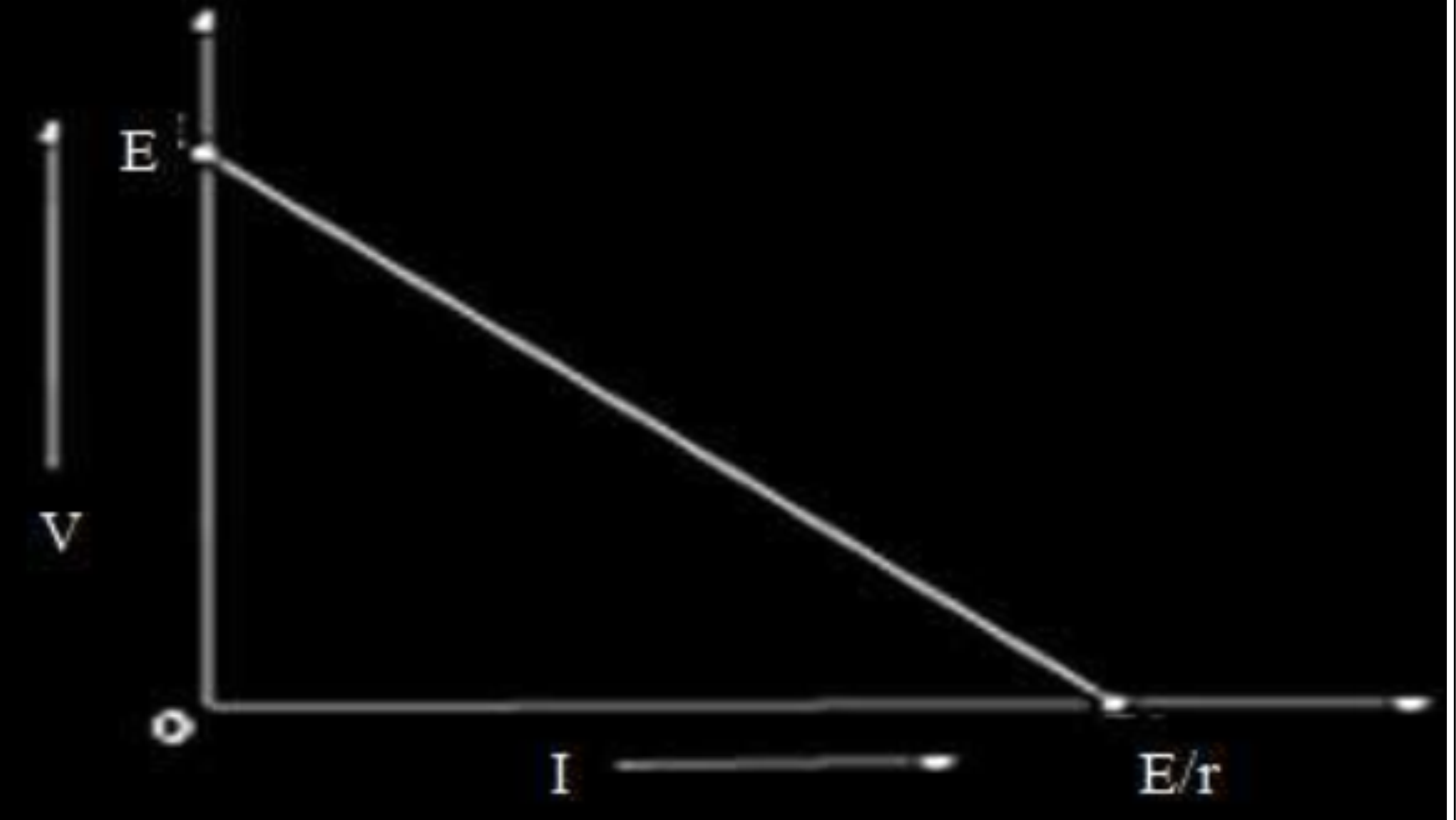
Obtain expressions for (i) current flowing in the circuit, and (ii) the terminal potential difference across the equivalent cell. **3**

(i)	
i Plot showing variation of V with R	(1)
ii Plot showing variation of V with I	(1)
(ii)	
(i) Obtaining expression for current flowing through circuit	$(1\frac{1}{2})$
(ii) Obtaining expression for terminal potential difference across the equivalent cell.	$(1\frac{1}{2})$

(i) i.



ii.



(ii)

$$\frac{1}{r_{eq}} = \frac{1}{r_1} + \frac{1}{r_2} + \frac{1}{r_3}$$

$$\frac{1}{r_{eq}} = \frac{1}{2r} + \frac{1}{3r} + \frac{1}{6r}$$

$$r_{eq} = r$$ ½

Given cells are of equal emf (E) and connected in parallel, so

$$\frac{E_{eq}}{r_{eq}} = \frac{E}{2r} + \frac{E}{3r} + \frac{E}{6r}$$

$$E_{eq} = E$$ ½

Current flowing $I = \dfrac{E_{eq}}{R + r_{eq}}$

$$I = \frac{E}{R + r}$$ ½

The terminal potential difference

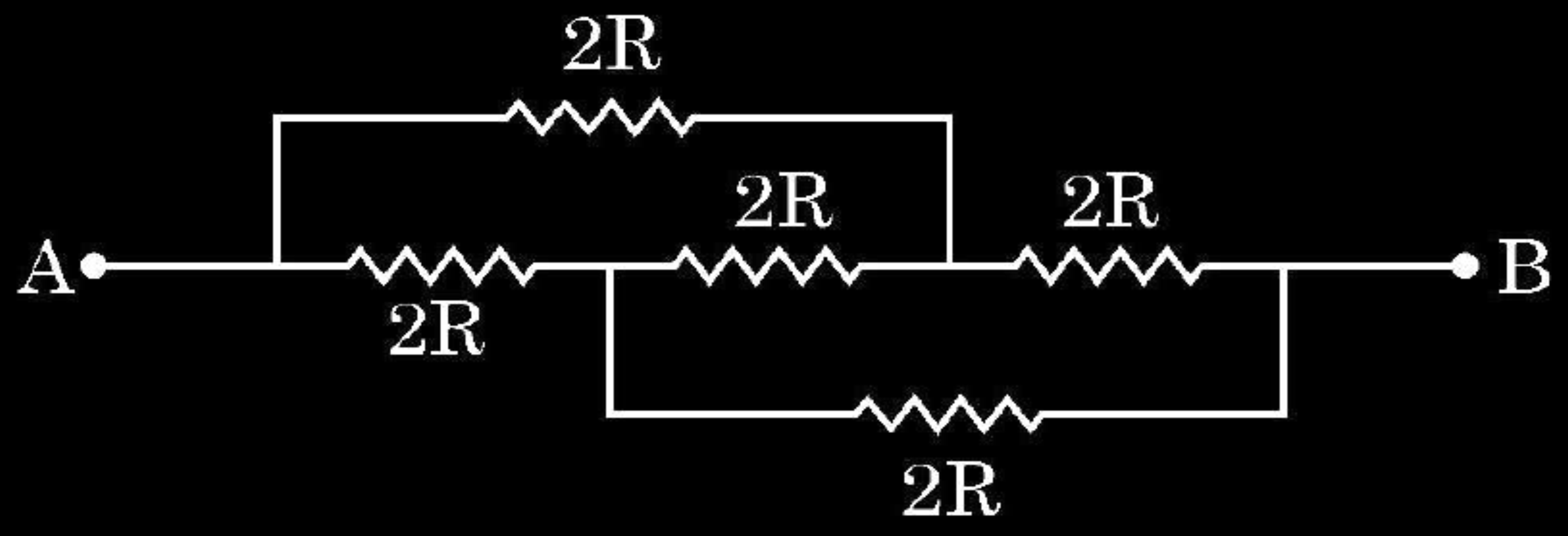
$$V = E_{eq} - Ir_{eq}$$ ½

$$V = E - \frac{E}{(R + r)} \times r$$ ½

$$V = \frac{ER}{R + r}$$ ½

17. **Assertion (A)** : The equivalent resistance between points A and B in the given network is 2R. **1**

Reason (R) : All the resistors are connected in parallel



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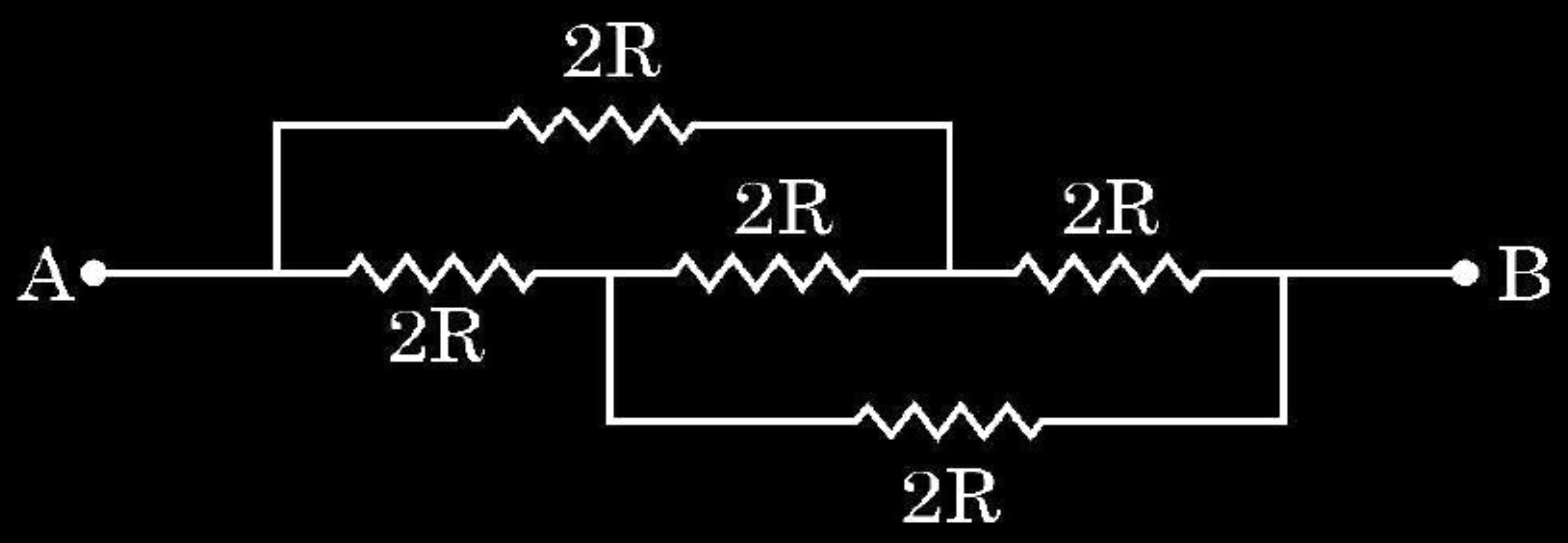


(C) Assertion (A) is true and reason (R) is false

24. A potential difference (V) is applied across a conductor of length 'L' and cross-sectional area 'A'.

How will the drift velocity of electrons and the current density be affected if another identical conductor of the same material were connected in series with the first conductor ? Justify your answers. **2**

Effect on drift velocity and current density	(½ + ½)
Justification	(½ + ½)

Drift velocity of electrons is halved $$v_d = \frac{V}{neAR}$$	½
When another identical conductor of same material were connected in series with the given conductor then total resistance becomes double. Therefore $$v_d' = \frac{v_d}{2}$$	½
Current density is halved	½
$$J = \frac{I}{A} = \frac{V}{RA}$$ $$J' = \frac{J}{2}$$	½

24. The potential difference applied across a given conductor is doubled. How will this affect (i) the mobility of electrons and (ii) the current density in the conductor ? Justify your answers. **2**

(i)	Effect of potential difference on mobility of electron	(½)
	Justification	(½)
(i)	Effect of potential difference on current density	(½)
	Justification	(½)

24. The potential difference applied across a given conductor is doubled. How will this affect (i) the mobility of electrons and (ii) the current density in the conductor ? Justify your answers. **2**

(i) No effect	½
$\mu = \frac{v_d l}{V} = \frac{l}{enAR} = \text{Constant}$	½
Alternatively: $\mu = \frac{v_d}{E} = \frac{e\tau}{m} = Constant$	
(ii) As $J \alpha V$, current density gets doubled	½
$J = \frac{I}{A} = \frac{V}{RA}$	½

(i)	Effect of potential difference on mobility of electron	(½)
	Justification	(½)
(i)	Effect of potential difference on current density	(½)
	Justification	(½)

3. The amount of charge flowing across a cross-section of a conductor, connected to a battery, in $4{\cdot}0$ s is 720 mC. The current in the circuit is :

(a) $0{\cdot}18$ A (b) $1{\cdot}8$ A

(c) 18 mA (d) 18 A

3. The amount of charge flowing across a cross-section of a conductor, connected to a battery, in 4·0 s is 720 mC. The current in the circuit is :

(a) 0·18 A (b) 1·8 A

(c) 18 mA (d) 18 A

(a) 0.18A

4. Two cells, one of emf 2E and internal resistance r and the other of emf E and internal resistance r/2 are connected in parallel, by connecting their positive terminals together and their negative terminals together. The equivalent emf of the combination is :

(a) 3E

(b) $\frac{4E}{3}$

(c) 2E

(d) $\frac{3E}{4}$

4. Two cells, one of emf 2E and internal resistance r and the other of emf E and internal resistance r/2 are connected in parallel, by connecting their positive terminals together and their negative terminals together. The equivalent emf of the combination is :

(a) 3E

(b) $\frac{4E}{3}$

(c) 2E

(d) $\frac{3E}{4}$

(b) $\frac{4E}{3}$

26. A cylindrical conductor of resistance R is connected to a battery. The drift velocity of electrons in this conductor is v_d. The conductor is disconnected from the battery and gradually stretched so that its length increases by 25%. It is again connected to the same battery. Find the new value of : *3*

(a) the drift speed of electrons, and

(b) the resistance of the conductor.

(a)Calculation of new value of drift speed	1½
(b) Calculate of new value of resistance of conductor	1 ½

(a)	
$V_d = (\frac{eV}{ml})\tau$	½
$V'_d = (\frac{eV}{ml'})\tau$	½
$l' = 1.25l$	
$V'_d = (\frac{V_d}{1.25}) = \frac{4}{5}V_d$	½
(b)	
$A \times l = A' \times l'$	½
$l' = 1.25l$	
$A' = \frac{A}{1.25}$	
$R' = \rho\frac{l'}{A'} = \rho\frac{1.25 \times 1.25l}{A}$	½
$R' = \frac{25}{16}R$	½

16. *Assertion (A) :* The temperature coefficient of resistance is positive for metals and negative for semi-conductors.

Reason (R) : The charge carriers in metals are negatively charged whereas in semiconductors they are positively charged.

16. *Assertion (A) :* The temperature coefficient of resistance is positive for metals and negative for semi-conductors.

Reason (R) : The charge carriers in metals are negatively charged whereas in semiconductors they are positively charged.

(c)Assertion (A) is true, but Reason (R) is false.

25. (a) When is more power delivered to a light bulb — just after it is turned on and the glow of the filament is increasing or after the glow becomes steady ? Why ? *2*

OR

(b) A battery is connected first across the series combination and then across the parallel combination, of three resistances R, 2R and 3R. In which of the three resistances will power dissipated be maximum in the two cases ? Justify your answer. *2*

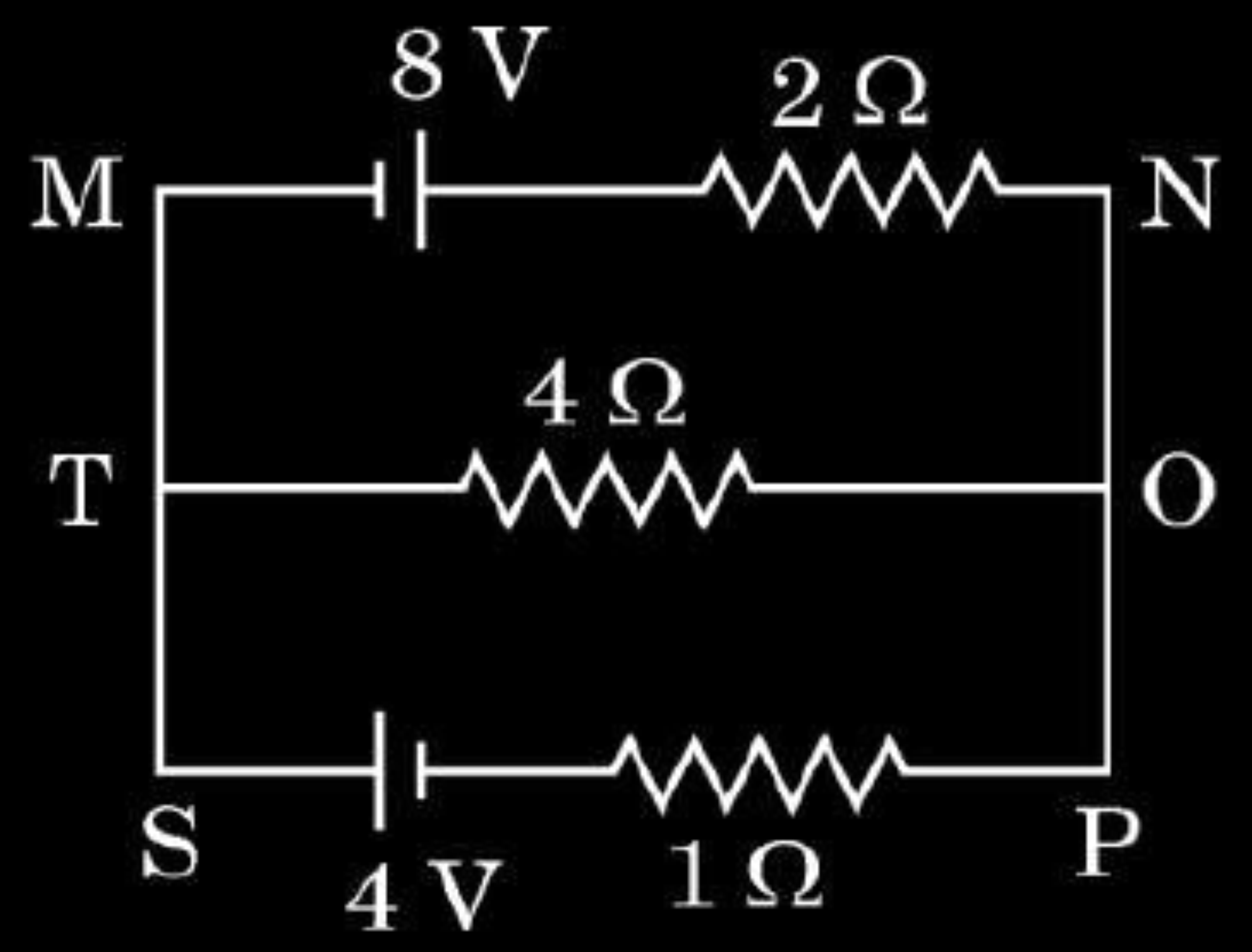
(a) For Answer	1
Reason	1

For series Answer	½
Justification	½
For parallel Answer	½
Justification	½

(a) Power delivered just after it is turned on, will be more because resistance of the bulb is low. After some time temperature of the bulb increases and resistance also increases and therefore power $(\frac{V^2}{R})$ becomes low.	1 1
For series Power dissipated will be maximum for 3R. Because current is same and power is proportional to resistance. For parallel Power dissipated will be maximum for R. Because voltage is same and power is inversely proportional to resistance.	½ ½ ½ ½

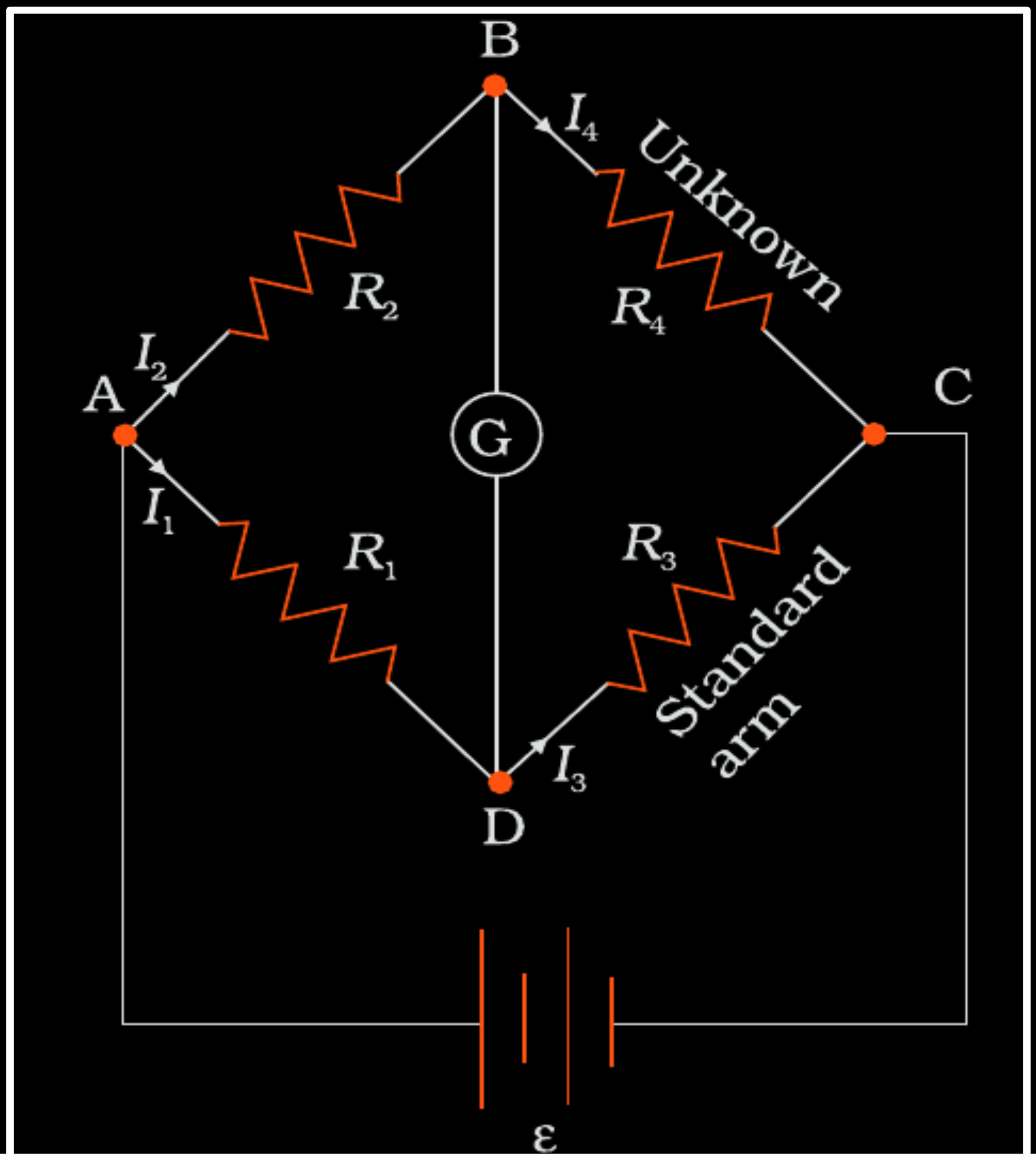
(b) (i) State Kirchhoff's rules. Use them to obtain the condition of balance for a Wheatstone Bridge.

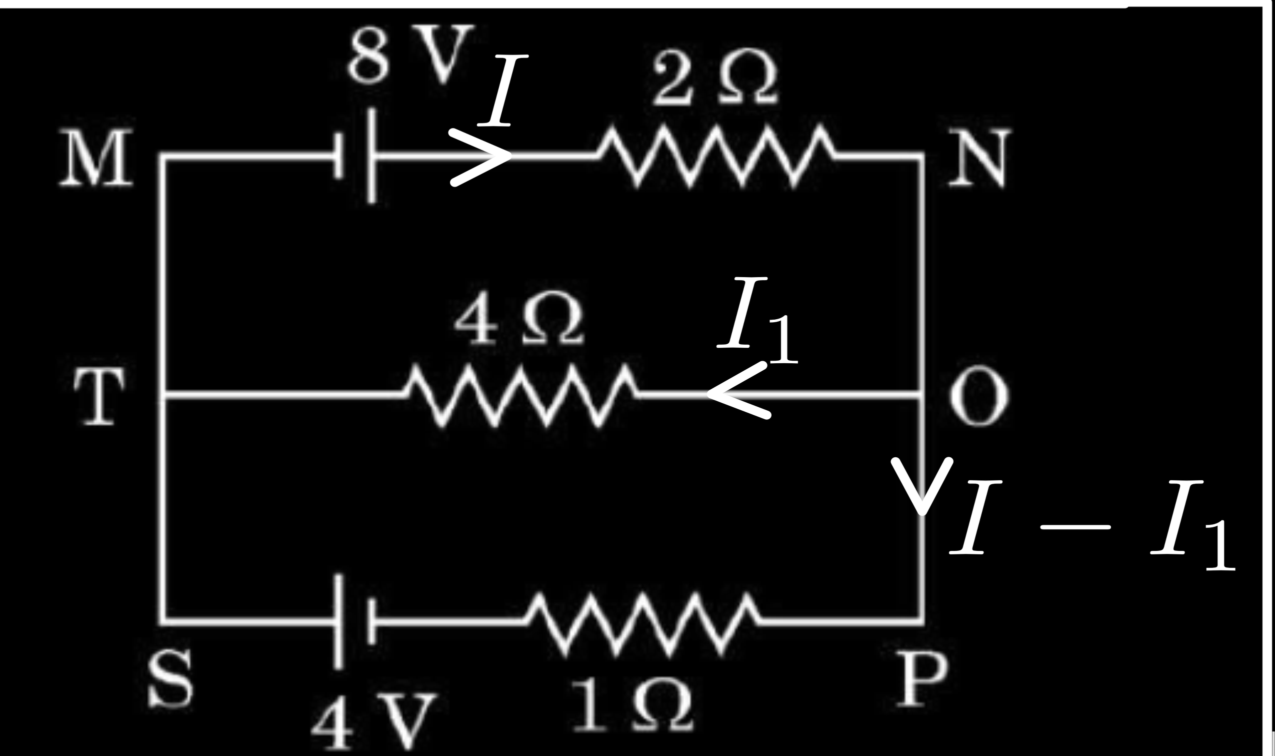
(ii) Use Kirchhoff's rule to determine the currents flowing through the branches MN, TO and SP in the circuit shown in the figure. *5*



(i) Statement of Kirchhoff two rules	½ + ½
Obtaining the balanced condition	2
(ii) Finding current in branches MN, T0 and SP	2

(i) Kirchhoff's junction rule - at any junction, the sum of the current entering the junction is equal to the sum of currents leaving the junction.	½
Kirchhoff second rule: The algebraic sum of changes in potential around any closed loop involving resistors and cells in the loop is zero.	½
In balanced fridge Ig=0, Hence $I_1 = I_3$ and $I_2 = I_4$ Using Kirchhoff's loop rule for closed loops ADBA and CBDC	
$- I_1R_1 + 0 + I_1R_1 = 0 \quad (Ig=0)$ -----(1)	½
In the second loop $I_3 = I_1$, $I_4 = I_2$	
$I_2R_4 + 0 - I_1R_3 = 0$ -----(2)	½
From equation (1) and (2) $\frac{I_1}{I_2} = \frac{R_2}{R_1}$ *and* $\frac{I_1}{I_2} = \frac{R_4}{R_3}$	
$\frac{R_2}{R_1} = \frac{R_4}{R_3}$ This is the condition for balanced Wheatstone bridge	½





(ii) In loop MNOTM	½
$2I + 4I_1 = 8$ ------(1)	
Loop OPSTO	
$-I + 5I_1 = -4$ ------(2)	½
On solving Current in MN, $I = 4A$ Current in TO, $I_1 = 0A$ Current in SP, $I - I_1 = 4A$	1

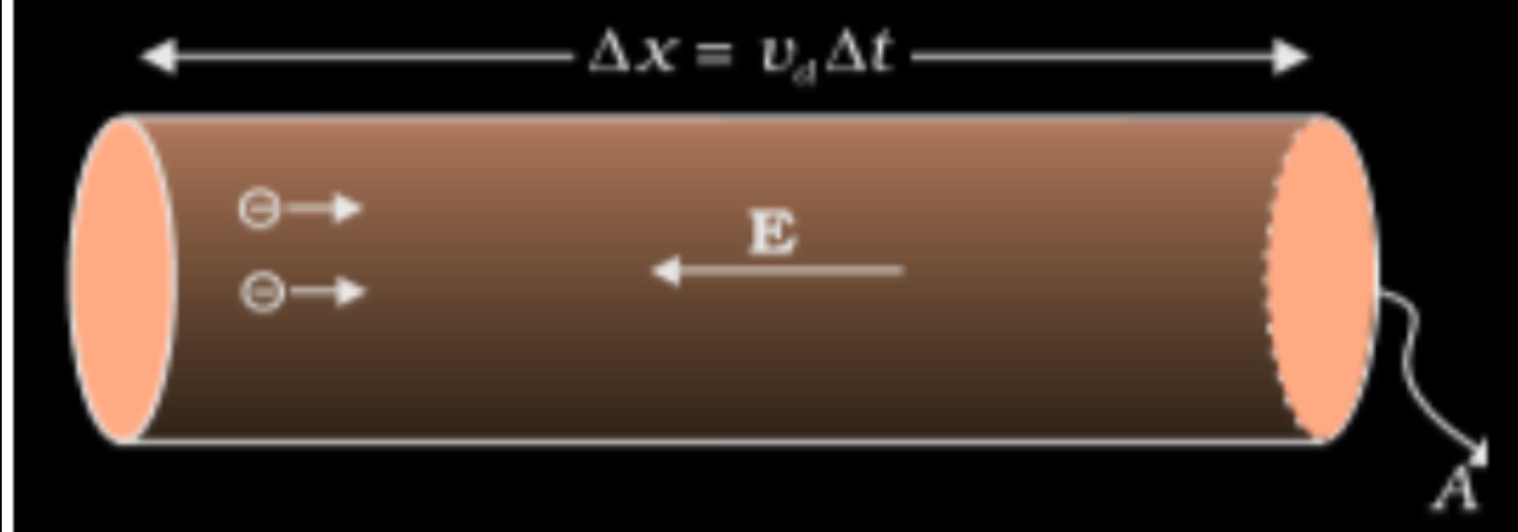
32. (a) (i) Derive the relation between the current and the drift velocity of free electrons in a conductor. Briefly explain the variation of resistance of a conductor with rise in temperature.

(ii) An ammeter, together with an unknown resistance in series is connected across two identical batteries, each of emf 1·5 V, connected (i) in series, and (ii) in parallel. If the current recorded in the two cases be $\left(\frac{1}{2}\right)$ A and $\left(\frac{1}{3}\right)$ A respectively, calculate the internal resistance of each battery. *5*

OR

(a) (i) Derivation of relation between I and v_d	2
Explanation	1
(ii) Calculating internal resistance of each battery	2

Total charge transported across the area A in time Δt is $\Delta Q = -neAv_d\,\Delta t$ ------(1)	½
Also the amount of charge crossing area 'A' in time Δt is $\Delta Q = I\,\Delta t$ ------(2)	½
Comparing equation (1) and (2) $I = neAv_d$	½
With increase in temperature, average speed of electrons increases resulting in more frequent collisions Hence relaxation time τ decreases As $R = \dfrac{ml}{ne^2\tau A}$ Resistance increases.	1
(ii) For series $I = \dfrac{E}{R+r}$ $\dfrac{1}{2} = \dfrac{3}{R+2r}$ $R + 2r = 6$ --------(1)	½
For parallel $\dfrac{1}{3} = \dfrac{1.5}{R+\dfrac{r}{2}}$ $2R + r = 9$ ----------(2)	½
After solving $r = 1\,\Omega$	1



CBSE - 2024

1. A heater coil rated as (P, V) is cut into two equal parts. One of the parts is then connected to a battery of V volt. The power consumed by it will be

(A) P

(B) $\frac{P}{2}$

(C) $\frac{P}{4}$

(D) 2P

CBSE - 2024

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(A) P

(B) $\frac{P}{2}$

(C) $\frac{P}{4}$

(D) 2P

(D) 2P

28. A potential difference of 1.0 V is applied across a conductor of length 5.0 m and area of cross-section 1.0 mm^2. When current of 4.25 A is passed through the conductor, calculate

(i) the drift speed and (ii) relaxation time, of electrons. (Given number density of electrons in the conductor, $n = 8.5 \times 10^{28}\ m^{-3}$).

(a) Calculating the drift speed	$1\frac{1}{2}$
(b) Calculation of Relaxation time	$1\frac{1}{2}$

28. A potential difference of 1.0 V is applied across a conductor of length 5.0 m and area of cross-section 1.0 mm^2. When current of 4.25 A is passed through the conductor, calculate
(i) the drift speed and (ii) relaxation time, of electrons. (Given number density of electrons in the conductor, $n = 8.5 \times 10^{28}\ m^{-3}$).

(i) $v_d = \dfrac{I}{enA}$	½
$= \dfrac{4.25}{1.6\times10^{-19}\times8.5\times10^{28}\times10^{-6}}$ m/s	½
$= 3.125\times10^{-4}$ m/s	½
(ii) $\tau = \dfrac{v_d ml}{eV}$	½
$= \dfrac{3.12\times10^{-4}\times9.1\times10^{-31}\times5}{1.6\times10^{-19}\times1}$ m/s	½
$= 88.72\times10^{-16}$ s	
$= 8.872\times10^{-15}$ s	½

(a) Calculating the drift speed	$1\frac{1}{2}$
(b) Calculation of Relaxation time	$1\frac{1}{2}$

24. (i) Define 'temperature coefficient of resistance' of a metal.

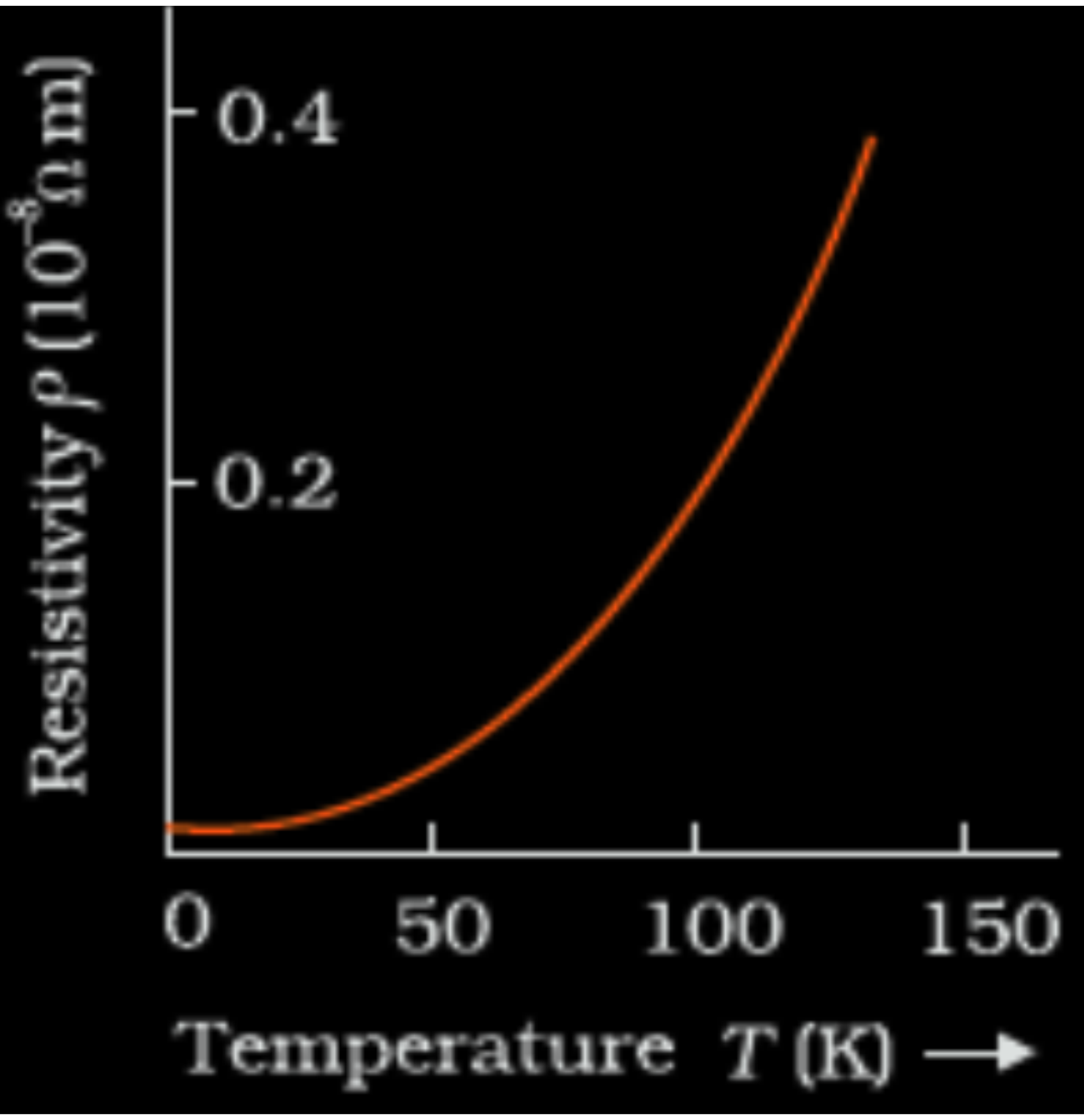
(ii) Show the variation of resistivity of copper with rise in temperature.

(iii) The resistance of a wire is 10 Ω at 27 °C. Find its resistance at –73 °C. The temperature coefficient of resistance of the material of the wire is 1.70×10^{-4} °C^{-1}.

(i) Defining temperature coefficient	1
(ii) Showing the variation of resistivity	1
(iii) Finding the resistance	1

(i) Change in resistance per unit original resistance per degree change in temperature is temperature coefficient of resistance. 1

(ii)



(iii) $R_2 = R_1(\theta_2 - \theta_1)\alpha + R_1$ ½

$= 10(-73-27)\times 1.70\times 10^{-4} + 10$

$= -0.170 + 10$

$R_2 = 9.83\,\Omega$ ½

Alternatively

$R_1 = R_0(1+\alpha t_1)$

$R_2 = R_0(1+\alpha t_2)$

$$\frac{R_1}{R_2} = \frac{(1+\alpha t_1)}{(1+\alpha t_2)}$$

$$R_2 = \frac{(1+\alpha t_1)}{(1+\alpha t_2)} R_1$$

$$R_2 = \left[\frac{1+1.70\times 10^{-4}\times(-73)}{1+1.70\times 10^{-4}\times 27}\right]\times 10$$ ½

$$R_2 = \frac{0.98759}{1.00459}\times 10\ \Omega$$

$R_2 = 9.83\ \Omega$ ½

1. A battery supplies 0.9 A current through a 2 Ω resistor and 0.3 A current through a 7 Ω resistor when connected one by one. The internal resistance of the battery is :

(A) 2 Ω

(B) 1.2 Ω

(C) 1 Ω

(D) 0.5 Ω

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20. Find the temperature at which the resistance of a conductor increases by 25% of its value at 27 °C. The temperature coefficient of resistance of the conductor is $2.0 \times 10^{-4}\ ^{\circ}C^{-1}$. **2**

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$R_2 = R_1 + 25\% \text{ of } R_1 = 1.25R_1$	½
Temperature coefficient of resistance	
$\alpha = \dfrac{R_2 - R_1}{R_1 . \Delta T}$	½
$T_2 - 27 = \dfrac{1.25R_1 - R_1}{R_1 \times 2 \times 10^{-4}}$	½
$T_2 = 1277\ ^0C$	½

14. **Assertion (A) :** When electrons drift in a conductor, it does not mean that all free electrons in the conductor are moving in the same direction.

Reason (R) : The drift velocity is superposed over large random velocities of electrons. **1**

14. **Assertion (A) :** When electrons drift in a conductor, it does not mean that all free electrons in the conductor are moving in the same direction.

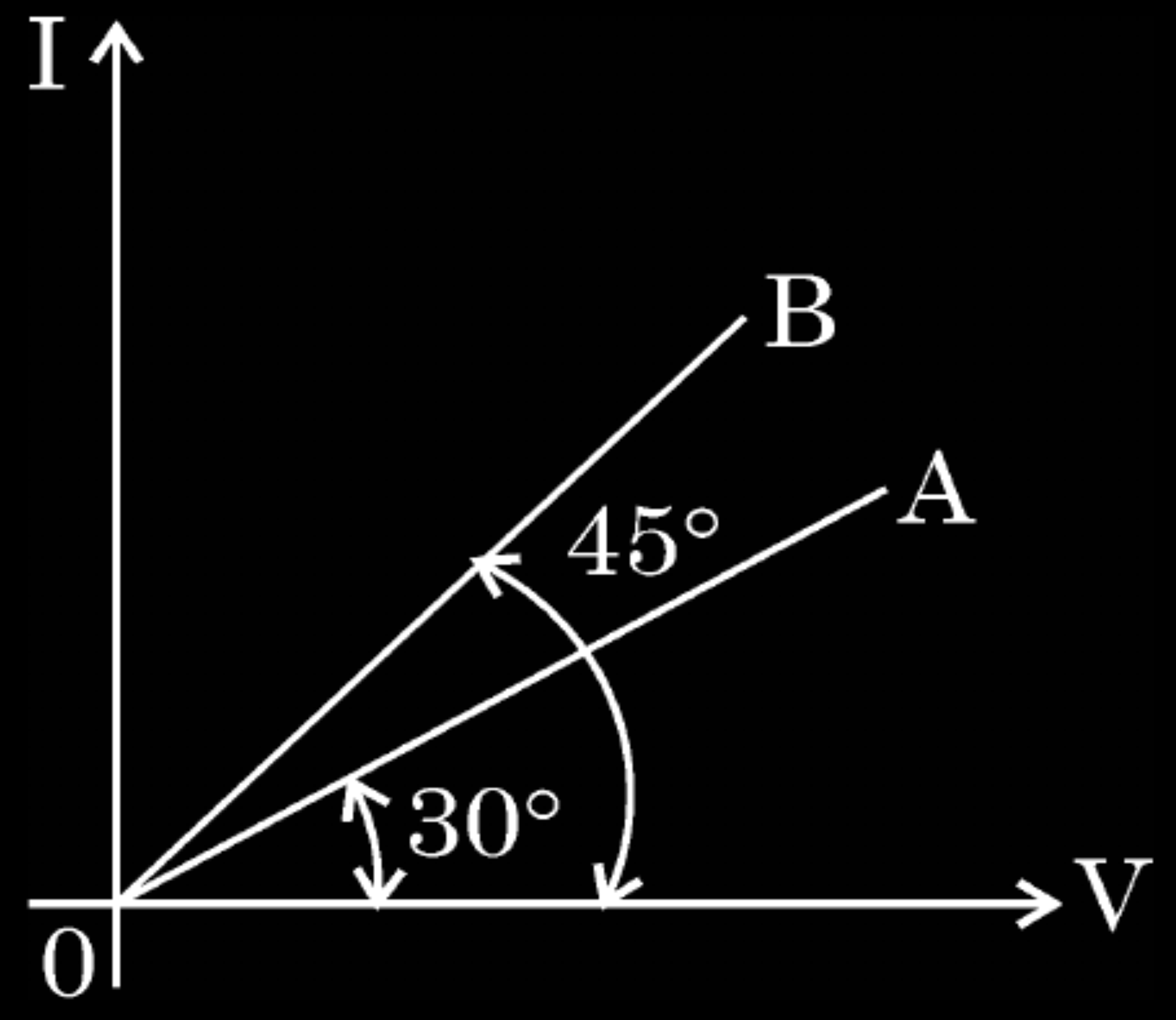
Reason (R) : The drift velocity is superposed over large random velocities of electrons. **1**

(A) Both assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion(A).

20. Two wires A and B of different metals have their lengths in ratio 1 : 2 and their radii in ratio 2 : 1 respectively. I-V graphs for them is shown in the figure. Find the ratio of their **2**

(i) Resistances (R_A/R_B) and

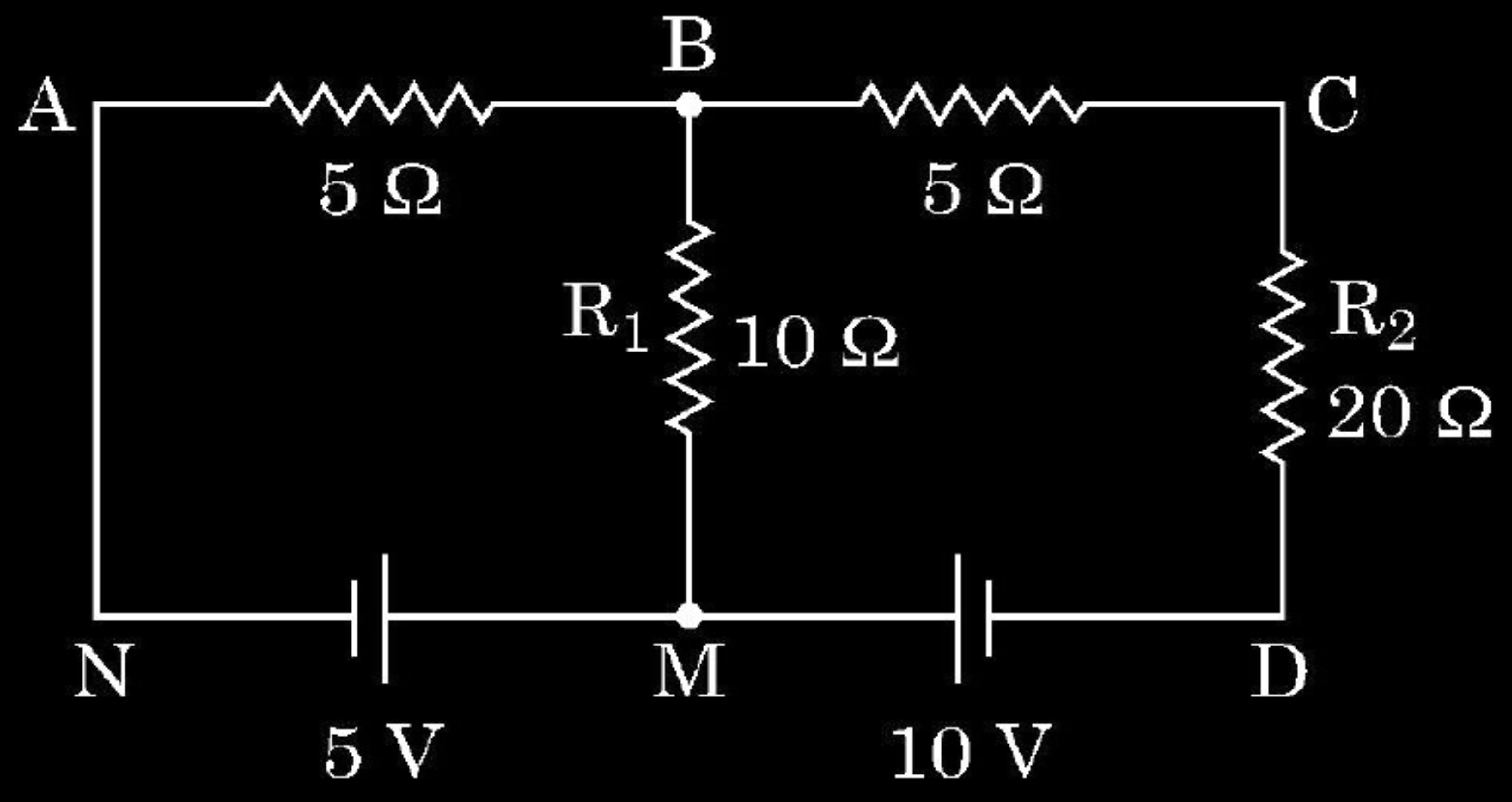
(ii) Resistivities (σ_A/σ_B)



(i) Finding resistance $\left(\frac{R_A}{R_B}\right)$	1
(ii) Finding resistivity $\left(\frac{\sigma_A}{\sigma_B}\right)$	1

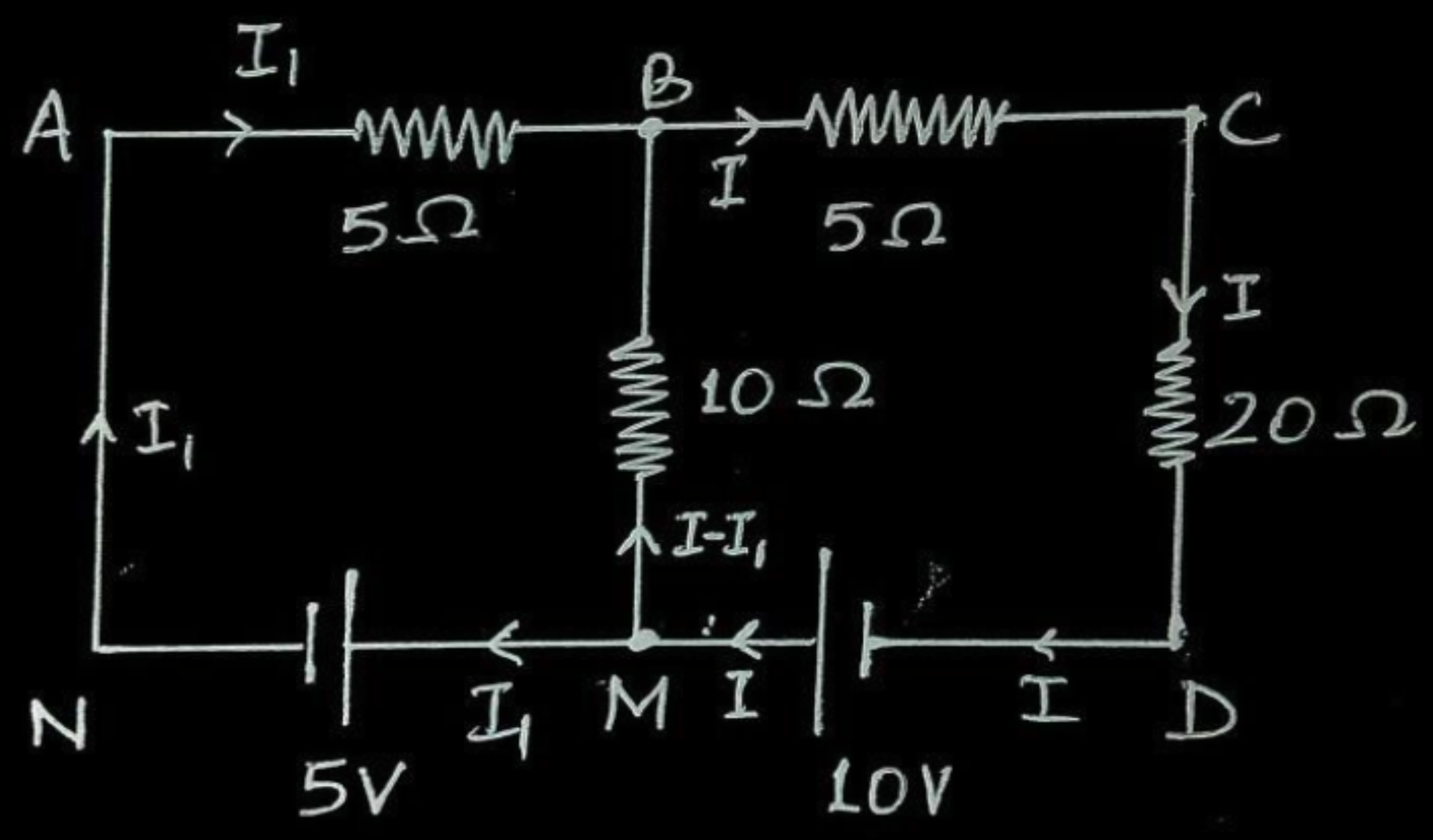
(i) Slope of I-V graph = $\left(\dfrac{\Delta I}{\Delta V}\right) = \dfrac{I}{R}$	½
$$\frac{R_A}{R_B} = \frac{\text{Slope of B}}{\text{Slope of A}}$$ $$= \frac{\tan 45^0}{\tan 30^0}$$ $$\frac{R_A}{R_B} = \sqrt{3}$$	½
(ii) $$\frac{\sigma_A}{\sigma_B} = \frac{R_A \dfrac{A_A}{l_A}}{R_B \dfrac{A_B}{l_B}}$$	½
$$= \frac{R_A}{R_B} \cdot \frac{A_A}{A_B} \cdot \frac{l_B}{l_A}$$ $$= \sqrt{3} \times \frac{4}{1} \times \frac{2}{1}$$ $$= 8\sqrt{3}$$	½

27. Find the currents flowing through the branches AB and BC in the network shown. **3**



Finding current in the arm AB	1 ½
Finding current in the arm BC	1 ½

Circuit diagram with distribution of current



1

Using Kirchhoff's voltage rule

In closed loop ABMNA,

$$-5I_1 + 10(I - I_1) - 5 = 0 \quad \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots \quad (1)$$

½

In closed loop ACDNA

$$-5I - 20I + 10 - 5 - 5I_1 = 0 \quad \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots \quad (2)$$

½

Solving eq (1) and (2)

$$I_1 = -\frac{3}{17}\ \text{A} \quad \text{and} \quad I = \frac{4}{17}$$

Magnitude of current in arm AB = $\frac{3}{17}$ A ½

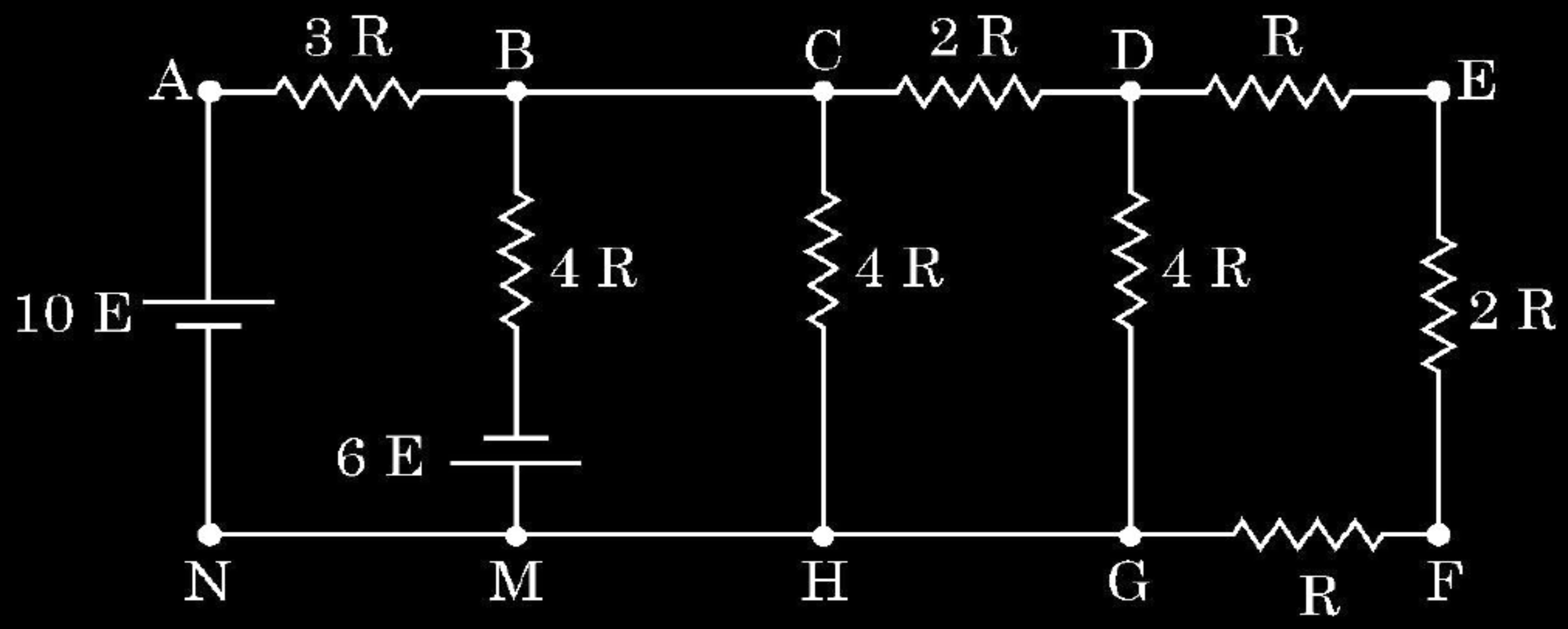
Magnitude of current in arm BC = $\frac{4}{17}$ A ½

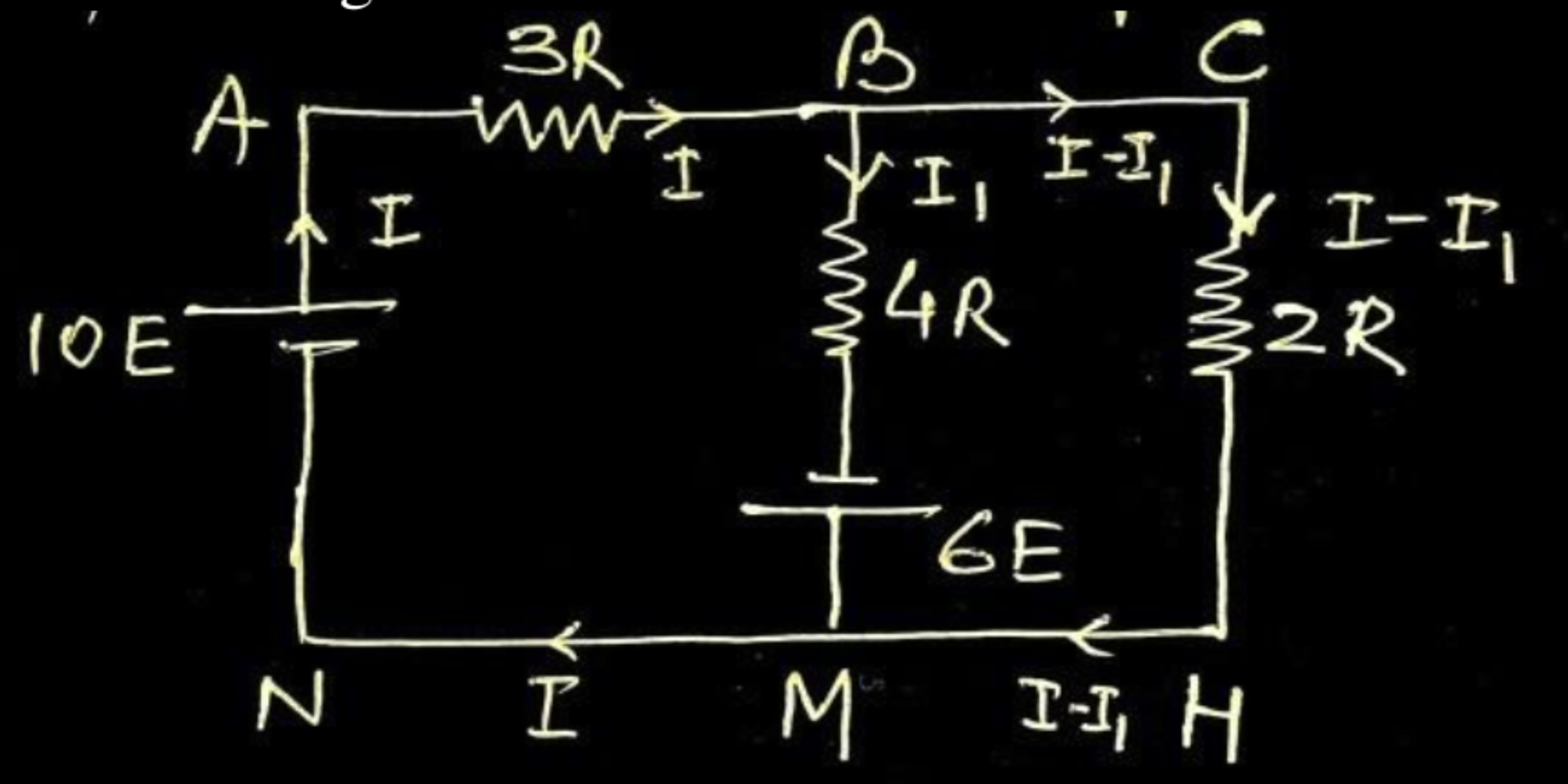
20. Two electric heaters have power ratings P_1 and P_2, at voltage V. They are connected in series to a dc source of voltage V. Find the power consumed by the combination. Will they consume the same power if connected in parallel across the same source ? **2**

Finding power consumed by two electric heaters in series combination	1 ½
Writing answer for parallel combination	½

$R_1 = \frac{V^2}{P_1}$ & $R_2 = \frac{V^2}{P_2}$	½
$R_{eq} = R_1 + R_2 = V^2\left(\frac{1}{P_1} + \frac{1}{P_2}\right)$	½
$P_{series} = \frac{V^2}{R_{eq}}$	
$P_{series} = \frac{V^2}{V^2\left(\frac{1}{P_1} + \frac{1}{P_2}\right)}$	
$\frac{1}{P_{series}} = \frac{1}{P_1} + \frac{1}{P_2}$	½
No	½

25. Find the current in branch BM in the network shown : 3



Finding equivalent resistance across CH , $R_{CH} = 2R$	½
Equivalent circuit diagram 	½
In closed loop ABMNA $-3IR - 4I_1R + 16E = 0 \qquad \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots(1)$	½
In closed loop BCHMB $-2R(I-I_1) - 6E + 4I_1R = 0 \qquad \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots(2)$	½
On solving equations (1) and (2) $$I_1 = \frac{25E}{13R}$$	1

17. (a) What is meant by 'relaxation time' of free electrons in a conductor ?

Show that the resistance of a conductor can be expressed by $R = \dfrac{m\,l}{ne^2 \tau A}$, where symbols have their usual meanings. *2*

OR

(b) Draw the circuit diagram of a Wheatstone bridge. Obtain the condition when no current flows through the galvanometer in it. *2*

Meaning of relaxation time	½
Derivation of R	1½

Circuit diagram of Wheatstone bridge	½
Obtaining the condition when current flows through galvanometer	1½

Average time between two successive collisions of electron in presence of electric field. ½

Drift velocity of an electron

$$\upsilon_d = \frac{eE}{m}\tau \qquad ---(i)$$ ½

Current flowing through a conductor of length *l* and area of cross section A

$$I = neA\upsilon_d \qquad ---(ii)$$

$$I = \frac{ne^2AE\tau}{m} = \frac{ne^2A\tau V}{ml}$$ ½

$$R = \frac{V}{I} = \frac{ml}{ne^2\tau A}$$ ½

By applying Kirchoff's loop rule to closed loops ADBA and CBDC

$$-I_1R_1 + 0 + I_2R_2 = 0 \qquad -----(i) \quad [I_g = 0]$$

$$I_2R_4 + 0 - I_1R_3 = 0 \qquad -----(ii)$$ ½

From eq (i) -

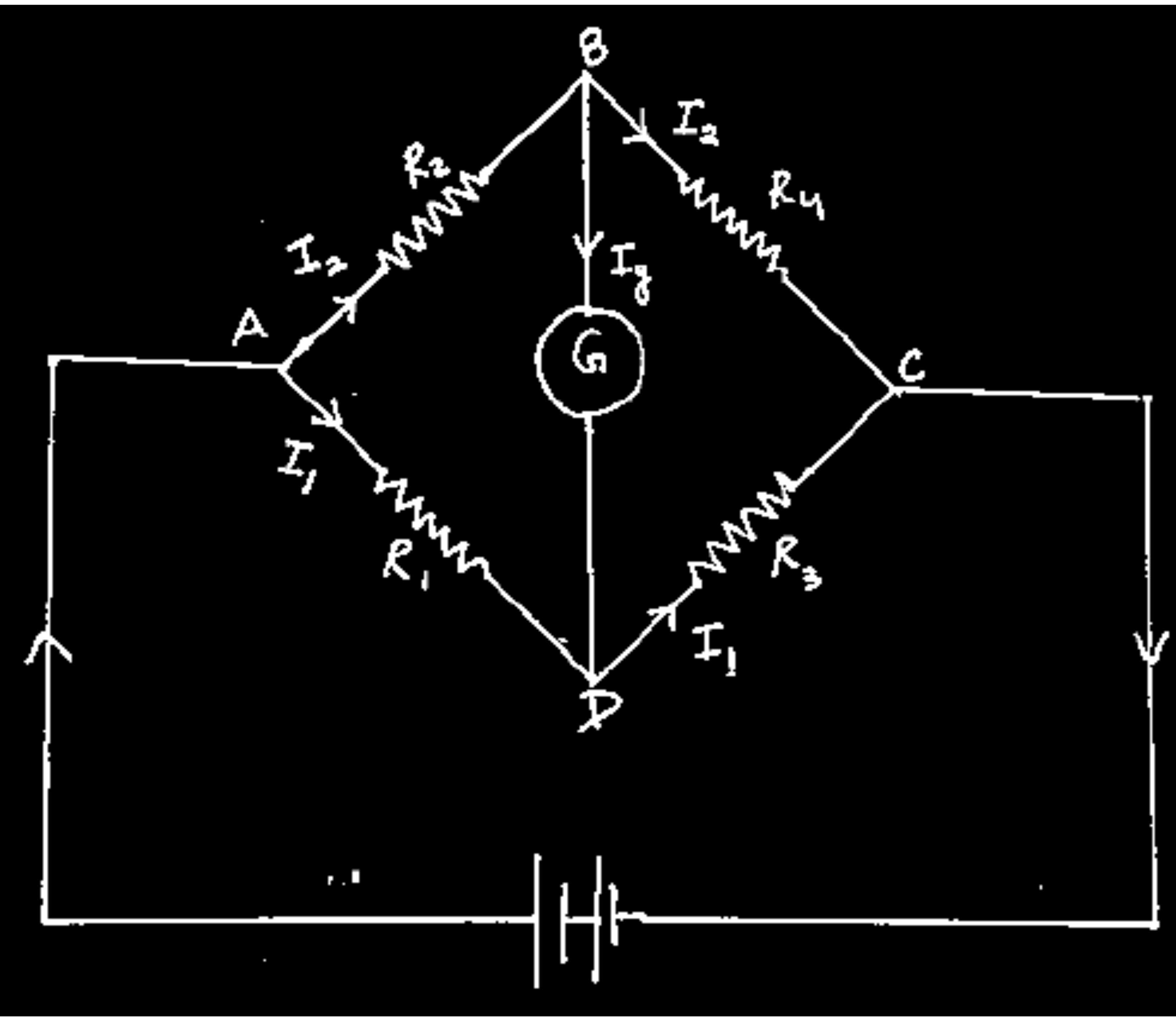
$$\frac{I_1}{I_2} = \frac{R_2}{R_1}$$ ½

From eq (ii) -

$$\frac{I_1}{I_2} = \frac{R_4}{R_3}$$

Hence,

$$\frac{R_2}{R_1} = \frac{R_4}{R_3}$$ ½



½

22. A current of 1·6 A flows through a wire when a potential difference of 1·0 V is applied across it. The length and cross-sectional area of the wire are 1·0 m and $1{\cdot}0 \times 10^{-7}$ m^2 respectively. Calculate : *3*

(a) Electric field across the wire

(b) Current density

(c) Average relaxation time (τ)

(Number density of free electrons in the wire is $9{\cdot}0 \times 10^{28}$ m^{-3})

Calculation of	
(a) Electric field across the wire	1
(b) Current density	1
(c) Average relaxation time (τ)	1

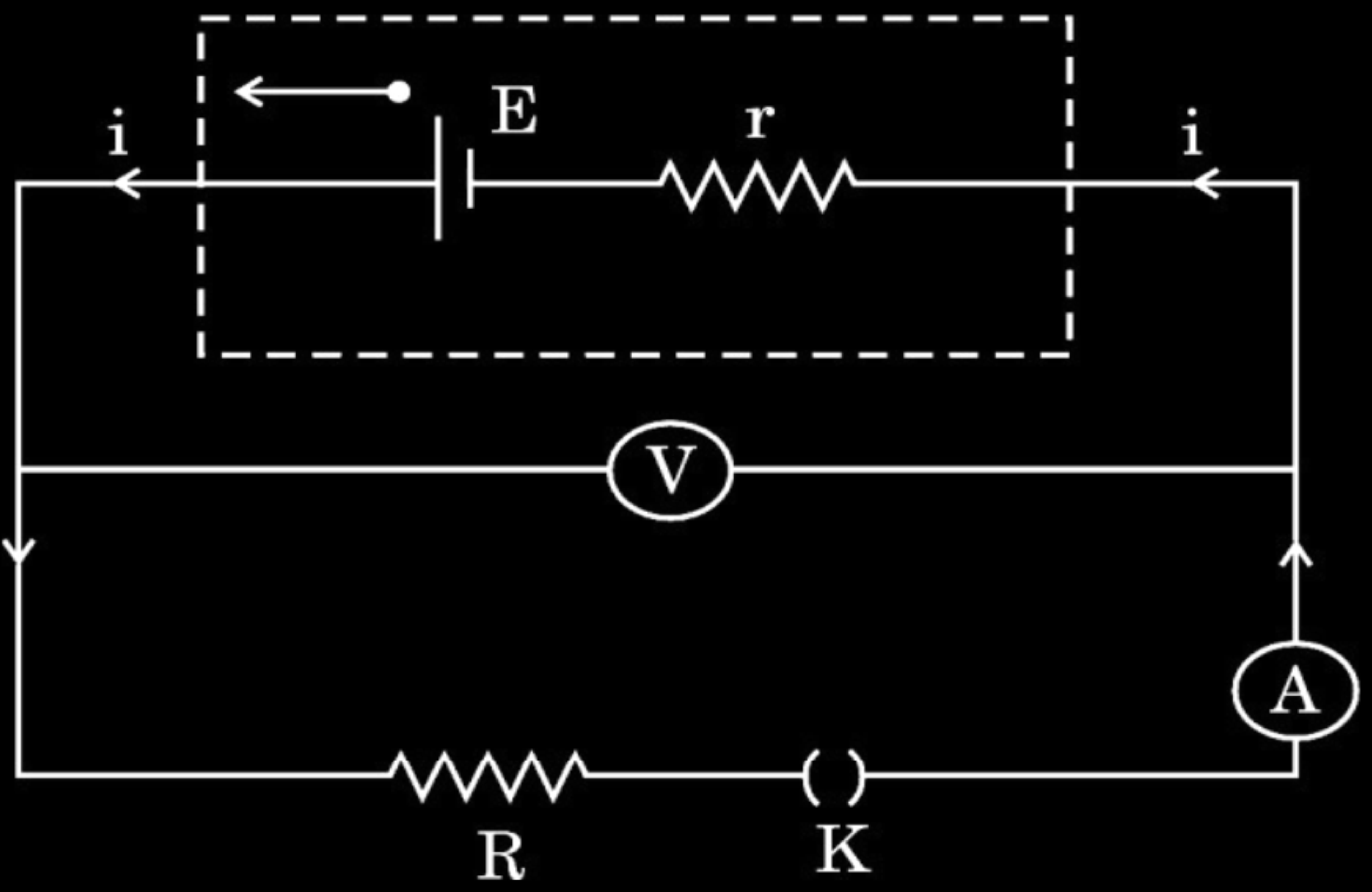
(a) $E = \dfrac{V}{l}$	½
$= \dfrac{1.0V}{1.0m} = 1.0$ V/m	½
(b) $J = {}^{I}\!/_{A}$	½
$J = \dfrac{1.6\,A}{1.0\times10^{-7}\,m^2} = 1.6\times10^{7}$ A/m^2	½
(c) $\tau = \dfrac{m}{ne^2}\dfrac{J}{E}$	½
$= \dfrac{9.1\times10^{-31}\times1\times1.6}{9\times10^{28}\times\left(1.6\times10^{-19}\right)^2}$	
$= 6.31\times10^{-14}$ s	½

22. A battery of unknown emf E and internal resistance r is connected in a circuit as shown in the figure. When the key (K) is open, the voltmeter reads 10·0 V and ammeter reads 0 A. In the closed circuit, the voltmeter reads 6·0 V and ammeter reads 2·0 A. Calculate : *3*

(a) emf of the battery,

(b) internal resistance of the battery (r), and

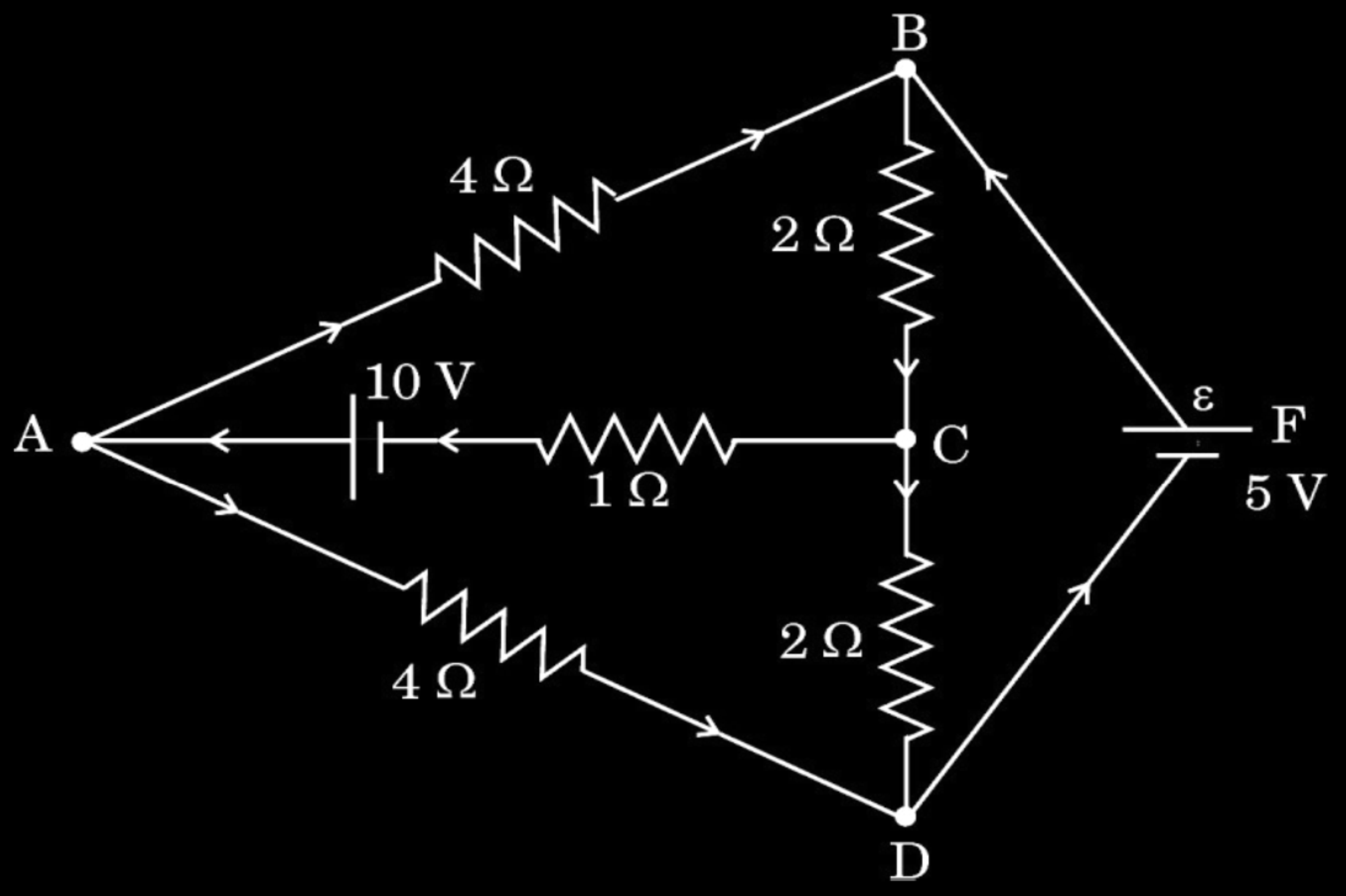
(c) external resistance (R).

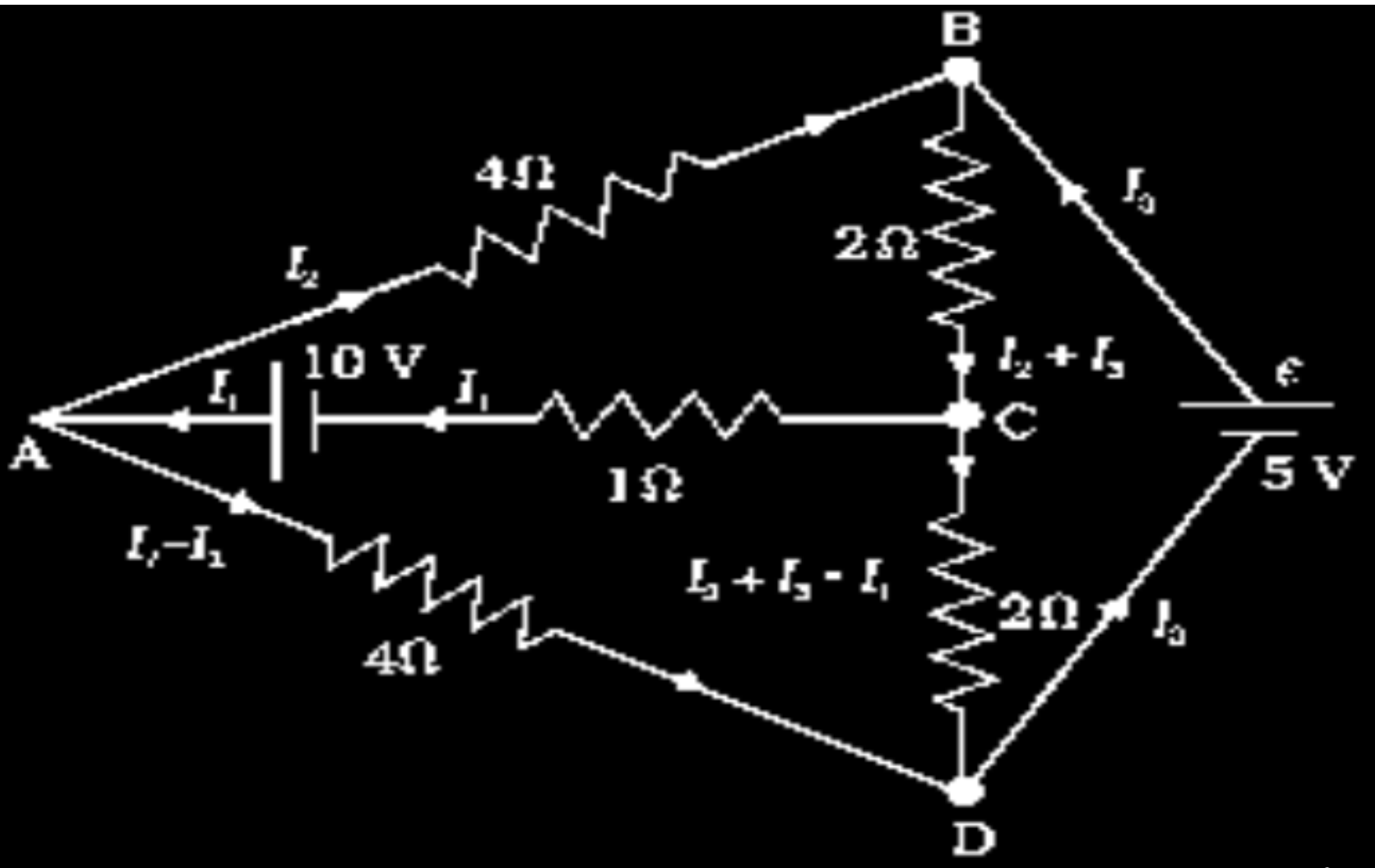


Calculation of	
(a) emf of battery	½
(b) Internal resistance of battery(r)	1½
(c) external resistance (R)	1

(a) $V = E = 10$ V(When key K is open and I=0 A)	½
(b) $V = E - Ir$ (When key K is closed and I=2 A)	½
$6 = 10 - 2r$	½
$r = 2\Omega$	½
(c) $E = I(r+R)$	½
$10 = 2(2+R)$	
$R = 3\ \Omega$	½

22. Determine the current in branches AB, AC and BC of the network shown in figure. *3*





For closed loop ADCA ,

$$10-4(I_1-I_2)+2(I_2+I_3-I_1)-I_1=0$$

$$7I_1-6I_2-2I_3=10 \quad -------(i)$$ ½

For closed loop ABCA ,

$$10-4I_2-2(I_2+I_3)-I_1=0$$

$$I_1+6I_2+2I_3=10 \quad -------(ii)$$ ½

For closed loop BCDED ,

$$5-2(I_2+I_3)-2(I_2+I_3-I_1)=0$$

$$2I_1-4I_2-4I_3=-5 \quad ------(iii)$$ ½

Current in branch AB = $I_2=\frac{5}{8}A$ ½

Current in branch AC $=I_1=2.5A$ ½

Current in branch BC $=I_2+I_3=2.5A$ ½

2. Electrons drift with speed v_d in a conductor with potential difference V across its ends. If V is reduced to $\left(\frac{V}{2}\right)$, their drift speed will become :

(A) $\frac{v_d}{2}$

(B) v_d

(C) $2\ v_d$

(D) $4\ v_d$

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(C) $2\ v_d$

(D) $4\ v_d$

(A) $\frac{v_d}{2}$

17. An electric field E is maintained in a wire of length 'l' and area of cross-section 'a'. Derive the relation between the current density 'σ' in the wire and the electric field E. *2*

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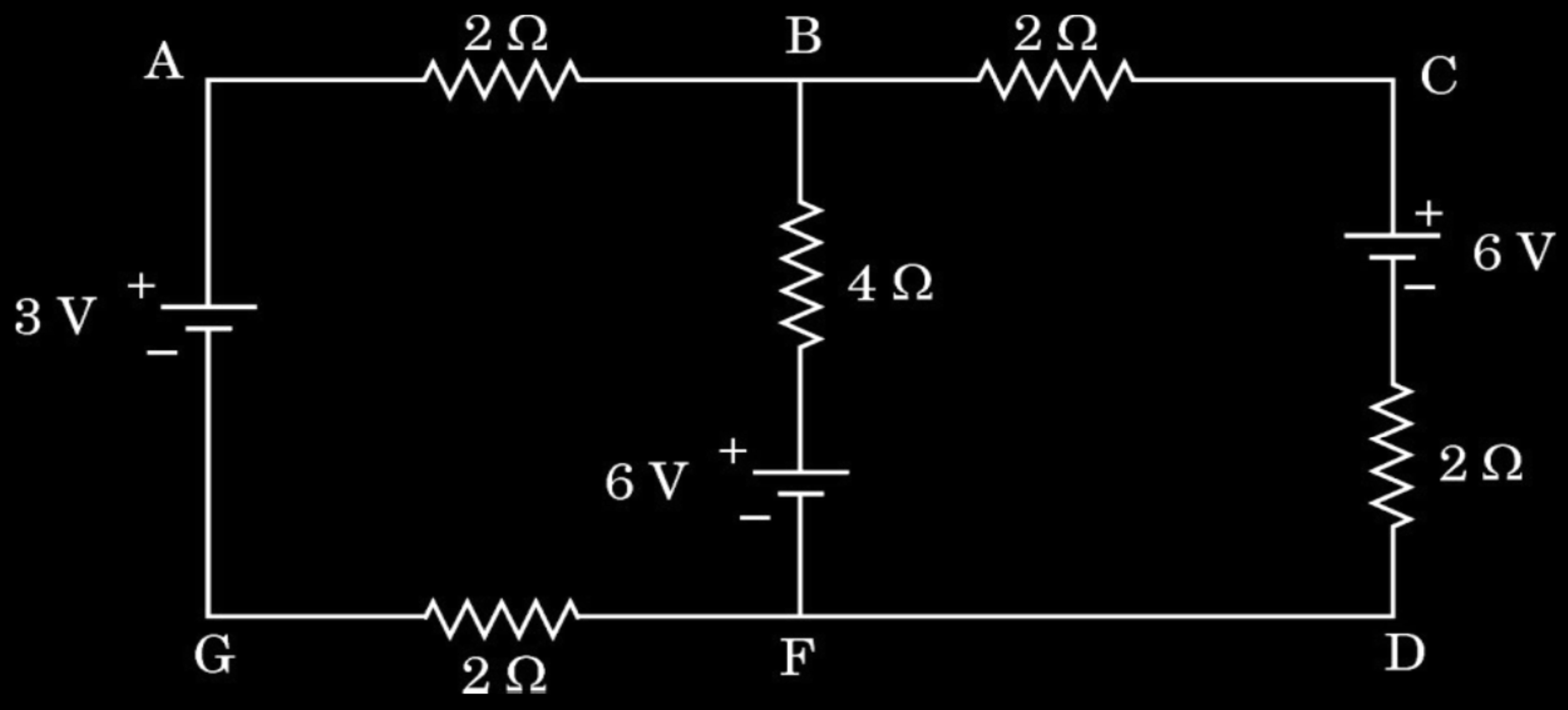
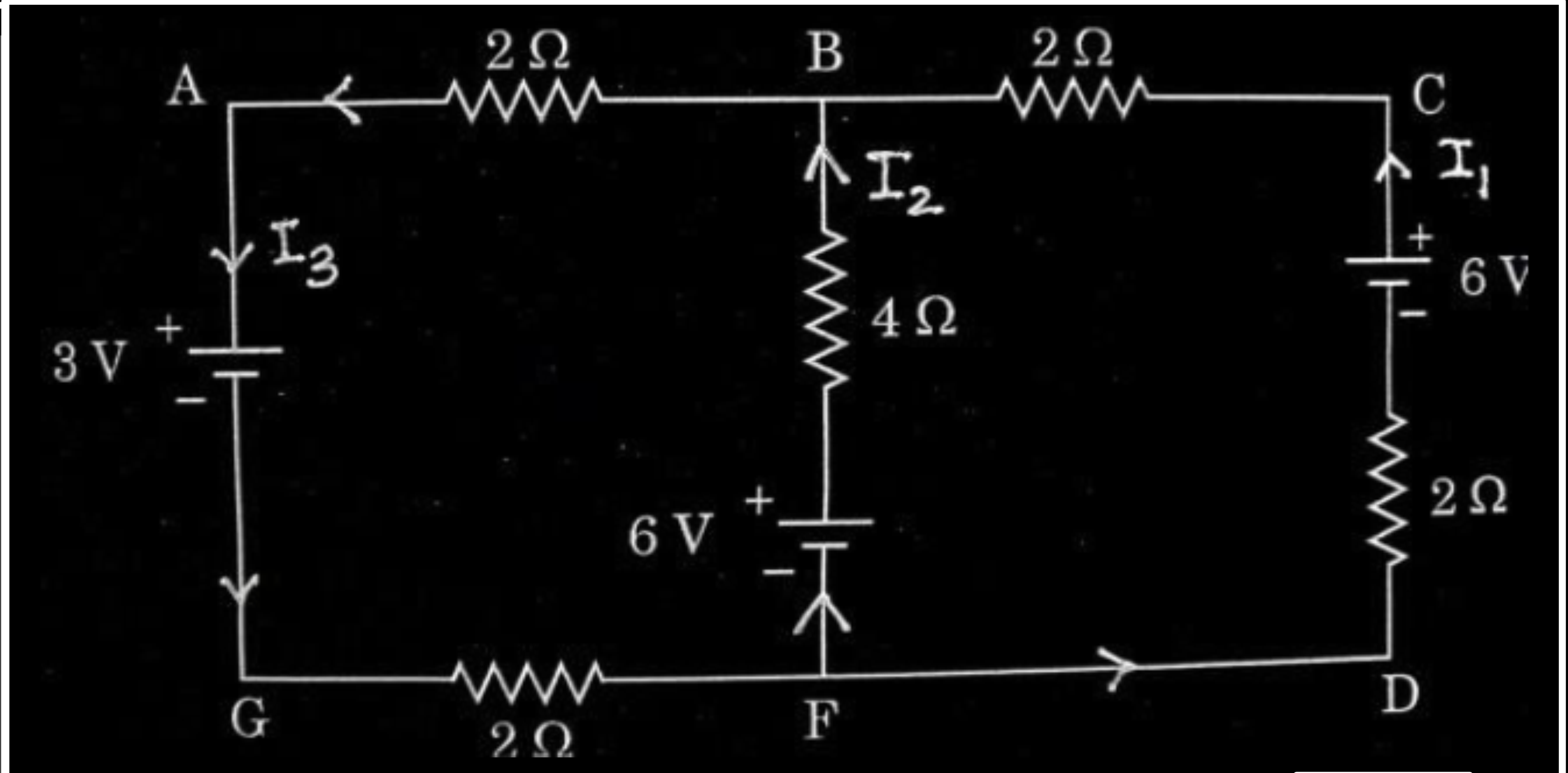
Deriving relation 2	
$V = IR$	½
$El = \frac{I\rho l}{A}$ (V= El, $R = \frac{\rho l}{A}$)	½
$E = \frac{I}{A}\rho$	½
$E = \sigma\rho$	½

25. The figure shows a circuit with three ideal batteries. Find the magnitude and direction of currents in the branches AG, BF and CD. *3*



Finding magnitude and direction of current in AG, BF and CD	1+1+1

By Kirchoff's Laws (at point B)	
$I_1 + I_2 = I_3$ ....(1)	
In the closed loop AGFBA	½
$3 + 2I_3 - 6 + 4I_2 + 2I_3 = 0$	
$I_2 + I_3 = \frac{3}{4}$ ....(2)	
From (i)	½
$2I_1 + I_2 = \frac{3}{4}$ ....(3)	
In closed loop BFDCB	½
$-4I_2 + 6 + 2I_1 - 6 + 2I_1 = 0$	
$I_2 - I_1 = 0$	
$I_2 = I_1$ ....(4)	
Putting in (3)	½
$I_1 = \frac{1}{4} A$	
From (4)	
$I_2 = \frac{1}{4} A$	
From (2) $I_3 = \frac{1}{2} A$	½



½

30. When the terminals of a cell are connected to a conductor of resistance R, an electric current flows through the circuit. The electrolyte of the cell also offers some resistance in the path of the current, like the conductor. This resistance offered by the electrolyte is called internal resistance of the cell (r). It depends upon the nature of the electrolyte, the area of the electrodes immersed in the electrolyte and the temperature. Due to internal resistance, a part of the energy supplied by the cell is wasted in the form of heat.

When no current is drawn from the cell, the potential difference between the two electrodes in known as emf of the cell (ε). With a current drawn from the cell, the potential difference between the two electrodes is termed as terminal potential difference (V).

(i) Choose the ***incorrect*** statement : *1*

(A) The potential difference (V) between the two terminals of a cell in a closed circuit is always less than its emf (ε), during discharge of the cell.

(B) The internal resistance of a cell decreases with the decrease in temperature of the electrolyte.

(C) When current is drawn from the cell then $V = \varepsilon - Ir$.

(D) The graph between potential difference between the two terminals of the cell (V) and the current (I) through it is a straight line with a negative slope.

(ii) Two cells of emfs 2·0 V and 6·0 V and internal resistances 0·1 Ω and 0·4 Ω respectively, are connected in parallel. The equivalent emf of the combination will be : *1*

(A) 2·0 V (B) 2·8 V

(C) 6·0 V (D) 8·0 V

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(A) 2·0 V

(B) 2·8 V

(C) 6·0 V

(D) 8·0 V

(iii) Dipped in the solution, the electrode exchanges charges with the electrolyte. The positive electrode develops a potential V_+ ($V_+ > 0$), and the negative electrode develops a potential $-(V_-)$ ($V_- \geq 0$), relative to the electrolyte adjacent to it. When no current is drawn from the cell then : *1*

(A) $\varepsilon = V_+ + V_- > 0$ (B) $\varepsilon = V_+ - V_- > 0$

(C) $\varepsilon = V_+ + V_- < 0$ (D) $\varepsilon = V_+ + V_- = 0$

(iv) (a) Five identical cells, each of emf 2 V and internal resistance 0·1 Ω are connected in parallel. This combination in turn is connected to an external resistor of 9·98 Ω. The current flowing through the resistor is : *1*

(A) 0·05 A (B) 0·1 A

(C) 0·15 A (D) 0·2 A

OR

(b) Potential difference across a cell in the open circuit is 6 V. It becomes 4 V when a current of 2 A is drawn from it. The internal resistance of the cell is : *1*

(A) 1·0 Ω (B) 1·5 Ω

(C) 2·0 Ω (D) 2·5 Ω

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(C) 0·15 A $\boxed{\text{(D)} \quad 0{\cdot}2\ \text{A}}$

OR

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$\boxed{\text{(A)} \quad 1{\cdot}0\ \Omega}$ (B) 1·5 Ω

(C) 2·0 Ω (D) 2·5 Ω

17. (a) "The electron drift speed is only a few mm/s for currents in the range of a few amperes for a given conductor." How then is current established almost the instant a circuit is closed ? Explain.

(b) 'V = IR is a statement of Ohm's Law' is not true. Explain. *2*

17. (a) "The electron drift speed is only a few mm/s for currents in the range of a few amperes for a given conductor." How then is current established almost the instant a circuit is closed ? Explain.

(b) 'V = IR is a statement of Ohm's Law' is not true. Explain. *2*

(a) Electric field is established throughout the circuit, almost instantly. It causes a local electron drift at every point, thus establishment of current does not have to wait for electrons from one end of the conductor to travel to other end.	**1**
(b) Ohm's law asserts that the plot of I versus V is linear i.e. R is independent of V, while equation V=IR defines resistance and it may be applied to all conducting devices whether they obey Ohm's law or not.	**1**

17. Define resistivity of a conductor. How does the resistivity of a conductor depend upon the following : *2*

(a) Number density of free electrons in the conductor (n)

(b) Their relaxation time (τ)

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(a) Number density of free electrons in the conductor (n)

(b) Their relaxation time (τ)

Defining resistivity	1
Dependence of resistivity on	
(a) Number density of free electron	½
(b) Relaxation time	½

Resistance offered by a material of unit length and having unit cross-sectional area is called resistivity.	**1**
$\rho = \frac{m}{ne^2\tau}$	
(a) $\rho \alpha \frac{1}{n}$	½
(b) $\rho \alpha \frac{1}{\tau}$	½

3. In a uniform straight wire, conduction electrons move along + x direction. Let $\vec{E}$ and $\vec{j}$ be the electric field and current density in the wire, respectively. Then :

(A) $\vec{E}$ and $\vec{j}$ both are along + x direction.

(B) $\vec{E}$ and $\vec{j}$ both are along – x direction.

(C) $\vec{E}$ is along + x direction, but $\vec{j}$ is along – x direction.

(D) $\vec{E}$ is along – x direction, but $\vec{j}$ is along + x direction.

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(C) $\vec{E}$ is along + x direction, but $\vec{j}$ is along – x direction.

(D) $\vec{E}$ is along – x direction, but $\vec{j}$ is along + x direction.

(B) $\vec{E}$ *and* $\vec{j}$ both are along –x direction

17. A uniform wire of length L and area of cross-section A has resistance R. The wire is uniformly stretched so that its length increases by 25%. Calculate the percentage increase in the resistance of the wire. *2*

17. A uniform wire of length L and area of cross-section A has resistance R. The wire is uniformly stretched so that its length increases by 25%. Calculate the percentage increase in the resistance of the wire. *2*

As the wire is uniformly stretched;	
$\Rightarrow Al = A'l' \Rightarrow A' = \dfrac{lA}{1.25l}$	½
$A' = \dfrac{4}{5}A$	
$R' = \dfrac{\rho l'}{A'} = \dfrac{\rho(1.25l)}{\frac{4}{5}A}$	
$R' = \dfrac{25}{16}R$	1
$\% \text{ increase} = \left(\dfrac{R'-R}{R}\right)\times 100$	
$= \left(\dfrac{25}{16}-1\right)\times 100 = 56.25\%$	½

23. (a) Define current density. Is it a scalar or a vector ? An electric field $\vec{E}$ is maintained in a metallic conductor. If n be the number of electrons (mass m, charge – e) per unit volume in the conductor and τ its relaxation time, show that the current density $\vec{j} = \alpha \vec{E}$, where $\alpha = \left(\frac{ne^2}{m}\right)\tau$. *3*

OR

(b) What is a Wheatstone bridge ? Obtain the necessary conditions under which the Wheatstone bridge is balanced. *3*

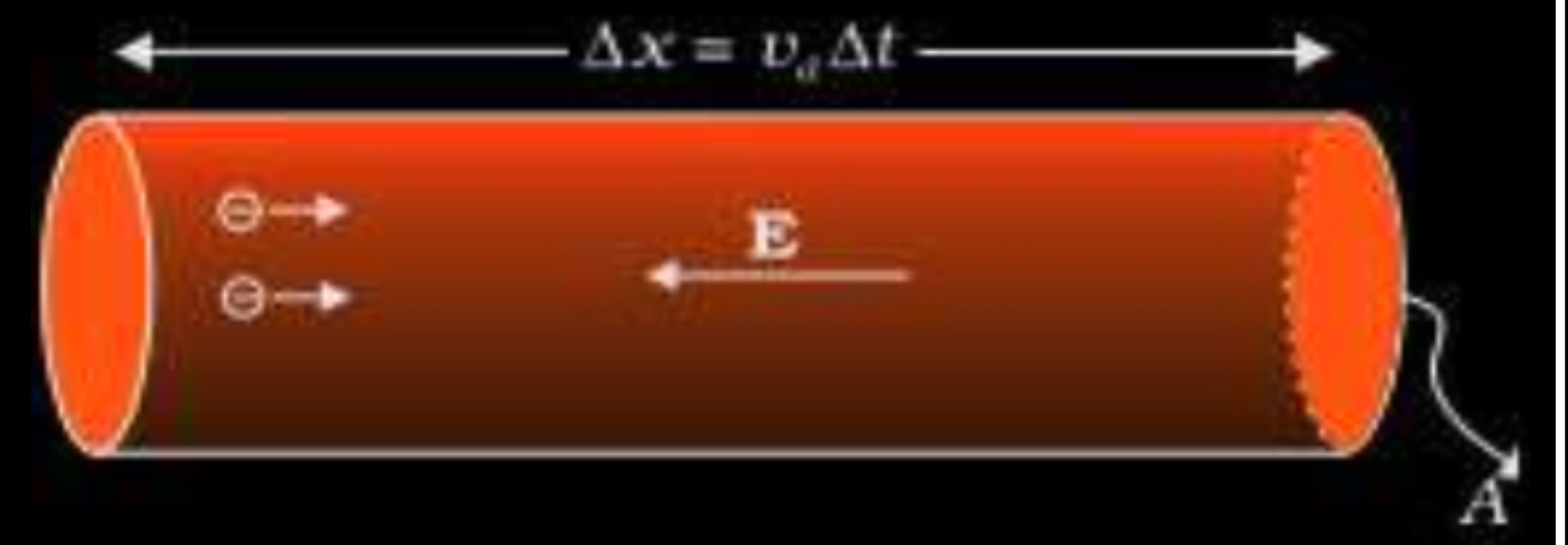
• Defining current density	½
• Whether scalar or vector	½
• Showing $\vec{j} = \alpha \vec{E}$	2

Defining Wheatstone bridge	1
Obtaining balancing conditions	2

Current density is the amount of charge flowing per second per unit area normal to the flow. ½
Alternatively:

$$j = \frac{I}{\text{A}}$$

It is a vector quantity. ½



The amount of charge crossing the area A in time Δt is I Δt, where I is the magnitude of the current. Hence,

$$\text{I}\,\Delta\,\text{t} = \text{ne A}\,|\text{v}_\text{d}|\,\Delta\text{t}$$ ½

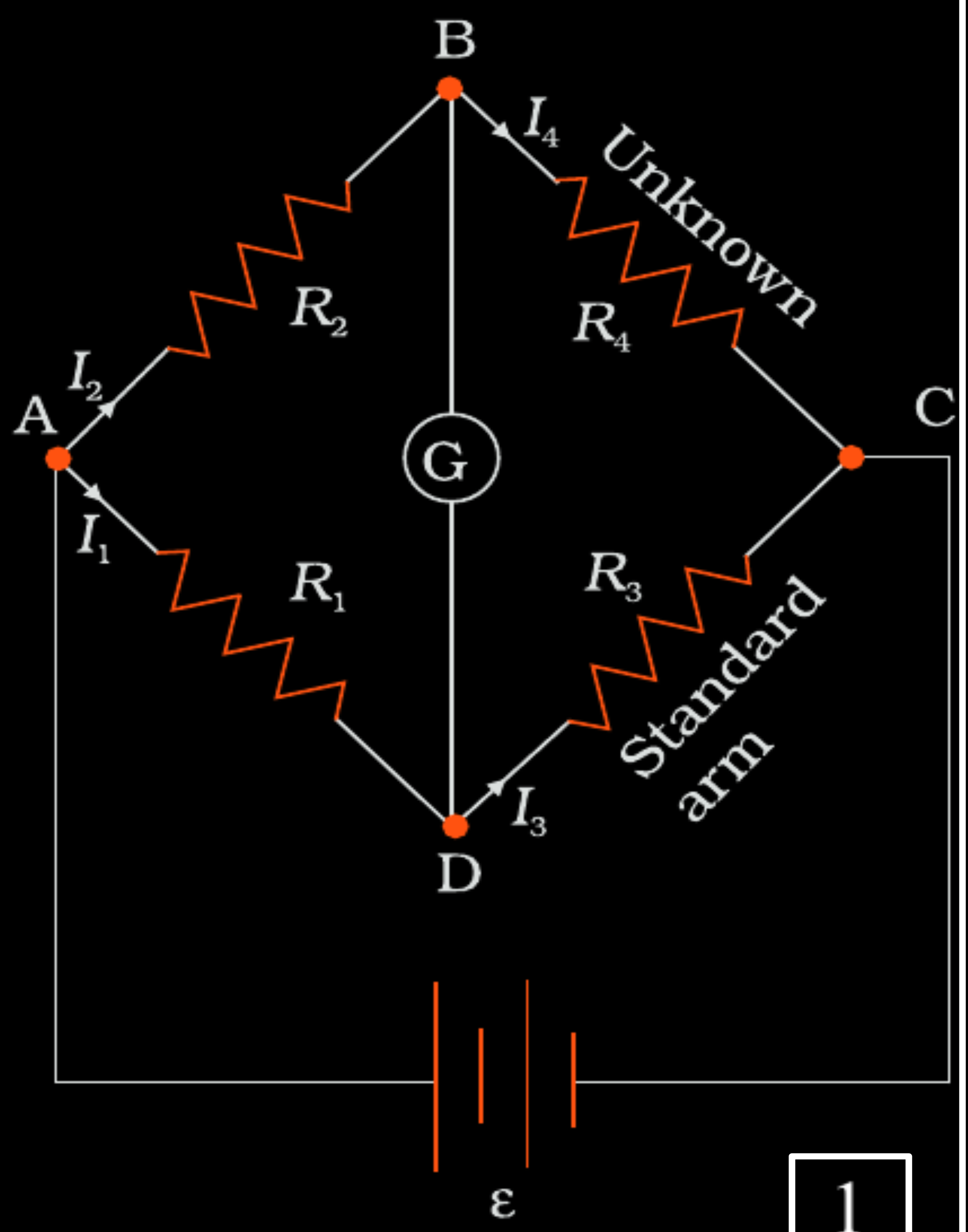
$$\text{I}\,\Delta\,\text{t} = \frac{e^2 A}{m}\,\tau\,\text{n}\,\Delta\text{t}\,|\text{E}|$$ ½

$$\text{I} = |\text{j}|\text{A}$$ ½

$$|\text{j}| = \frac{ne^2}{m}\,\tau\,|\text{E}|$$ ½

$$\vec{j} = \alpha\,\vec{E}$$

Alternatively:
If the figure is explained in words full credit to be given.



For loop ADBA:
$-I_1 R_1 + I_2 R_2 + I_g G = 0$ (1) ½

For loop CBDC:
$I_4 R_4 - I_3 R_3 - I_g G = 0$ (2) ½

For balanced wheatstone bridge, $I_g = 0$ ½
And by applying Kirchoff's junction rule to junction D and B,

$I_1 = I_3$ & $I_2 = I_4$

From eqn (1) and (2)

$$\frac{I_1}{I_2} = \frac{R_2}{R_1} \text{ and } \frac{I_1}{I_2} = \frac{R_4}{R_3}$$

$$\Rightarrow \frac{R_2}{R_1} = \frac{R_4}{R_3}$$ ½

3. A student is asked to connect four cells, each of emf E and internal resistance r, in series. But she/he connects one cell wrongly in series with the other cells. The equivalent emf and the equivalent internal resistance of the combination will be :

(A) 4E and 2r

(B) 4E and 3r

(C) 3E and 4r

(D) 2E and 4r

3. A student is asked to connect four cells, each of emf E and internal resistance r, in series. But she/he connects one cell wrongly in series with the other cells. The equivalent emf and the equivalent internal resistance of the combination will be :

(A) 4E and 2r

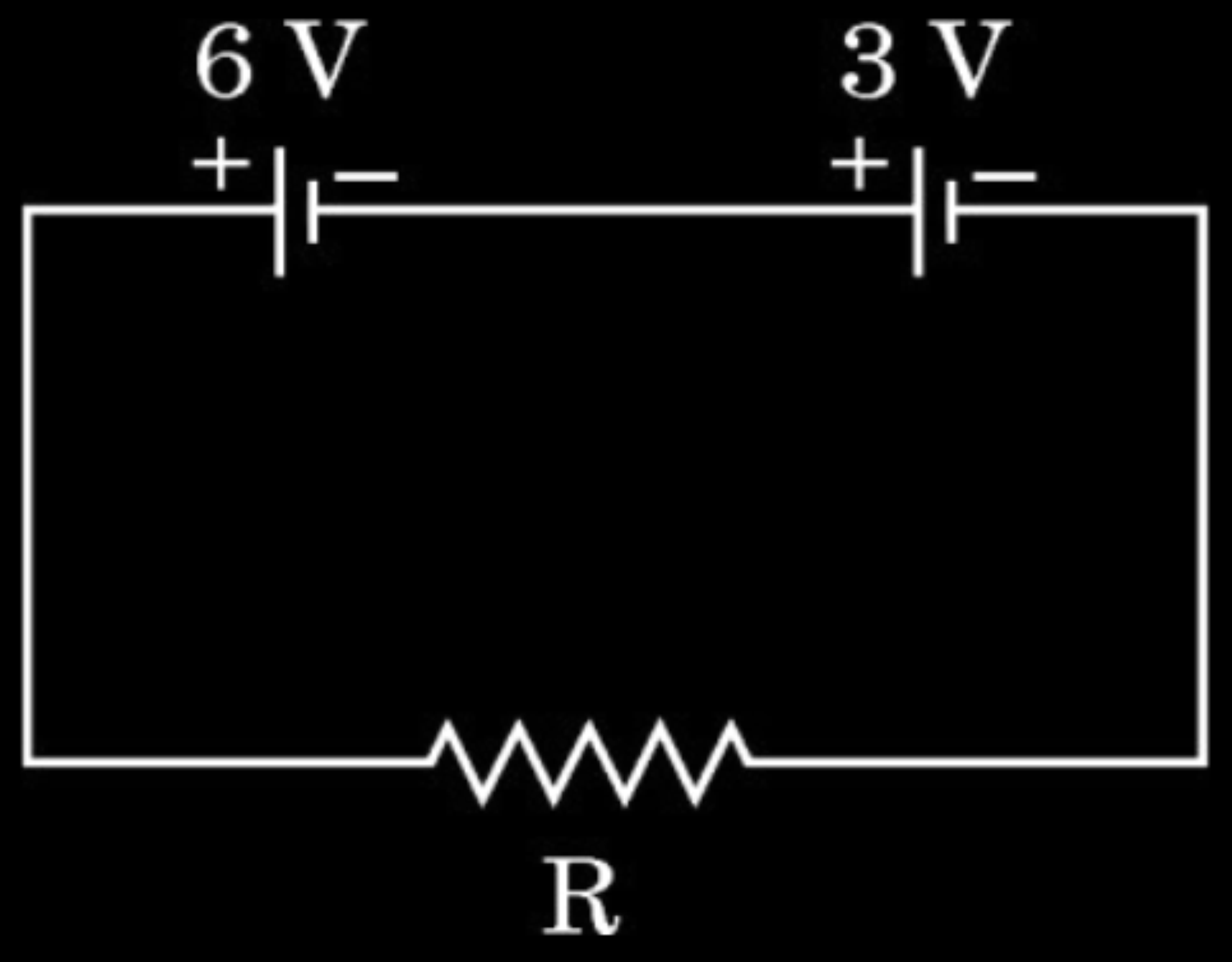
(B) 4E and 3r

(C) 3E and 4r

(D) 2E and 4r

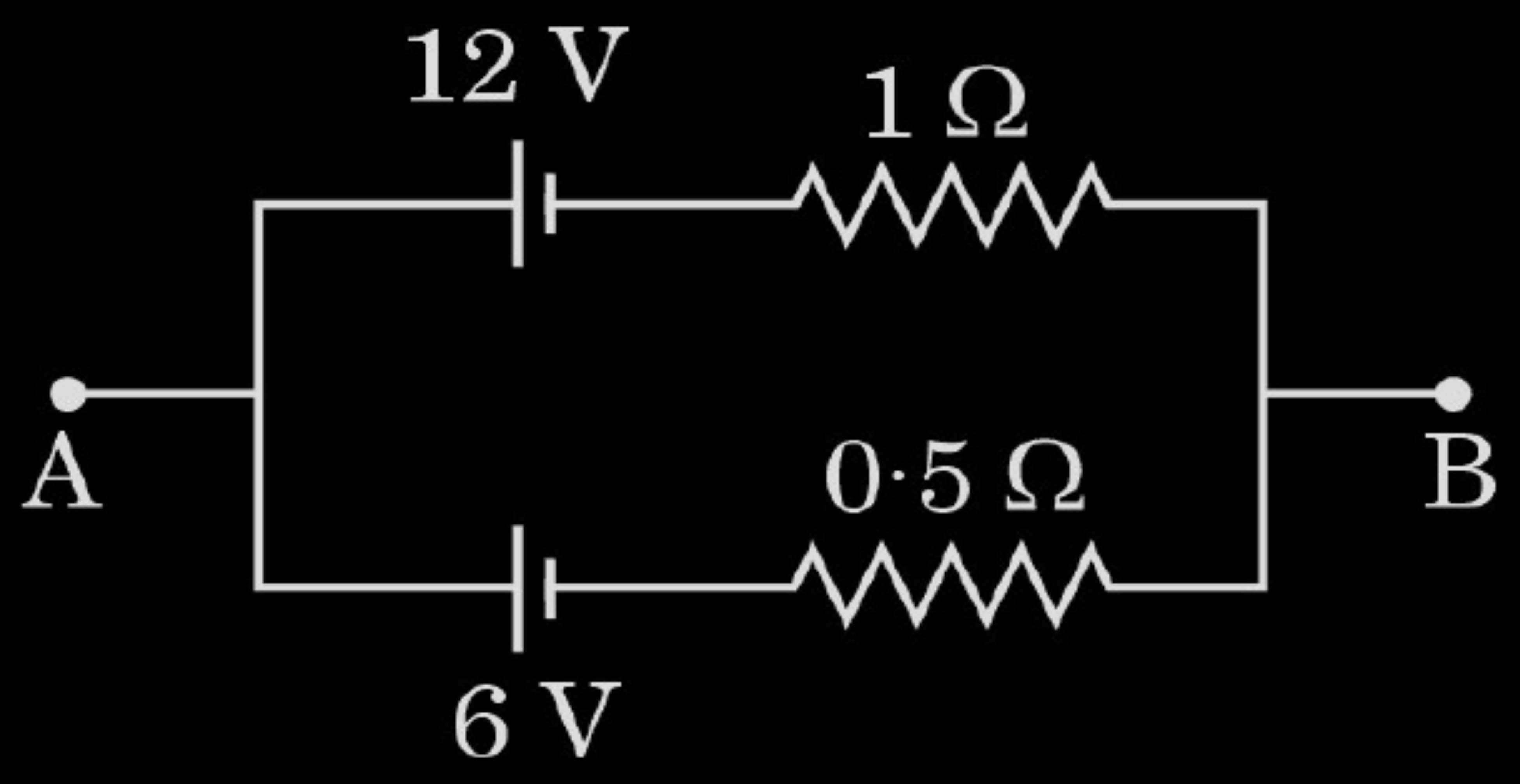
(D) 2E and 4r

17. Two batteries of emfs 6 V and 3 V and internal resistances 0·8 Ω and 0·2 Ω respectively are connected in series to an external resistance R, as shown in figure. Find the value of R so that the potential difference across the 6 V battery be zero. *2*



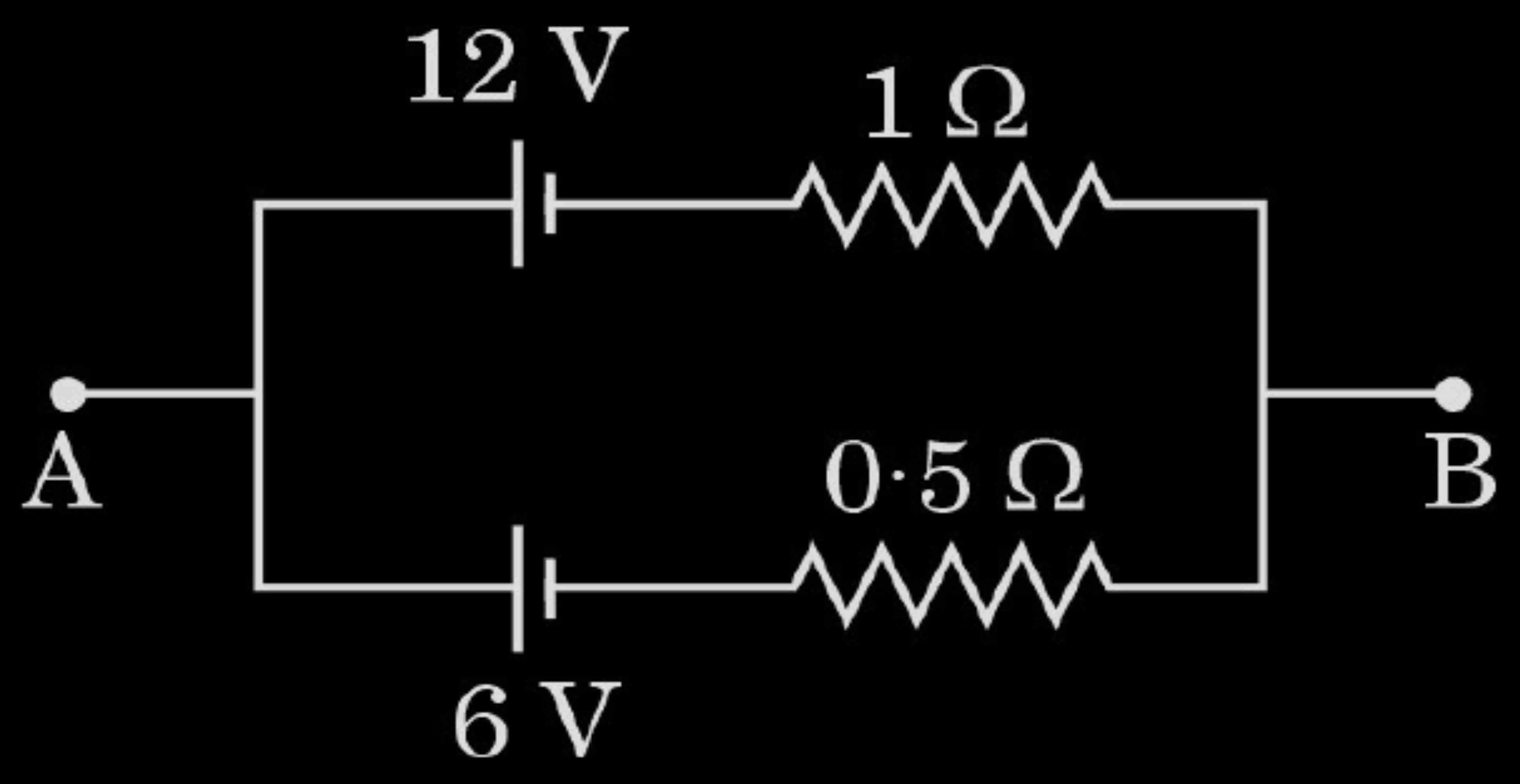
Finding the value of R 2		
$$i = \frac{9}{R+1}$$		½
As potential difference across 6V is zero; $6 - \text{ir} = 0 \Rightarrow 6 - \left(\frac{9}{R+1}\right)(0.8) = 0$		½
On solving; R= 0.2Ω		1

3. Consider the circuit shown in the figure. The potential difference between points A and B is :



(A) 6 V (B) 8 V (C) 9 V (D) 12 V

3. Consider the circuit shown in the figure. The potential difference between points A and B is :



(A) 6 V (B) 8 V (C) 9 V (D) 12 V

(B) 8V

17. Find the temperature at which the resistance of a wire made of silver will be twice its resistance at 20°C. Take 20°C as the reference temperature and temperature coefficient of resistance of silver at 20°C = $4 \cdot 0 \times 10^{-3}\ K^{-1}$. *2*

Finding the temperature 2	
$R = R_\circ\left[1+\alpha\left(T - T_\circ\right)\right]$	½
$R = 2R_\circ$ [Given]	
$2\ R_\circ = R_\circ\left[1+\alpha\left(T - T_\circ\right)\right]$	½
On solving $T = T_\circ + 250$	
$T = 270°C$ or $543\,K$	1

2. The emf and internal resistance of a cell are 3 V and $0 \cdot 2\ \Omega$ respectively. It is connected to an external resistor of $5 \cdot 8\ \Omega$. The potential difference across the cell will be :

(A) $1 \cdot 2$ V (B) $2 \cdot 0$ V

(C) $2 \cdot 9$ V (D) $3 \cdot 2$ V

2. The emf and internal resistance of a cell are 3 V and $0 \cdot 2\ \Omega$ respectively. It is connected to an external resistor of $5 \cdot 8\ \Omega$. The potential difference across the cell will be :

(A) $1 \cdot 2$ V (B) $2 \cdot 0$ V

(C) $2 \cdot 9$ V (D) $3 \cdot 2$ V

(C) 2.9 V

17. Two identical cells, each of emf E and internal resistance r, are connected in series with a resistor of 8 Ω. A current of 1·5 A flows in the circuit. When these cells are connected in parallel, they send 1·0 A current through the same resistor. Calculate the internal resistance, r and emf, E of the cell. *2*

Calculation of (i) Internal Resistance	1
(ii) EMF	1

<u>In Series Combination</u> $2E = I(2r+8)$ $= 1.5(2r+8)$ ....(1) <u>In Parallel Combination</u> $E = I\left(\frac{r}{2}+8\right)$ $E = 1\left(\frac{r}{2}+8\right)$ ....(2) Solving equation (1) & (2)	½ ½
$2 = \frac{1.5(2r+8)}{\left(\frac{r}{2}+8\right)}$ $r+16 = 3r+12$ $2r = 4$ $r = 2\Omega$ Substituting the value of r in equation(2) $E = \left(\frac{2}{2}+8\right)$ $E = 9V$	 ½ ½

22. (a) Define the terms – 'temperature coefficient of resistance' and 'electrical conductivity' of a conductor. Why are constantan and manganin used for making standard resistances ? Explain. *3*

OR

(b) Define 'drift velocity'. Show that resistivity of the material of a conductor is inversely proportional to the relaxation time for the free electrons in the conductor. *3*

Definition of (i) Temperature coefficient of resistance	1
(ii) Electrical Conductivity	1
Explanation of choosing constantan and manganin for making standard resistances.	1

Definition of Drift velocity	1
Showing resistivity is inversely proportional to relaxation time	2

Temperature coefficient of resistance is defined as the fractional change in resistance per unit increase in temperature. Electrical conductivity is defined as current density per unit electric field. Alternatively: Electrical conductivity is the reciprocal of electrical resistivity. <u>Reason</u>: Alloys have high resistivity Low temperature coefficient of resistance Ability to achieve more resistance value in small size (Any Two)	1 1 ½+ ½

Drift velocity is the average velocity acquired by electrons under the application of external electric field.	1
Current flowing through a conductor,	
$I = neAv_d$................(1)	½
$v_d = \frac{eE}{m}\tau$................(2)	½
Substituting (2) in (1)	
$I = \frac{ne^2EA}{m}\tau$	
$\frac{I}{A} = J = \frac{ne^2E}{m}\tau$	½
$\therefore \sigma = \frac{ne^2}{m}\tau$	
$\rho = \frac{1}{\sigma} = \frac{m}{ne^2\tau}$	½

19. (a) A cell is connected across an external resistance 12 Ω and supplies 0.25 A current. When the external resistance is increased by 4 Ω, the current reduces to 0.2 A. Calculate (i) the emf, and (ii) the internal resistance, of the cell. *2*

19. (a) A cell is connected across an external resistance 12 Ω and supplies 0.25 A current. When the external resistance is increased by 4 Ω, the current reduces to 0.2 A. Calculate (i) the emf, and (ii) the internal resistance, of the cell. *2*

(i) Calculating e.m.f of cell	1
(ii) Calculating internal resistance of cell	1

$$I = \frac{\varepsilon}{R+r}$$

$$0.25 = \frac{\varepsilon}{12+r} \qquad \text{------}(1)$$ ½

$$0.2 = \frac{\varepsilon}{16+r} \qquad \text{--------}(2)$$ ½

On solving eq (1) and (2)

$r = 4\Omega$ ½

$\varepsilon = 4\,\text{V}$ ½

15. *Assertion (A) :* The current density ($\vec{J}$) at a point in a conducting wire is in the direction of electric field ($\vec{E}$) at that point.

Reason (R) : A conducting wire obeys Ohm's law.

15. *Assertion (A) :* The current density ($\vec{J}$) at a point in a conducting wire is in the direction of electric field ($\vec{E}$) at that point.

Reason (R) : A conducting wire obeys Ohm's law.

(B) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of the Assertion (A).

28. (a) The radius of a conducting wire AB uniformly decreases from its one end A to another end B. It is connected across a battery. How will (i) electric field, (ii) current density, and (iii) mobility of electrons change from end A to end B ? Justify your answer in each case. *3*

28. (a) The radius of a conducting wire AB uniformly decreases from its one end A to another end B. It is connected across a battery. How will (i) electric field, (ii) current density, and (iii) mobility of electrons change from end A to end B ? Justify your answer in each case. *3*

(i) Variation of electric field and justification	½ + ½
(ii) Variation of current density and justification	½ + ½
(iii) Variation of mobility of electrons and justification	½ + ½

With the decrease in area of cross-section.

(i) $E = \frac{I}{A}\rho$, electric field increases	½+½
(ii) $j = \frac{I}{A}$, current density increases	½+½
(iii) $\mu_e = \frac{e\tau}{m}$, mobility remains same	½+½

22. (a) Define mobility of free electrons in a conductor. How does the mobility of electrons in a metal change when :

(i) temperature of the conductor is decreased at constant potential difference ?

(ii) applied potential difference is doubled at a constant temperature ? *3*

OR

(b) Obtain an expression for emf and internal resistance of a cell which is equivalent to parallel combination of two cells of emf and internal resistance (E_1, r_1) and (E_2, r_2), respectively. *3*

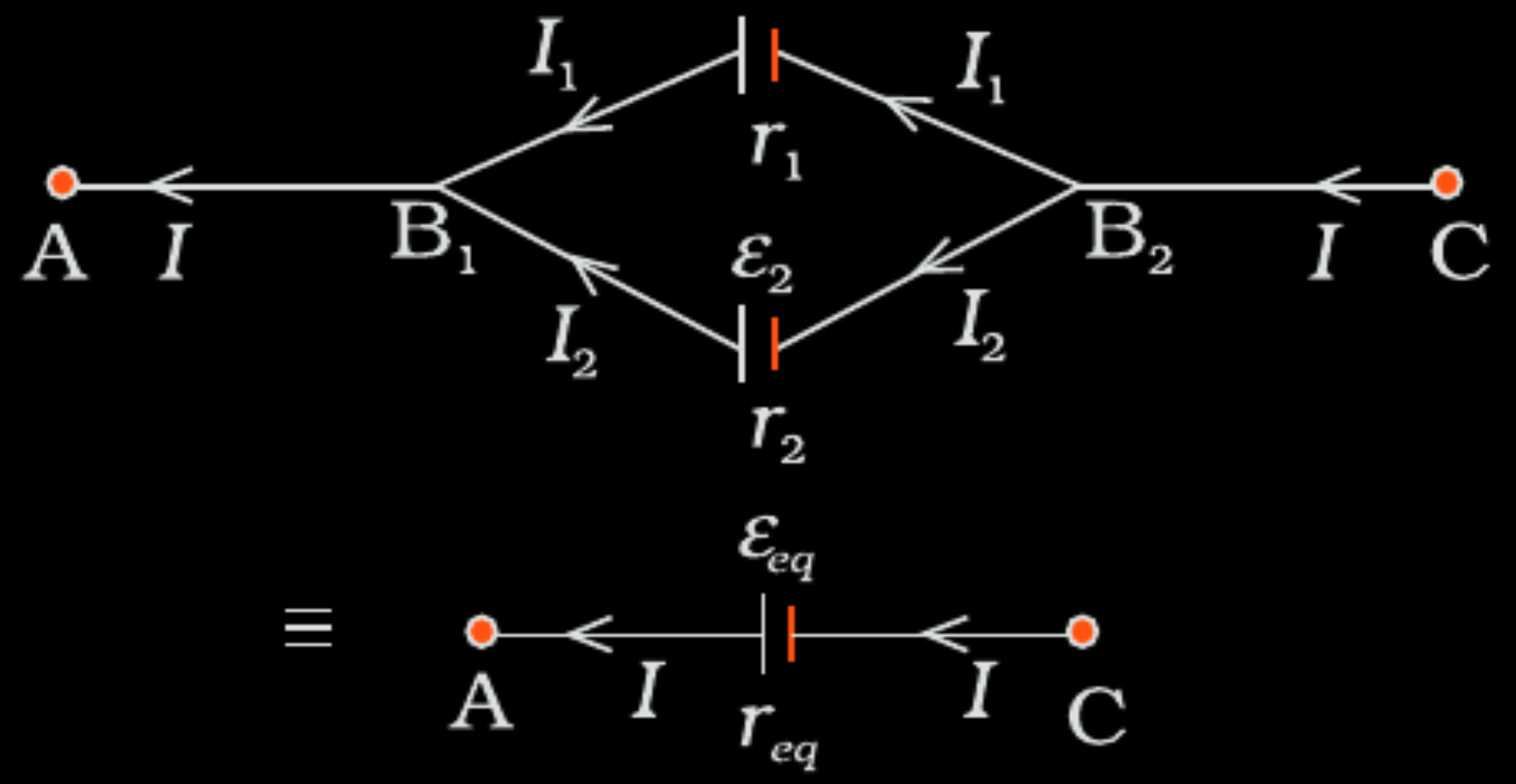
•	Definition of mobility	1
•	Change in mobility when	
(i)	Temperature of conductor is decreased	1
(ii)	Applied voltage is doubled	1

Obtaining expression for equivalent emf	2
Equivalent resistance	1

Mobility of free electrons in a conductor is defined as the magnitude of the drift velocity per unit electric field. Alternatively:- $\mu = \frac{\lvert \vec{v}_d \rvert}{E}$	1
(i) When the temperature of the conductor is decreased at constant potential difference the mobility will increase.	1
(ii) When the applied potential difference is doubled at a constant temperature the mobility will be unchanged. Alternatively:- (i) $\mu = \frac{e\tau}{m}$ As temperature decreases τ will increase and μ will also increase. (ii) $\mu = \frac{e\tau}{m}$ As μ is independent of applied voltage hence μ remains unchanged.	1

When two cells are connected in parallel the total current

$I = I_1 + I_2$ --------------(1)



$$I = \frac{\varepsilon_1 - V}{r_1} + \frac{\varepsilon_2 - V}{r_2}$$ ½

$$I = \left(\frac{\varepsilon_1}{r_1} + \frac{\varepsilon_2}{r_2}\right) - V\left(\frac{1}{r_1} + \frac{1}{r_2}\right)$$ ½

$$V = \frac{\varepsilon_1 r_2 + \varepsilon_2 r_1}{r_1 + r_2} - I\left(\frac{r_1 r_2}{r_1 + r_2}\right)$$ ½

Hence

$$\varepsilon_{eq} = \frac{\varepsilon_1 r_2 + \varepsilon_2 r_1}{r_1 + r_2}$$ ½

$$r_{eq} = \frac{r_1 r_2}{r_1 + r_2}$$ ½

½

17. Two conducting wires of same material have radii r and $\frac{r}{2}$. The current flowing through them is I and 2I, respectively. Find the ratio of drift velocity of free electrons in the wires. *2*

17. Two conducting wires of same material have radii r and $\frac{r}{2}$. The current flowing through them is I and 2I, respectively. Find the ratio of drift velocity of free electrons in the wires. *2*

For the first wire $v_d = \frac{I}{enA}$ $v_d = \frac{I}{en\pi r^2}$ ----------------------------(1)	½
For the second wire $v'_d = \frac{2I}{\pi en\left(\frac{r^2}{4}\right)} = \frac{8I}{\pi enr^2}$ ---------(2)	½
From eq (1) and (2) $\frac{v'_d}{v_d} = \frac{8}{1}$	1

3. A steady current I is passed through a conductor at room temperature for time t. It is observed that its temperature rises by 0·5°C. If 2I current is passed through the conductor (at room temperature) for the same duration, the rise in its temperature will be approximately :

(A) 1·0°C

(B) 1·5°C

(C) 2·0°C

(D) 4·0°C

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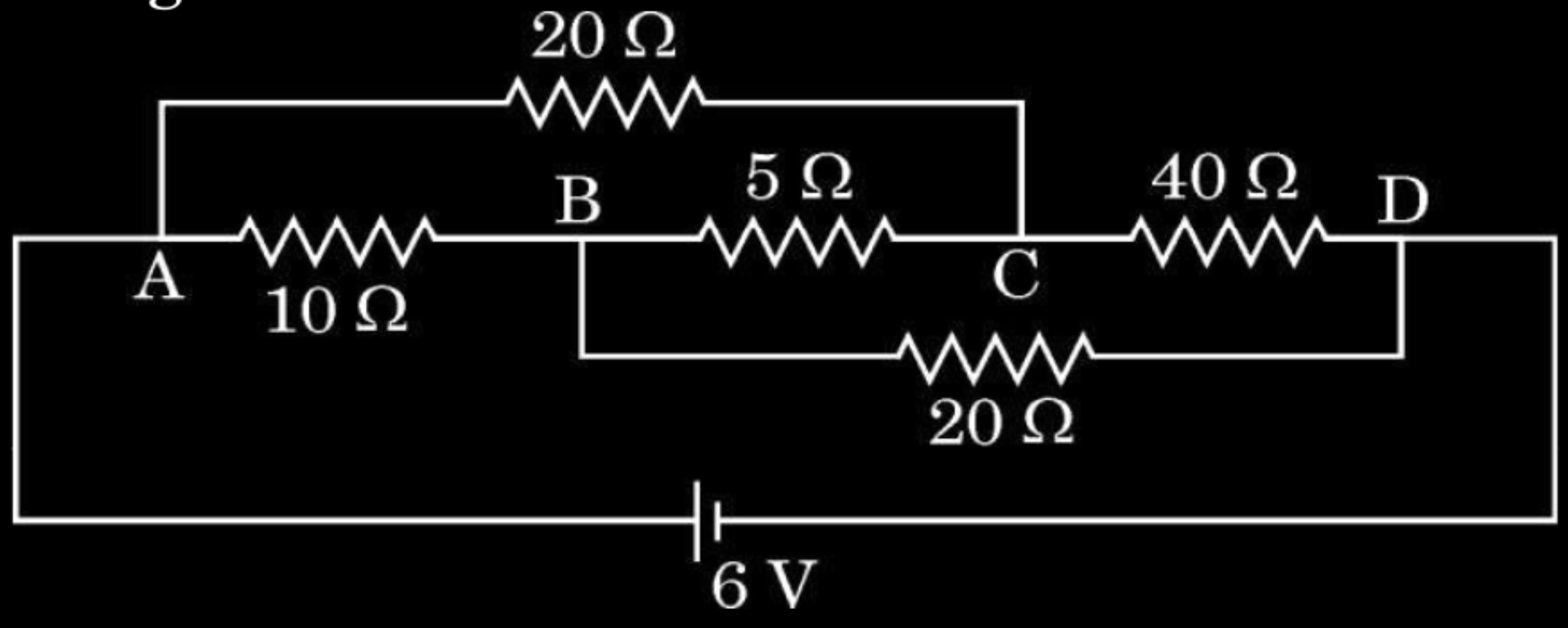
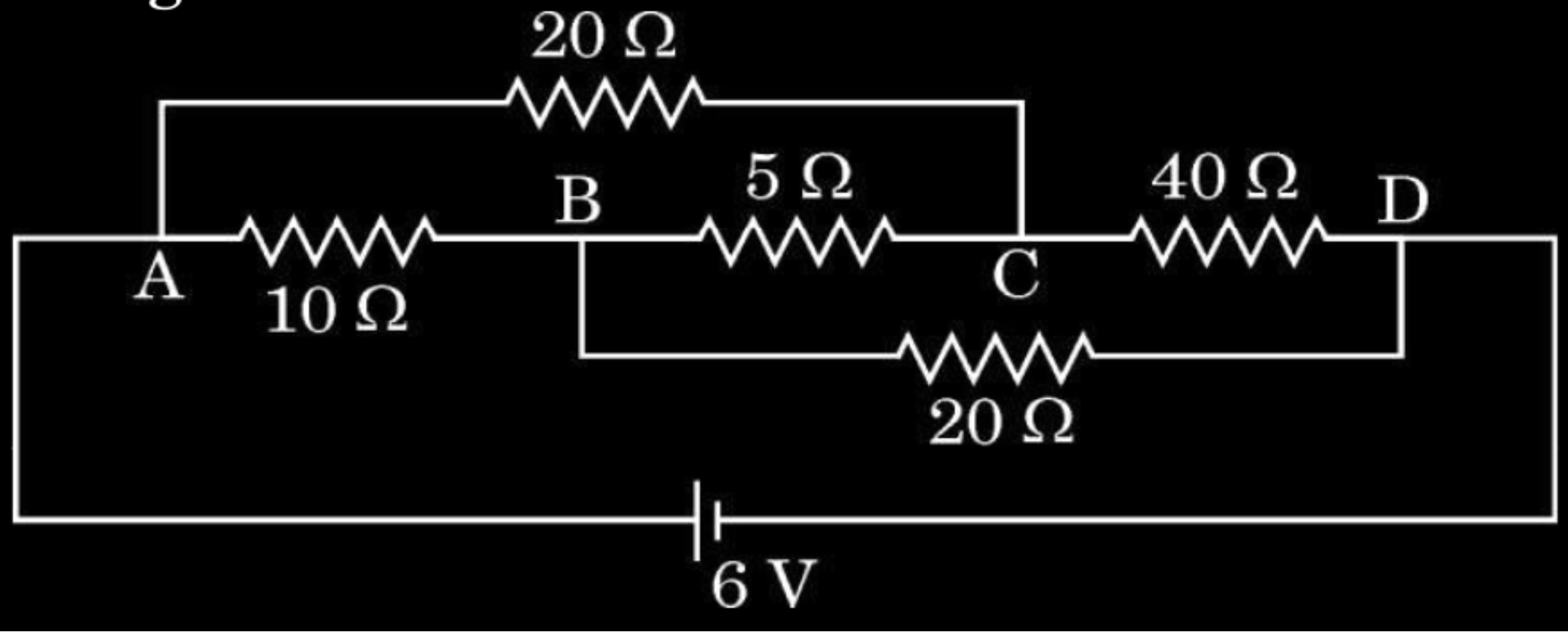
(A) 1·0°C
(B) 1·5°C
(C) 2·0°C
(D) 4·0°C

(C) 2.0 ^{0}C

17. Calculate the value of the current passing through the battery in the given circuit diagram. *2*



17. Calculate the value of the current passing through the battery in the given circuit diagram. *2*



The given circuit forms a balanced Wheatstone bridge so no current flows through 5Ω resistor.	½
$R_{eq} = \frac{30\times 60}{30+60}$	½
$= 20\Omega$	
$I = \frac{V}{R_{eq}}$	½
$= \frac{6}{20} = \frac{3}{10} A$	½

29. In a metallic conductor, an electron, moving due to thermal motion, suffers collisions with the heavy fixed ions but after collision, it will emerge out with the same speed but in random directions. If we consider all the electrons, their average velocity will be zero. When an electric field is applied, electrons move with an average velocity, known as drift velocity (v_d). The average time between successive collisions is known as relaxation time (τ). The magnitude of drift velocity per unit electric field is called mobility (μ).

An expression for current through the conductor can be obtained in terms of drift velocity, number of electrons per unit volume (n), electronic charge (– e), and the cross-sectional area (A) of the conductor. This expression leads to an expression between current density ($\hat{j}$) and the electric field ($\vec{E}$). Hence, an expression for resistivity (ρ) of a metal is obtained. This expression helps us to understand increase in resistivity of a metal with increase in its temperature, in terms of change in the relaxation time (τ) and change in the number density of electrons (n).

(i) (a) Consider the contribution of the following two factors I and II in resistivity of a metal :

I. Relaxation time of electrons

II. Number of electrons per unit volume

The resistivity of a metal increases with increase in its temperature because :

(A) I decreases and II increases.

(B) I increases and II is almost constant.

(C) Both I and II increase.

(D) I decreases and II is almost constant.

OR

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The resistivity of a metal increases with increase in its temperature because :

(A) I decreases and II increases.

(B) I increases and II is almost constant.

(C) Both I and II increase.

$\boxed{\text{(D) I decreases and II is almost constant.}}$

OR

(b) A steady current flows in a copper wire of non-uniform cross-section. Consider the following three physical quantities :

I. Electric field

II. Current density

III. Drift speed

Then at the different points along the wire : *1*

(A) II and III change, but I is constant.

(B) I and II change, but III is constant.

(C) I and III change, but II is constant.

(D) All I, II and III change.

(b) A steady current flows in a copper wire of non-uniform cross-section. Consider the following three physical quantities :

I. Electric field

II. Current density

III. Drift speed

Then at the different points along the wire : *1*

(A) II and III change, but I is constant.

(B) I and II change, but III is constant.

(C) I and III change, but II is constant.

$\boxed{\text{(D) All I, II and III change.}}$

(ii) The temperature coefficient of resistance of nichrome is $1{\cdot}70 \times 10^{-4}\ {}^{\circ}C^{-1}$. In order to increase resistance of a nichrome wire by $8{\cdot}5\%$, the temperature of the wire should be increased by : *1*

(A) 250°C

(B) 500°C

(C) 850°C

(D) 1000°C

(ii) The temperature coefficient of resistance of nichrome is $1 \cdot 70 \times 10^{-4}\ {}^\circ C^{-1}$. In order to increase resistance of a nichrome wire by $8 \cdot 5\%$, the temperature of the wire should be increased by : *1*

(A) $250^\circ C$

$\boxed{\text{(B)} \quad 500^\circ C}$

(C) $850^\circ C$

(D) $1000^\circ C$

(iii) A wire of length 0·5 m and cross-sectional area $1{\cdot}0 \times 10^{-7}$ m^2 is connected to a battery of 2 V that maintains a current of 1·5 A in it. The conductivity of the material of the wire (in Ω^{-1} m^{-1}) is : *1*

(A) $2{\cdot}5 \times 10^4$

(B) $3{\cdot}0 \times 10^5$

(C) $3{\cdot}75 \times 10^6$

(D) $5{\cdot}0 \times 10^7$

(iii) A wire of length 0·5 m and cross-sectional area $1 \cdot 0 \times 10^{-7}$ m^2 is connected to a battery of 2 V that maintains a current of 1·5 A in it. The conductivity of the material of the wire (in Ω^{-1} m^{-1}) is : *1*

(A) $2 \cdot 5 \times 10^4$

(B) $3 \cdot 0 \times 10^5$

$\boxed{\text{(C)} \quad 3 \cdot 75 \times 10^6}$

(D) $5 \cdot 0 \times 10^7$

(iv) Consider two cylindrical conductors A and B, made of the same metal connected in series to a battery. The length and the radius of B are twice that of A. If μ_A and μ_B are the mobility of electrons in A and B respectively, then $\frac{\mu_A}{\mu_B}$ is : *1*

(A) $\frac{1}{2}$

(B) $\frac{1}{4}$

(C) 2

(D) 1

(iv) Consider two cylindrical conductors A and B, made of the same metal connected in series to a battery. The length and the radius of B are twice that of A. If μ_A and μ_B are the mobility of electrons in A and B respectively, then $\frac{\mu_A}{\mu_B}$ is : *1*

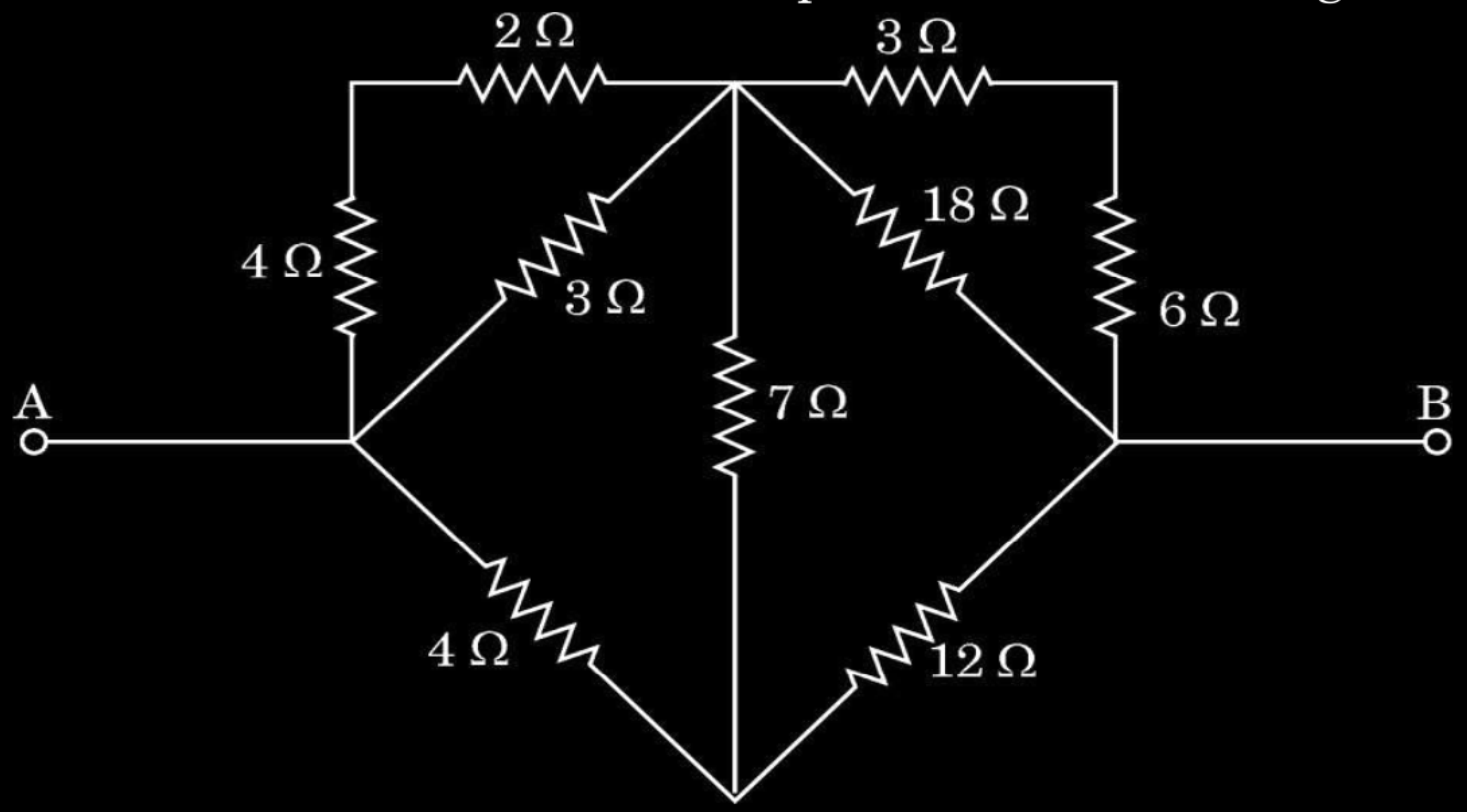
(A) $\frac{1}{2}$

(B) $\frac{1}{4}$

(C) 2

(D) 1

7. The effective resistance between points A and B in the given circuit is :



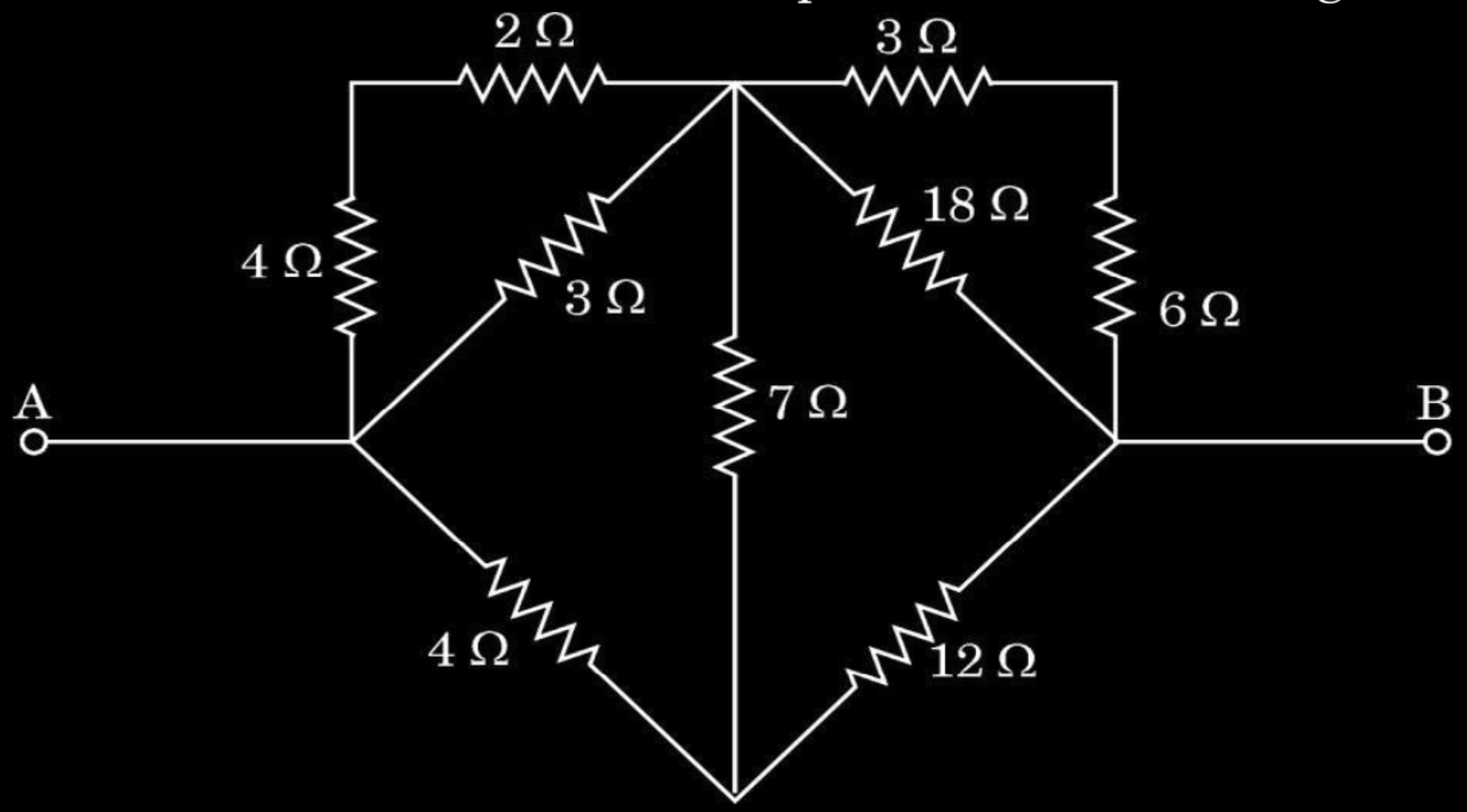
(A) 6 Ω

(B) $\frac{8}{3}$ Ω

(C) $\frac{16}{3}$ Ω

(D) 2 Ω

7. The effective resistance between points A and B in the given circuit is :



(A) 6 Ω

(B) $\frac{8}{3}$ Ω

(C) $\boxed{\frac{16}{3}\ \Omega}$

(D) 2 Ω

17. Two wires made of the same material have the same length (l) but different cross-sectional areas A_1 and A_2. They are connected together with a cell of voltage V. Find the ratio of the drift velocities of free electrons in the two wires when they are joined in (i) series, and (ii) parallel. *2*

17. Two wires made of the same material have the same length (l) but different cross-sectional areas A_1 and A_2. They are connected together with a cell of voltage V. Find the ratio of the drift velocities of free electrons in the two wires when they are joined in (i) series, and (ii) parallel. *2*

(i) Series	
$I = neAv_d$	½
$\frac{v_{d_1}}{v_{d_2}} = \frac{A_2}{A_1}$	½
(ii) Parallel	
$v_d = \frac{V}{RneA} = \frac{V}{\rho l ne}$	½
$\frac{v_{d_1}}{v_{d_2}} = \frac{1}{1}$	½

17. A cell of emf E and internal resistance r is connected across a resistor of variable resistance R. Show graphically the variation of

(a) the terminal voltage across the cell,

(b) the current supplied by the cell,

with R as it is increased from 0 to the maximum value. *2*

17. A cell of emf E and internal resistance r is connected across a resistor of variable resistance R. Show graphically the variation of

(a) the terminal voltage across the cell,

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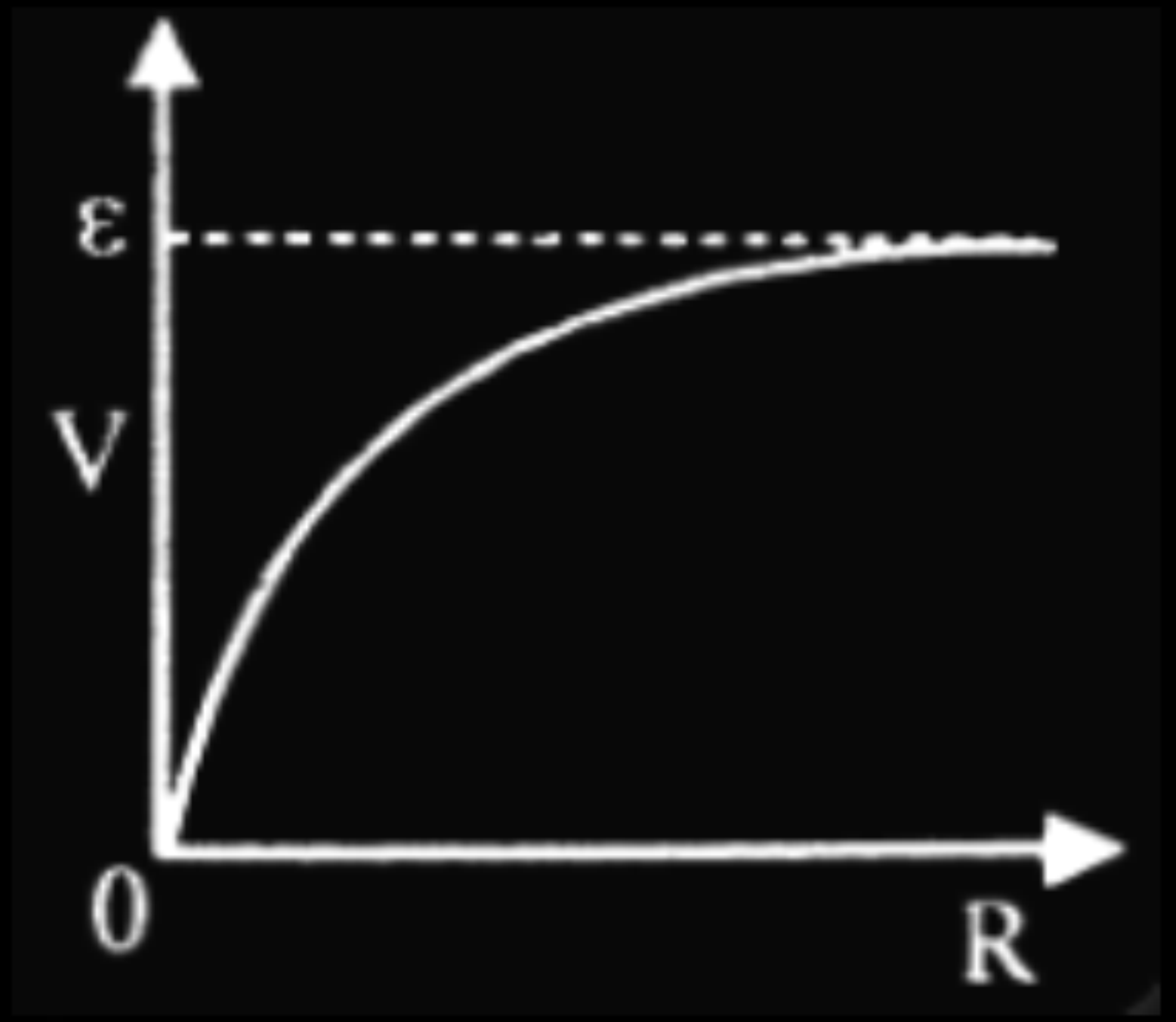
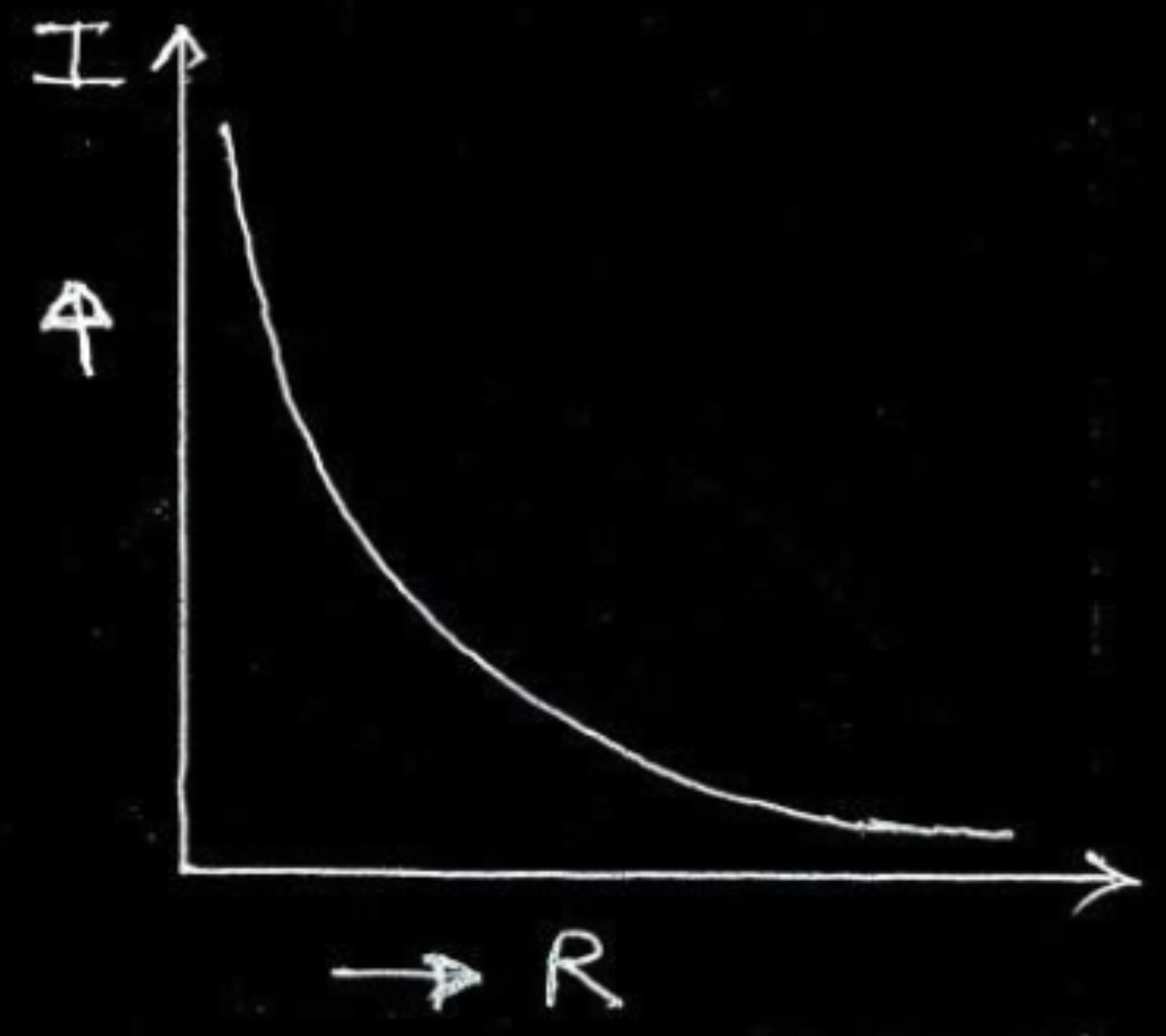
with R as it is increased from 0 to the maximum value. *2*

(a)



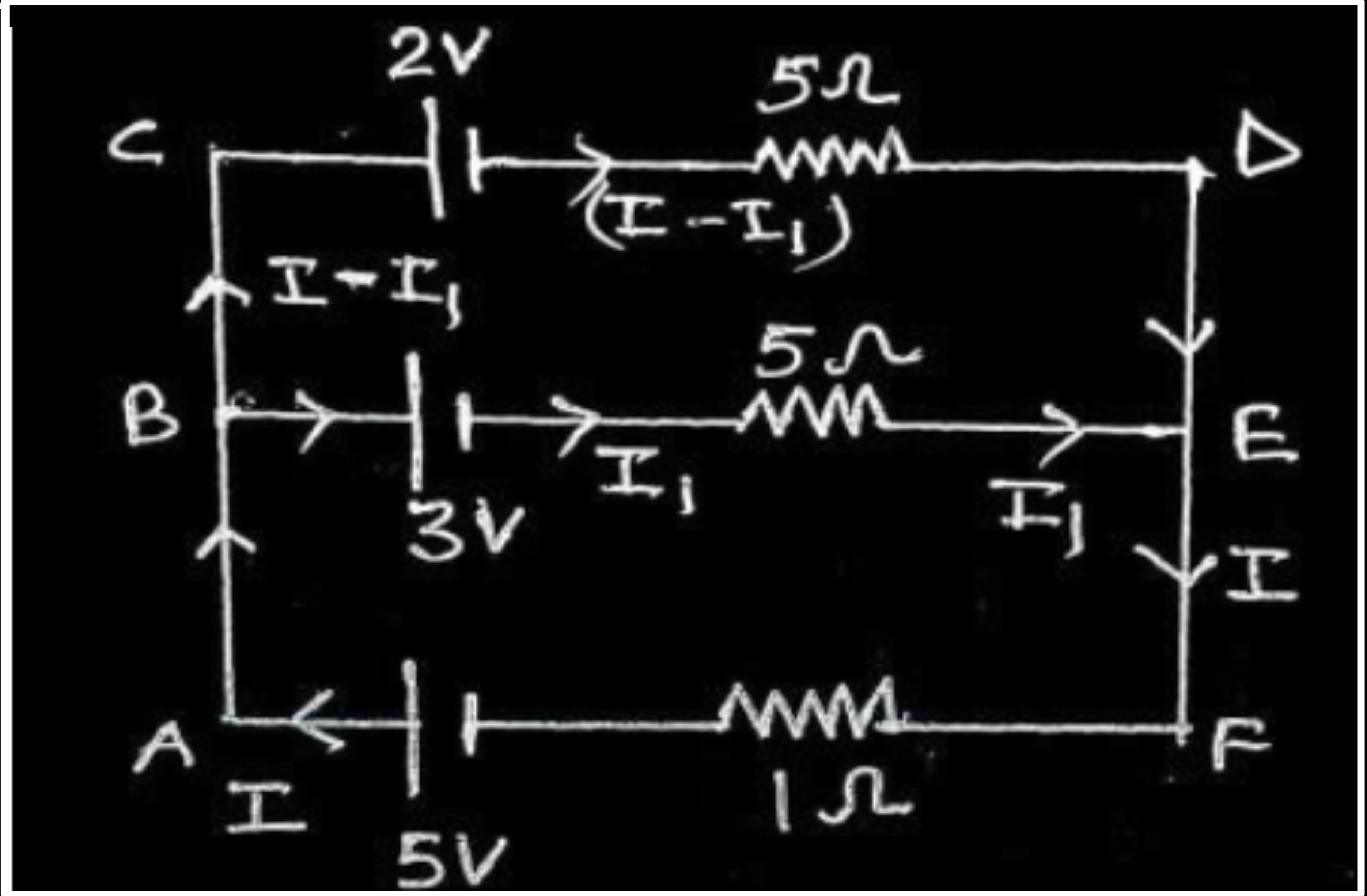
(b)



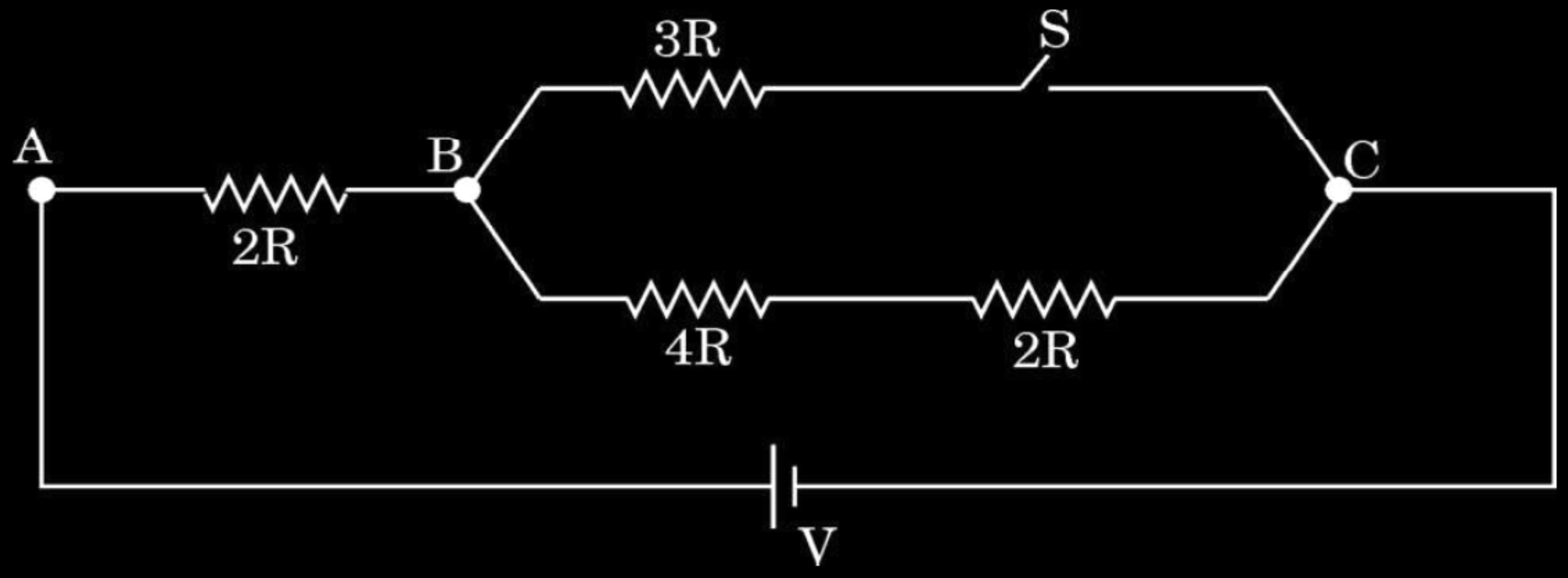
22. Three cells A, B and C of emfs 2 V, 3 V and 5 V respectively are connected in parallel to each other. Their internal resistances are 5 Ω, 5 Ω and 1 Ω respectively. Calculate the currents flowing through the cells A, B and C. *3*

22. Three cells A, B and C of emfs 2 V, 3 V and 5 V respectively are connected in parallel to each other. Their internal resistances are 5 Ω, 5 Ω and 1 Ω respectively. Calculate the currents flowing through the cells A, B and C. *3*

In loop ABEFA, $5-3-5I_1-I=0$	½
$2=5I_1+I \qquad ------(1)$	
In loop CBEDC, $2-3-5I_1+5I-5I_1=0$	½
$-1=10I_1-5I \qquad ------(2)$	
Solving equation (1) and (2)	
$I=\frac{5}{7}A$ in arm AF/through the cell of 5V(A)	½
$I_1=\frac{9}{35}A$ in arm BE/through the cell of 3V(B)	½
$I-I_1=\frac{16}{35}A$ in arm CD/through the cell of 2V(C)	½



1. The ratio of potential difference across AB in the circuit shown for the case (i) when switch S is closed and (ii) when S is open is :



(A) $\frac{1}{4}$

(B) $\frac{1}{2}$

(C) 1

(D) 2

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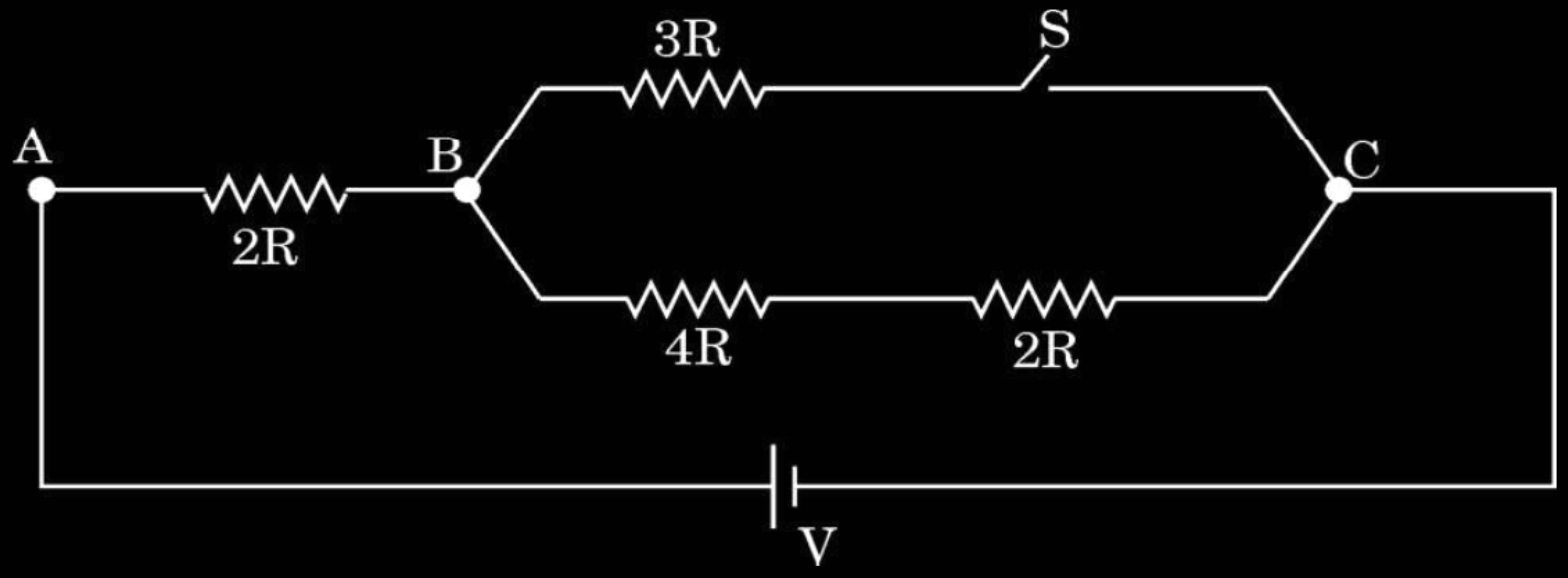

(A) $\frac{1}{4}$

(B) $\frac{1}{2}$

(C) 1

(D) $\boxed{2}$

18. (a) What is the difference between 'velocity' and 'drift velocity' of electrons in a current-carrying conductor.

(b) A copper wire of uniform cross-sectional area carries a current of 3·4 A. The drift velocity of conduction electrons is 0·2 mm/s. If the number density of electrons in copper is $8{\cdot}5 \times 10^{28}$ m^{-3}, find the area of cross-section of the wire. *2*

18. (a) What is the difference between 'velocity' and 'drift velocity' of electrons in a current-carrying conductor.

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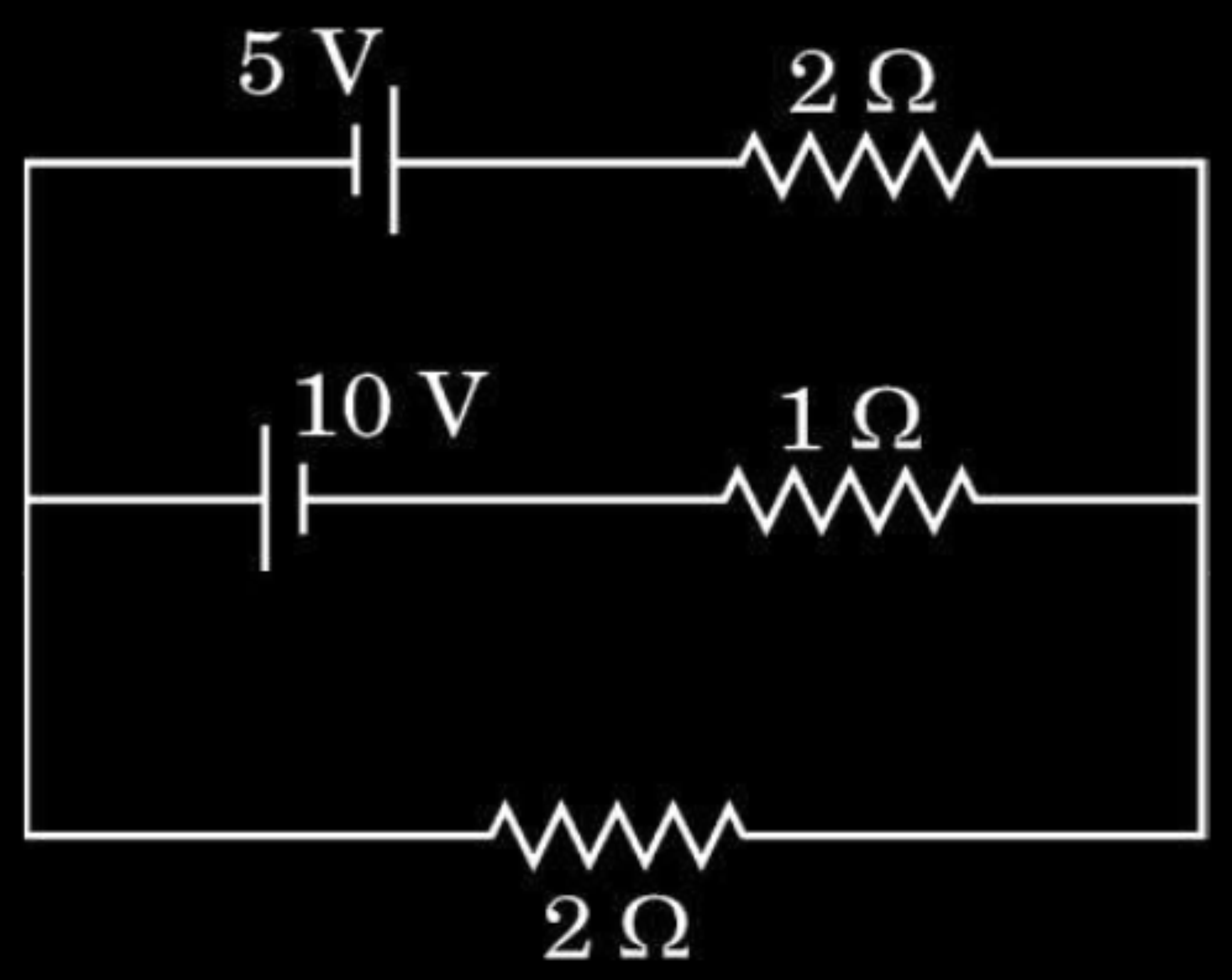
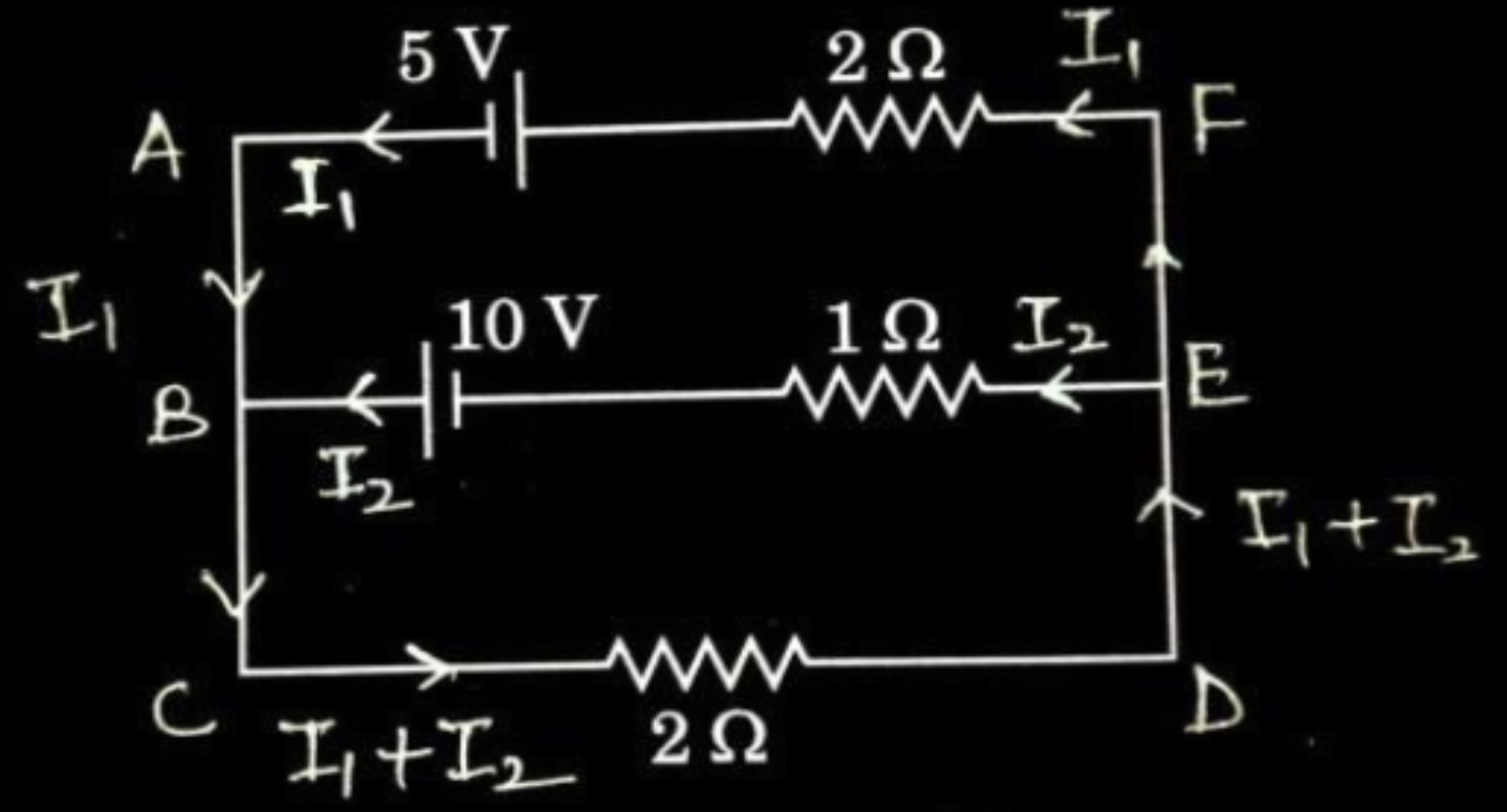
(a) **Velocity** is the rate of change of displacement with time.	½
Drift velocity is the average velocity of free electrons under the influence of external electric field.	½
(b) $I = neAv_d$	½
$A = \dfrac{I}{nev_d}$	
$A = \dfrac{3.4}{8.5 \times 10^{28} \times 1.6 \times 10^{-19} \times 0.2 \times 10^{-3}}$	
$A = 1.25 \times 10^{-6}\ m^2$	½

26. State Kirchhoff's laws. Apply these laws to find the values of current flowing in the three branches of the given circuit. *3*



Stating Kirchoff's laws	½ + ½
Finding the values of current in all the three branches	2

Junction rule: At any junction, the sum of the currents entering the junction is equal to the sum of the currents leaving the junction.	½
Loop rule: the algebraic sum of changes in potential around any closed loop involving resistors and cells in the loop is zero.	½
In closed loop $BCDEB$ $\quad 2I_1+3I_2=10 \quad$ ------(1) In closed loop $ABCDEFA$ $\quad 4I_1+2I_2=-5 \quad$ ------(2)	½
On solving eq. (1) and (2) $I_1=-\frac{35}{8}\text{A} \quad in\, arm\, AF$	½
$I_2=\frac{25}{4}\text{A} \quad in\, arm\, BE$	½
$I_1+I_2=-\frac{15}{8}\text{A} \quad in\, arm\, CD$  	½

1. A wire of resistance R, connected to an ideal battery consumes a power P. If the wire is gradually stretched to double its initial length, and connected across the same battery, the power consumed will be :

(A) $\frac{P}{4}$

(B) $\frac{P}{2}$

(C) P

(D) 2P

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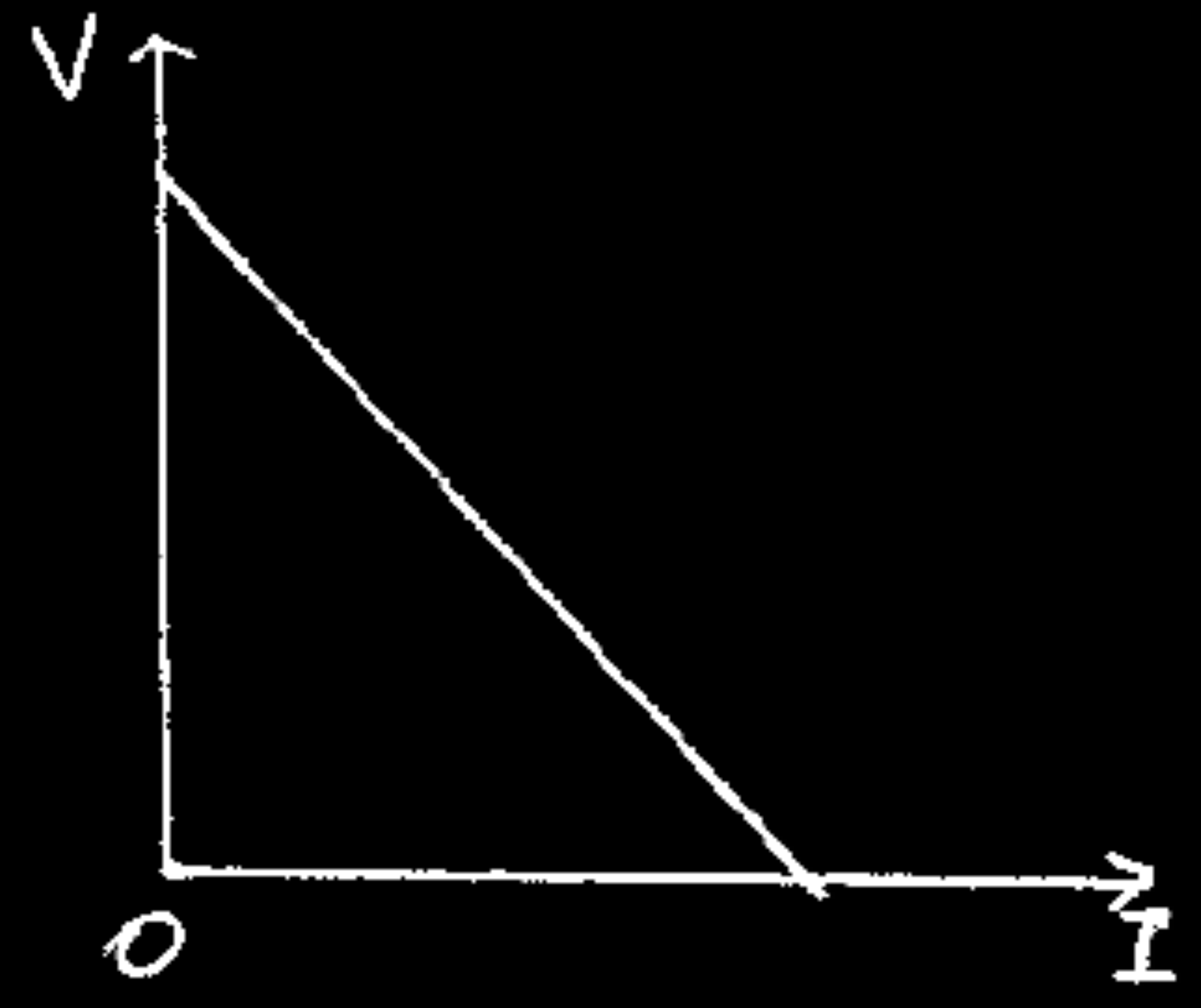
(C) P

(D) 2P

(A) $\frac{P}{4}$

17. A cell of emf E and internal resistance r is connected to an external variable resistance R. Plot a graph showing the variation of terminal voltage V of the cell as a function of current I, supplied by the cell. Explain how the emf of the cell and its internal resistance can be found from it. *2*

17. A cell of emf E and internal resistance r is connected to an external variable resistance R. Plot a graph showing the variation of terminal voltage V of the cell as a function of current I, supplied by the cell. Explain how the emf of the cell and its internal resistance can be found from it. *2*



1

$V = E - Ir$

E = intercept on y-axis (i.e. V-axis) ½

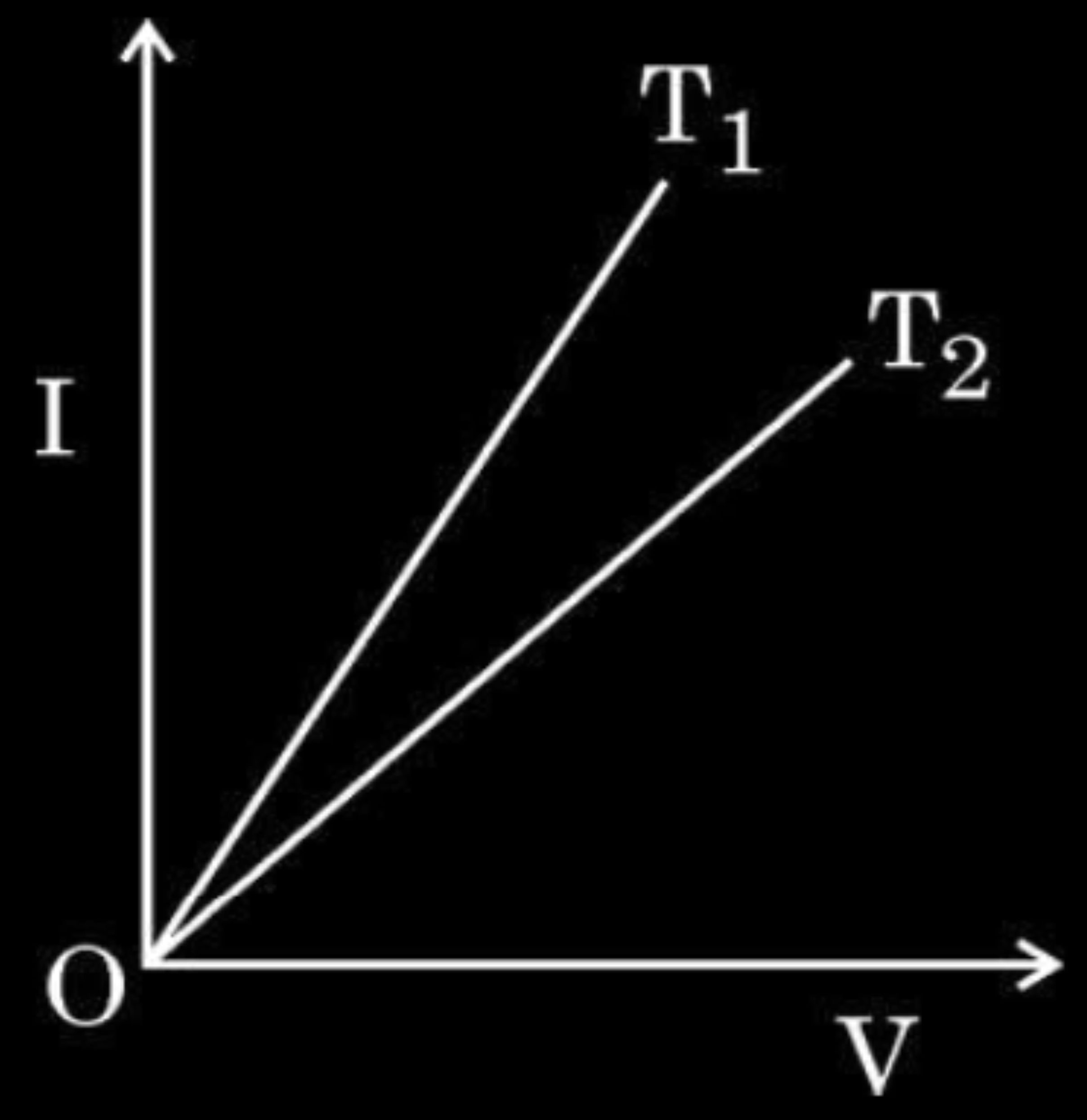
r = |slope of graph| ½

22. (a) Define the term 'drift velocity' of conduction electrons in a conductor.

(b) A conductor of length l and area of cross-section A is connected across an ideal battery of emf V. Derive the formula for the current density in terms of relaxation time τ. *3*

(a) Drift velocity is the average velocity with which the free electrons move under external electric field in a conductor.	1
(b) Current density	
$j = \dfrac{I}{A}$	½
$\because I = neA\mathrm{v}_d$	½
$\therefore j = ne\mathrm{v}_d$	
$But\ \mathrm{v}_d = \dfrac{eE\tau}{m}$	½
$\therefore j = \dfrac{ne^2\tau E}{m} = \dfrac{ne^2\tau}{m}\left(\dfrac{V}{l}\right)$	½

1. The figure shows the voltage (V) versus the current (I) graphs for a wire at two temperatures T_1 and T_2. One can conclude that :



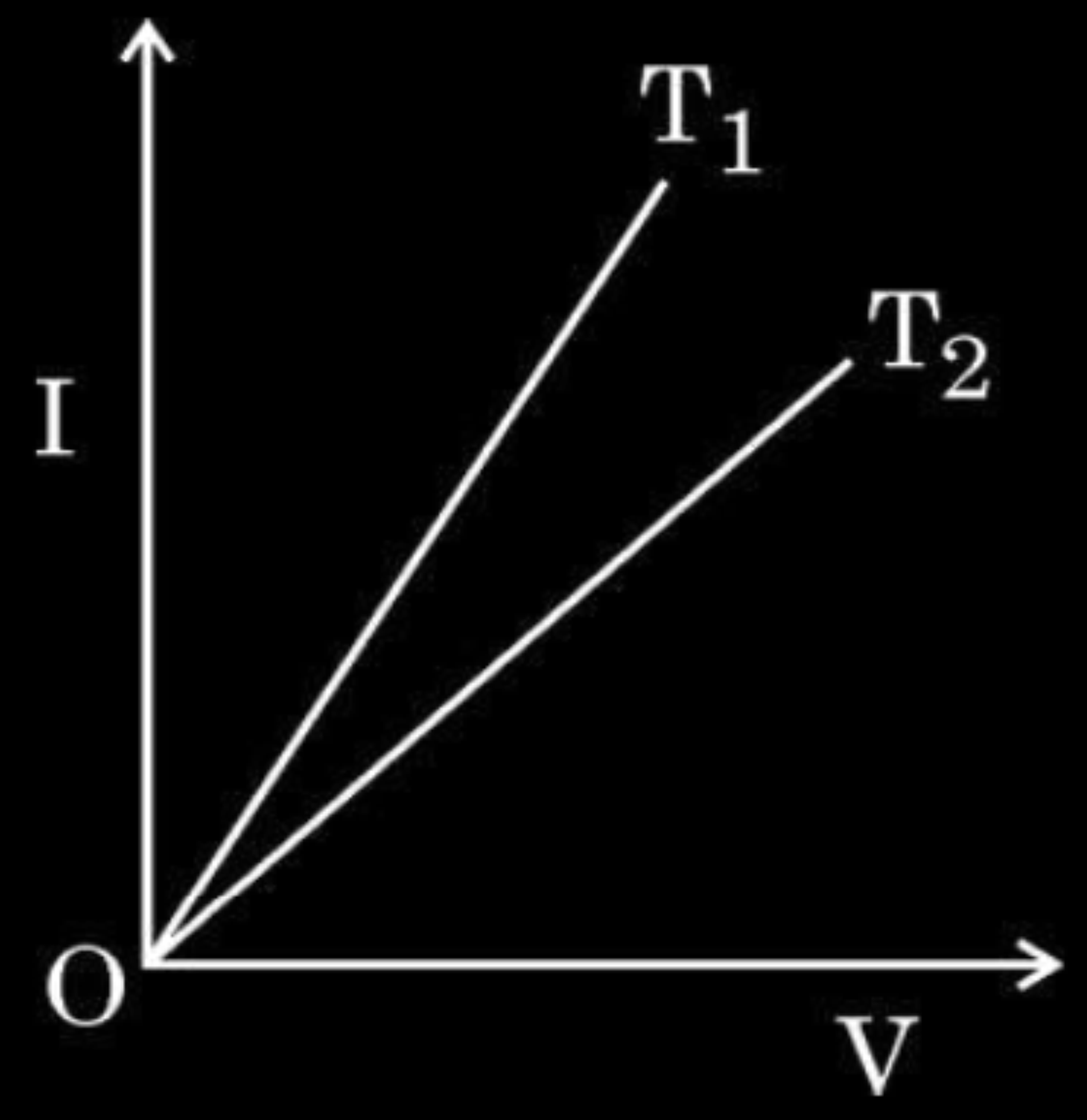
(A) $T_2 = 2T_1$

(B) $T_1 > T_2$

(C) $T_1 = T_2 / 3$

(D) $T_1 < T_2$

1. The figure shows the voltage (V) versus the current (I) graphs for a wire at two temperatures T_1 and T_2. One can conclude that :



(A) $T_2 = 2T_1$

(B) $T_1 > T_2$

(C) $T_1 = T_2 / 3$

(D) $\boxed{T_1 < T_2}$

2. If R_s and R_p are the equivalent resistances of n resistors, each of value R, in series and parallel combinations respectively, then the value of $(R_s - R_p)$ is :

(A) $\left(\frac{n^2-1}{n^2}\right)R$

(B) $\left(\frac{n^2+1}{n^2-1}\right)R$

(C) $\left(\frac{n^2-1}{n}\right)R$

(D) $\frac{(n^2+1)\,R}{n^2}$

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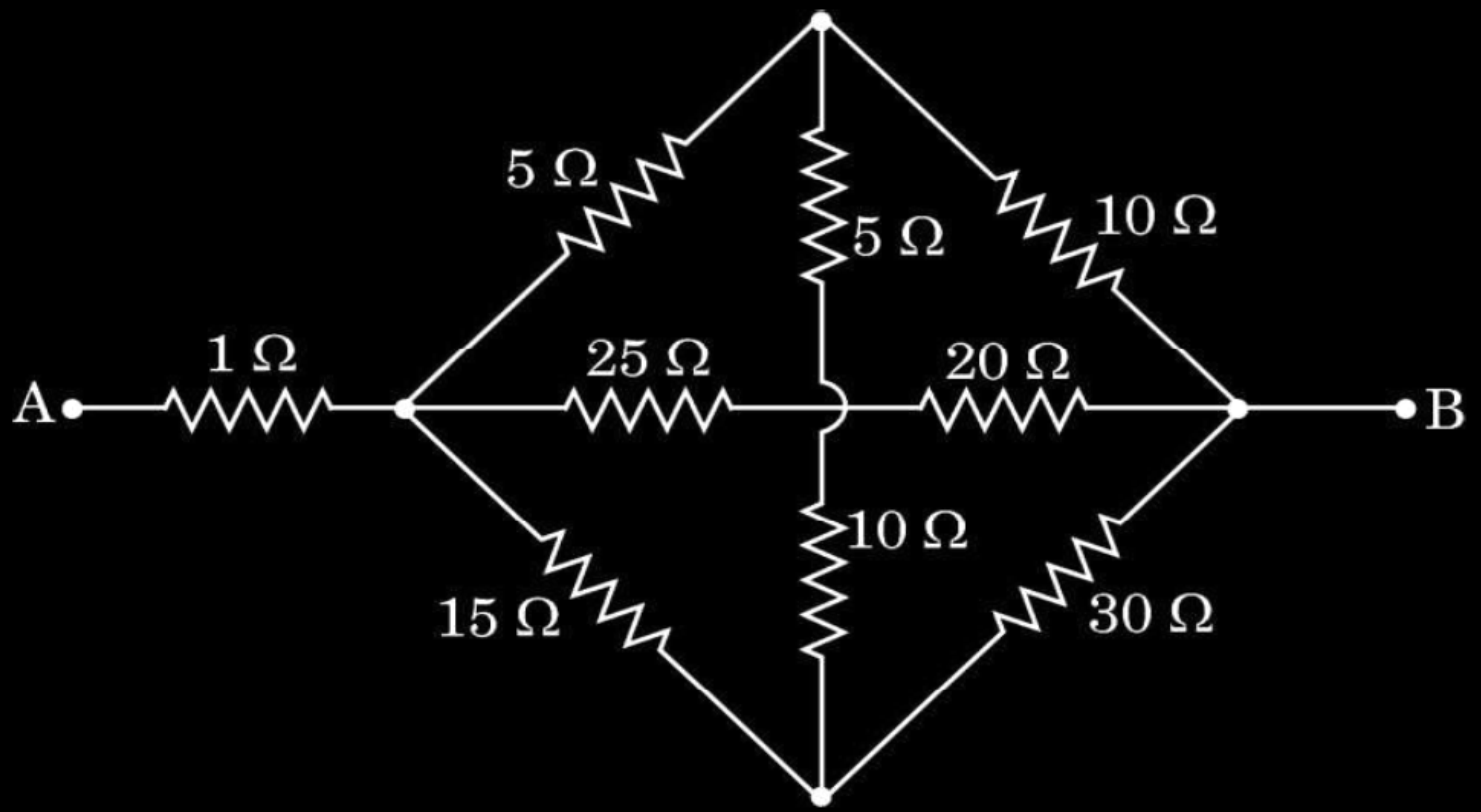
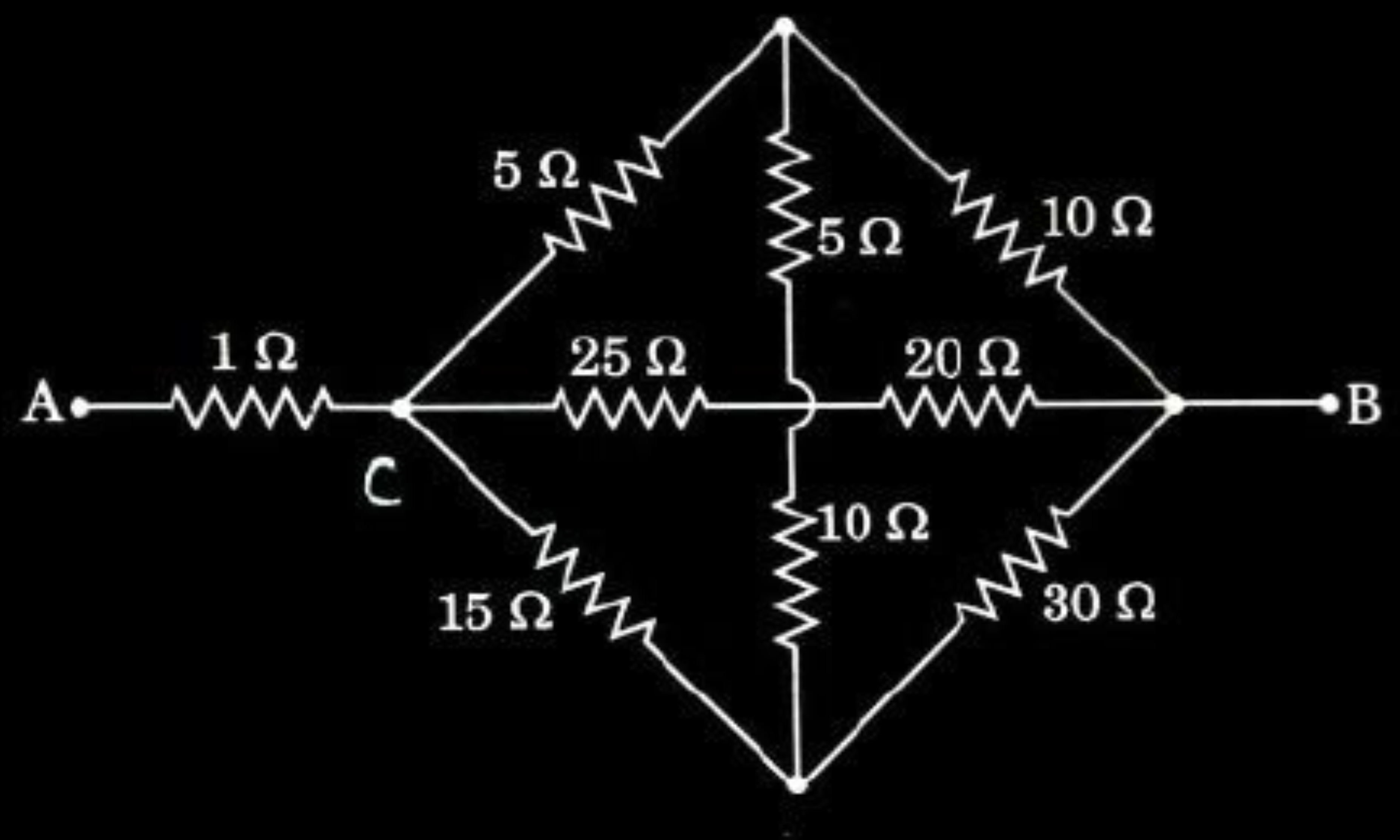
$\boxed{\text{(C)} \quad \left(\frac{n^2-1}{n}\right)R}$

(D) $\frac{(n^2+1)R}{n^2}$

17. Find the equivalent resistance between points A and B for the network shown in the figure. *2*



Resistance between points C and B 	
$\frac{1}{R}=\frac{1}{R_1}+\frac{1}{R_2}+\frac{1}{R_3}$	½
$\frac{1}{R}=\frac{1}{15}+\frac{1}{45}+\frac{1}{45}$	
R = 9 Ω	
Equivalent resistance between points A and B	½
$R_{eq}=R_1+R_2$	½
$R_{eq} = 1 + 9$	
$= 10\ \Omega$	½

2. The resistance of a wire at 20 °C is 50 Ω. When it is heated to 120 °C, the resistance becomes 51 Ω. The temperature coefficient of resistance of the material of the wire is **1**

(A) 2×10^{-4} °C^{-1} (B) 1×10^{-4} °C^{-1}

(C) 5×10^{-4} °C^{-1} (D) 1×10^{-3} °C^{-1}

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(A) $\boxed{2 \times 10^{-4}\ {}^\circ C^{-1}}$ (B) $1 \times 10^{-4}\ {}^\circ C^{-1}$

(C) $5 \times 10^{-4}\ {}^\circ C^{-1}$ (D) $1 \times 10^{-3}\ {}^\circ C^{-1}$

17. A 20.0 cm long wire is connected across an ideal battery of 4 V. The drift velocity of conduction electrons in the wire is 0.8 mm/s. Calculate the relaxation time of the electrons. **2**

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Drift velocity, $v_d = \dfrac{eE\tau}{m}$	½
using $E = \dfrac{V}{\ell}$ and $\tau = \dfrac{mv_d l}{eV}$	½
$\tau = \dfrac{0.8\times10^{-3}\times9.1\times10^{-31}\times0.20}{1.6\times10^{-19}\times4}$	½
$\tau = 2.28\times10^{-16}\,s$	½

22. (a) Define the terms 'emf' and 'terminal voltage' of a cell. Can the terminal voltage of a cell ever exceed its emf ? Explain. **3**

OR

(b) Define the term current density. Write its SI unit. Derive the equivalent form of Ohm's law $\vec{j} = \sigma\vec{E}$. **3**

• Defining emf and terminal voltage	1+1
• Explaining terminal voltage exceeding the emf	1

Defining current density and Writing its SI unit	1+ ½
Deriving $\vec{J} = \sigma\vec{E}$	1 ½

emf- It is the potential difference across the terminal of the cell, when no current is being drawn from it.	1
Terminal Voltage - It is the potential difference between the terminals of the cell in a closed circuit.	1
Alternatively $V = \varepsilon - Ir$	
Yes,	½
During charging, $V = \varepsilon - (-I)\,r = \varepsilon + Ir$	½
Current density – It is defined as the current through unit area normal to the current.	1
Alternatively, $\vec{J} = \frac{I}{\vec{A}}$	
S.I unit A/m^2	½
Derivation	
$V = IR$	½
$\vec{E}l = I\frac{\rho l}{\vec{A}}$	½
$\vec{E} = \vec{J}\rho$	
$\vec{J} = \frac{1}{\rho}\vec{E} = \sigma\vec{E}$	½

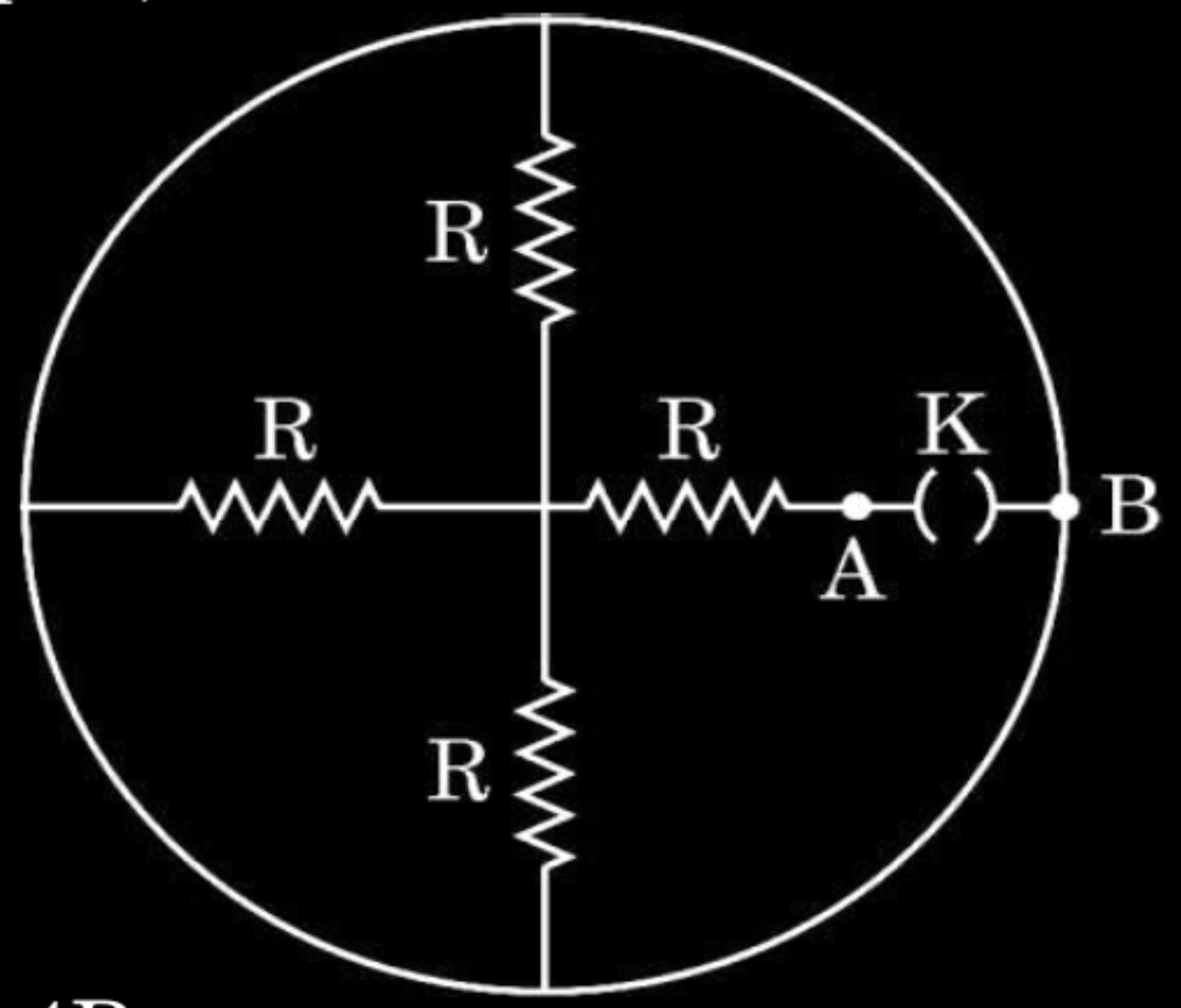
Alternatively

$$I = ne\vec{A}v_d$$

$$\frac{I}{\vec{A}} = \frac{ne^2\tau\vec{E}}{m}$$

$$\vec{J} = \frac{\vec{E}}{\rho} = \sigma\vec{E}$$

5. Four resistors, each of resistance R and a key K are connected as shown in the figure. The equivalent resistance between points A and B when key K is open, will be :



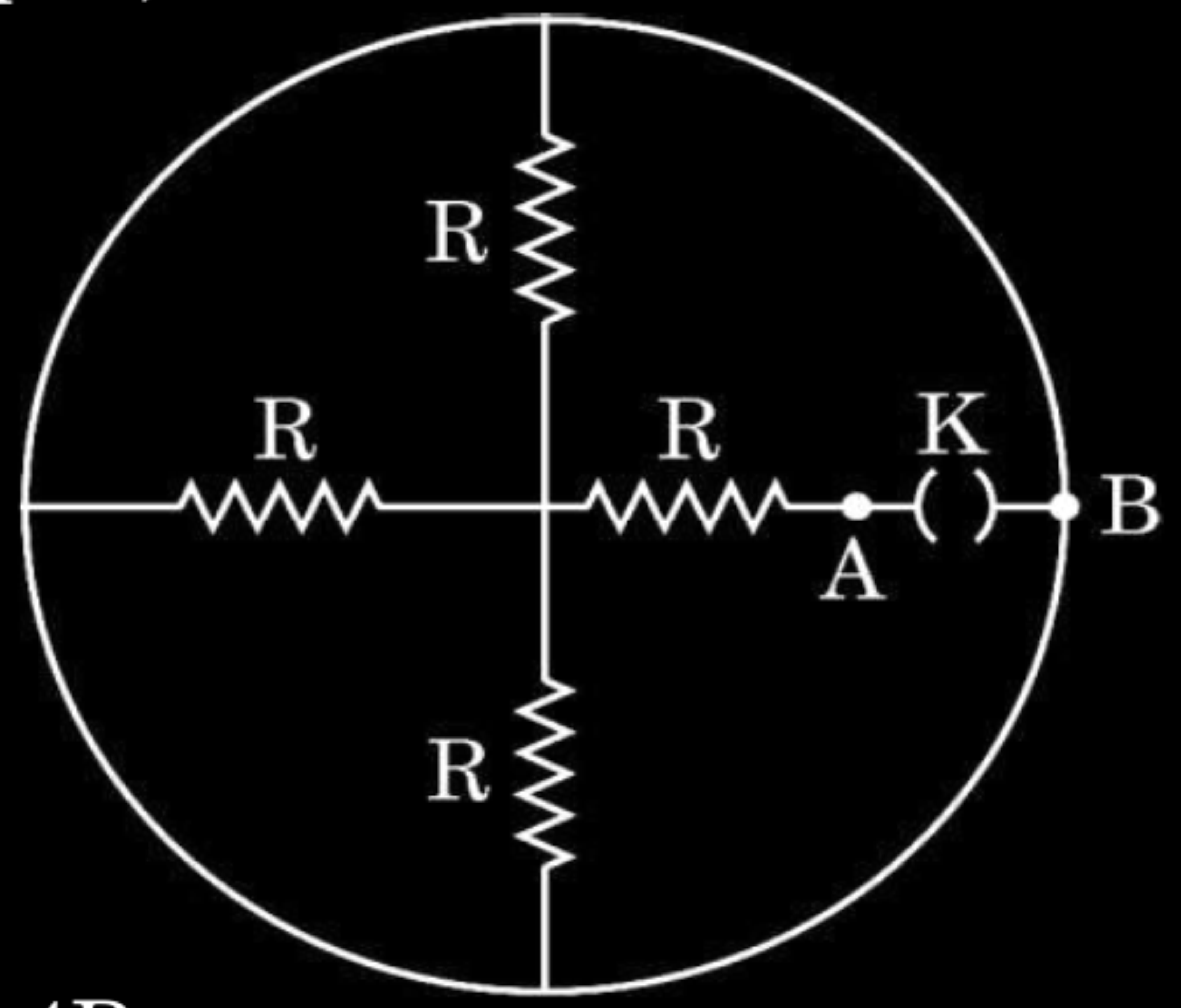
(A) 4R

(B) ∞

(C) $\frac{R}{4}$

(D) $\frac{4R}{3}$

5. Four resistors, each of resistance R and a key K are connected as shown in the figure. The equivalent resistance between points A and B when key K is open, will be :



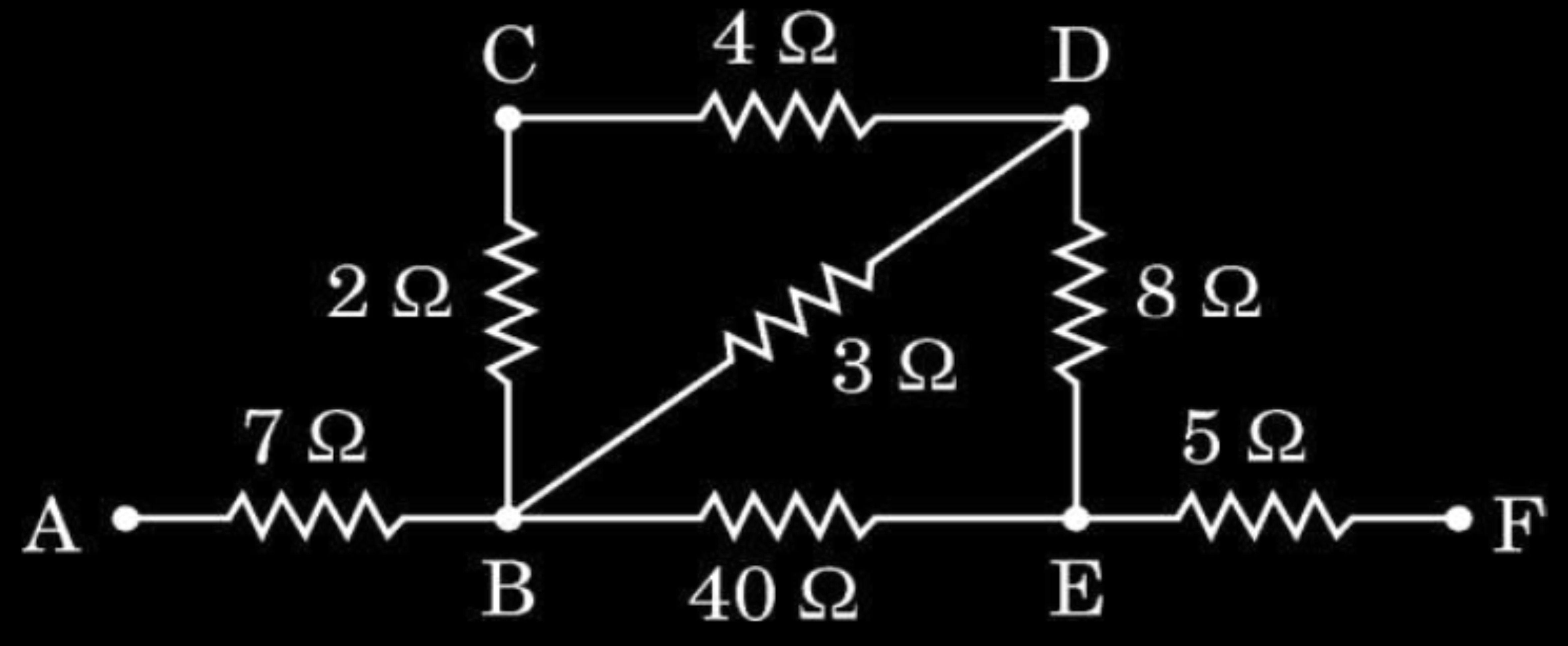
(A) 4R

(B) ∞

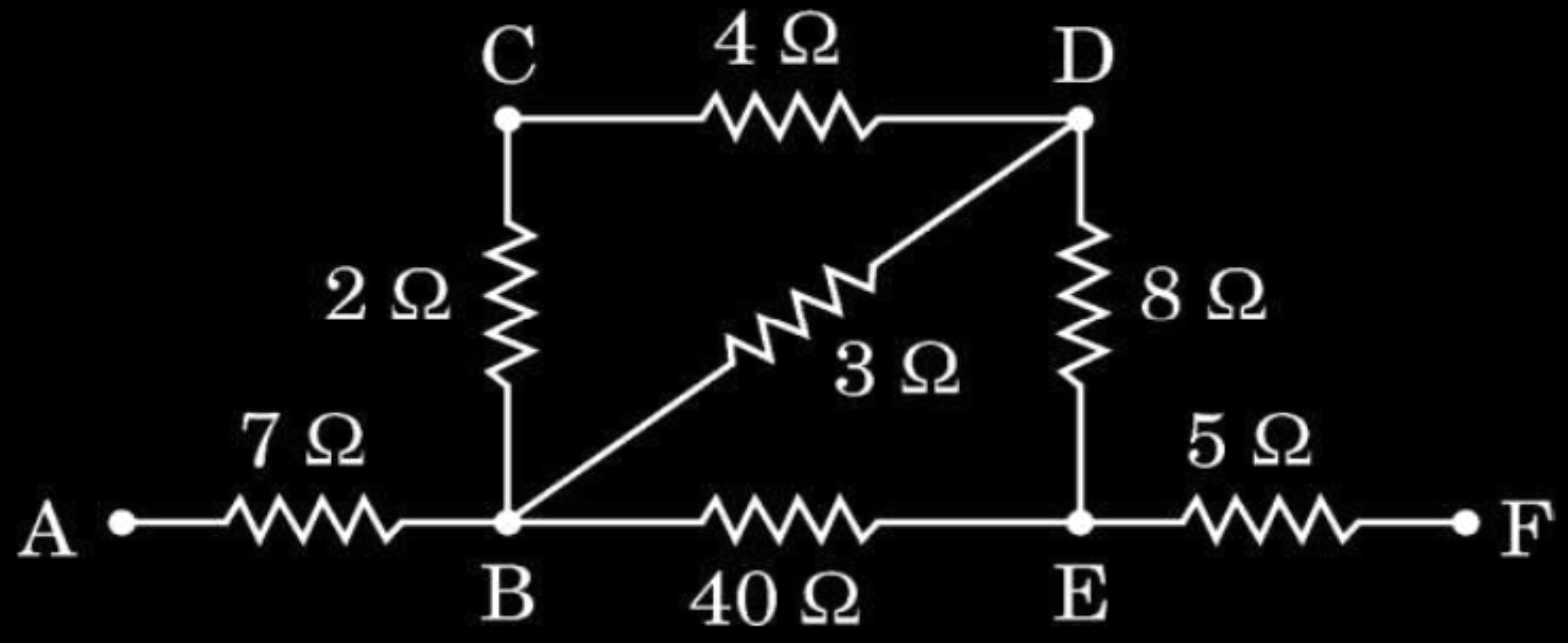
(C) $\frac{R}{4}$

(D) $\boxed{\frac{4R}{3}}$

17. Find the effective resistance of the network of resistors between points A and F as shown in the figure. *2*

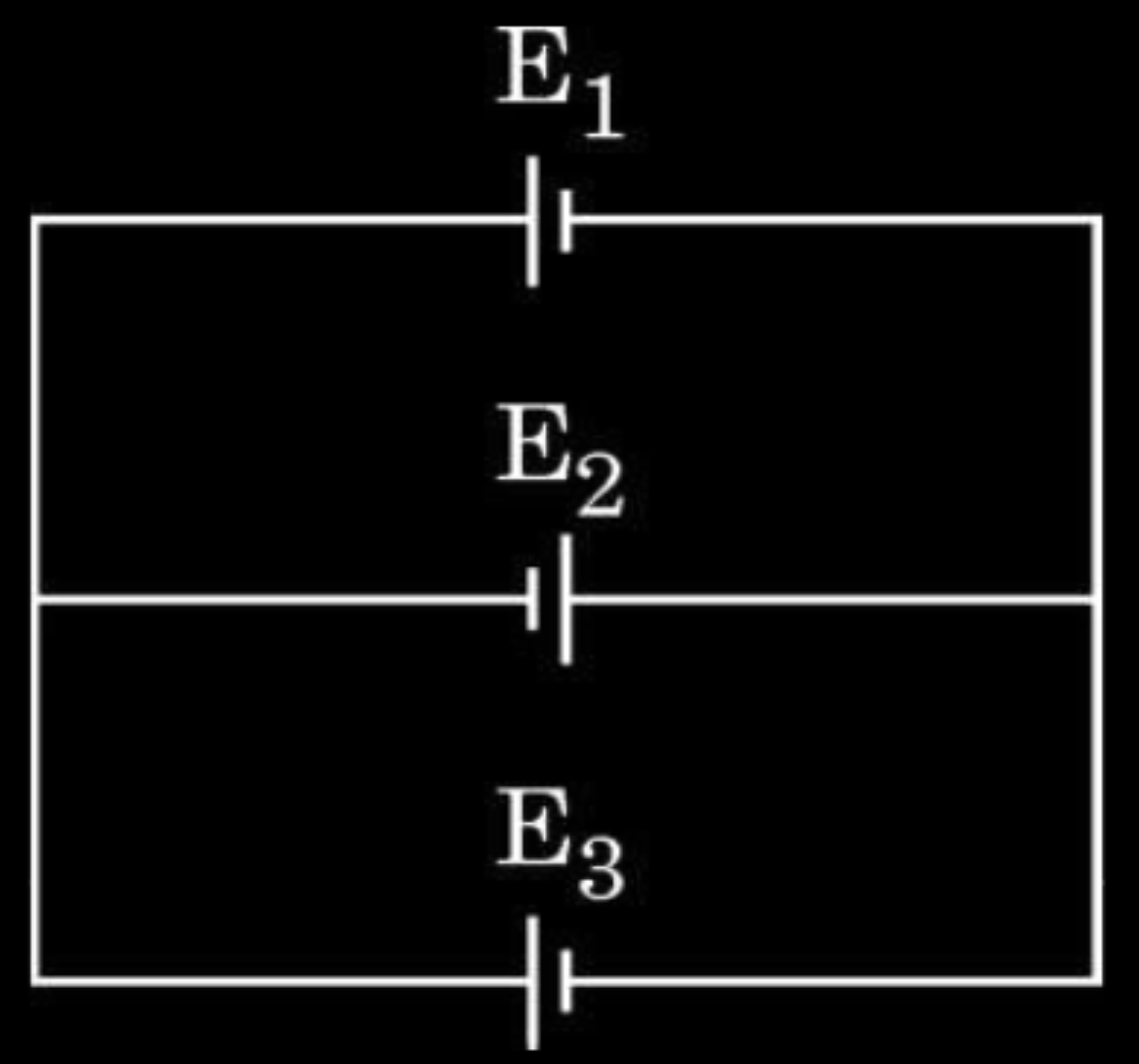


17. Find the effective resistance of the network of resistors between points A and F as shown in the figure. *2*

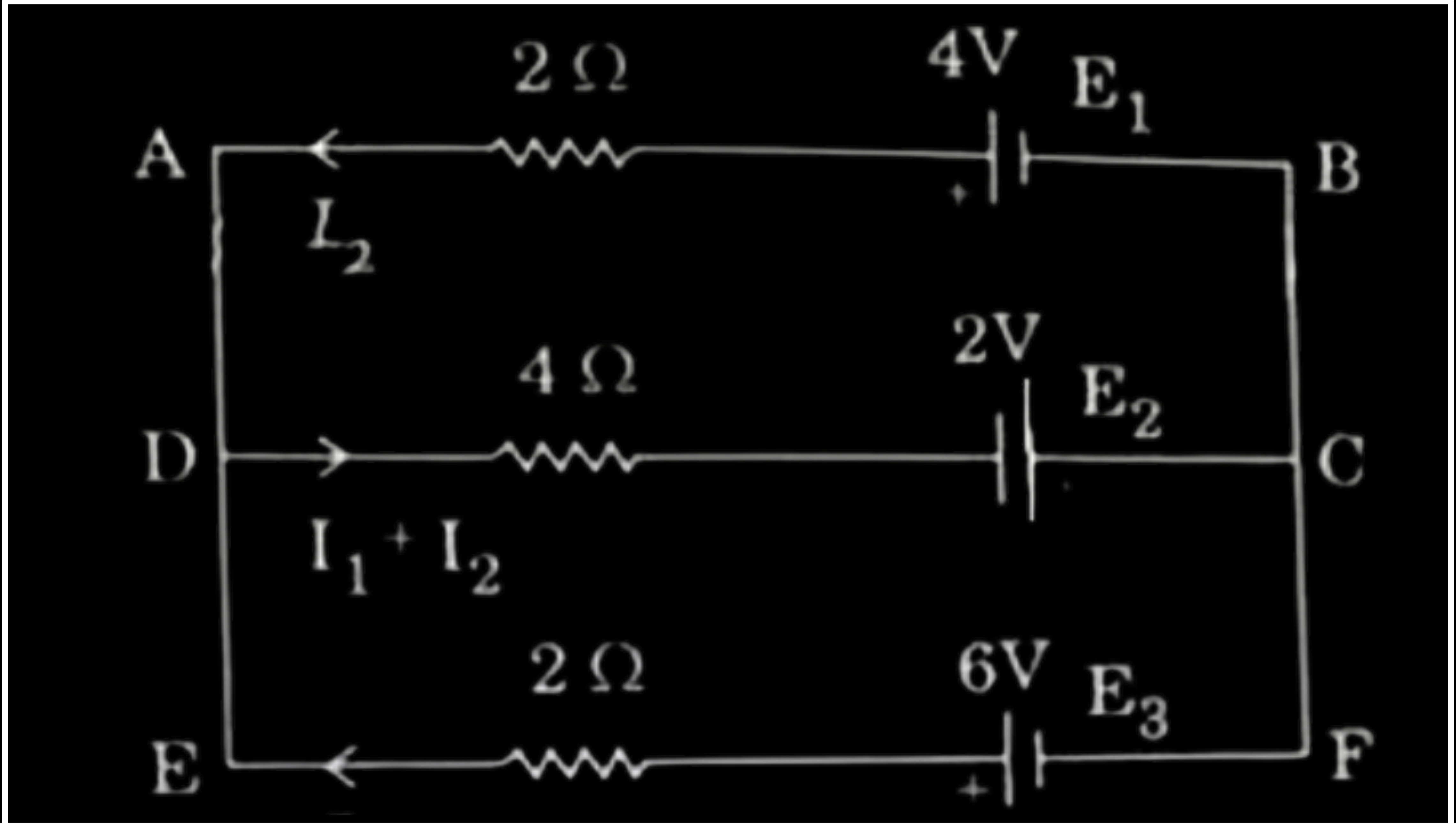


$R_{BCD} = 2+4 = 6\Omega$	
$R_{BD} = \dfrac{3\times6}{6+3}$	½
$= 2\Omega$	
$R_{BDE} = 2+8$	
$= 10\Omega$	½
$R_{BE} = \dfrac{40\times10}{40+10}$	
$= 8\Omega$	½
$R_{AF} = 7+8+5$	½
$= 20\Omega$	

32. (a) (i) Three batteries E_1, E_2 and E_3 of emfs and internal resistances (4 V, 2 Ω), (2 V, 4 Ω) and (6 V, 2 Ω) respectively are connected as shown in the figure. Find the values of the currents passing through batteries E_1, E_2 and E_3.

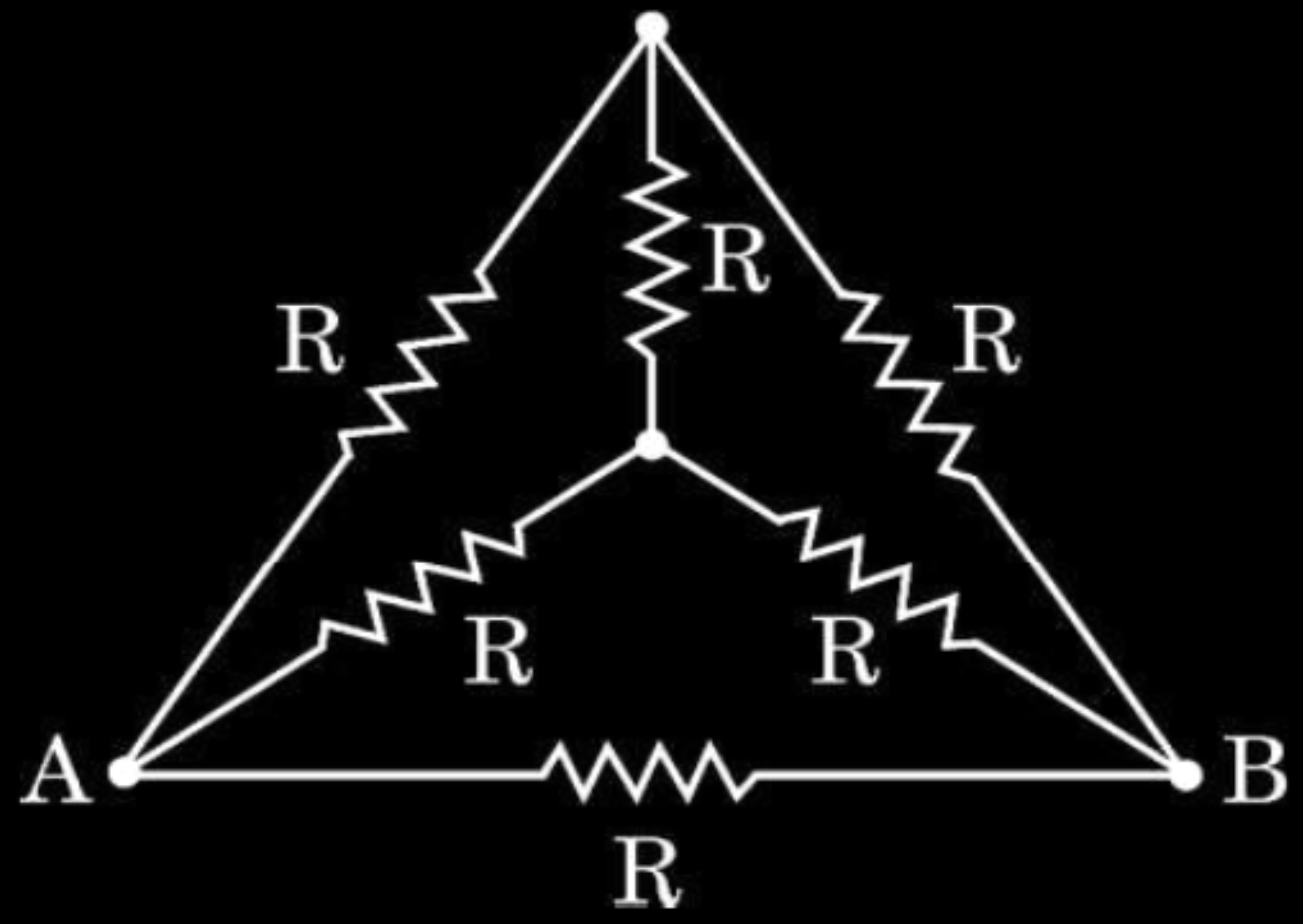


(i)	Finding current through batteries E_1, E_2 and E_3	3
(ii)	Finding effective resistance	2

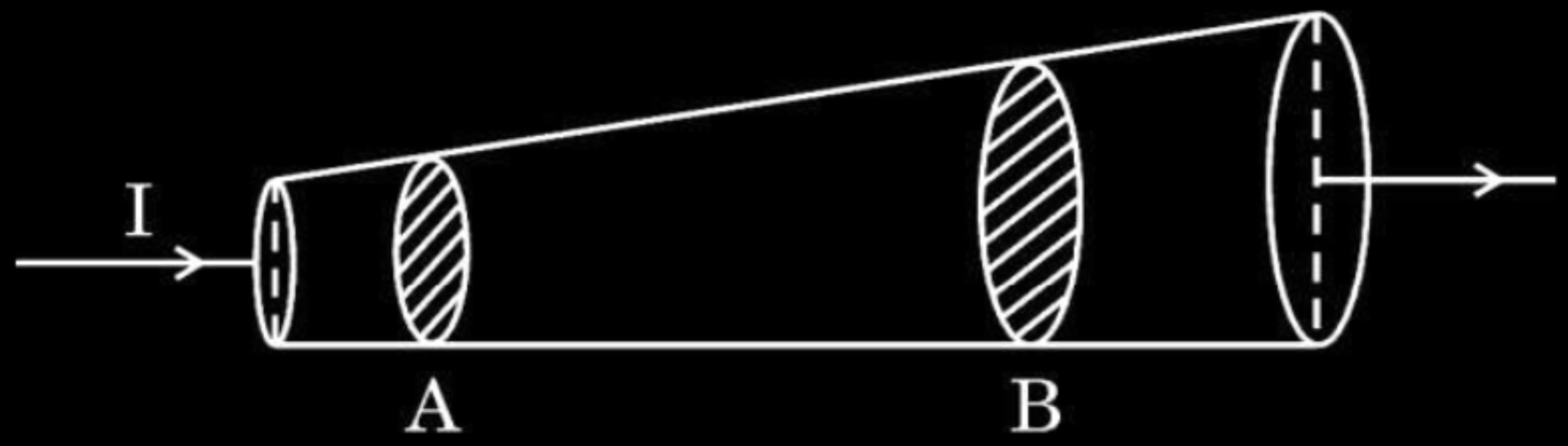


i) In closed loop ABCD, using Kirchhoff's loop law $4I_1 + 6I_2 = 6$ ……………………………(1) Similarly In closed loop CDFE $6I_1 + 4I_2 = 8$ ……………………………(2) Solving eqn. (1) and (2) $I_2 = \frac{1}{5}A$ $I_1 = \frac{6}{5}A$ $I_1 + I_2 = \frac{7}{5}A$	 ½ ½ ½ ½ ½

(ii) The ends of six wires, each of resistance R (= 10 Ω) are joined as shown in the figure. The points A and B of the arrangement are connected in a circuit. Find the value of the effective resistance offered by it to the circuit. *5*



(b) (i) Current I (= 1 A) is passing through a copper rod ($n = 8{\cdot}5 \times 10^{28}\ m^{-3}$) of varying cross-sections as shown in the figure. The areas of cross-section at points A and B along its length are $1{\cdot}0 \times 10^{-7}\ m^2$ and $2{\cdot}0 \times 10^{-7}\ m^2$ respectively. Calculate :



(I) the ratio of electric fields at points A and B.

(II) the drift velocity of free electrons at point B.

(i) Calculating		
(I)	ratio of electric fields at points A & B	1 ½
(II)	drift velocity of free electrons at point B	1 ½
(ii) Finding net electric field at point $\vec{r}$		2

(i) (I) $\vec{j} = \sigma \vec{E}$	½
$\dfrac{j_A}{j_B} = \dfrac{E_A}{E_B}$	
$= \dfrac{I/A_A}{I/A_B}$	½
$= \dfrac{A_B}{A_A}$	½
$= \dfrac{2}{1}$	
(II) $v_d = \dfrac{I}{neA}$	½
$= \dfrac{1}{8.5\times10^{28}\times1.6\times10^{-19}\times2\times10^{-7}}$	½
$= 3.6\times10^{-4}\ m/s$	½

18. A wire of resistance X ohm is gradually stretched till its length becomes twice its original length. If its new resistance becomes 40 Ω, find the value of X. *2*

18. A wire of resistance X ohm is gradually stretched till its length becomes twice its original length. If its new resistance becomes 40 Ω, find the value of X. *2*

$R = n^2 x$	
$40 = (2)^2 x$	½
	½
$x = 10\ \Omega$	**1**
Alternatively Volume of wire before and after stretch will remain same	
$A_1 l_1 = A_1 l_1$	½
$A_2 = \frac{Al}{2l}$	
$= \frac{A}{2}$	½
$R = \rho \frac{l_2}{A_2}$	½
$40 = 4\rho \frac{2l}{A/2}$	
$\therefore\ x = 10\Omega$	½

21. The resistance of a wire at 25°C is 10·0 Ω. When heated to 125°C, its resistance becomes 10·5 Ω. Find (i) the temperature coefficient of resistance of the wire, and (ii) the resistance of the wire at 425°C. *2*

21. The resistance of a wire at 25°C is 10·0 Ω. When heated to 125°C, its resistance becomes 10·5 Ω. Find (i) the temperature coefficient of resistance of the wire, and (ii) the resistance of the wire at 425°C. *2*

Finding (i) temperature coefficient of resistance 1 ½ (ii) resistance of wire at 425 °C ½	
(i) $R_2 = R_1(1+\alpha(t_2-t_1))$	½
$10.5 = 10(1+\alpha\times100)$	½
$\alpha = 5\times10^{-4} / {}^{o}C$	
(ii) $R_{425} = R_{25}(1+\alpha(425-25))$	½
$= 10(1+5\times10^{-4}\times400)$	½
$= 12\,\Omega$	

2. Three wires A, B and C of the same material have lengths and area of cross-sections as $(2l, \frac{A}{2})$, (l, A) and $(\frac{l}{2}, 2A)$, respectively. If the resistances of these wires are R_A, R_B and R_C respectively, then :

(A) $R_A > R_B > R_C$

(B) $R_B > R_C > R_A$

(C) $R_C > R_A > R_B$

(D) $R_A > R_C > R_B$

2. Three wires A, B and C of the same material have lengths and area of cross-sections as $(2l, \frac{A}{2})$, (l, A) and $(\frac{l}{2}, 2A)$, respectively. If the resistances of these wires are R_A, R_B and R_C respectively, then :

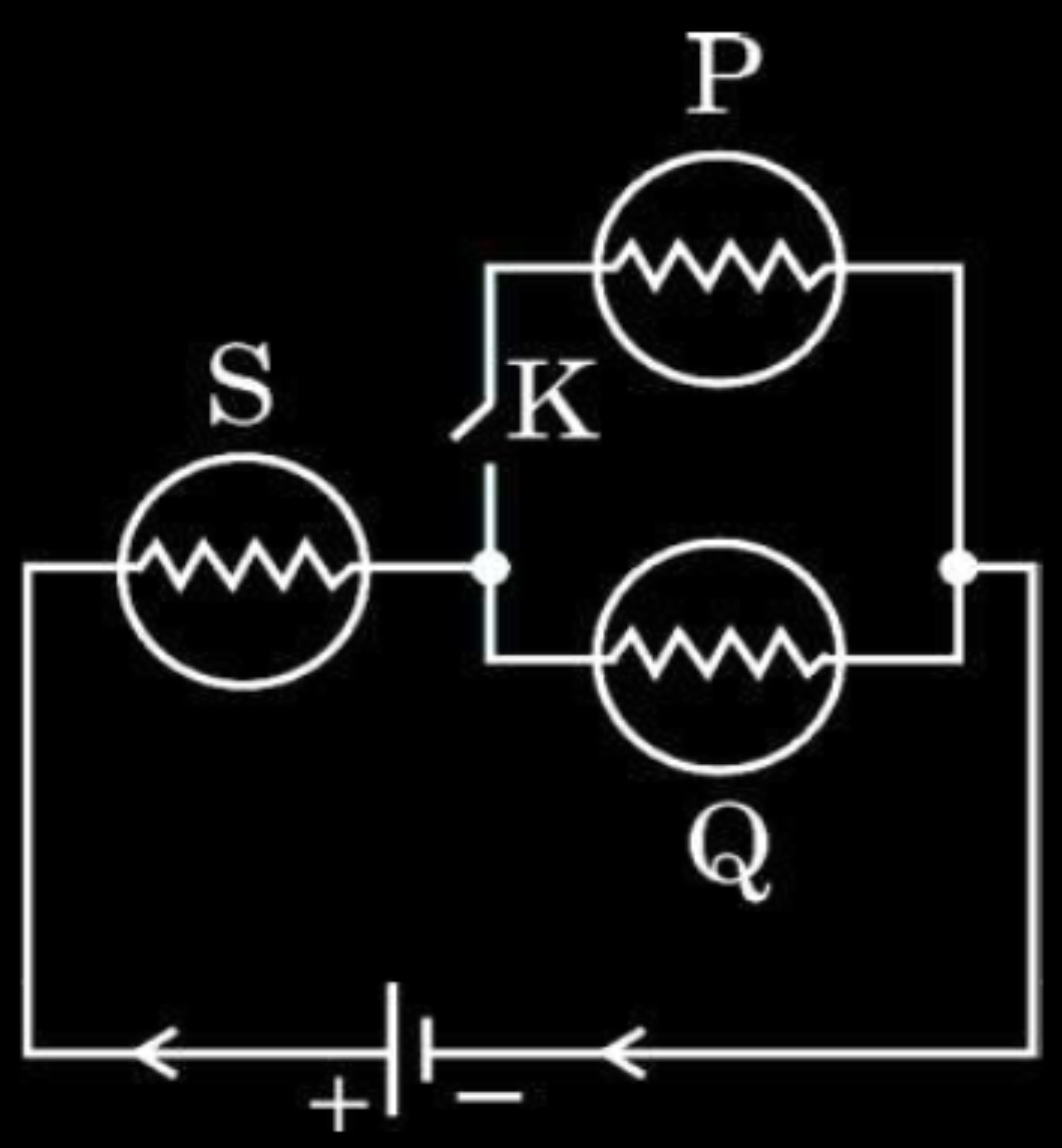
$\boxed{\text{(A)} \quad R_A > R_B > R_C}$

(B) $R_B > R_C > R_A$

(C) $R_C > R_A > R_B$

(D) $R_A > R_C > R_B$

21. (a) In the given figure, three identical bulbs P, Q and S are connected to a battery.



(i) Compare the brightness of bulbs P and Q with that of bulb S when key K is closed.

(ii) Compare the brightness of the bulbs S and Q when the key K is opened.

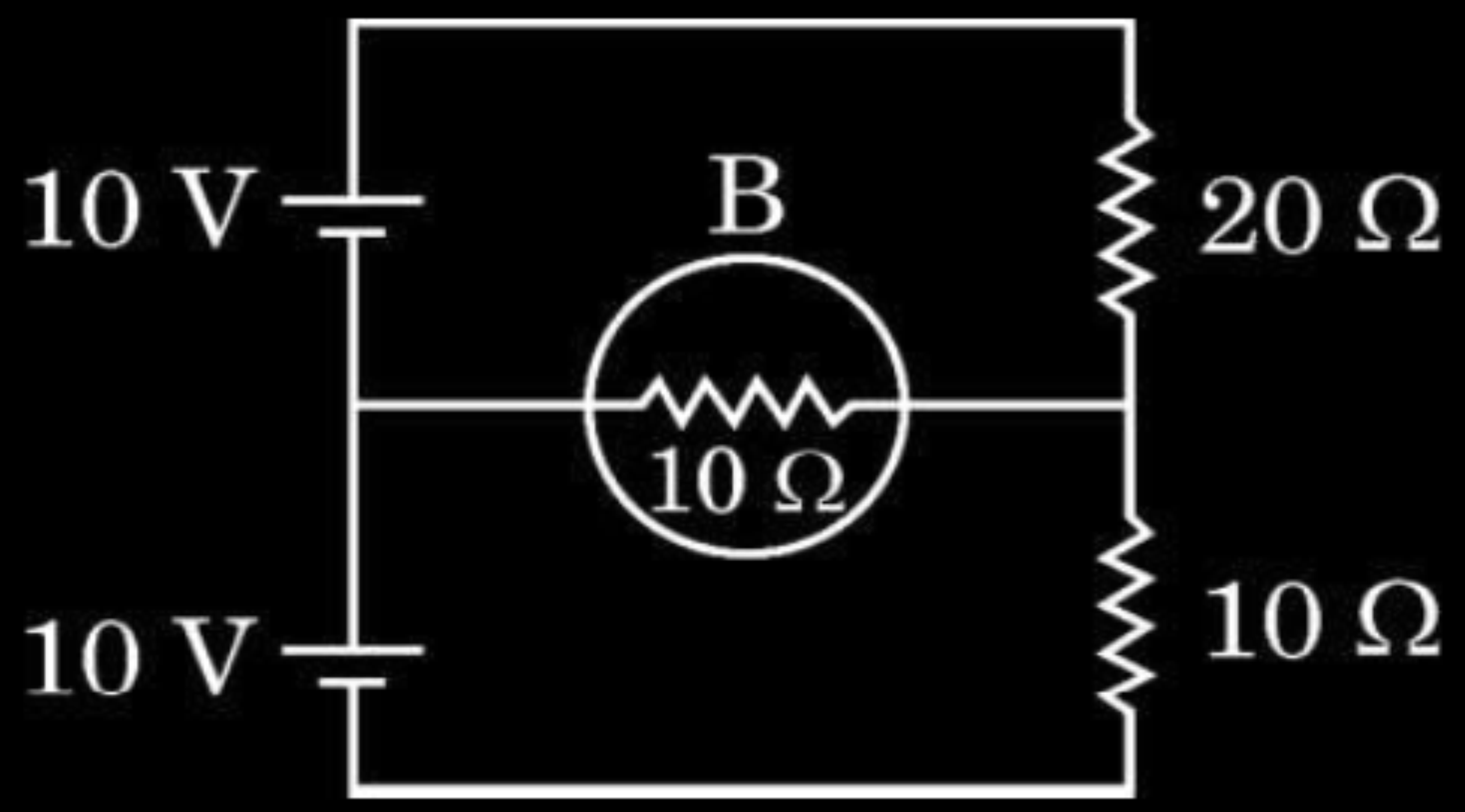
Justify your answer in both cases. *2*

(i) Comparison of brightness of bulbs P and Q with bulb S	½
Justification	½
(ii) Comparison of brightness of bulb S with Q	½
Justification	½

(i) Brightness of the bulb ‘S’ will be more than bulbs ‘P’ and ‘Q’	½
The current flowing through the bulb ‘S’ is twice of the current in bulbs ‘P’ and ‘Q’.	½
(ii) Brightness of the bulb ‘S’ and ‘Q’will be same	½
The current flowing through both bulbs is same.	½
Alternatively-	
(i) Brightness of the bulb ‘S’ will be more than bulbs ‘P’ and ‘Q’	
The potential difference across ‘S’ is twice than the potential difference across bulbs ‘P’ and ‘Q’	
(ii) Brightness of both bulbs ‘S’ and ‘Q’ is same.	
The potential difference across ‘S’ and ‘Q’ will be same.	

OR

(b) Two cells of emf 10 V each, two resistors of 20 Ω and 10 Ω and a bulb B of 10 Ω resistance are connected together as shown in the figure. Find the current that flows through the bulb. *2*



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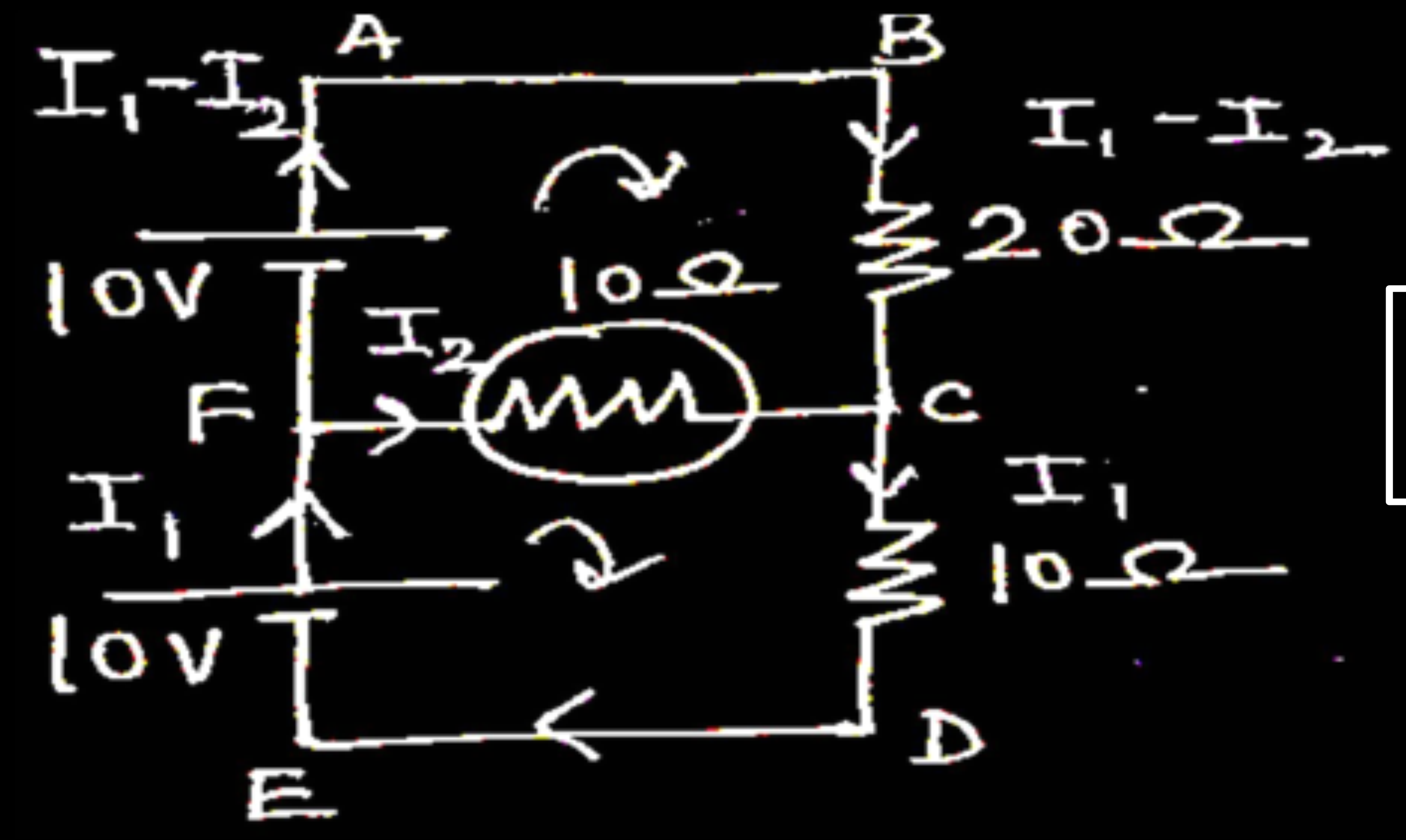
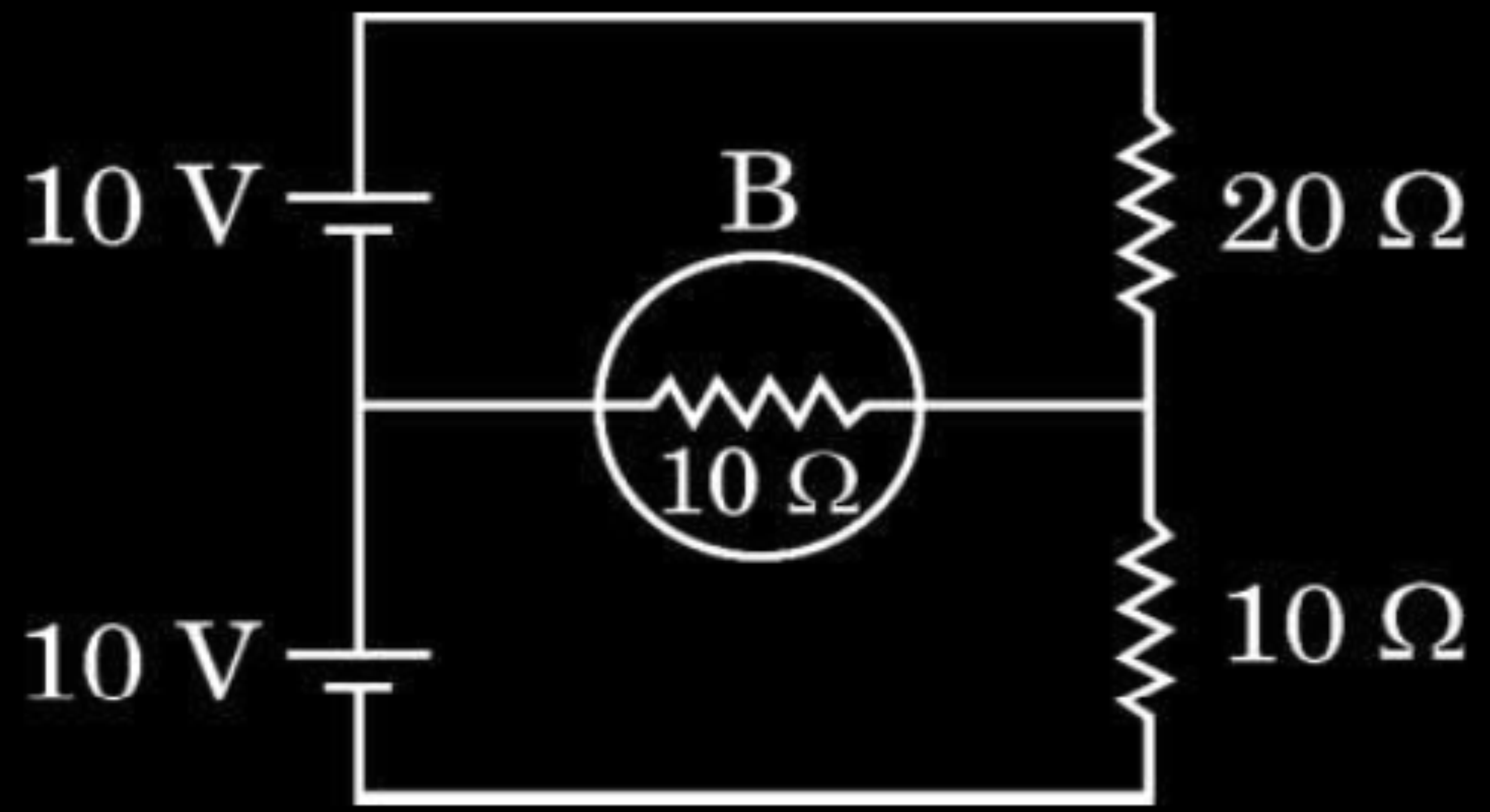


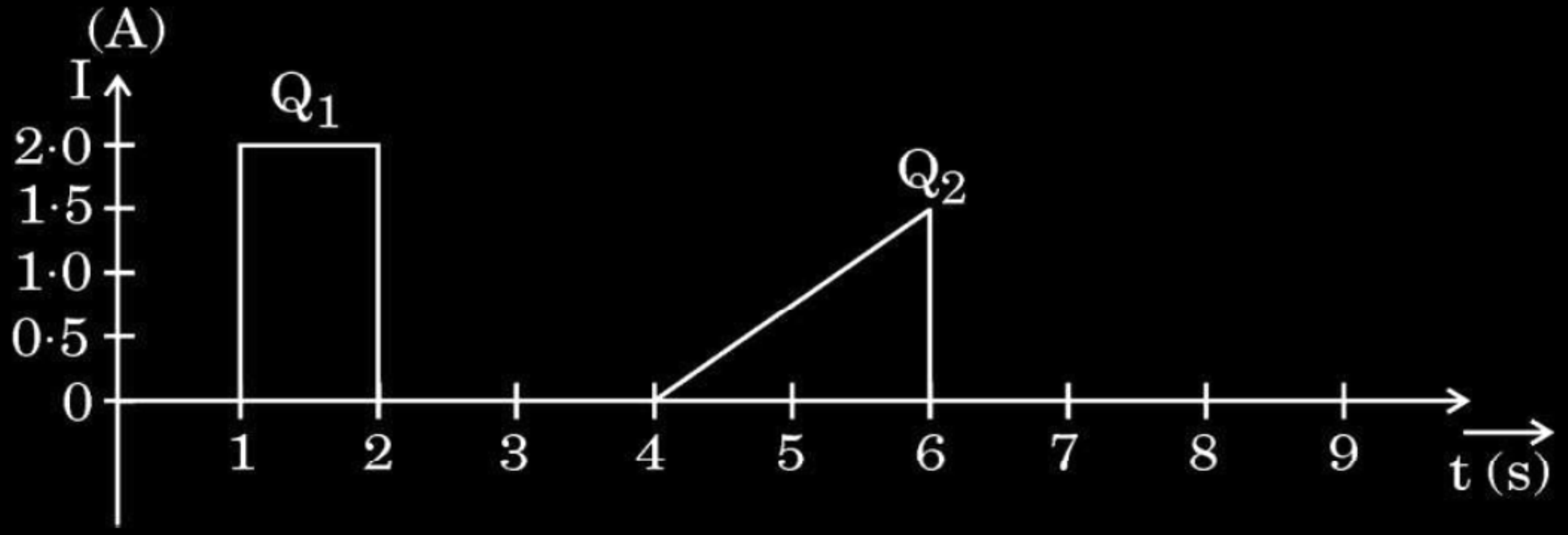
½



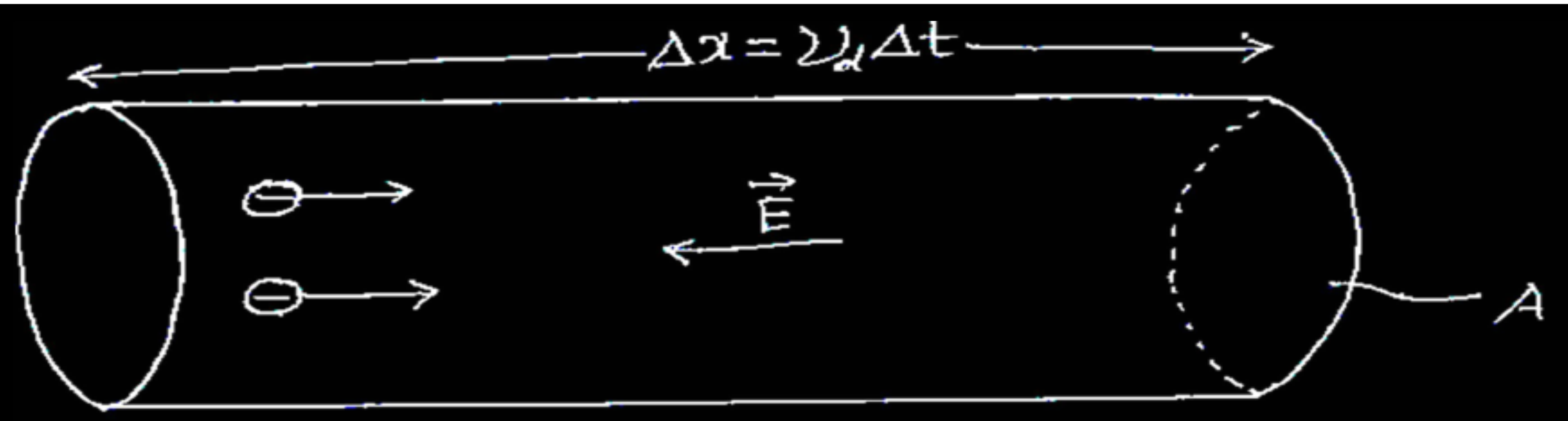
By applying Kirchhoff's loop rule to closed loops ABCFA and FCDEF $2I_1-3I_2=1$----(1) $I_1+I_2=1$-----(2) On solving,	½ ½
Current through the bulb, $I_2=\frac{1}{5}$ A	½

27. (a) (i) Derive an expression for the resistivity of a conductor in terms of number density of free electrons and relaxation time.

(ii) The figure shows the plot of current through a cross-section of wire over two different time intervals. Compare the charges (Q_1 and Q_2) that pass through the cross-section during these time intervals. *3*



(i) Deriving the expression for resistivity of a conductor	2
(ii) Comparison of charges Q_1 and Q_2	1



Total charge transported along E is

$$I\Delta t = \frac{e^2 A}{m}\tau\, n\Delta t E$$ ½

$$\frac{I}{A} = \frac{ne^2}{m}\,\tau E$$ ½

$$J = \frac{1}{\rho}E$$ ½

$$\rho = \frac{m}{ne^2\tau}$$ ½

(ii) From given graph

$$\frac{Q_1}{Q_2} = \frac{A_1\,(\text{Area of rectangle})}{A_2\,(\text{Area of triangle})}$$ ½

$$\frac{Q_1}{Q_2} = \frac{2}{3/2}$$

$$Q_1 > Q_2$$ ½

OR

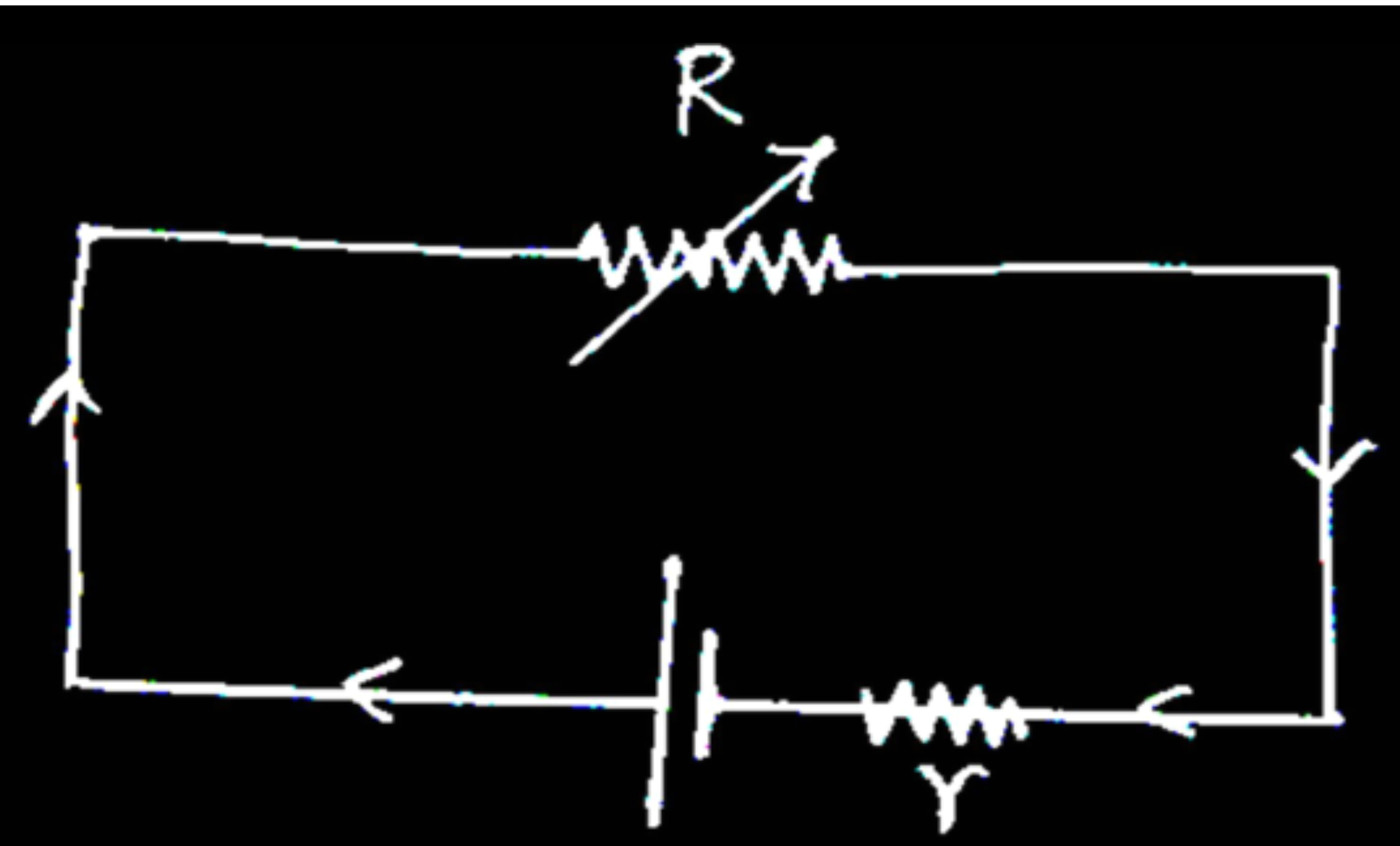
(b) (i) A battery of emf E and internal resistance r is connected to a variable external resistance R.

(I) Obtain the expression for current I in the circuit and the value of maximum current the battery can supply.

(II) Obtain the terminal voltage V across the battery and its maximum possible value.

(ii) The above battery sends a current I_1 when $R = R_1$ and a current I_2 when $R = R_2$. Obtain the internal resistance of the battery in terms of I_1, I_2, R_1 and R_2. *3*

(i) (I)	
• Obtaining the expression for current	½
• Finding value of maximum current	½
(II) Obtaining terminal voltage V and its maximum possible value	1
(ii) Obtaining the internal resistance of the battery	1



- $V = E - Ir$

  $IR = E - Ir$

  $I = \dfrac{E}{R + r}$ ½

  For maximum value of current R=0

$I_{max} = \dfrac{E}{r}$ ½

(II) $V = V_{+} + V_{-} - Ir$

$V = E - Ir$ ½

$V_{max} = E$, when I=0 ½

(ii) $I_1R_1 + I_1r = I_2R_2 + I_2r$ ½

$r = \dfrac{I_2R_2 - I_1R_1}{I_1 - I_2}$ ½

2. When the switch of the circuit is turned on, the filament of the bulb glows instantaneously because :

(A) the electrons coming from the power source move fast through the initially empty filament.

(B) the filament may be old having low resistance.

(C) electric field is established instantaneously across the filament which pushes the electrons.

(D) free electrons in the filament travel with the speed of light.

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(D) free electrons in the filament travel with the speed of light.

2. A current flows through a cylindrical conductor of radius R. The current density at a point in the conductor is $j = \alpha r$ (along its axis), here α is a constant and r is distance from the axis of the conductor. The current flowing through the portion of the conductor from $r = 0$ to $r = \frac{R}{2}$ is proportional to :

(A) R

(B) R^2

(C) R^3

(D) R^4

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(A) R

(B) R^2

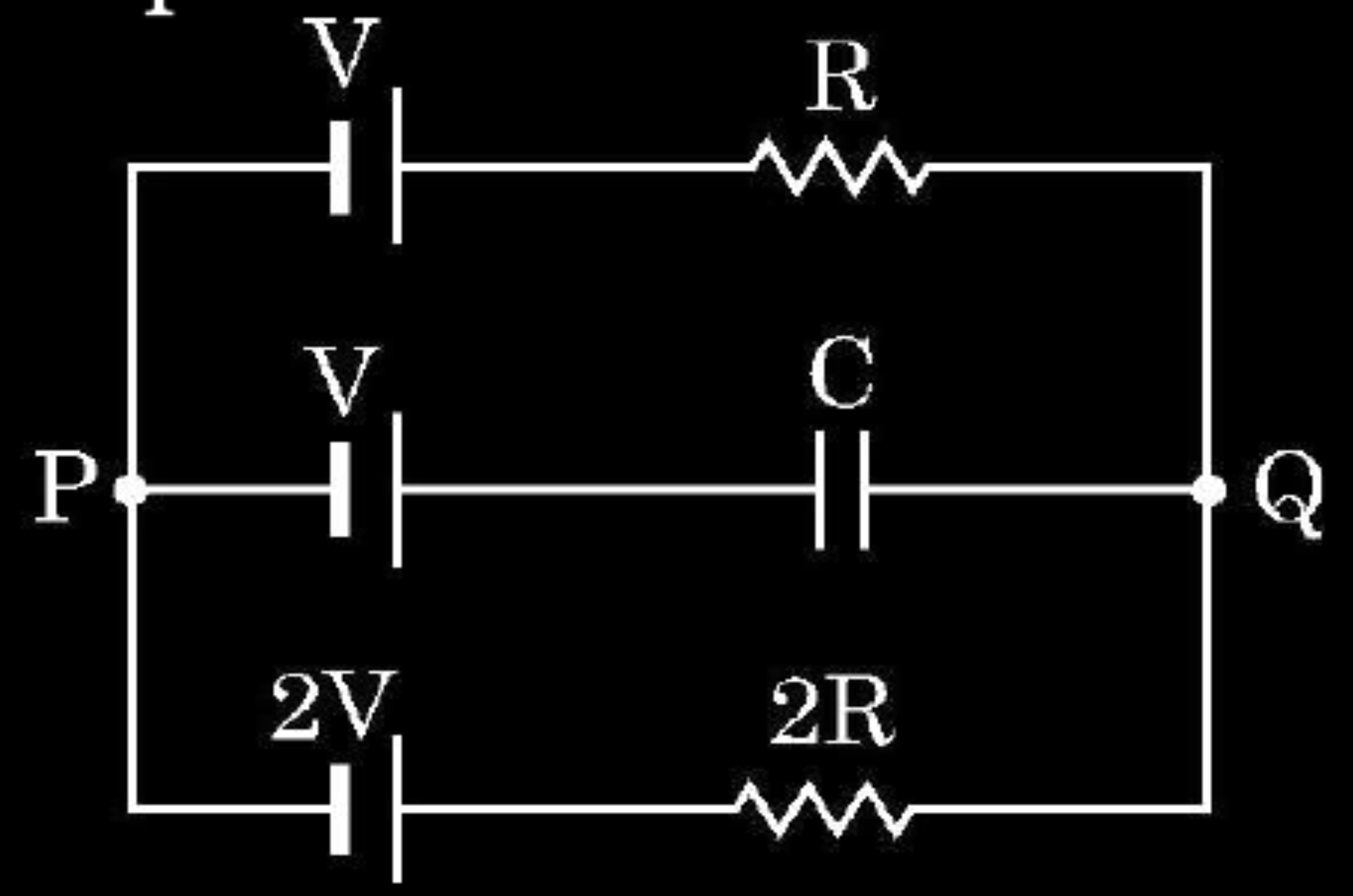
$\boxed{\text{(C)} \quad R^3}$

(D) R^4

21. (a) Two wires of the same material and the same radius have their lengths in the ratio 2 : 3. They are connected in parallel to a battery which supplies a current of 15 A. Find the current through the wires. **2**

OR

(b) In the circuit three ideal cells of e.m.f. V, V and 2V are connected to a resistor of resistance R, a capacitor of capacitance C and another resistor of resistance 2R as shown in figure. In the steady state find (i) the potential difference between P and Q and (ii) potential difference across capacitor C.



(a)	
Finding current 2	
$R_1 = \frac{\rho l_1}{A}$; $R_2 = \frac{\rho l_2}{A}$	½
$\frac{l_1}{l_2} = \frac{2}{3} => \frac{R_1}{R_2} = \frac{2}{3}$	
$I \alpha \frac{1}{R}$	
$=> \frac{I_1}{I_2} = \frac{3}{2}$	½
$=> I_1 = \frac{3}{5}$ x 15 = 9A	½
$=> I_2 = \frac{2}{5}$ x 15 = 6A	½

(b)

Finding the potential difference	
(i) between P and Q	1
(ii) across capacitor C	1

In steady state,

$$2V - V = i(2R+R)$$

$$i = \frac{V}{3R}$$

(i) $V_P - V_Q = -V - iR$

$$= -V - \frac{V}{3}$$

$$V_P - V_Q = -\frac{4V}{3}$$ 1

(ii) $V_P - V_Q = -V + V_C$

$$-\frac{4V}{3} = -V + V_C$$ 1

$$V_C = -\frac{V}{3}$$

22. (a) Define Electrical conductivity. Obtain the expression of electrical conductivity of a conductor in terms of number density and relaxation time of free electrons. **3**

(b) Explain qualitative change in resistivity of a conductor with temperature using expression obtained in (a).

(a) Defining Electrical conductivity	1
Obtaining expression of electrical conductivity	1
(b) Explaining qualitative change in resistivity with temperature	1

(a) Electrical conductivity: - It is the reciprocal of the resistivity. 1

Alternatively: -

$\sigma = 1/\rho$

$I\triangle t = +neA|\vec{v}_d|\triangle t$ ½

Substituting $|\vec{v}_d| = \dfrac{e|\vec{E}|}{m}\tau$

$I\triangle t = \dfrac{e^2 A}{m}\tau\, n \triangle t|\vec{E}|$

$I = |\vec{j}|A$

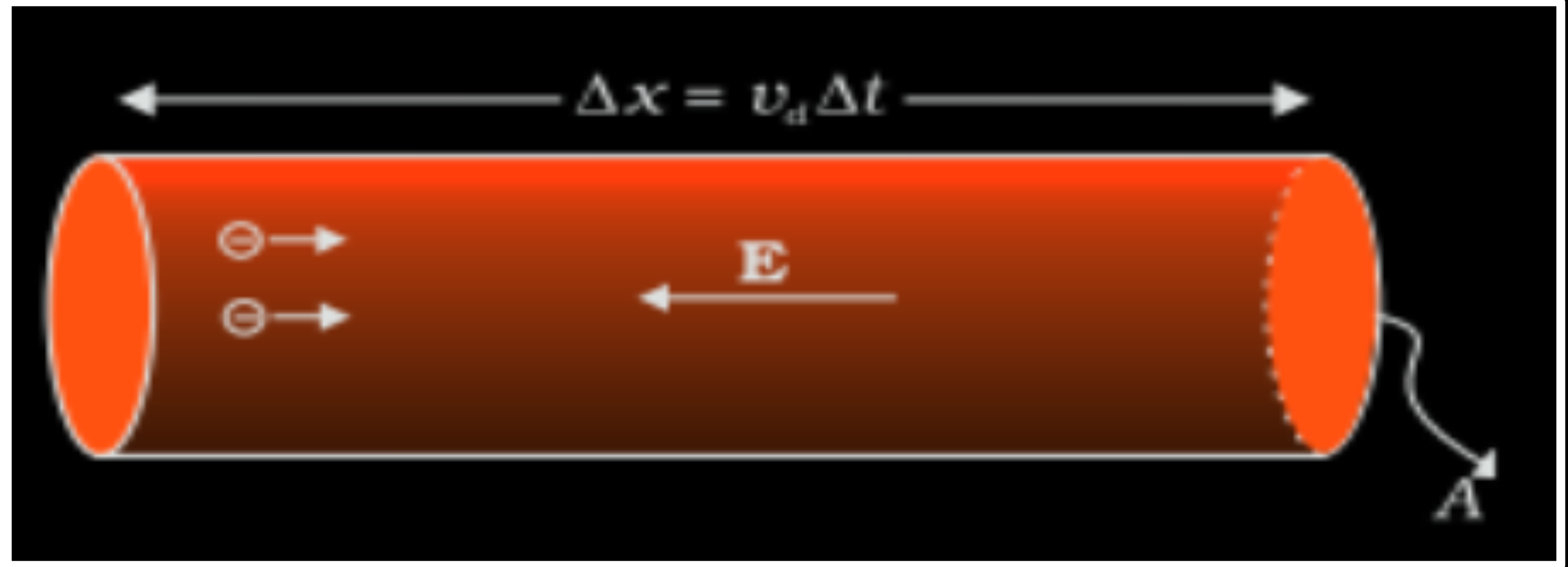
$\Rightarrow |\vec{j}| = \dfrac{ne^2}{m}\tau|\vec{E}|$.........(1)

$\vec{j} = \sigma\vec{E}$.......................(2)

Comparing eq. (1) & (2): - ½

$$\sigma = \frac{ne^2}{m}\tau$$

(b) On increasing the temperature, the value of τ decreases as a consequence conductivity decreases and hence resistivity increases. 1



2. A battery of e.m.f. 12 V and internal resistance 0.5 Ω is connected to a 9.5 Ω resistor through a key. The ratio of potential difference between the two terminals of the battery, when the key is open to that when the key is closed, is 1

(A) 1.05 (B) 1

(C) 0.95 (D) 1.1

2. A battery of e.m.f. 12 V and internal resistance 0.5 Ω is connected to a 9.5 Ω resistor through a key. The ratio of potential difference between the two terminals of the battery, when the key is open to that when the key is closed, is **1**

$\boxed{\text{(A) } 1.05}$ (B) 1

(C) 0.95 (D) 1.1

22. (a) Define resistivity of a conductor. Discuss its dependence on temperature of the conductor and draw a plot of resistivity of copper as a function of temperature. **3**

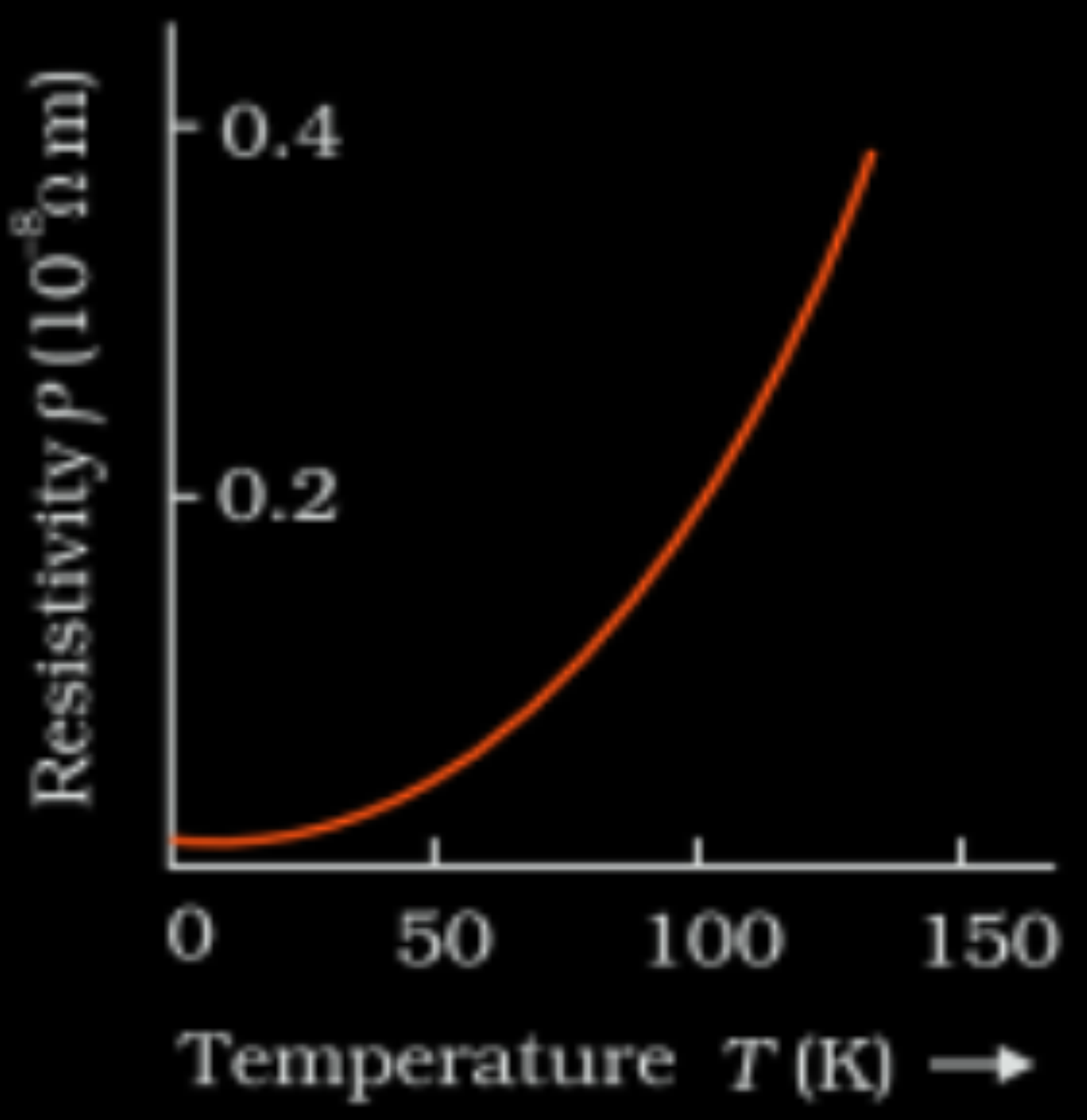
(b) (i) "A low voltage battery from which high current is required must have low internal resistance." Justify.

(ii) "A high voltage battery must have a large internal resistance." Justify.

(a) Defining resistivity	1
Discussing its dependence on temperature	½
Plotting graph of resistivity with temperature for copper	½
(b) (i) Justification	½
(ii) Justification	½

(a) Resistivity is the resistance of a material of unit length having unit area of cross-section. 1

On increasing the temperature of a conductor, the resistivity increases. ½



½

<u>Note</u>: Full credit to be given if values are not shown on the graph.

(b) (i) A low internal resistance allows large current to be drawn even at a low voltage. ½

(ii) To limit the current. ½

2. A student has three resistors, each of resistance R. To obtain a resistance of $\frac{2}{3}$ R, she should connect **1**

(A) all the three resistors in series.

(B) all the three resistors in parallel.

(C) two resistors in series and then this combination in parallel with the third resistor.

(D) two resistors in parallel and then this combination in series with the third resistor.

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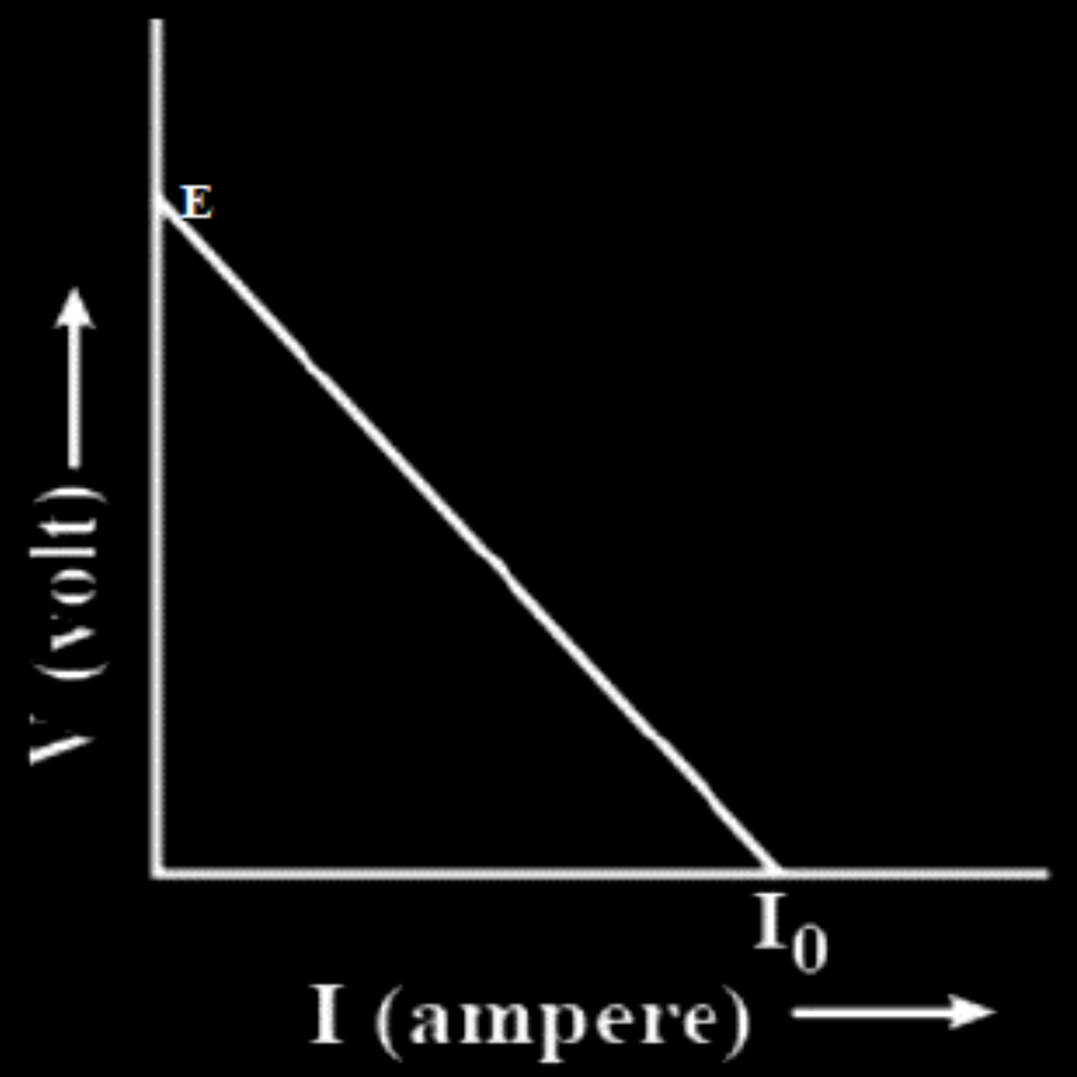
22. (a) A cell of e.m.f. E and internal resistance r is connected with a variable external resistance R and a voltmeter showing potential drop V across R. Obtain the relationship between V, E, R and r. **3**

(b) Draw the shape of the graph showing the variation of terminal voltage V of the cell as a function of current I drawn from it. How one can determine the e.m.f. of the cell and its internal resistance from this graph ?

22. (a) A cell of e.m.f. E and internal resistance r is connected with a variable external resistance R and a voltmeter showing potential drop V across R. Obtain the relationship between V, E, R and r. **3**

(b) Draw the shape of the graph showing the variation of terminal voltage V of the cell as a function of current I drawn from it. How one can determine the e.m.f. of the cell and its internal resistance from this graph ?

(a) Current drawn from the cell $(I) = \dfrac{E}{r+R}$	½
Potential difference (V)=IR	
$V = \dfrac{ER}{r+R}$	½
(b) 	1
Slope of the graph is equal to internal resistance of the cell.	½
The intercept on Y-axis gives the emf of the cell.	½

2. Two conductors A and B of the same material have their lengths in the ratio 1:2 and radii in the ratio 2:3. If they are connected in parallel across a battery, the ratio $\frac{v_A}{v_B}$ of the drift velocities of electrons in them will be - **1**

(A) 2 (B) $\frac{1}{2}$

(C) $\frac{3}{2}$ (D) $\frac{8}{9}$

2. Two conductors A and B of the same material have their lengths in the ratio 1:2 and radii in the ratio 2:3. If they are connected in parallel across a battery, the ratio $\frac{v_A}{v_B}$ of the drift velocities of electrons in them will be - **1**

$\boxed{\text{(A)} \quad 2}$ (B) $\frac{1}{2}$

(C) $\frac{3}{2}$ (D) $\frac{8}{9}$

17. n identical cells, each of e.m.f. E and internal resistance r, are connected in series. Later on it was found out that two cells 'X' and 'Y' are connected in reverse polarities. Calculate the potential difference across the cell 'X'. **2**

17. n identical cells, each of e.m.f. E and internal resistance r, are connected in series. Later on it was found out that two cells 'X' and 'Y' are connected in reverse polarities. Calculate the potential difference across the cell 'X'. **2**

Calculating the potential difference 2	
Net e.m.f = (n-4)E $\therefore I = \frac{(n-4)E}{nr}$	½
Potential difference across' X' V= E +Ir	½

22. (a) Two batteries of emf's 3V & 6V and internal resistances 0.2 Ω & 0.4 Ω are connected in parallel. This combination is connected to a 4 Ω resistor. Find : **3**

(i) the equivalent emf of the combination

(ii) the equivalent internal resistance of the combination

(iii) the current drawn from the combination

OR

22. (a) Two batteries of emf's 3V & 6V and internal resistances 0.2 Ω & 0.4 Ω are connected in parallel. This combination is connected to a 4 Ω resistor. Find : **3**

(i) the equivalent emf of the combination

(ii) the equivalent internal resistance of the combination

(iii) the current drawn from the combination

(i) Because $E_{eq} = \dfrac{E_1 r_2 + E_2 r_1}{r_1 + r_2}$	½
$E_{eq} = \dfrac{3 \times 0.4 + 6 \times 0.2}{0.6} = 4\ \text{V}$	½
(ii) $r_{eq} = \dfrac{r_1 r_2}{r_1 + r_2}$	½
$r_{eq} = \dfrac{0.2 \times 0.4}{0.2 + 0.4} = 0.133\Omega$	½
(iii) $I = \dfrac{E}{R + r_{eq}}$	½
$I = \dfrac{4}{4 + 0.13} = \dfrac{4}{4.13} A$	½
$I = 0.9A$	

(b) (i) A conductor of length l is connected across an ideal cell of emf E. Keeping the cell connected, the length of the conductor is increased to $2l$ by gradually stretching it. If R and R' are initial and final values of resistance and ν_d and ν_d' are initial and final values of drift velocity, find the relation between (i) R' and R and (ii) ν_d' and ν_d.

(ii) When electrons drift in a conductor from lower to higher potential, does it mean that all the 'free electrons' of the conductor are moving in the same direction ?

(i) $l' = 2l$ $Al = A'l'$ = volume of the wire $Al = A'(2l)$ $\frac{A}{2} = A'$ $R = \frac{\rho l}{A}$ $R' = \frac{\rho l'}{A'}$ $R' = \frac{\rho(2l)}{A/2}$ $\frac{R'}{R} = 4$	**Alternatively** $R' = n^2 R$ n= 2 $R' = 4R$ (ii) $v_d = \frac{eE}{m}\tau$ $v_d = \frac{eV}{ml}\tau$ $v_d' = \frac{eV}{ml'}\tau$ $\frac{v_d'}{v_d} = \frac{l}{l'} = \frac{1}{2}$	½ ½ ½ ½

(ii) No	1

2. The resistance of a wire of length L and radius r is R. Which one of the following would provide a wire of the same material of resistance $\frac{R}{2}$? **1**

(A) Using a wire of same radius and twice the length
(B) Using a wire of same radius and half length
(C) Using a wire of same length and twice the radius
(D) Using a wire of same length and half the radius

2. The resistance of a wire of length L and radius r is R. Which one of the following would provide a wire of the same material of resistance $\frac{R}{2}$? **1**

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(B) $\boxed{\text{Using a wire of same radius and half length}}$

(C) Using a wire of same length and twice the radius

(D) Using a wire of same length and half the radius

17. Show that $\vec{E} = \rho \vec{j}$ leads to Ohm's law. Write a condition in which the Ohm's law is not valid for a material. **2**

17. Show that $\vec{E} = \rho\vec{j}$ leads to Ohm's law. Write a condition in which the Ohm's law is not valid for a material. **2**

Obtaining Ohm's law from $\vec{E} = \rho\vec{j}$	1 ½
Writing the condition	½

$\vec{E} = \rho\vec{j}$	½
$\dfrac{V}{l} = \rho\dfrac{I}{A}$	½
$\dfrac{V}{I} = \rho\dfrac{l}{A} =$ constant (Ohm's Law)	½
Condition for non- validity of Ohm's Law High temperature / in semiconductor	½

17. A battery of emf E and internal resistance r is connected to a rheostat. When a current of 2A is dawn from the battery, the potential difference across the rheostat is $5V$. The potential difference becomes $4V$ when a current of 4A is drawn from the battery. Calculate the value of E and r. **2**

17. A battery of emf E and internal resistance r is connected to a rheostat. When a current of 2A is dawn from the battery, the potential difference across the rheostat is $5V$. The potential difference becomes $4V$ when a current of 4A is drawn from the battery. Calculate the value of E and r. **2**

Calculating the value of E 1 Calculating the value of r 1	
$E = V + Ir$	
In first case	½
E = 5 + 2r	
In second case	
E = 4 + 4r	½
After solving	
E= 6 V	½
r = 0.5 Ω	½

CBSE 2020

12. A copper wire of non-uniform area of cross-section is connected to a d.c. battery. The physical quantity which remains constant along the wire is ________ **1**

12. A copper wire of non-uniform area of cross-section is connected to a d.c. battery. The physical quantity which remains constant along the wire is Electric current 1

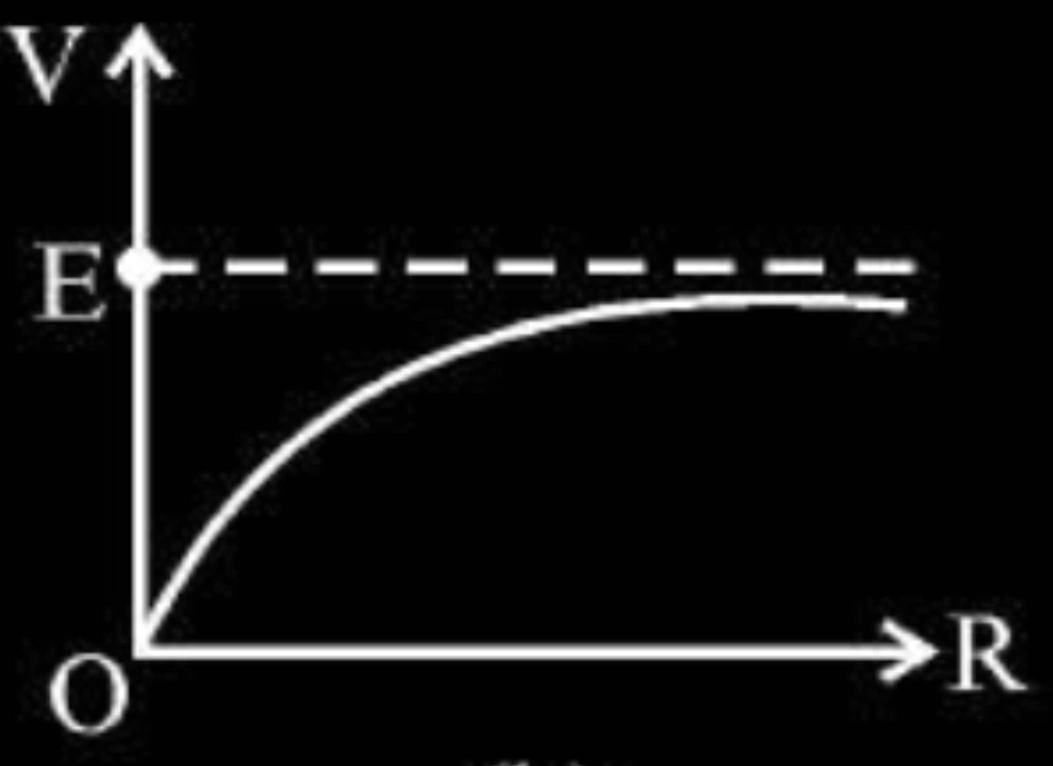
28. (a) Differentiate between electrical resistance and resistivity of a conductor.

(b) Two metallic rods, each of length L, area of cross A_1 and A_2, having resistivities ρ_1 and ρ_2 are connected in parallel across a d.c. battery. Obtain the expression for the effective resistivity of this combination. **3**

(a) Electrical resistance (R) of a conductor equals the ratio of the potential difference (V) applied across it, to the resulting current (I) flowing through it. (Alternatively: $R = \frac{V}{I}$)	1
The resistivity of a conductor equals the resistance of a wire of unit length and unit area of cross section, drawn from the material of that conductor. (Alternatively: $R = \rho \frac{l}{A}$ or $\rho = \frac{RA}{l}$)	1
(or any other one relevant difference)	
(b) For the parallel combination equivalent resistance is given by $\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2}$	½
$\frac{A_1 + A_2}{\rho_{eq} L} = \frac{A_1}{\rho_1 L} + \frac{A_2}{\rho_2 L}$	
Where $(A_1 + A_2)$ is the effective area of cross section of combined rod in parallel combination of the rods.	
$\frac{\rho_1 \rho_2}{(\rho_2 A_1 + \rho_1 A_2)} = \frac{\rho_{eq}}{(A_1 + A_2)}$	
$\Rightarrow \rho_{eq} = \frac{\rho_1 \rho_2 (A_1 + A_2)}{(\rho_2 A_1 + \rho_1 A_2)}$	½
(**Note :**If a student just writes the expression of equivalent resistance, award half mark of this part)	

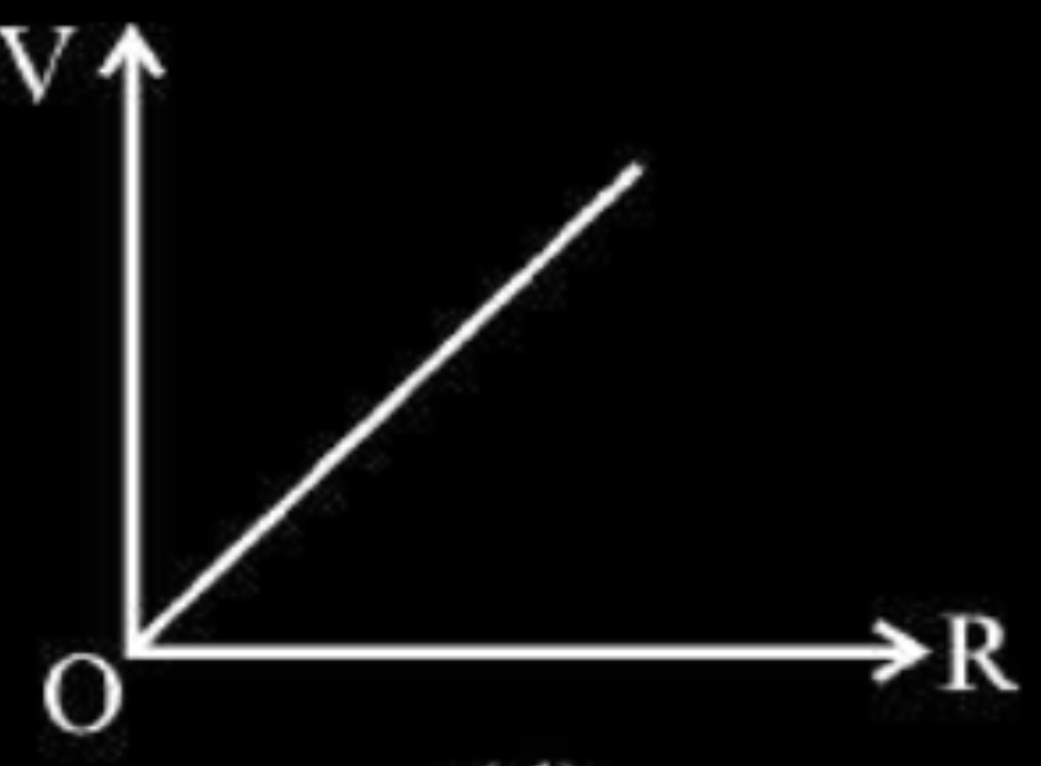
1. A cell of emf (E) and internal resistance r is connected across a variable external resistance R. The graph of terminal potential difference V as a function of R is –

(a)



(b)

(c)

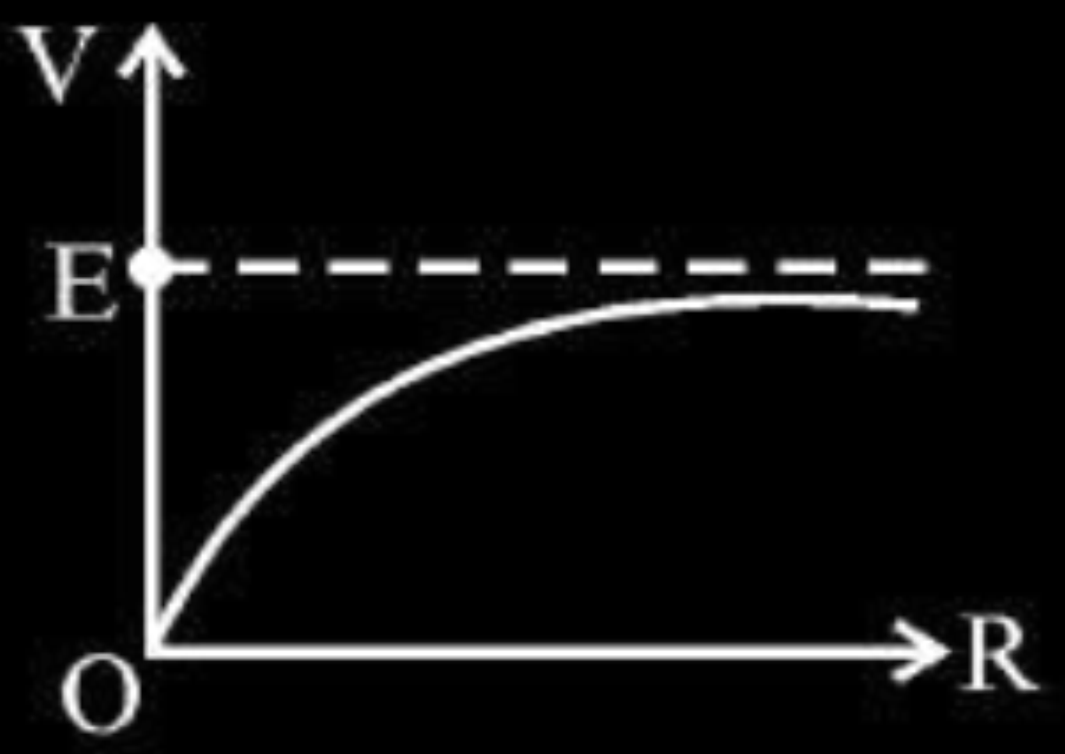


(d)

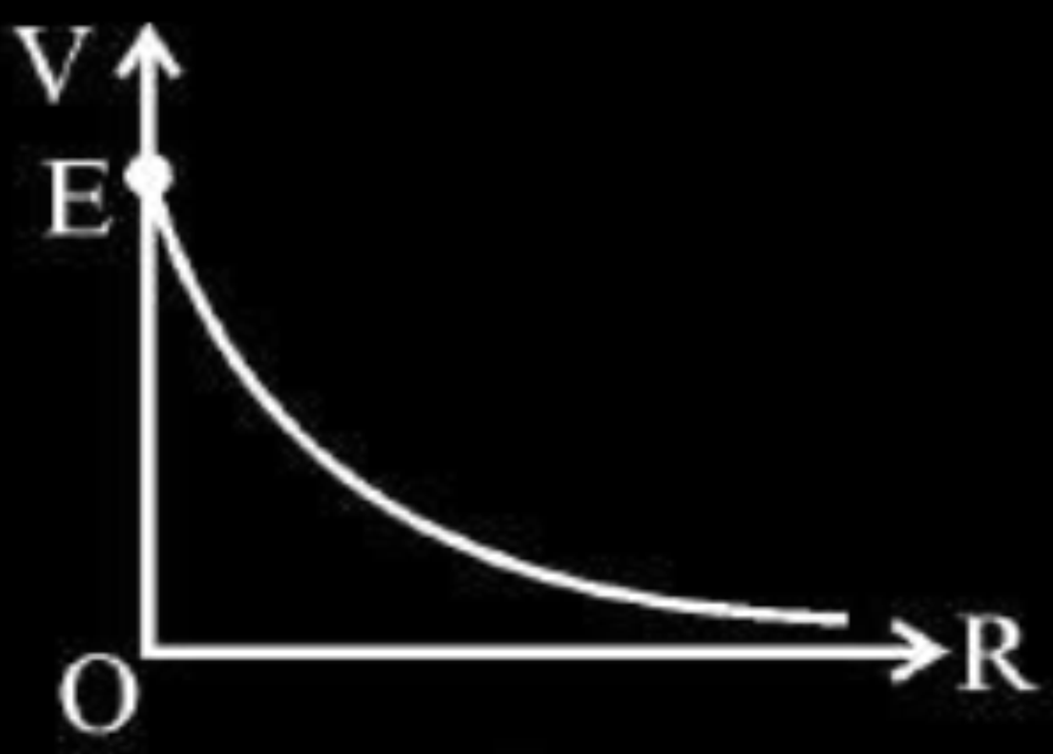
1

1. A cell of emf (E) and internal resistance r is connected across a variable external resistance R. The graph of terminal potential difference V as a function of R is –

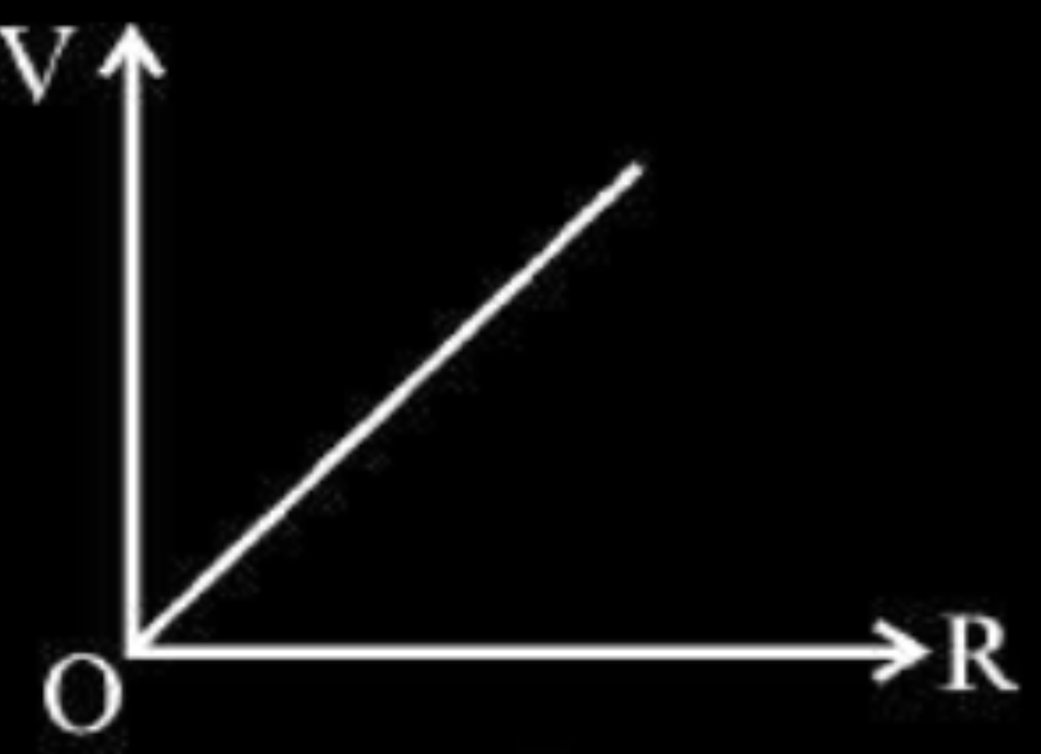
(a)



(b)



(c)



(d) 1

2. A uniform wire of resistance 2R is bent in the form of a circle. The effective resistance between the ends of any diameter of the circle is :

(a) 2R (b) R (c) $\frac{R}{2}$ (d) $\frac{R}{4}$ **1**

2. A uniform wire of resistance 2R is bent in the form of a circle. The effective resistance between the ends of any diameter of the circle is :

(a) 2R (b) R $\boxed{\text{(c)} \quad \frac{R}{2}}$ (d) $\frac{R}{4}$ **1**

22. A wire of length L_o has a resistance R_o. It is gradually stretched till its length becomes $2\,L_o$.

(a) Plot a graph showing variation of its resistance R with its length l during stretching.

(b) What will be its resistance when its length becomes $2\,L_o$? **2**

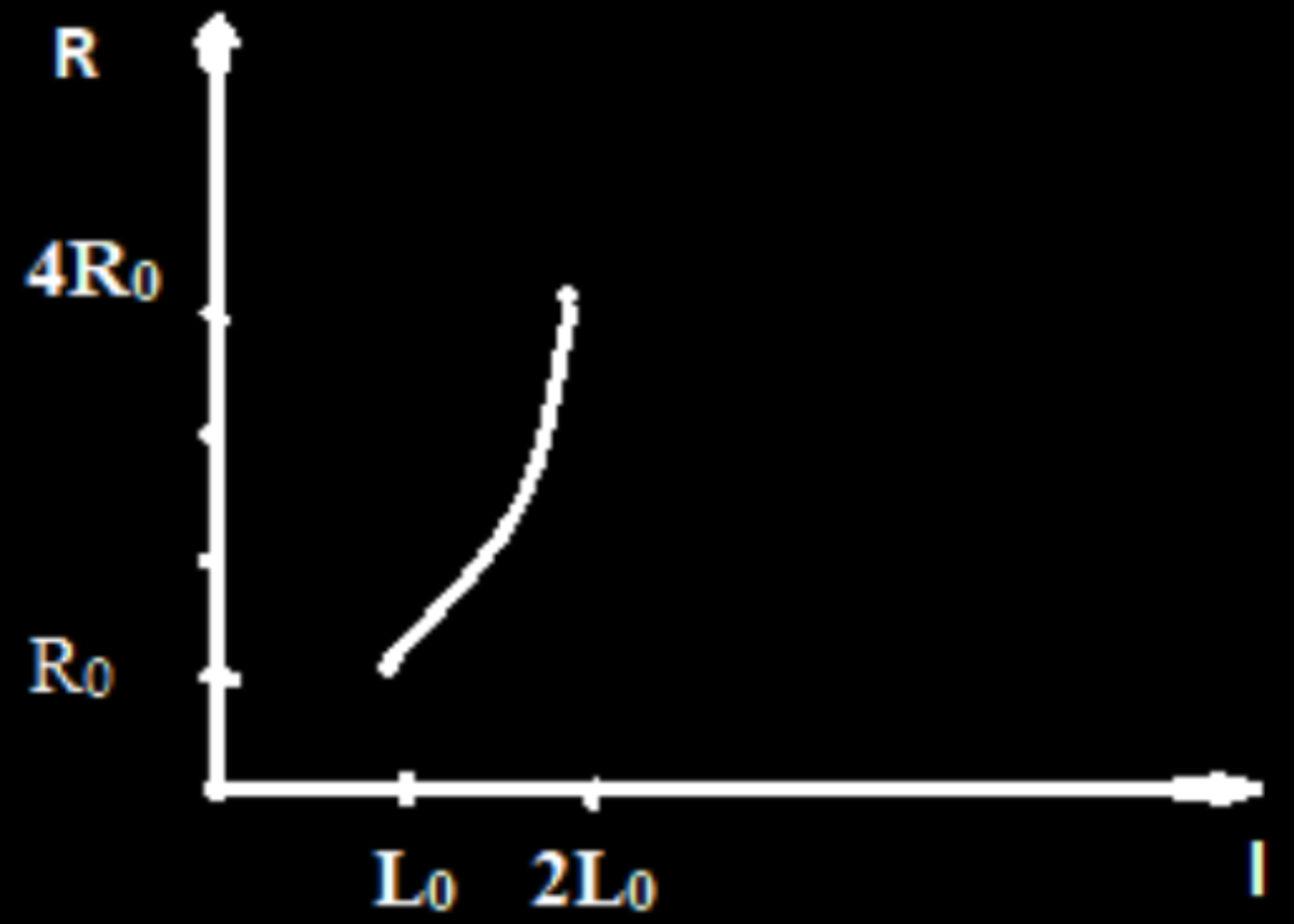
22. A wire of length L_o has a resistance R_o. It is gradually stretched till its length becomes $2\ L_o$.

(a) Plot a graph showing variation of its resistance R with its length l during stretching.

(b) What will be its resistance when its length becomes $2\ L_o$? **2**

(a) Graph	1
(b) Final resistance	1

(a) $R = S\frac{l}{A} = S\frac{l^2}{V}$ ½



½

[Note if a student draws only the graph correctly, award full 1 mark]

(b) Resistance becomes 4 R_0 1

(Hint :- As $R = \frac{\rho l}{A} = \frac{\rho l^2}{Al} = \frac{\rho l^2}{V}$) (V= Volume)

$R \propto l^2$

4. The electrical resistance of a conductor *1*

(A) varies directly proportional to its area of cross-section.

(B) decreases with increase in its temperature.

(C) decreases with increase in its conductivity.

(D) is independent of its shape but depends only on its volume.

4. The electrical resistance of a conductor *1*

(A) varies directly proportional to its area of cross-section.

(B) decreases with increase in its temperature.

(C) decreases with increase in its conductivity.

(D) is independent of its shape but depends only on its volume.

5. $m^2V^{-1}s^{-1}$ is the SI unit of which of the following ? *1*

(A) Drift velocity

(B) Mobility

(C) Resistivity

(D) Potential gradient

5. $m^2V^{-1}s^{-1}$ is the SI unit of which of the following ? *1*

(A) Drift velocity

$\boxed{\text{(B) Mobility}}$

(C) Resistivity

(D) Potential gradient

6. The element of a heater is rated (P, V). If it is connected across a source of voltage $\frac{V}{2}$, then the power consumed by it will be *1*

(A) P

(B) 2P

(C) $\frac{P}{2}$

(D) $\frac{P}{4}$

6. The element of a heater is rated (P, V). If it is connected across a source of voltage $\frac{V}{2}$, then the power consumed by it will be *1*

(A) P

(B) 2P

(C) $\frac{P}{2}$

(D) $\boxed{\frac{P}{4}}$

28. (a) Differentiate between the random velocity and the drift velocity of electrons in an electrical conductor. Give their order of magnitudes.

(b) A conductor of uniform cross-sectional area is connected across a dc source of variable voltage. Draw a graph showing variation of drift velocity of electrons (v_d) as a function of current density (J) in it. *3*

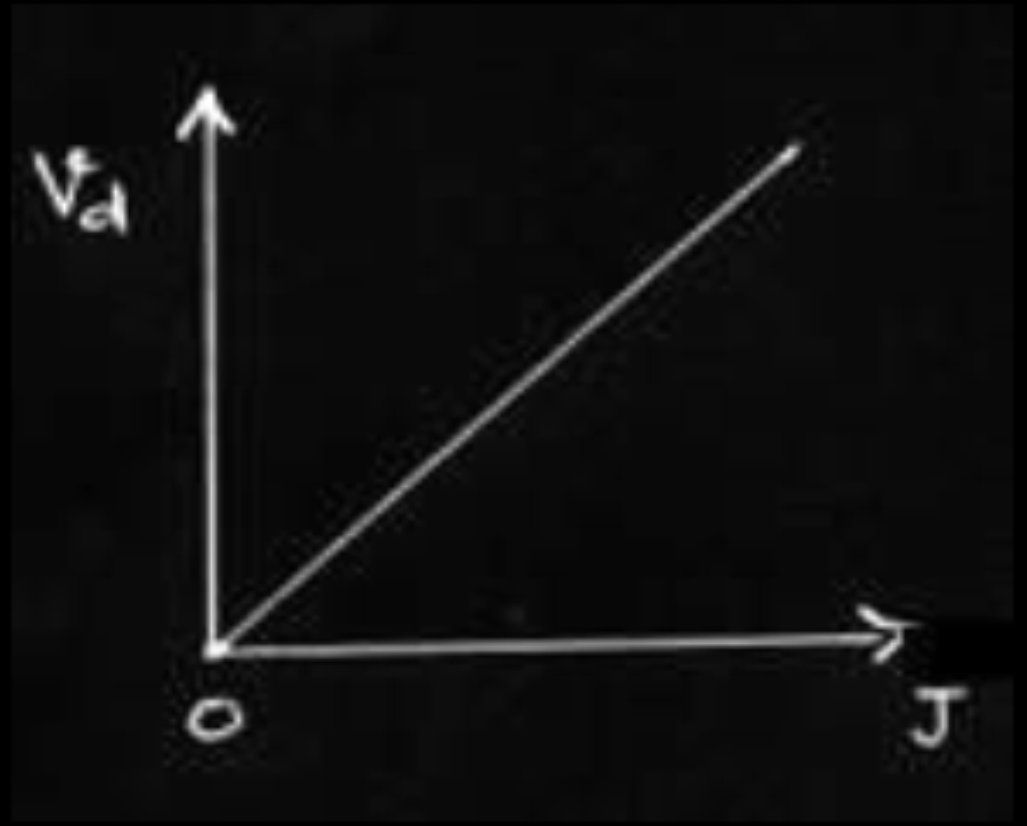
a) differentiating between random velocity and drift velocity	1mark
Order of magnitude	1 mark
b) drawing the graph showing the variation of drift velocity as a function of Current density	1 mark

Random Velocity v	Drift Velocity v_d
1. The velocity acquired by the free electrons in the absence of electric field.	1. The average velocity acquired by the free electrons in presence of electric field.
2. The average random velocity is zero.	2. The average drift velocity is not zero.
3. Has quite a large value	3. Has a very small value

1

Order of magnitude of random velocity is 10^2m/s. ½
Order of magnitude of drift velocity is 10^{-3}m/s. ½

[Note: If the student writes drift speed is nearly 10^{-5} times smaller than random velocity ,award the last 1 mark]



1

[if a student writes $J = \frac{I}{A} = \frac{n\ e\ A\ v_d}{A} = nev_d$ but does not draw the graph award ½ mark only]

1. A cell of internal resistance r connected across an external resistance R can supply maximum current when *1*

(A) $R = r$

(B) $R > r$

(C) $R = \frac{r}{2}$

(D) $R = 0$

1. A cell of internal resistance r connected across an external resistance R can supply maximum current when *1*

(A) $R = r$

(B) $R > r$

(C) $R = \frac{r}{2}$

(D) $\boxed{R = 0}$

2. In a current carrying conductor, the ratio of the electric field and the current density at a point is called *1*

(A) Resistivity

(B) Conductivity

(C) Resistance

(D) Mobility

2. In a current carrying conductor, the ratio of the electric field and the current density at a point is called *1*

$\boxed{\text{(A) Resistivity}}$

(B) Conductivity

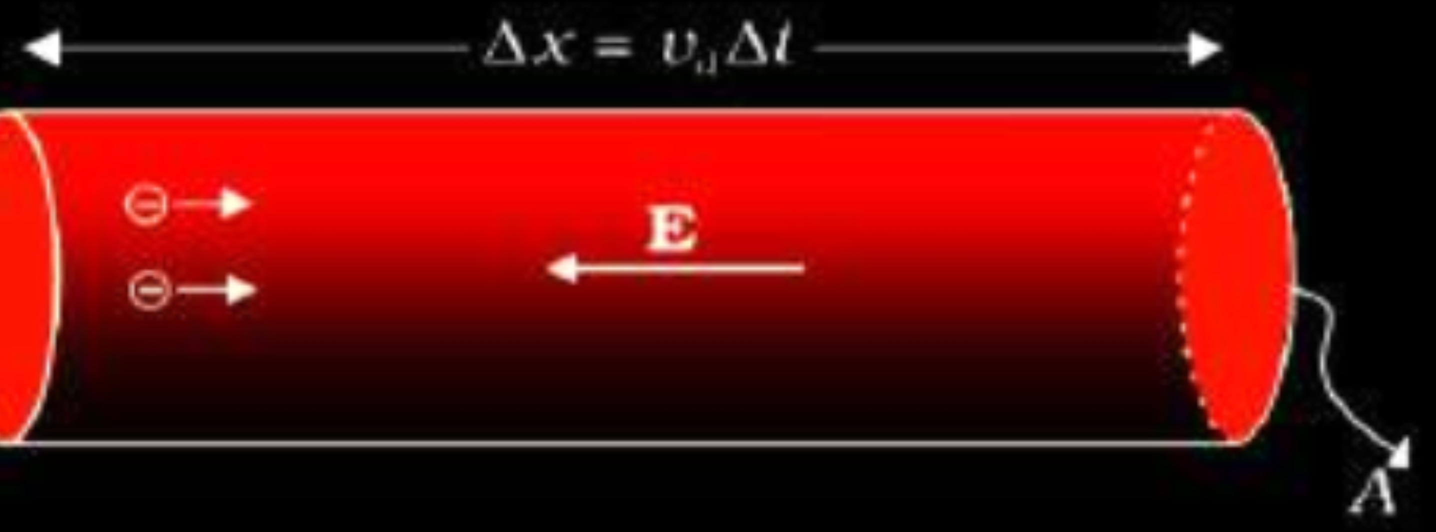
(C) Resistance

(D) Mobility

21. Define the term 'mobility' of charge carriers in a current carrying conductor. Obtain the relation for mobility in terms of relaxation time. *2*

OR

Define the term 'drift velocity' of electrons in a current carrying conductor. Obtain the relationship between the current density and the drift velocity of electrons. *2*

Mobility is defined as the magnitude of drift velocity per unit electric field. $\mu = \frac{\lvert\vec{V_d}\rvert}{E}$ [Even if a student writes only the mathematical relation award ½ mark] Given $V_d = \frac{e\tau E}{m}$ Hence, $\mu = \frac{V_d}{E} = \frac{e\tau}{m}$	1 ½ ½
The average speed with which electrons move when an electric field or potential difference is applied is called drift velocity. $\vec{V_d} = \frac{-e\vec{E}\tau}{m}$ [Award 1/2mark if student writes the formulae]  The amount of charge crossing the area A in time Δt $I\Delta t = ne\,A\,\lvert\vec{V_d}\rvert\,\Delta t$ Hence current density $j = \frac{I}{A} = ne\,V_d$	1 ½ ½

4. Two resistors R_1 and R_2 of 4 Ω and 6 Ω are connected in parallel across a battery. The ratio of power dissipated in them, $P_1 : P_2$ will be *1*

(A) 4 : 9

(B) 3 : 2

(C) 9 : 4

(D) 2 : 3

4. Two resistors R_1 and R_2 of 4 Ω and 6 Ω are connected in parallel across a battery. The ratio of power dissipated in them, $P_1 : P_2$ will be *1*

(A) 4 : 9

$\boxed{\text{(B)} \quad 3 : 2}$

(C) 9 : 4

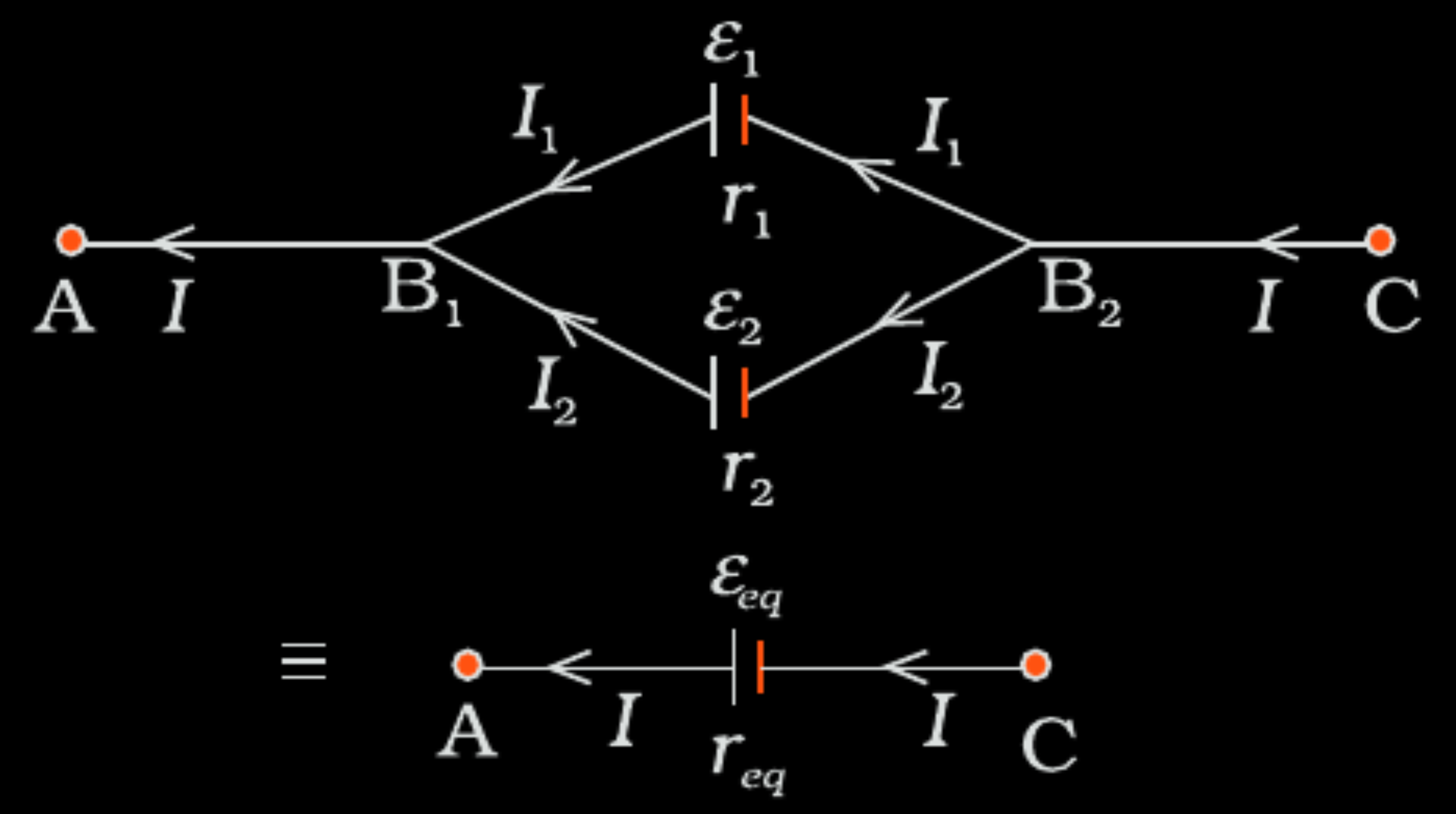
(D) 2 : 3

28. (a) Two cells of emf E_1 and E_2 have their internal resistances r_1 and r_2, respectively. Deduce an expression for the equivalent emf and internal resistance of their parallel combination when connected across an external resistance R. Assume that the two cells are supporting each other.

(b) In case the two cells are identical, each of emf E = 5 V and internal resistance r = 2 Ω, calculate the voltage across the external resistance R = 10 Ω. *3*

a) Internal resistance	1 ½ mark
b) Voltage across R	1 ½ mark

Current drawn from cell -1 $$I_1 = \frac{E_1 - V}{r_1}$$	½
Current drawn from cell -2 $$I_2 = \frac{E_2 - V}{r_2}$$	
Resultant current $\quad I = I_1 + I_2$	
On solving $$\therefore I = \frac{E_1 r_2 + E_2 r_1}{r_1 r_2} - V\left(\frac{r_2 + r_1}{r_1 r_2}\right)$$ $$\therefore V = \frac{E_1 r_2 + E_2 r_1}{r_1 r_2} - I\left(\frac{r_1 r_2}{r_2 + r_1}\right)$$ $$V = E_{eq} - I r_{eq}$$	
$$E_{eq} = \frac{E_1 r_2 + E_2 r_1}{r_1 + r_2}$$	½
$$r_{eq} = \frac{r_1 r_2}{r_2 + r_1}$$	½



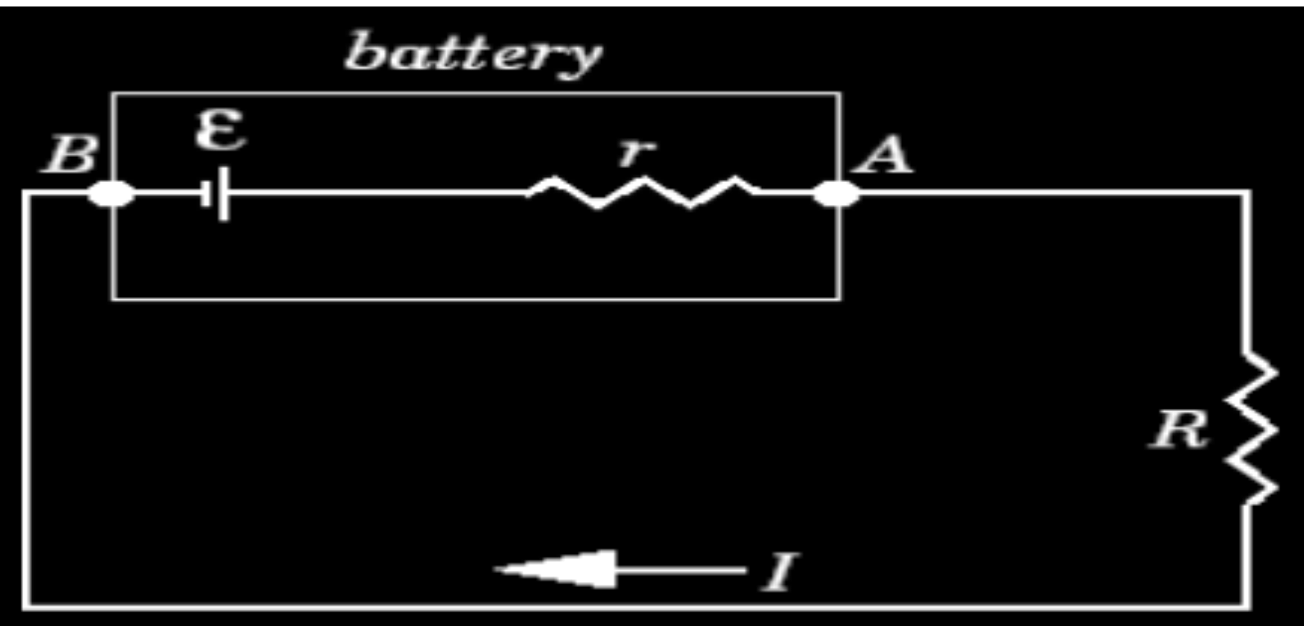
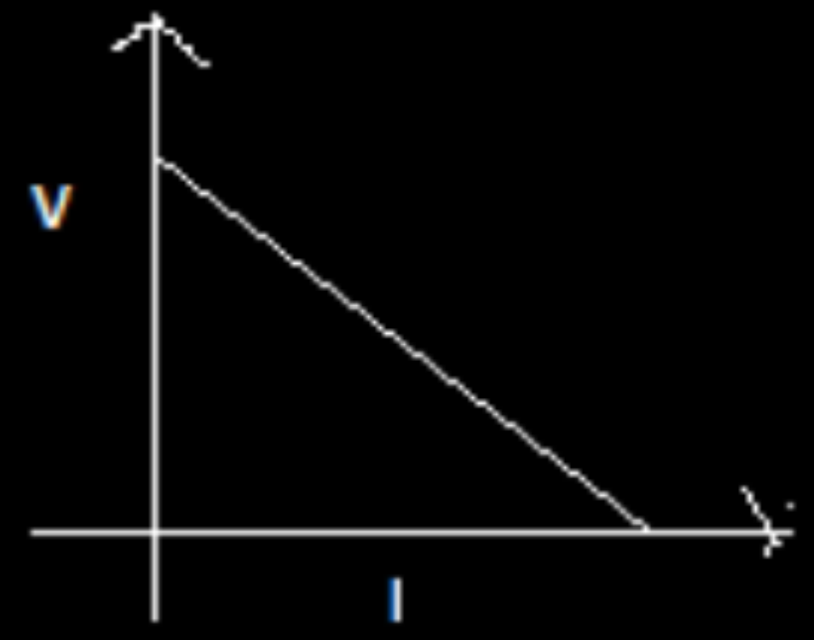
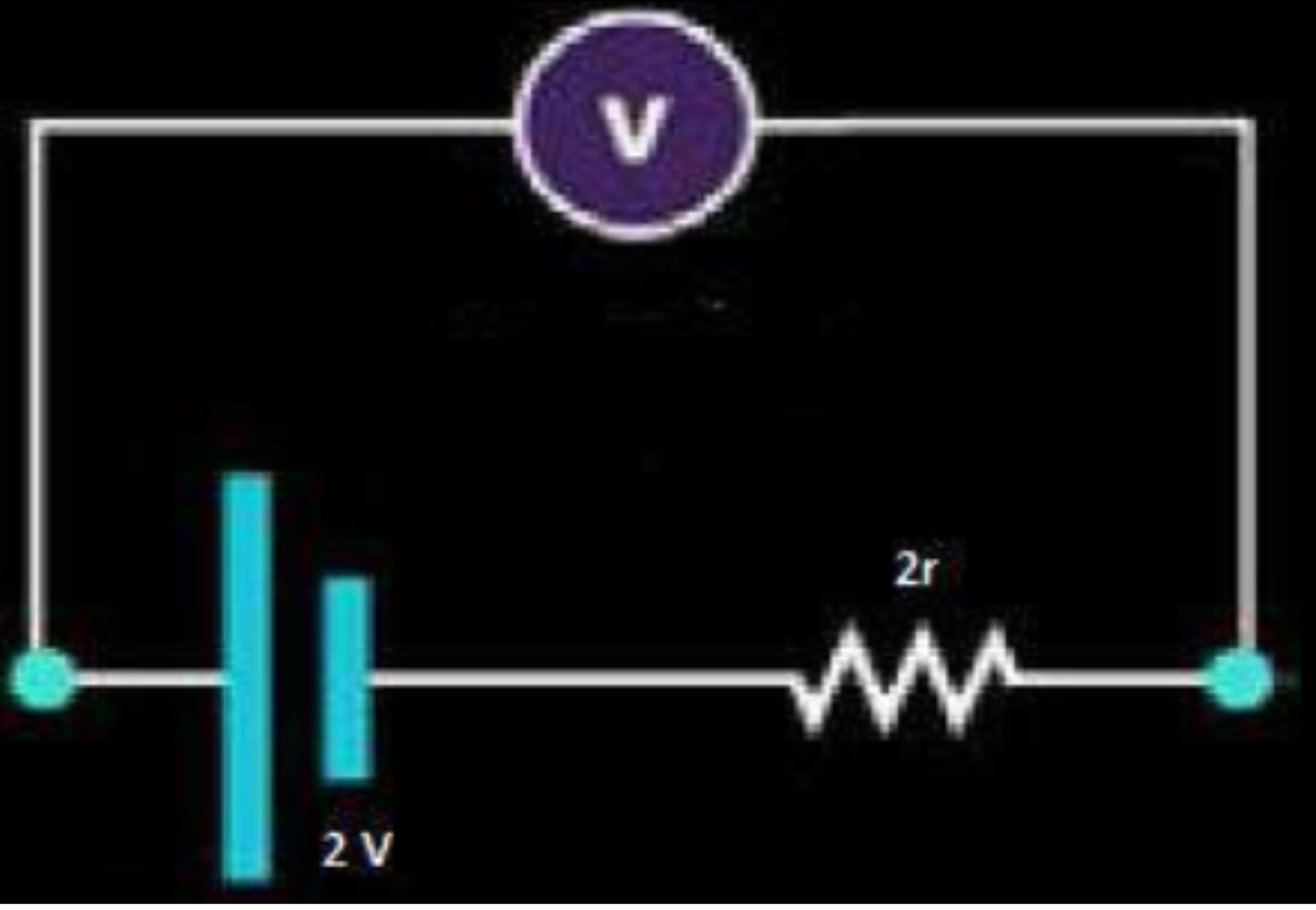
$$r_{eff} = \frac{r_1 r_2}{r_1 + r_2} = \frac{2 \times 2}{2 + 2} = 1\Omega$$	½
Current through R $$I = \frac{E_{effect}}{R + r_{eff}} = \frac{5}{10 + 1} = \frac{5}{11} A$$	½
P.D across R $$= \frac{5}{11} \times 10 = 4.54\ volt$$	½

CBSE 2019

(a) Derive a relation between the internal resistance, emf and terminal potential difference of a cell from which current I is drawn. Draw V vs I graph for a cell and explain its significance.

(b) A voltmeter of resistance 998 Ω is connected across a cell of emf 2 V and internal resistance 2 Ω. Find the potential difference across the voltmeter and also across the terminals of the cell. Estimate the percentage error in the reading of the voltmeter.

a) Relation between E, V, r	1
Graph V v/s I	1
Significance of graph	1
b) Current in voltmeter	1
Potential difference across voltmeter	½
Percentage Error	½

(a)	 E- IR – rI = 0 E – V – Ir = 0 E = V + Ir [Award 1 mark even if student writes the relation directly]  Significance of Graph – To find cmf and internal resistance of cell.	½ ½ 1
(b)	V = E – Ir 998 ×I = 2 – 2I 1000 ×I = 2 $I = \frac{2}{1000} = .002$ A V = .002 x 998 V = 1.996 V % error = $\frac{.004}{2} \times 100 = 0.2\ \%$ 	½ ½ ½

(a) Two cells of different emfs and internal resistances are connected in parallel with one another. Derive the expression for the equivalent emf and equivalent internal resistance of the combination.

(b) Two identical cells of emf 1·5 V and internal resistance r are each connected in parallel providing a supply to an external circuit consisting of two resistances of 17 Ω each joined in parallel. A very high resistance voltmeter reads the terminal voltage of the cell to be 1·4 V. Calculate the internal resistance of each cell.

a)	Expression of current in terms of E and V	1
	Expression of voltage in terms of emf and internal resistance	1
	Expression for Eeq and req	½ + ½
b)	Value of current & value of internal resistance	1+1

$I = I_1 + I_2$

Potential difference across $B_1 B_2$

$V = E_1 - I_1 r_1 \quad \Rightarrow \quad I_1 = \dfrac{E_1 - V}{r_1}$ ½

$V = E_2 - I_2 r_2 \quad \Rightarrow \quad I_2 = \dfrac{E_2 - V}{r_2}$

$I = I_1 + I_2$

$= \dfrac{E_1 - V}{r_1} + \dfrac{E_2 - V}{r_2}$ ½

$= \left(\dfrac{E_1}{r_1} + \dfrac{E_2}{r_2}\right) - V\left(\dfrac{1}{r_1} + \dfrac{1}{r_2}\right)$

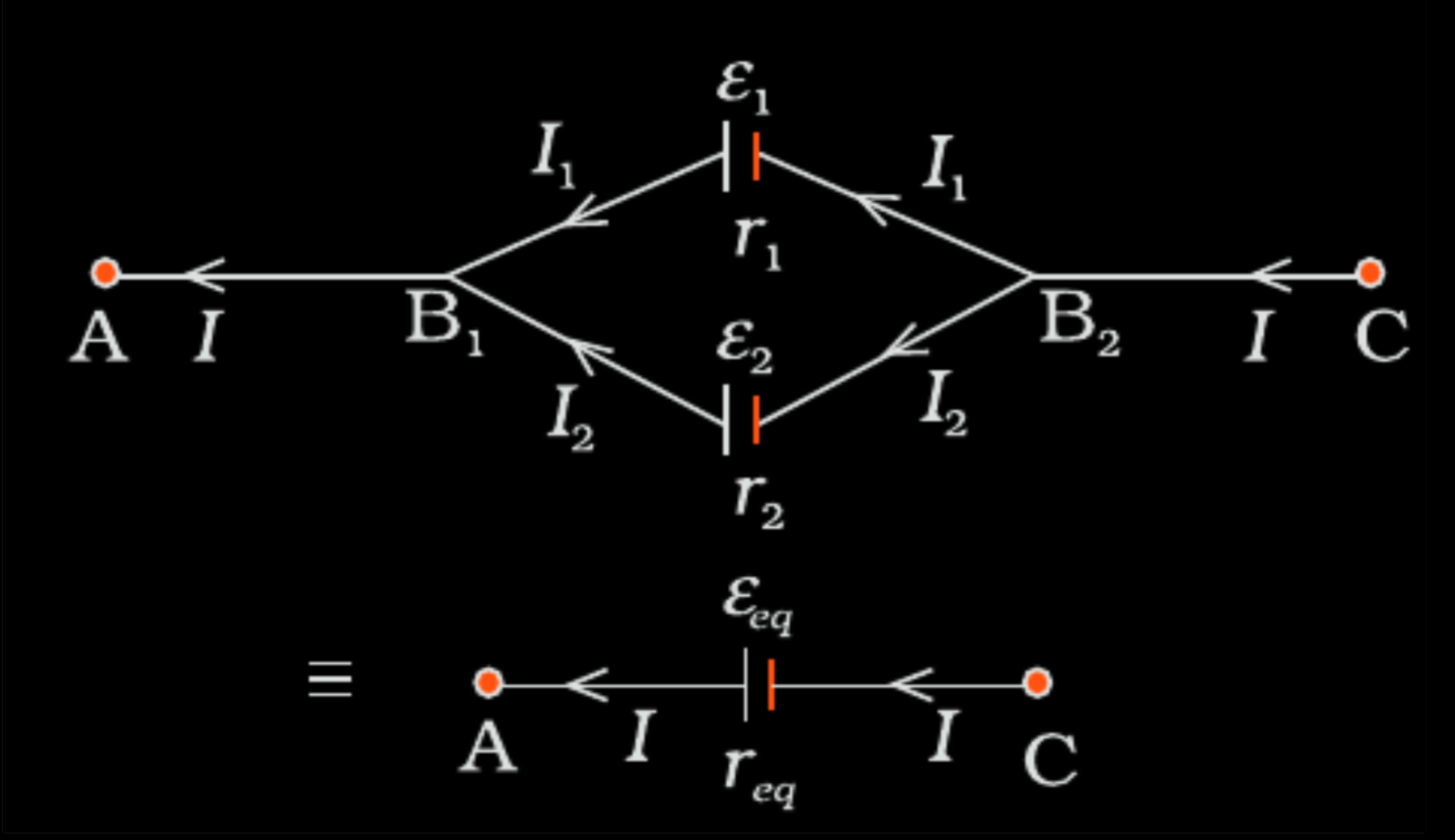
$V = \dfrac{E_1 r_2 + E_2 r_1}{r_1 + r_2} - \dfrac{I r_1 r_2}{r_1 + r_2}$ ½

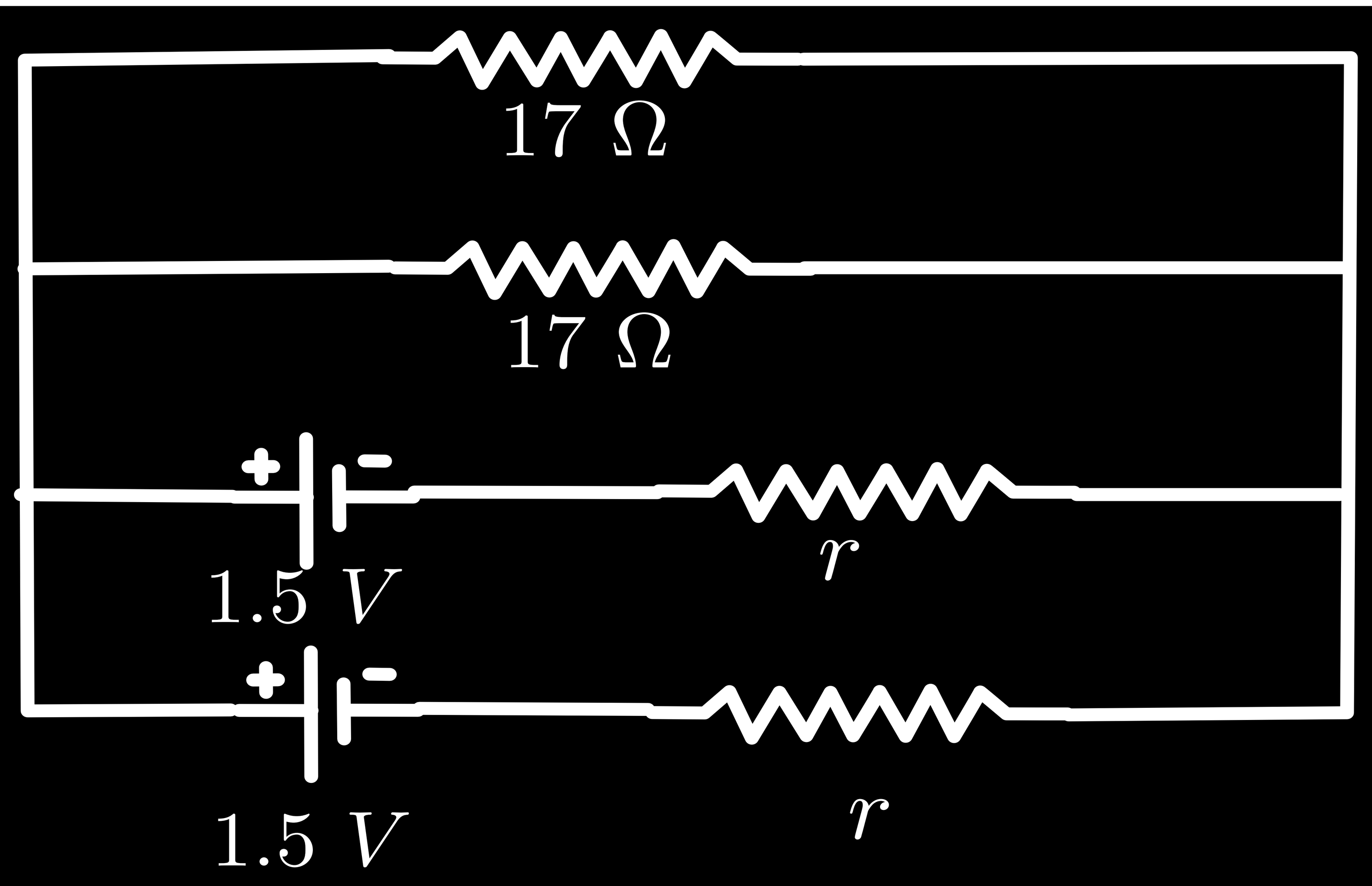
Compare with

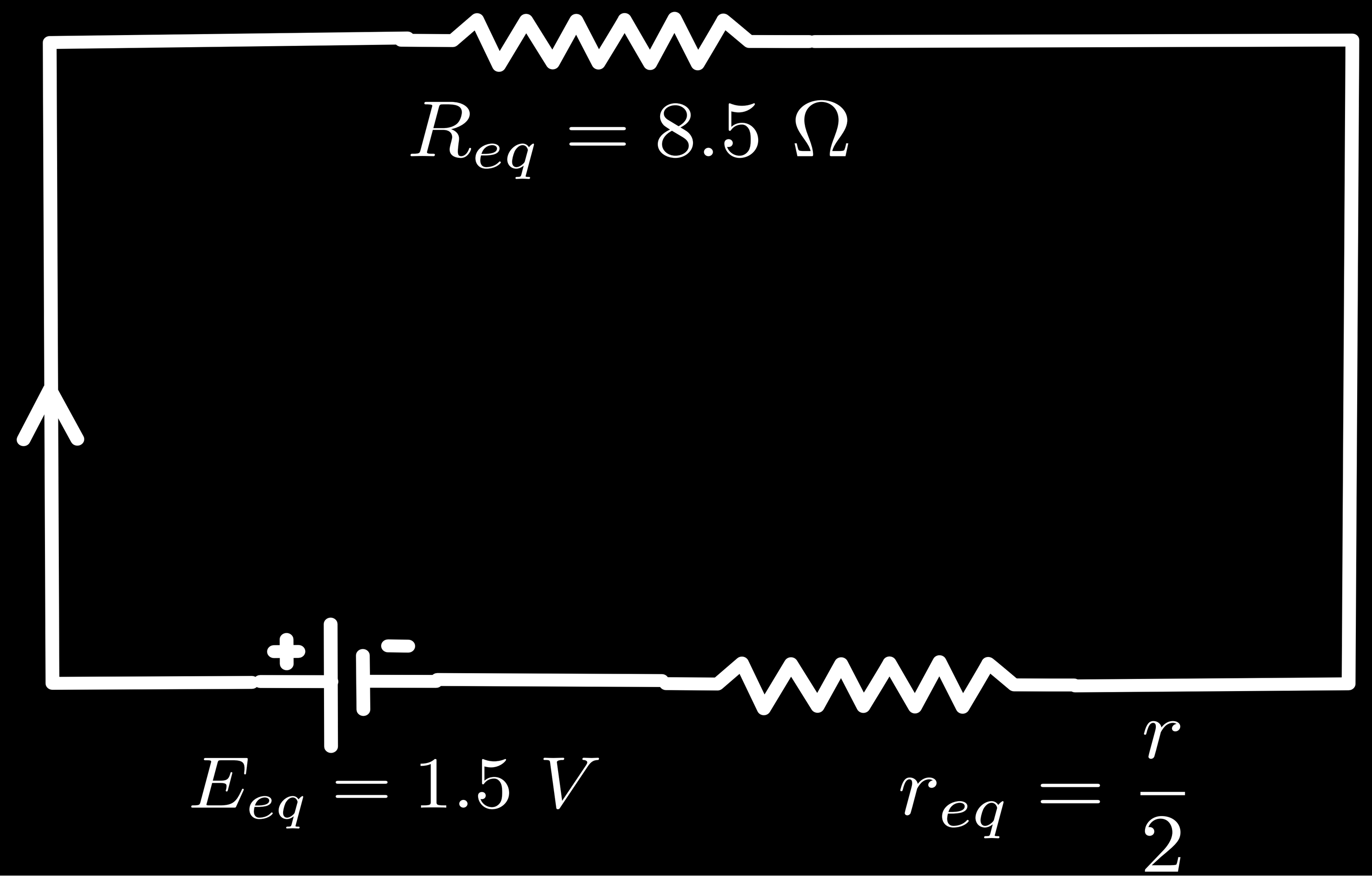
$V = E\,eq - I\,r_{eg}$ ½

$$E_{eq} = \frac{E_1 r_2 + E_2 r_1}{r_1 + r_2}$$ ½

$$r_{eq} = \frac{r_1 r_2}{r_1 + r_2}$$







Equivalent circuit 1

$$I = \frac{V}{R_{eq}} = \frac{1.4}{8.5}\ A$$

And

$$V = E_{eq} - I r_{eq}$$

$$\Rightarrow 1.4 = 1.5 - \frac{1.4}{8.5} X \frac{r}{2}$$ 1

$$\Rightarrow r = 1.21\Omega$$

Under what condition will the current in a wire be the same when connected in series and in parallel of n identical cells each having internal resistance r and external resistance R ?

Under what condition will the current in a wire be the same when connected in series and in parallel of n identical cells each having internal resistance r and external resistance R ?

When $R+nr = nR+r$ **Alternatively** $r = R$	1

(a) Draw a graph showing the variation of current versus voltage in an electrolyte when an external resistance is also connected.

(b) (i) The graph between resistance (R) and temperature (T) for Hg is shown in the figure (a). Explain the behaviour of Hg near 4 K.

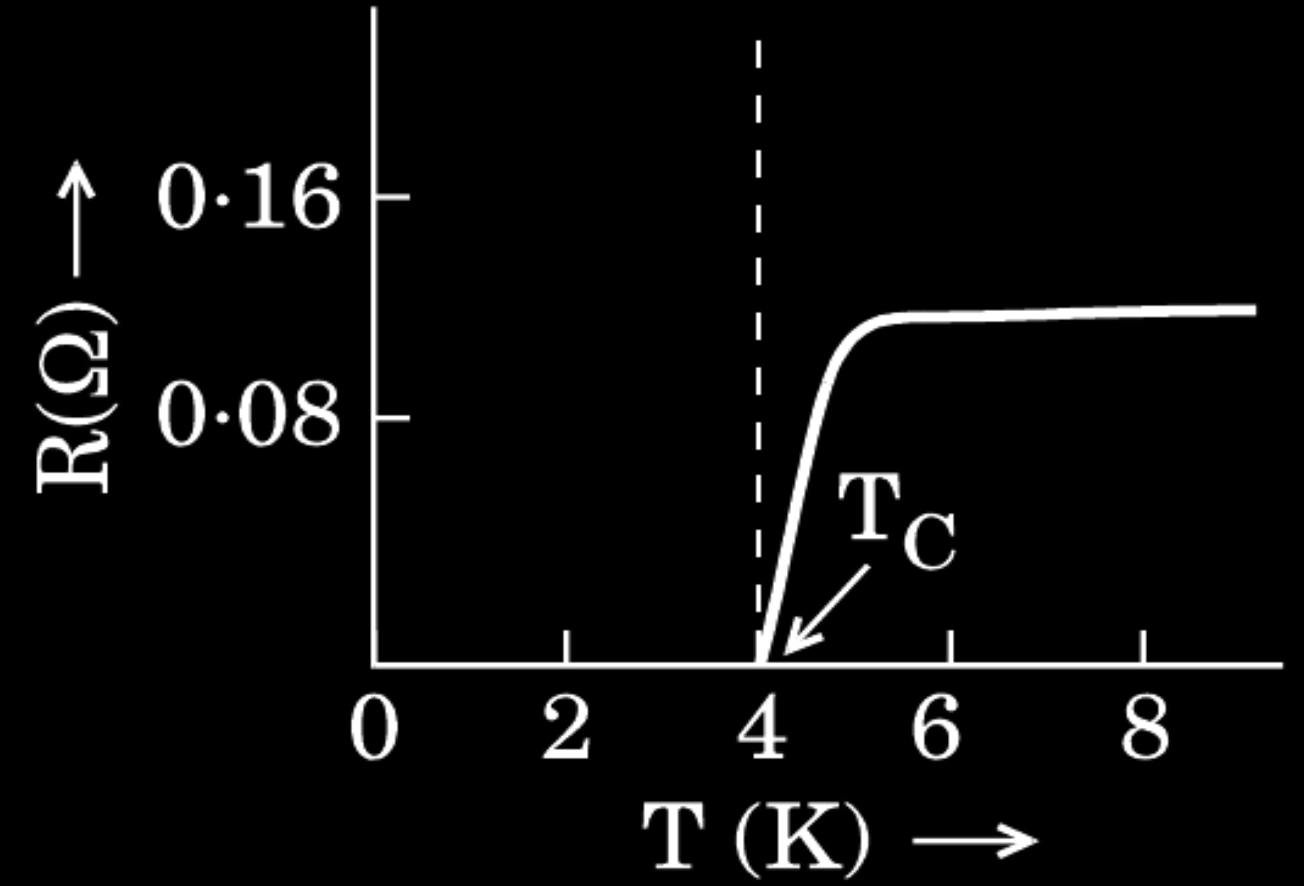


Figure (a)

(ii) In which region of the graph shown in the figure (b) is the resistance negative and why ?

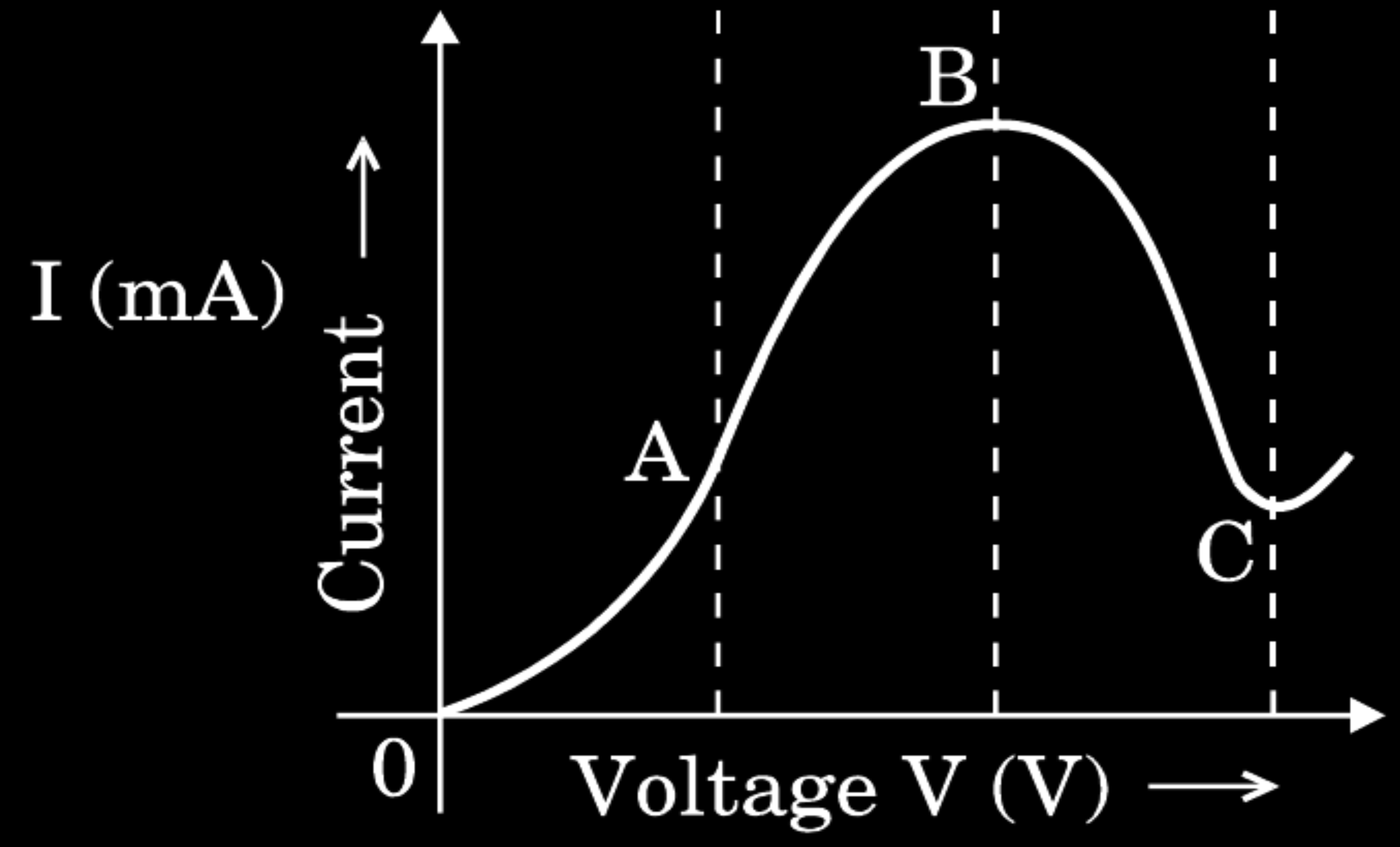
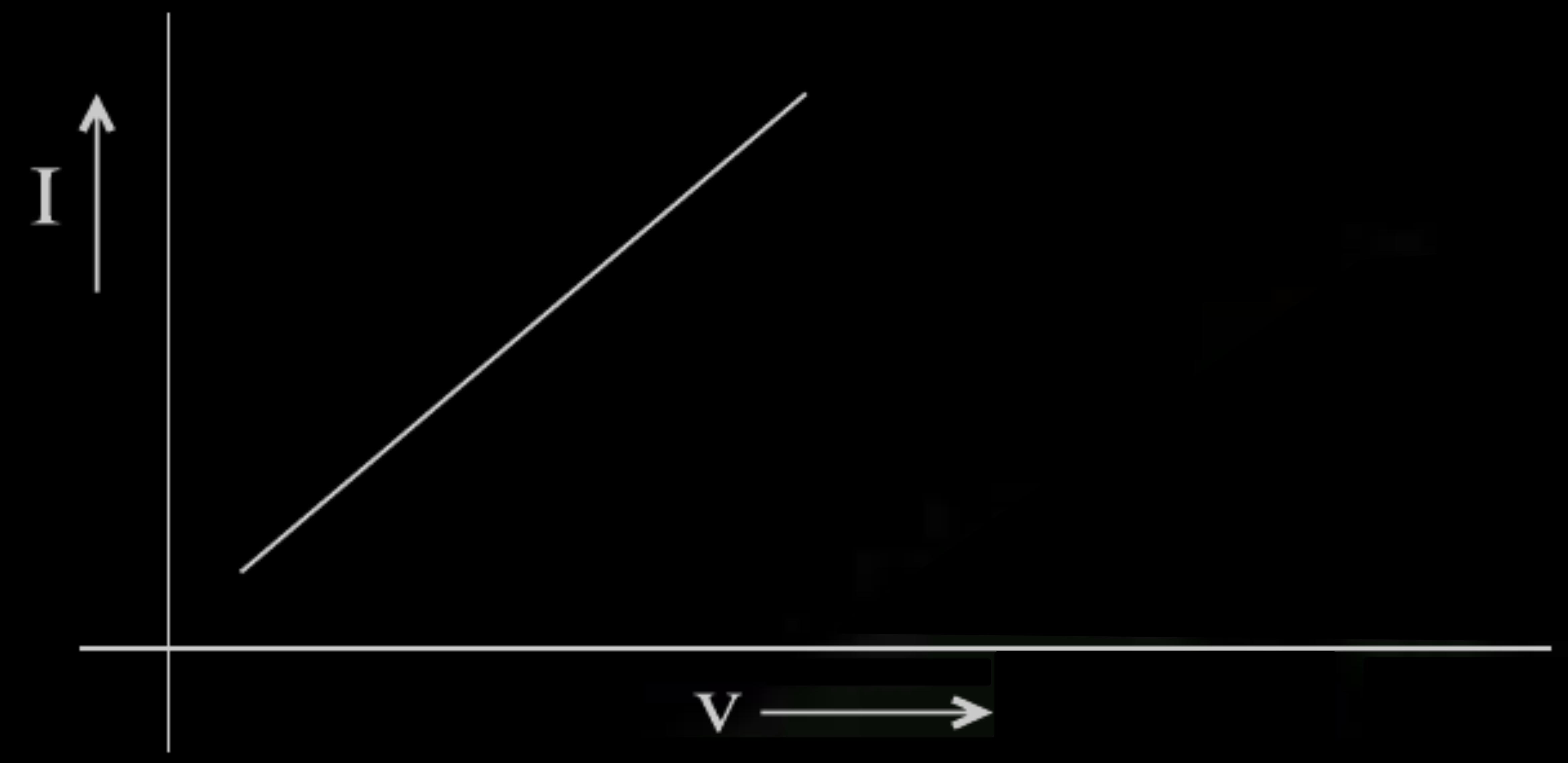
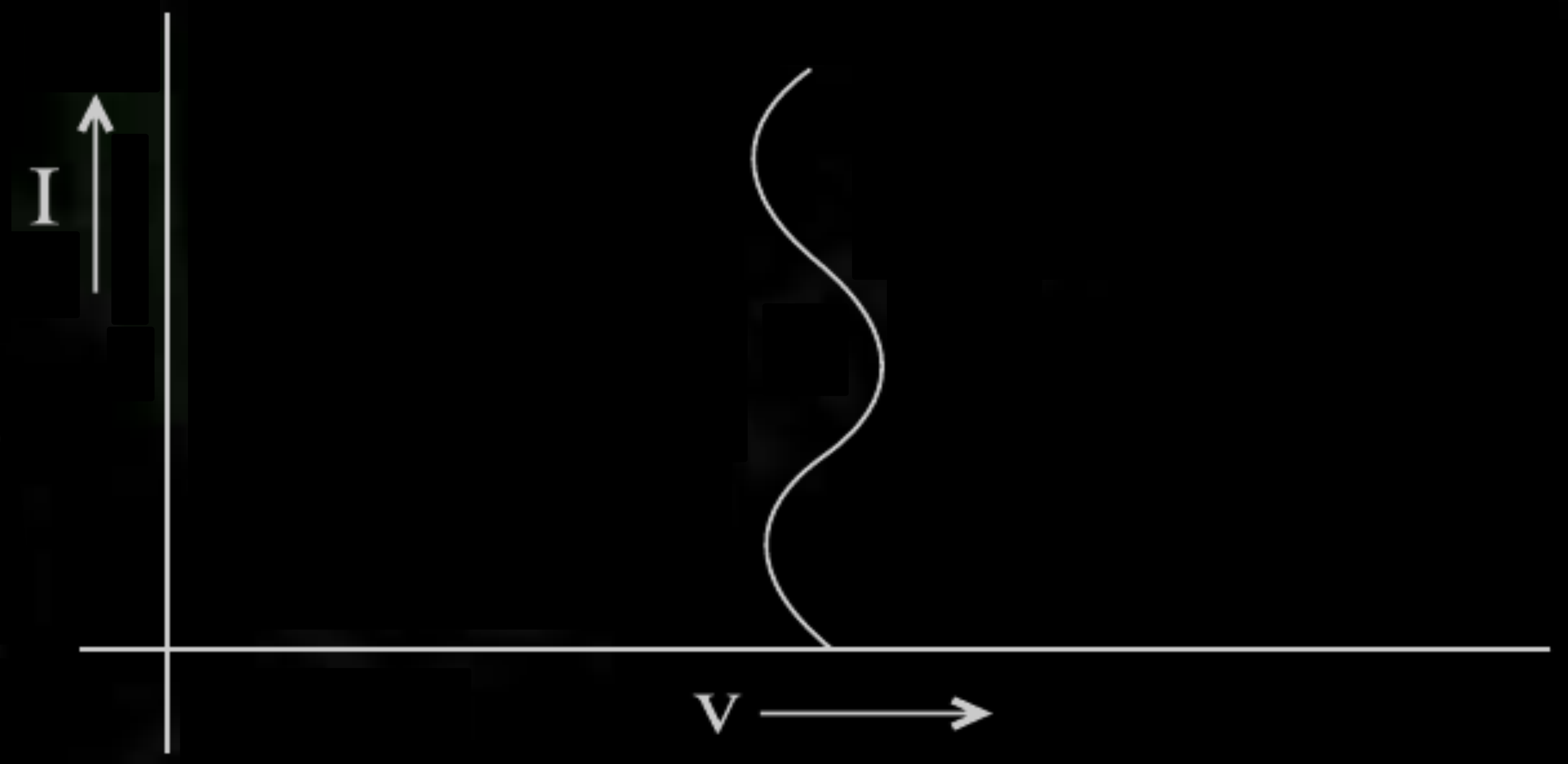


Figure (b)

a) Note : Award this 1 mark even if the student draw the following or some other non linear graph — 1



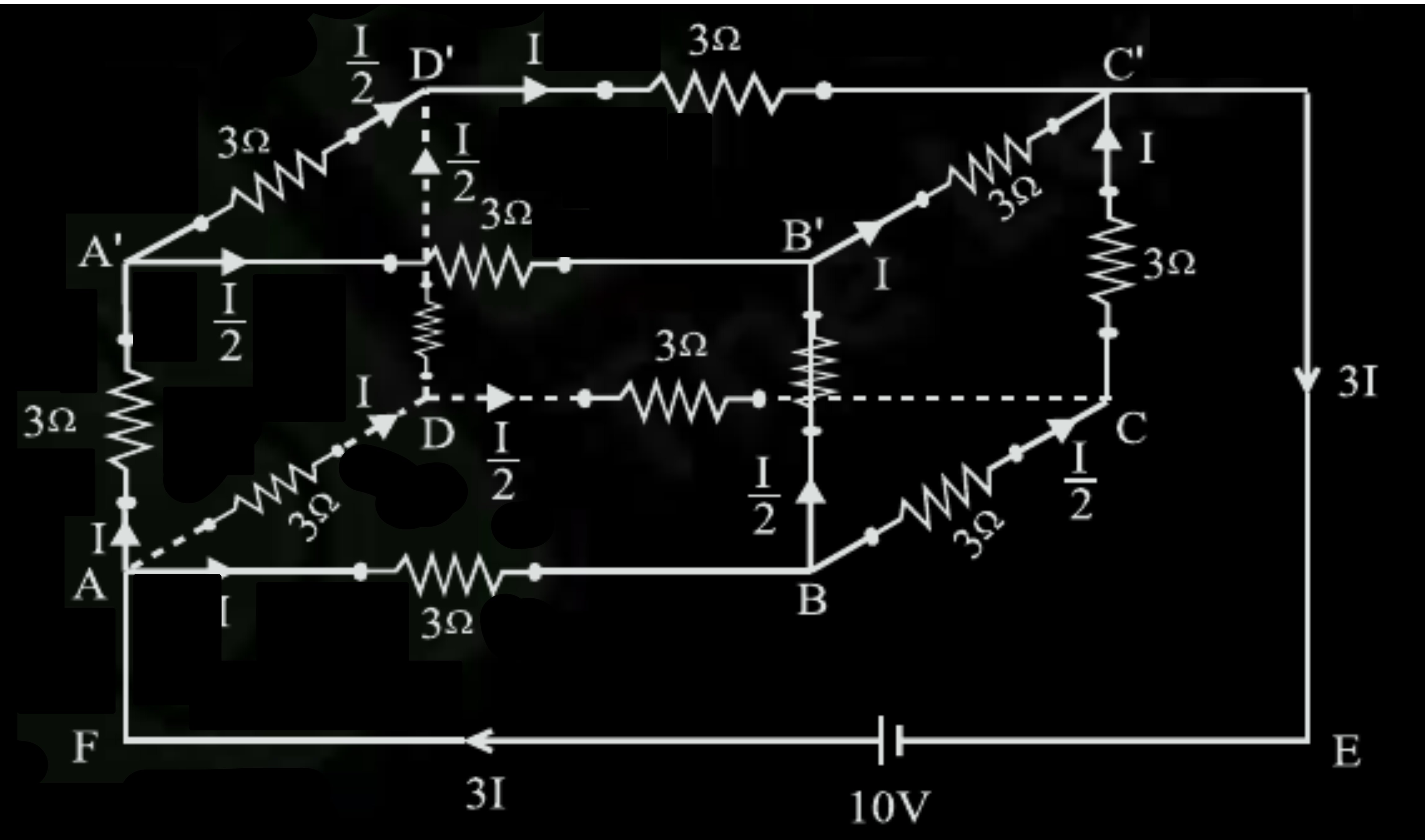
OR



b) At a temperature of 4 K, the Resistance of Hg becomes zero. — 1
Note - Award this 1 mark if the student writes that Hg becomes a super conductor at temperature of 4 K

c) Region BC ; since current is decreasing with increasing voltage — 1
Alternately :The graph has negative slope for this region.

Twelve wires each having a resistance of 3 Ω are connected to form a cubical network. A battery of 10 V and negligible internal resistance is connected across the diagonally opposite corners of this network. Determine its equivalent resistance and the current along each edge of the cube.



½

Applying loop rule to ABCC′EFA

$$3I + 3\frac{I}{2} + 3I - 10 = 0$$ ½

$$\frac{15}{2}I = 10$$

$$I = \frac{2\times10}{15} = \frac{20}{15}\,A = \frac{4}{3}A$$ ½

$$R_{tot} = \frac{V}{3I} = \frac{10\times15}{3\times20} = 2.5\Omega$$ ½

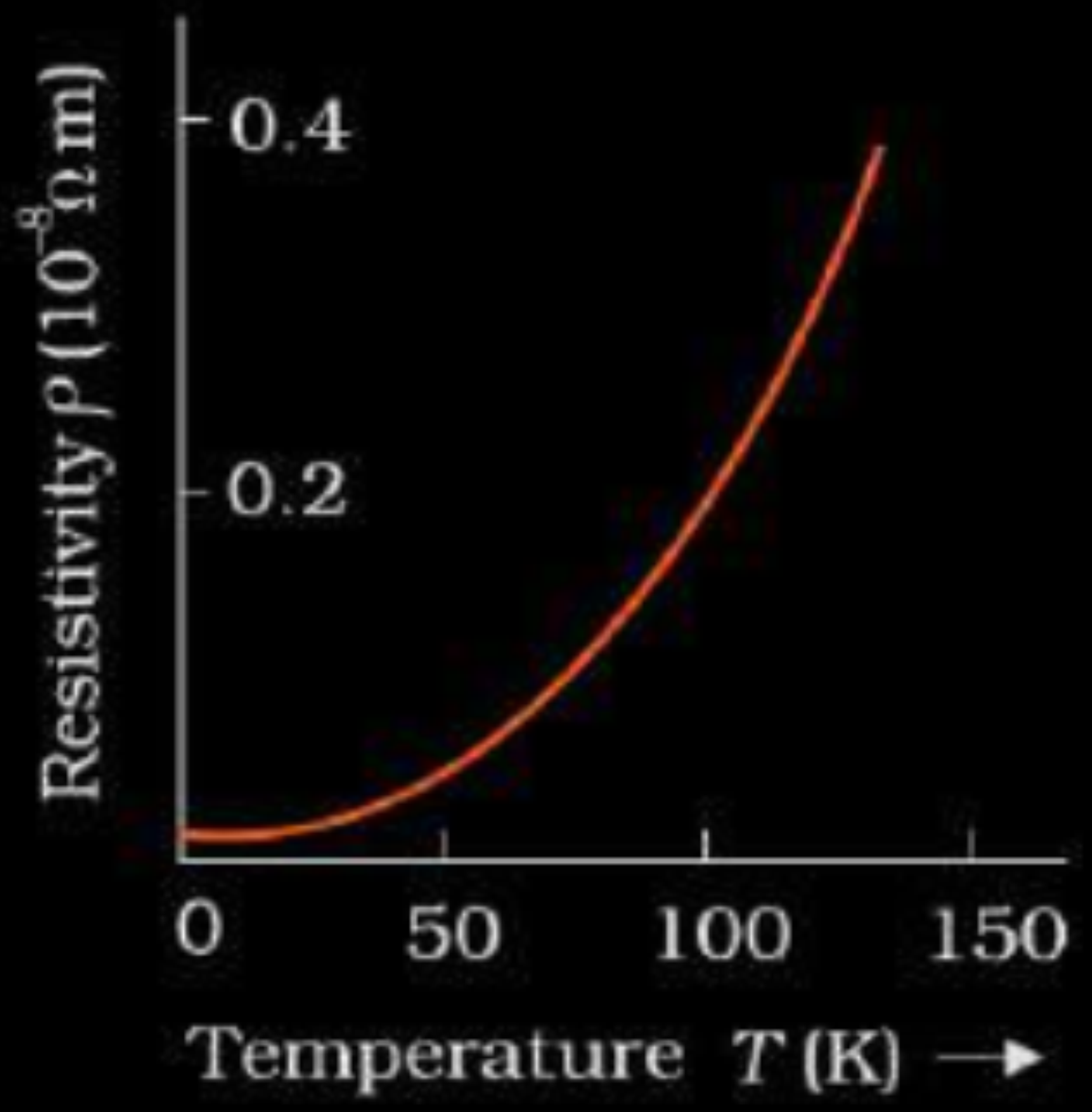
$$\text{Current} = I_{AB}(= I_{AA'} = I_{AD} = I_{D'C'} = I_{B'C'} = I_{CC'}) = \frac{4}{3}A$$ ½

$$I_{DD'}(= I_{A'B'} = I_{A'D'} = I_{DC} = I_{BC} = I_{BB'}) = \frac{2}{3}A$$ ½

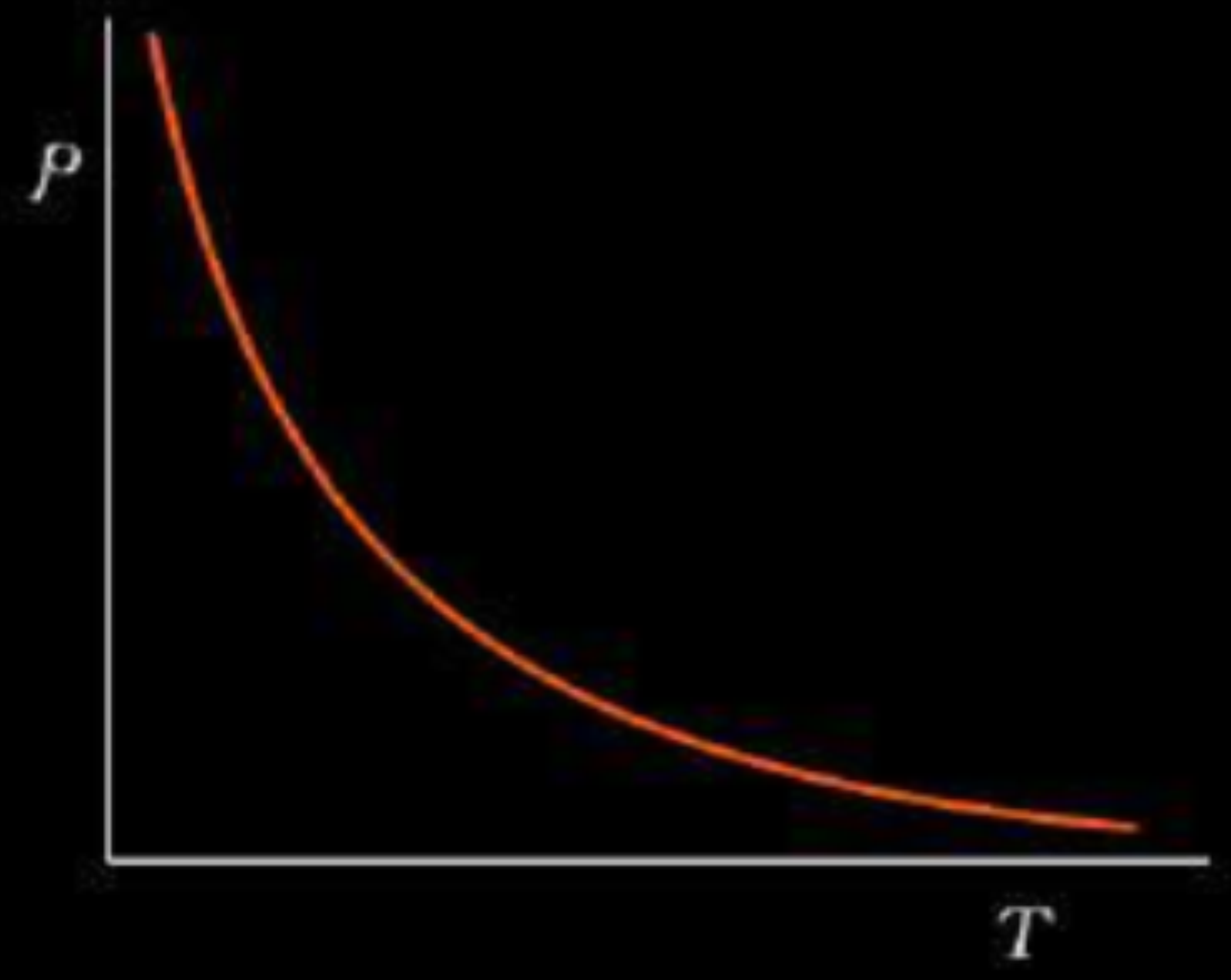
Show, on a plot, variation of resistivity of (i) a conductor, and (ii) a typical semiconductor as a function of temperature.

Using the expression for the resistivity in terms of number density and relaxation time between the collisions, explain how resistivity in the case of a conductor increases while it decreases in a semiconductor, with the rise of temperature.

• Showing the plot of variation of resistivity	½ + ½
• Expression for resistivity	1
• Explaining variation of resistivity for conductor and semiconductor	½ + ½



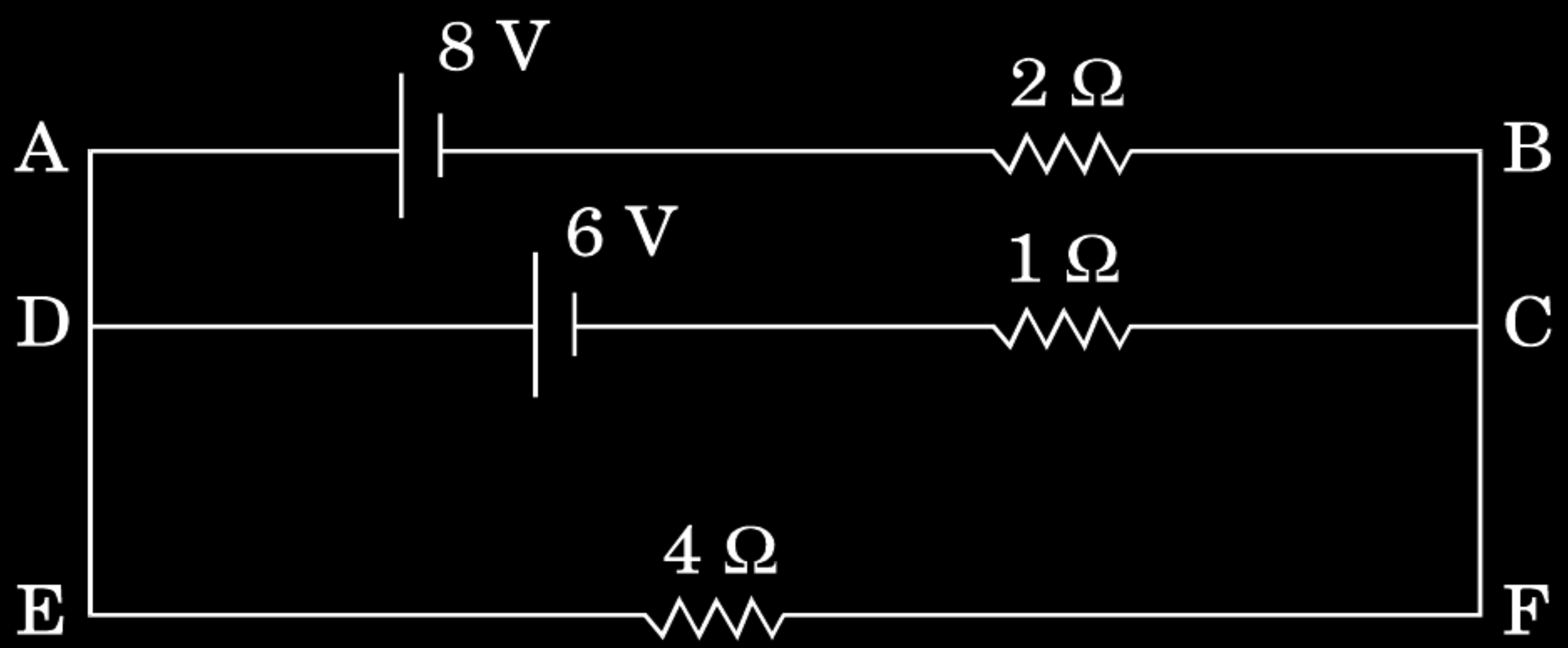
½

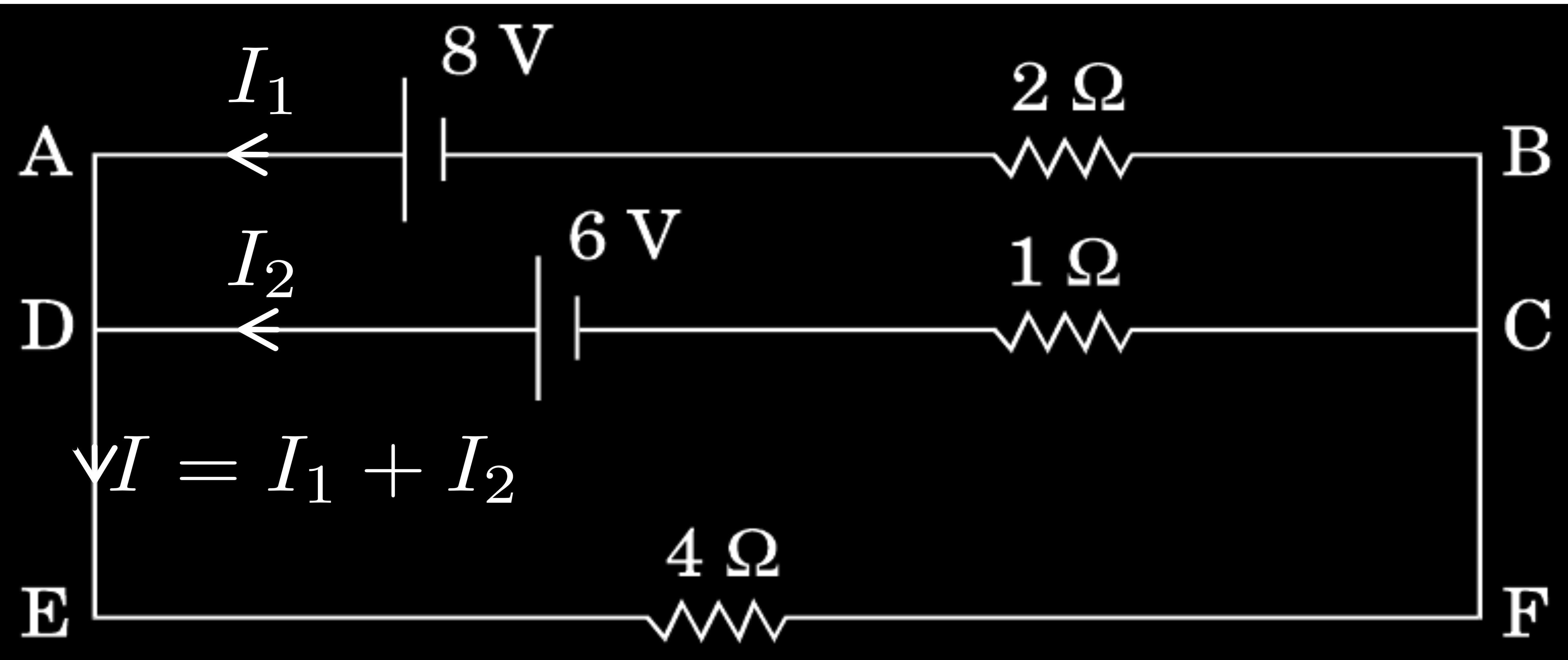


½

- $$\rho = \frac{m}{ne^2\tau}$$ 1
- In case of conductors with increase in temperature, relaxation time decreases, so resistivity increases. ½
- In case of semiconductors with increase in temperature number density (n) of free electrons increases, hence resistivity increases. ½

(c) Calculate the potential difference across the 4 Ω resistor in the given electrical circuit, using Kirchhoff's rules.





1

$I = I_1 + I_2 \qquad (1)$

In loop ABCDA

$-8 + 2I_1 - 1 \times I_2 + 6 = 0 \qquad (2)$

1

In loop DEFCD

$-4I - 1 \times I_2 + 6 = 0$ ½

$4I + I_2 = 6$

$4(I_1 + I_2) + I_2 = 6$

$4I_1 + 5I_2 = 6 \qquad (3)$ ½

From equations (1) *and* (2) *we get*

$I_1 = \frac{8}{7}A,\ \ I_2 = \frac{2}{7}A,\ \ I = \frac{10}{7}$

Potential difference across resistor 4Ω is:

$V = \frac{10}{7} \times 4 = \frac{40}{7} volt$

How does the mobility of electrons in a conductor change, if the potential difference applied across the conductor is doubled, keeping the length and temperature of the conductor constant ?

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No Change	1

Two bulbs are rated (P_1, V) and (P_2, V). If they are connected (i) in series and (ii) in parallel across a supply V, find the power dissipated in the two combinations in terms of P_1 and P_2.

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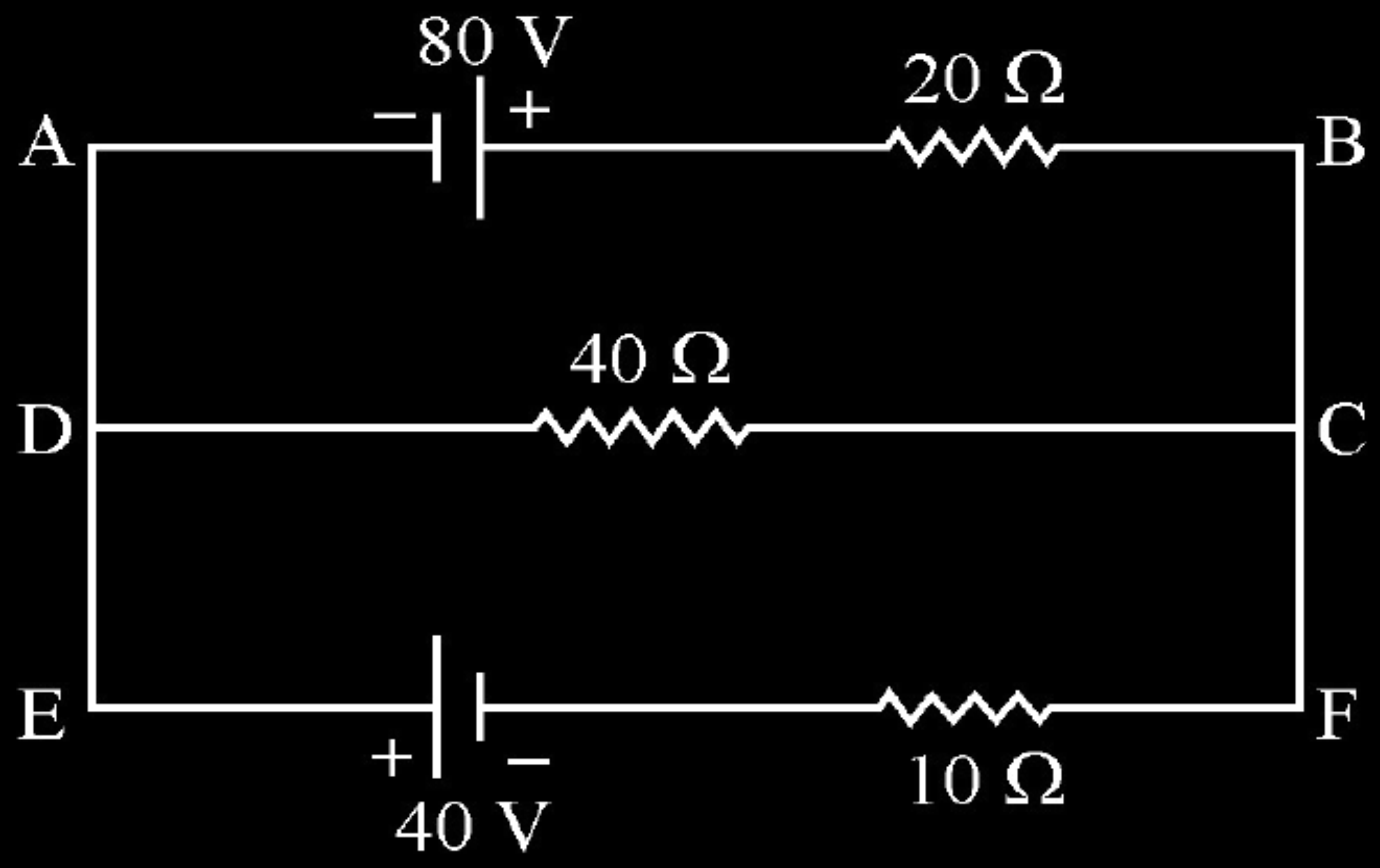
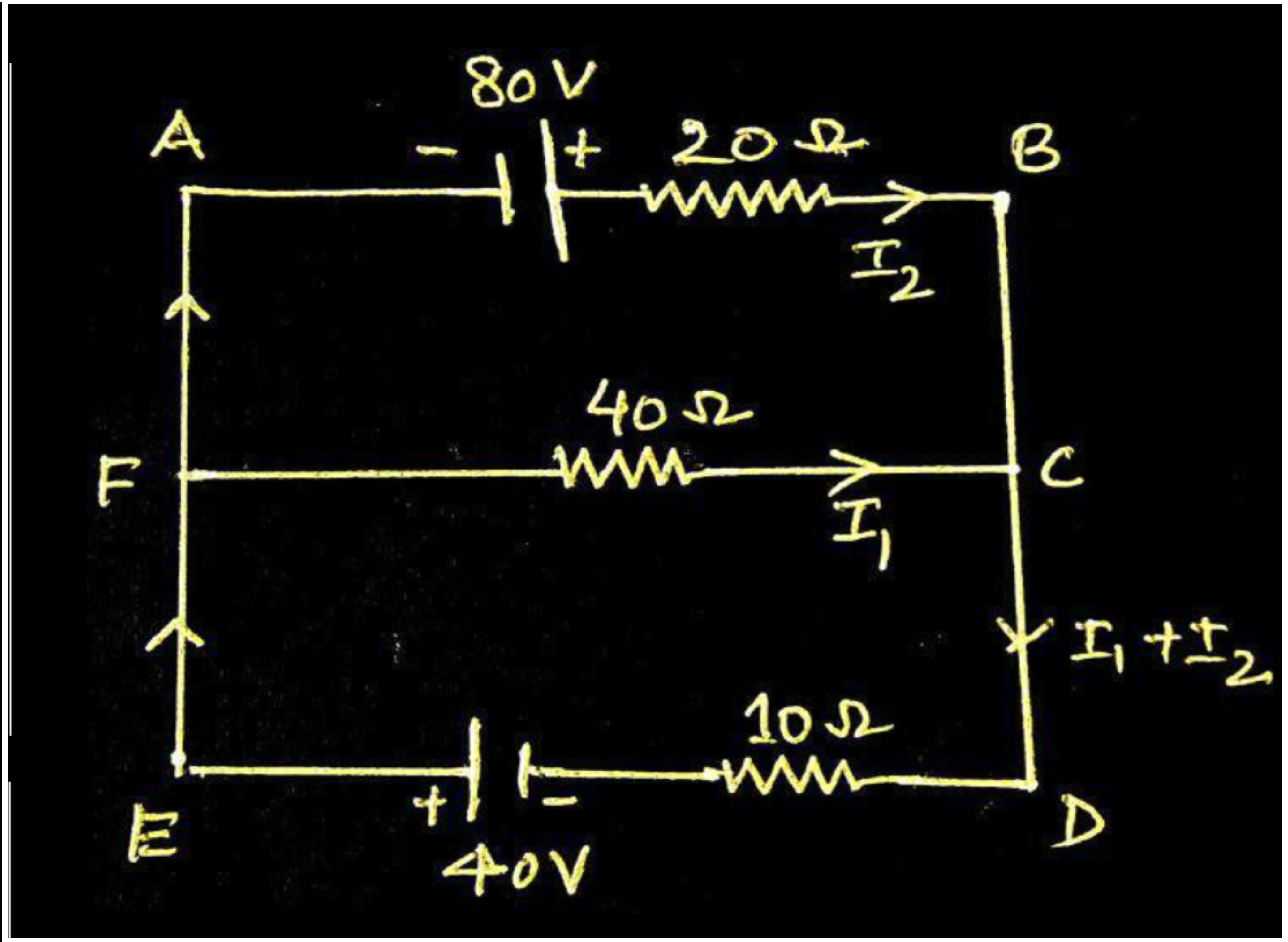
$R_1 = \frac{V^2}{P_1}$, $R_2 = \frac{V^2}{P_2}$,	½
$P_s = \frac{V^2}{Rs} = \frac{P_1 P_2}{P_1+P_2}$	
$\frac{1}{Ps} = \frac{1}{P_1} + \frac{1}{P_2}$	½
$\frac{1}{Rp} = \frac{1}{R_1} + \frac{1}{R_2} = \frac{P_1+P_2}{V^2}$	½
$\therefore P_p = \frac{V^2}{R_P} = P_1 + P_2$	½

Using Kirchhoff's rules, calculate the current through the 40 Ω and 20 Ω resistors in the following circuit :



In loop ABCFA $+80 - 20\,I_2 + 40\,I_1 = 0$ $4 = I_2 - 2\,I_1$ In loop FCDEA $-40\,I_1 - 10(I_1 + I_2) + 40 = 0$	1
$-50\,I_1 - 10\,I_2 + 40 = 0$ $5\,I_1 + I_2 = 4$ Solving these two equations	1
$I_1 = 0A$	½
$\&\ I_2 = 4A$	½



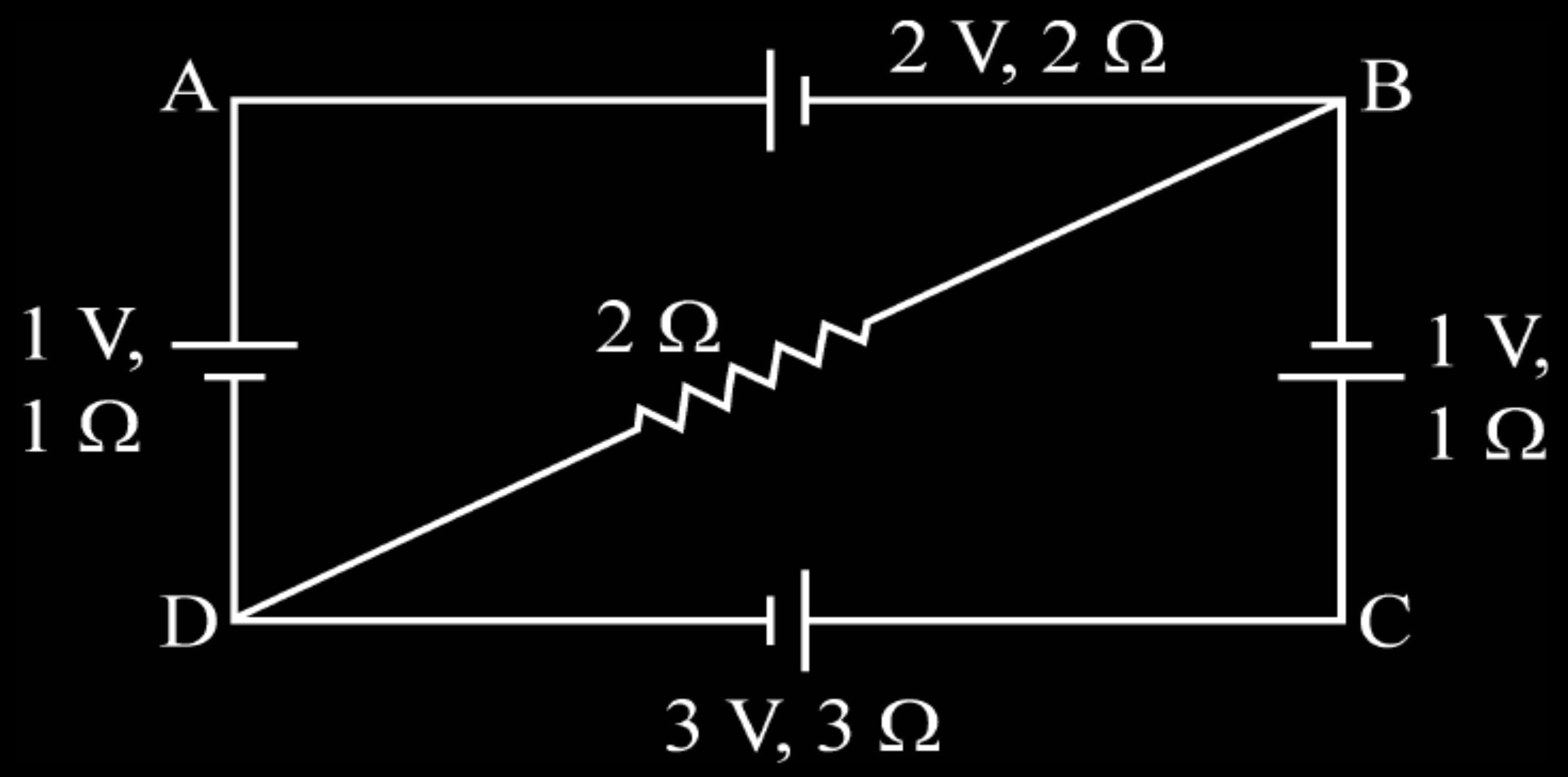
CBSE 2018

Two cells of emfs ε_1 & ε_2 and internal resistances r_1 & r_2 respectively are connected in parallel. Obtain expressions for the equivalent.

(i) resistance and

(ii) emf of the combination

Using Kirchhoff's rules, calculate the potential difference between B and D in the circuit diagram as shown in the figure.

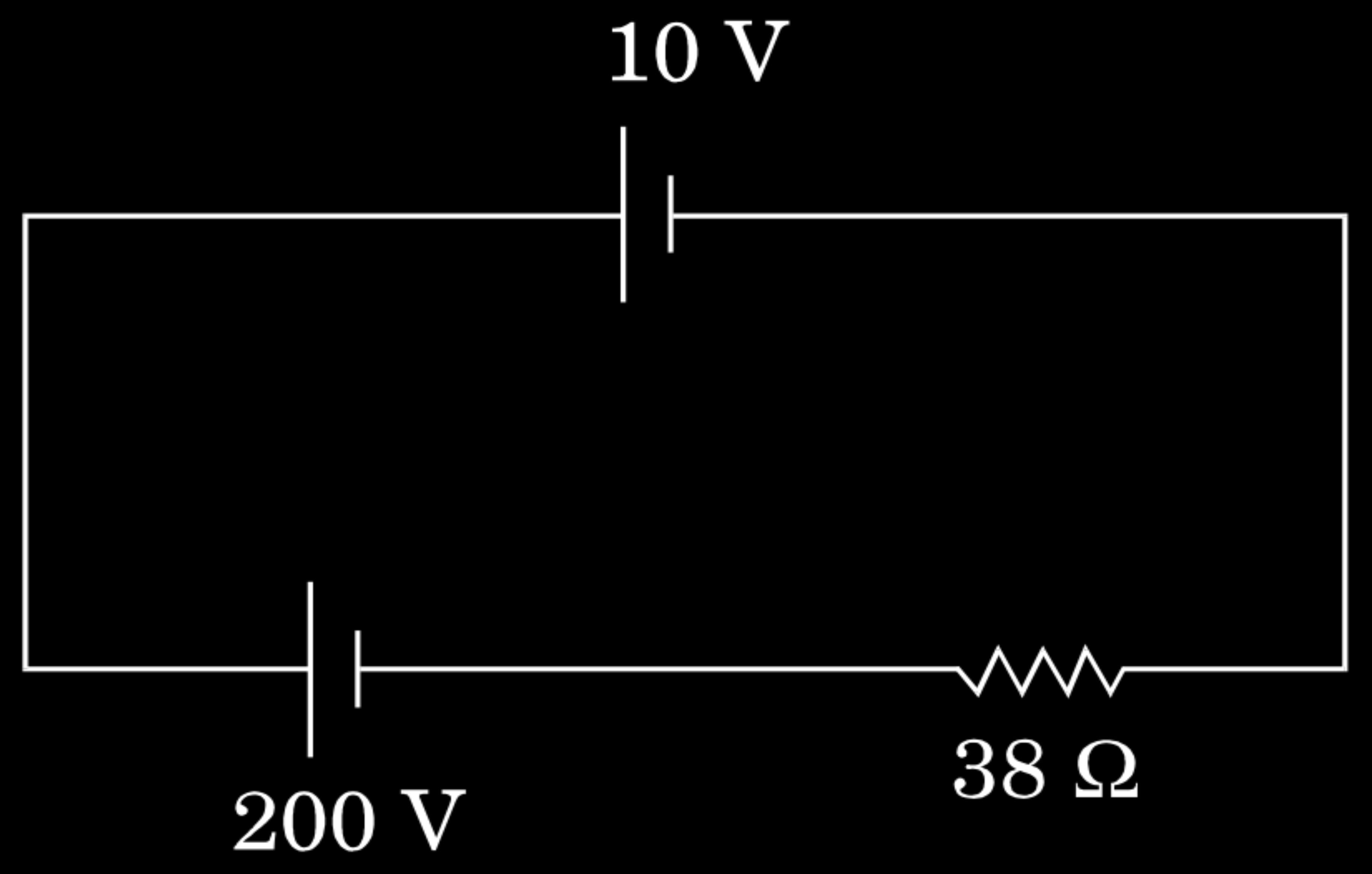


Two electric bulbs P and Q have their resistances in the ratio of 1 : 2. They are connected in series across a battery. Find the ratio of the power dissipation in these bulbs.

Two electric bulbs P and Q have their resistances in the ratio of 1 : 2. They are connected in series across a battery. Find the ratio of the power dissipation in these bulbs.

Power = $I^2 R$	
The current, in the two bulbs, is the same as they are connected in series.	½ ½
$\therefore \frac{P_1}{P_2} = \frac{I^2 R_1}{I^2 R_2} = \frac{R_1}{R_2}$	½
$= \frac{1}{2}$	½

A 10 V cell of negligible internal resistance is connected in parallel across a battery of emf 200 V and internal resistance 38 Ω as shown in the figure. Find the value of current in the circuit.



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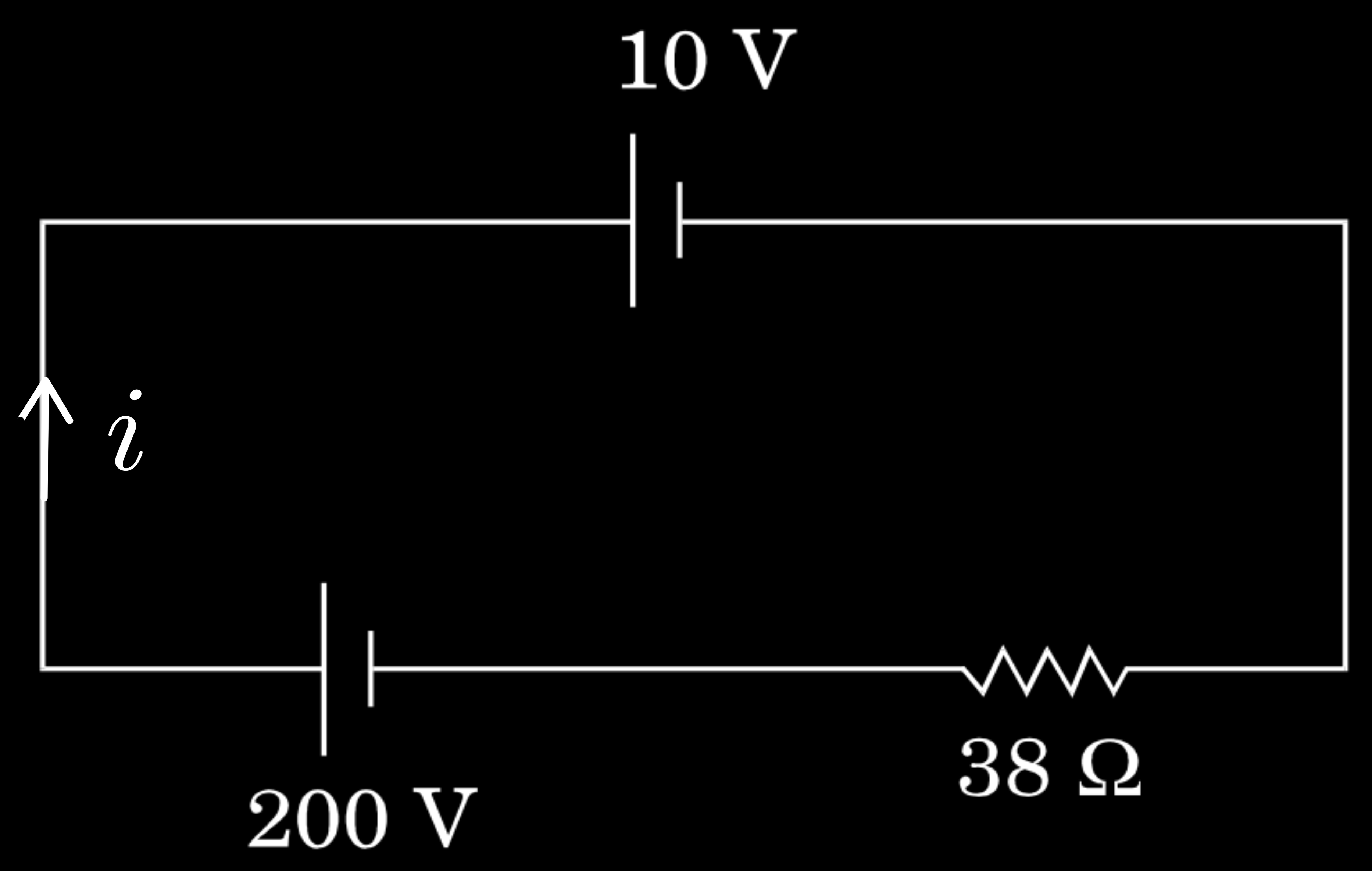


By Kirchoff's law, we have, for the loop ABCD, +200 – 38i– 10 = 0	**1**
$\therefore i = \frac{190}{38}$ A= 5A	**1**

CBSE 2017

Nichrome and copper wires of same length and same radius are connected in series. Current I is passed through them. Which wire gets heated up more ? Justify your answer.

CBSE 2017

Nichrome and copper wires of same length and same radius are connected in series. Current I is passed through them. Which wire gets heated up more ? Justify your answer.

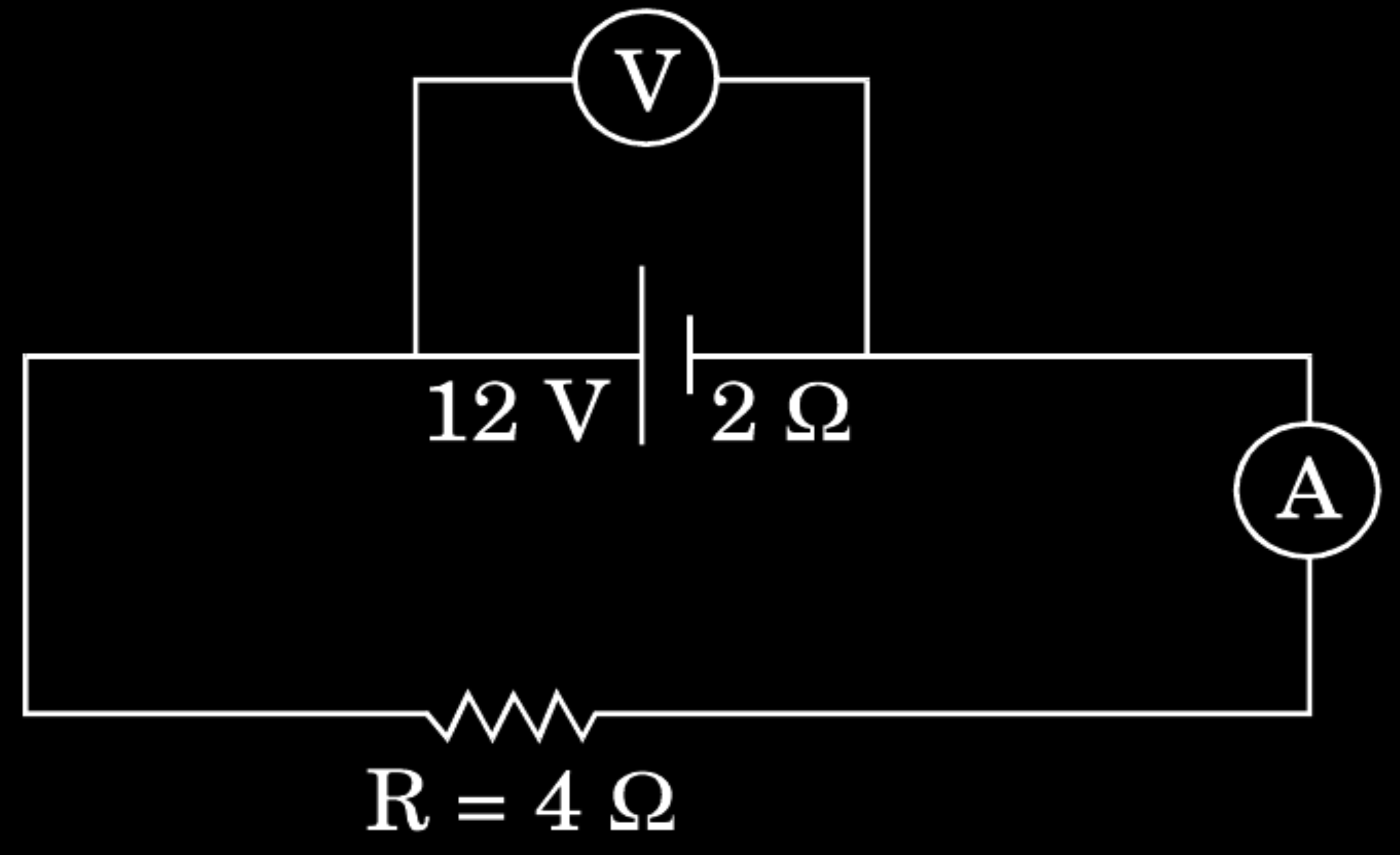
i.	Nichrome	½
ii.	$R_{Ni} > R_{Cu}$ (or $\text{Resistivity}_{Ni} > \text{Resistivity}_{Cu}$)	½

(a) The potential difference applied across a given resistor is altered so that the heat produced per second increases by a factor of 9. By what factor does the applied potential difference change ?

(a) The potential difference applied across a given resistor is altered so that the heat produced per second increases by a factor of 9. By what factor does the applied potential difference change ?

a) $H = \frac{V^2}{R}$	½
$\therefore V$ increases by a factor of $\sqrt{9} = 3$	½

(b) In the figure shown, an ammeter A and a resistor of 4 Ω are connected to the terminals of the source. The emf of the source is 12 V having an internal resistance of 2 Ω. Calculate the voltmeter and ammeter readings.



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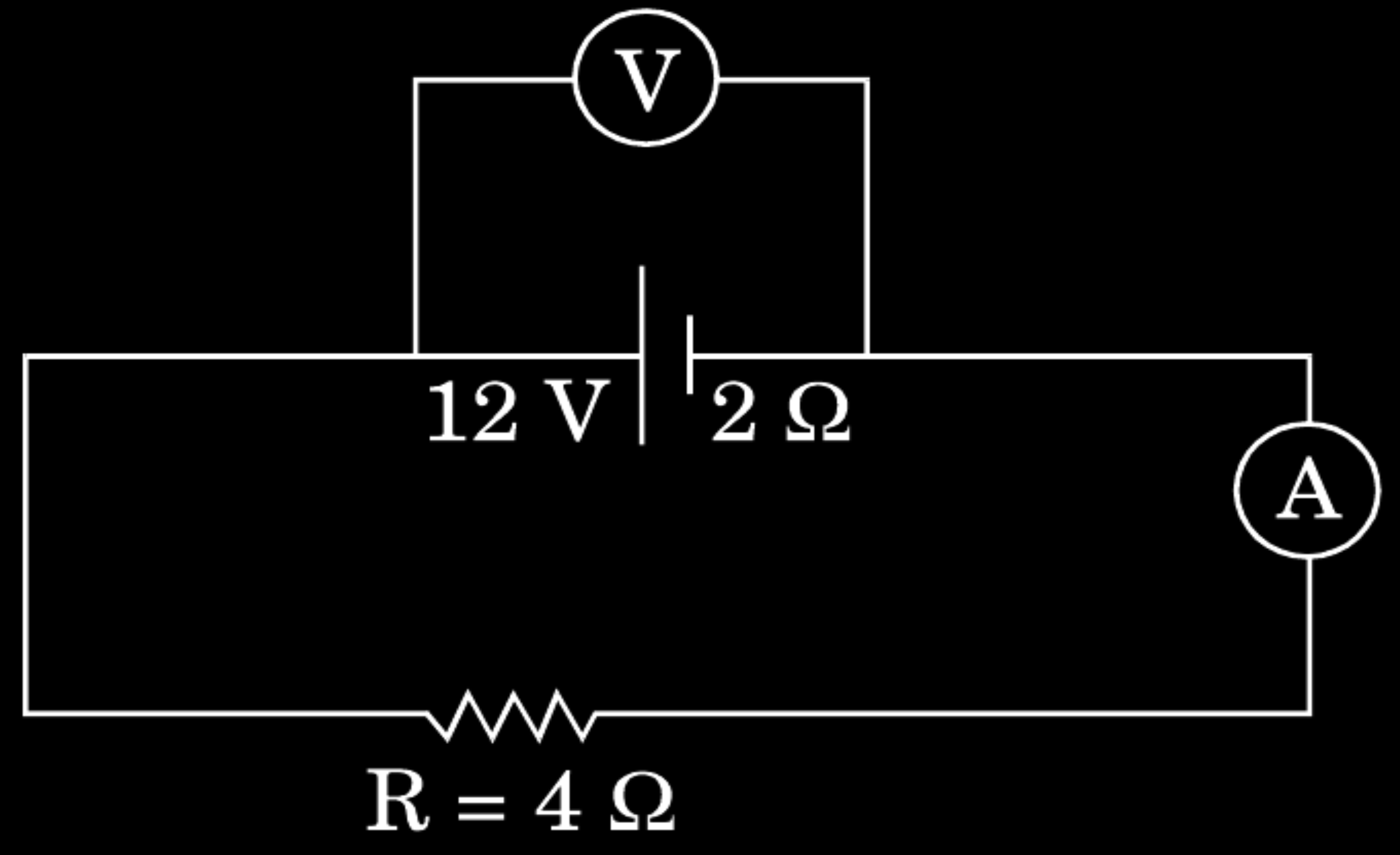
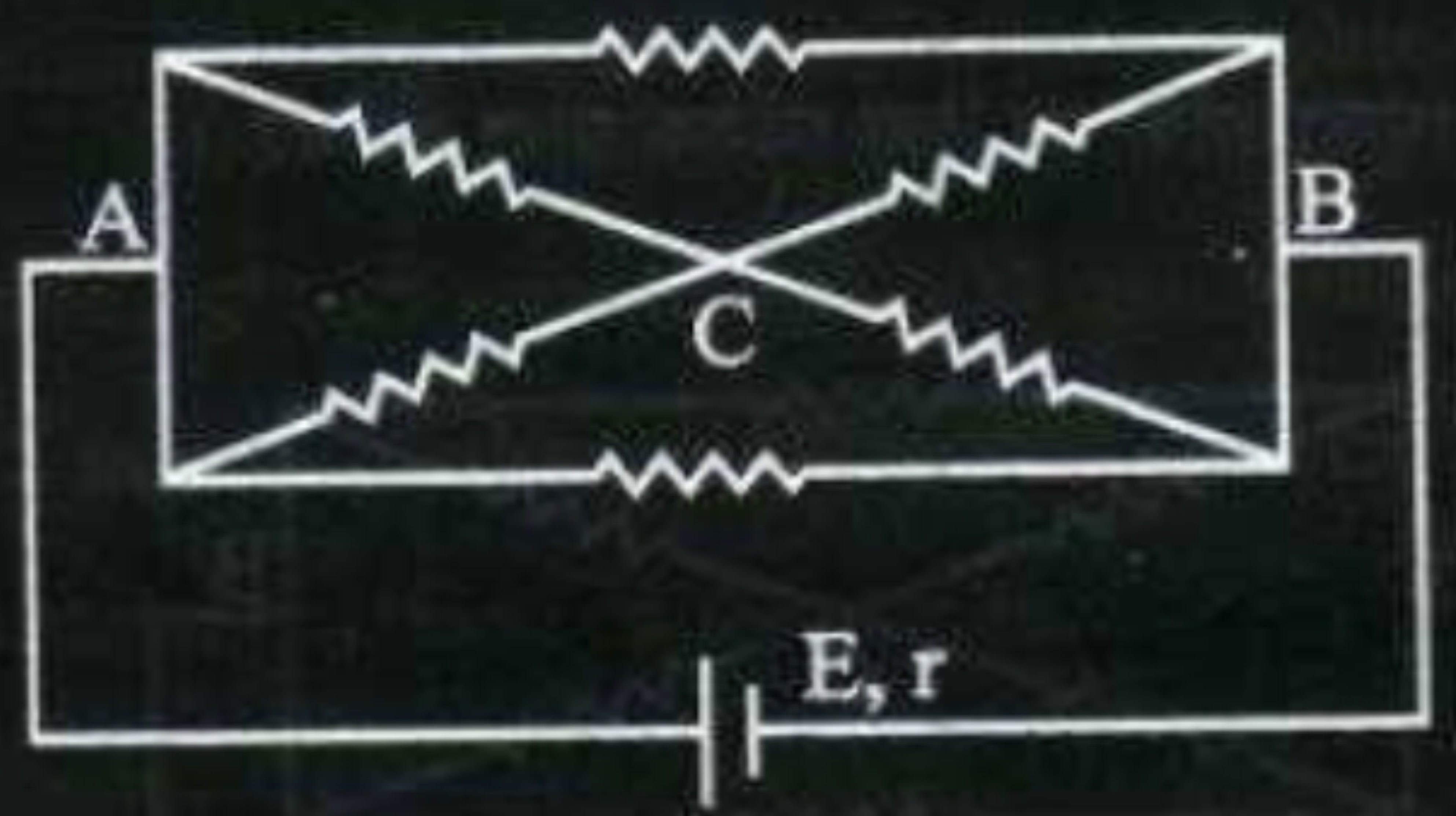


b) Ammeter Reading $I = \frac{V}{R+r}$	½
$= \frac{12}{4+2} \text{A} = 2\text{A}$	½
Voltmeter Reading $V = E - Ir$	½
$= [12 - (2 \times 2)]\ \text{V} = 8\text{V}$ (Alternatively, $V = iR = 2 \times 4\text{V} = 8\text{V}$)	½

(i) State the two Kirchhoff's laws. Explain briefly how these rules are justified.

(ii) The current is drawn from a cell of emf E and internal resistance r connected to the network of resistors each of resistance r as shown in the figure. Obtain the expression for (i) the current draw from the cell and (ii) the power consumed in the network.



(i) Junction Rule: At any Junction, the sum of currents, entering the junction, is equal to the sum of currents leaving the junction.	1
Loop Rule: The Algebraic sum, of changes in potential, around any closed loop involving resistors and cells, in the loop is zero. $$\sum(\Delta V) = 0$$	1
Justification: The first law is in accord with the law of conservation of charge.	½
The Second law is in accord with the law of conservation of energy.	½
(ii) Equivalent resistance of the loop $$R = {}^{r}/_{3}$$	½
Hence current drawn from the cell	
$$I = \frac{E}{{}^{r}/_{3} + r} = \frac{3E}{4r}$$	½
Power consumed $P = I^2\ ({}^{r}/_{3})$	½
$$= \frac{9E^2}{16r^2} \text{ x } {}^{4r}/_{3} = \frac{3E^2}{4r}$$	½
[Note: Award the last 1 ½ marks for this part, if the calculations, for these parts, are done by using (any other) value of equivalent resistance obtained by the student.)	

(i) Derive an expression for drift velocity of electrons in a conductor. Hence deduce Ohm's law.

(ii) A wire whose cross-sectional area is increasing linearly from its one end to the other, is connected across a battery of V volts. Which of the following quantities remain constant in the wire ?

(a) drift speed

(b) current density

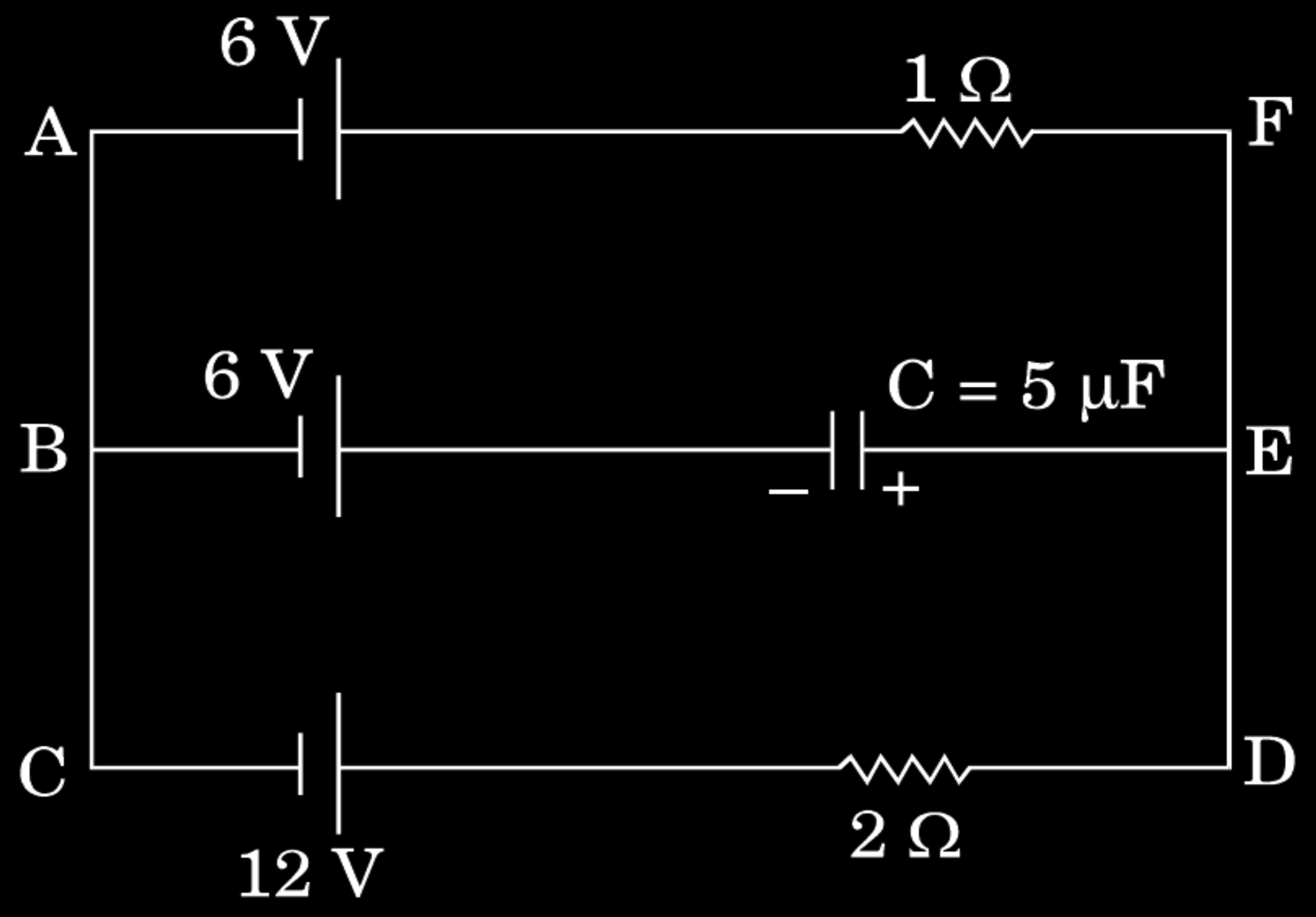
(c) electric current

(d) electric field

Justify your answer.

Let an electric field E be applied the conductor. Acceleration of each electron is $a = -\frac{eE}{m}$	½
Velocity gained by the electron $v = -\frac{eE}{m}t$	½
Let the conductor contain n electrons per unit volume. The average value of time $'t'$, between their successive collisions, is the relaxation time, $'\tau'$.	½
Hence average drift velocity $v_d = \frac{-eE}{m}\tau$	½
The amount of charge, crossing area A, in time Δt,is $\equiv neAv_d\Delta t = I\Delta t$	½
Substituting the value of v_d, we get $I\Delta t = neA\left(\frac{eE\tau}{m}\right)\Delta t$ $\therefore I = \left(\frac{e^2 A\tau n}{m}\right)E = \sigma E, \left(\sigma = \frac{e^2\tau n}{m} \text{ is the conductivity}\right)$	½
But I = JA, where J is the current density $\Rightarrow$ J = $\left(\frac{e^2\tau n}{m}\right)E$	½
$\Rightarrow$ J = σE This is Ohm's law [Note : Credit should be given if the student derives the alternative form of Ohm's law by substituting E = $\frac{V}{\ell}$]	½
ii) Electric current well remain constant in the wire.	½
All other quantities, depend on the cross sectional area of the wire.	½

In the given circuit, with steady current, calculate the potential difference across the capacitor and the charge stored in it.



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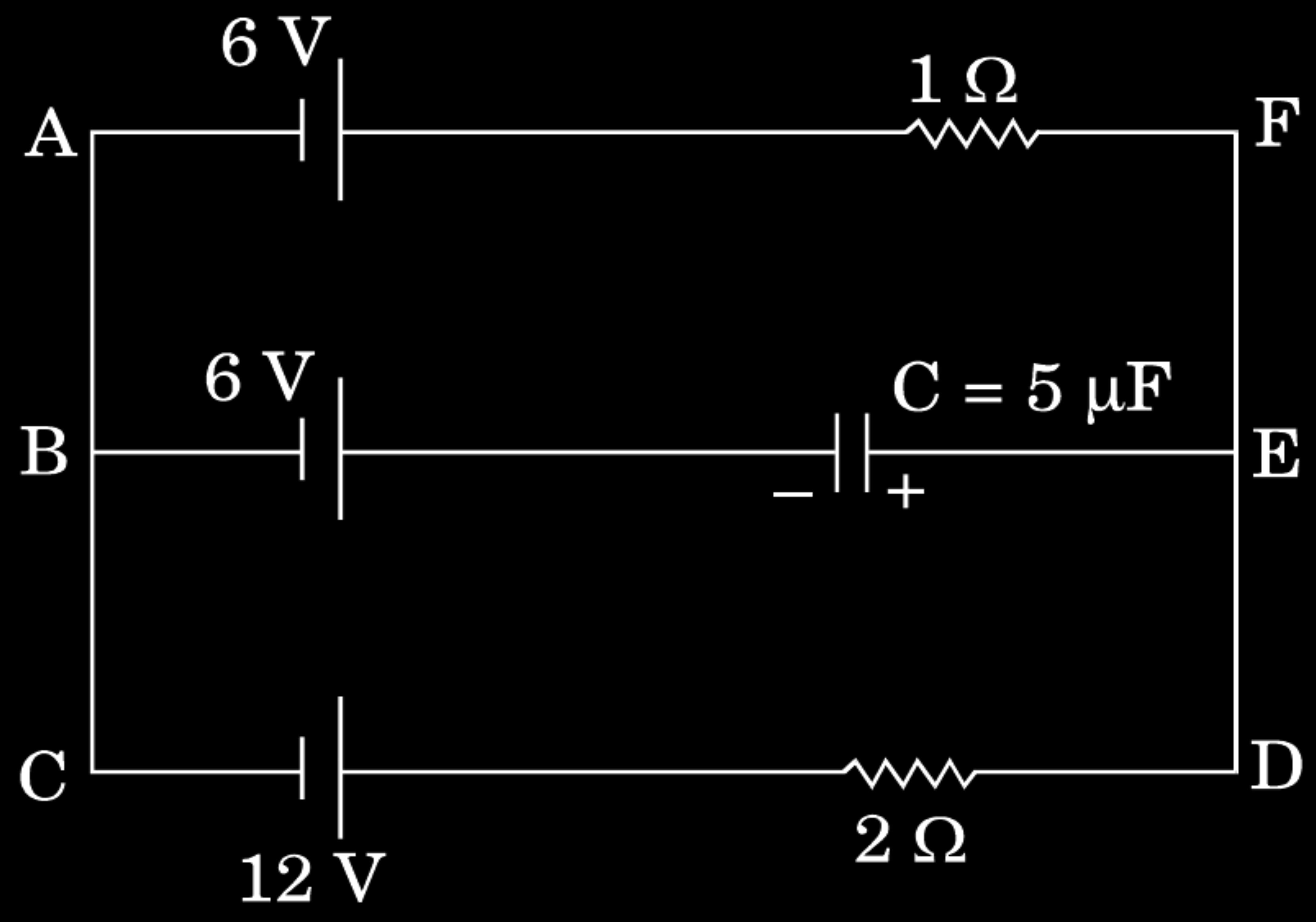
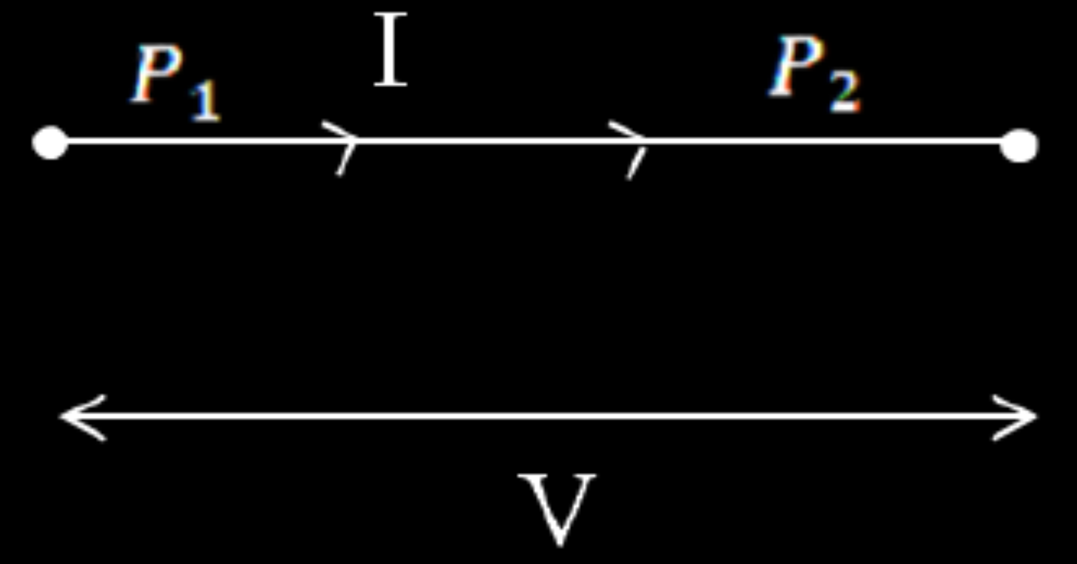


In loop ACDFA $I = \frac{12-6}{(1+2)} = 2\text{A}$	1
$V_{AF} = V_{BE}$ $\Rightarrow 6 + 2 = 6 + V_c$	½
$\Rightarrow V_c = 2V$	½
Charge $Q = CV_c = 5\mu F \times 2V = 10\mu C$	1

(b) Two metallic wires, P_1 and P_2 of the same material and same length but different cross-sectional areas, A_1 and A_2 are joined together and connected to a source of emf. Find the ratio of the drift velocities of free electrons in the two wires when they are connected (i) in series, and (ii) in parallel.

(b) Two metallic wires, P_1 and P_2 of the same material and same length but different cross-sectional areas, A_1 and A_2 are joined together and connected to a source of emf. Find the ratio of the drift velocities of free electrons in the two wires when they are connected (i) in series, and (ii) in parallel.

b) In series, the current remains the same.	½
	
$I = neA_1V_{d1} = neA_2V_{d2}$	½
$\therefore \frac{V_{d1}}{V_{d2}} = \frac{A_2}{A_1}$	½
In parallel potential difference is same but currents are different.	
$V = I_1R_1 = neA_1V_{d1}\frac{\varrho l}{A_1} = ne\varrho V_{d1}l$	½
Similarly, $V = I_2R_2 = ne\varrho V_{d2}l$	
$I_1R_1 = I_2R_2$	½
$\therefore \frac{V_{d1}}{V_{d2}}=1$	

CBSE 2016

Two cells of emfs 1.5 V and 2.0 V having internal resistances 0.2 Ω and 0.3 Ω respectively are connected in parallel. Calculate the emf and internal resistance of the equivalent cell.

CBSE 2016

Two cells of emfs 1.5 V and 2.0 V having internal resistances 0.2 Ω and 0.3 Ω respectively are connected in parallel. Calculate the emf and internal resistance of the equivalent cell.

$\text{emf} = \frac{E_1 r_2 + E_2 r_1}{r_1 + r_2}$	½
$= \frac{1.5 \times 0.3 + 2 \times 0.2}{0.2 + 0.3}\ \text{V}$	
$= \frac{0.45 + 0.40}{0.5}\ \text{V} = 1.7\ \text{V}$	½
$r = \frac{r_1 r_2}{r_1 + r_2}$	½
$= \frac{0.2 \times 0.3}{0.2 + 0.3}\ \Omega$	
$= \frac{0.06}{0.5}\ \Omega$	
$= 0.12\ \Omega$	½

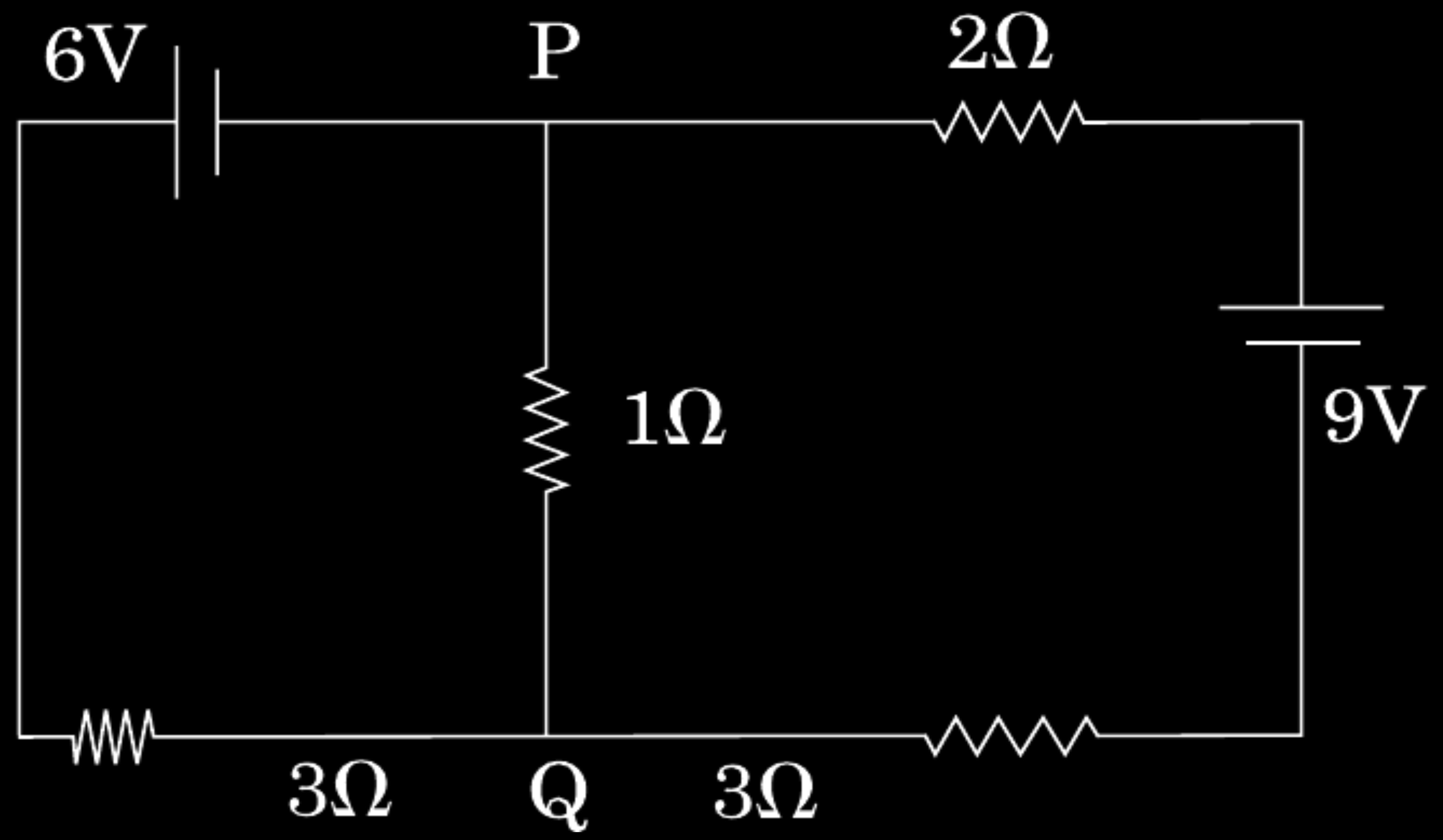
(i) Define the term drift velocity.

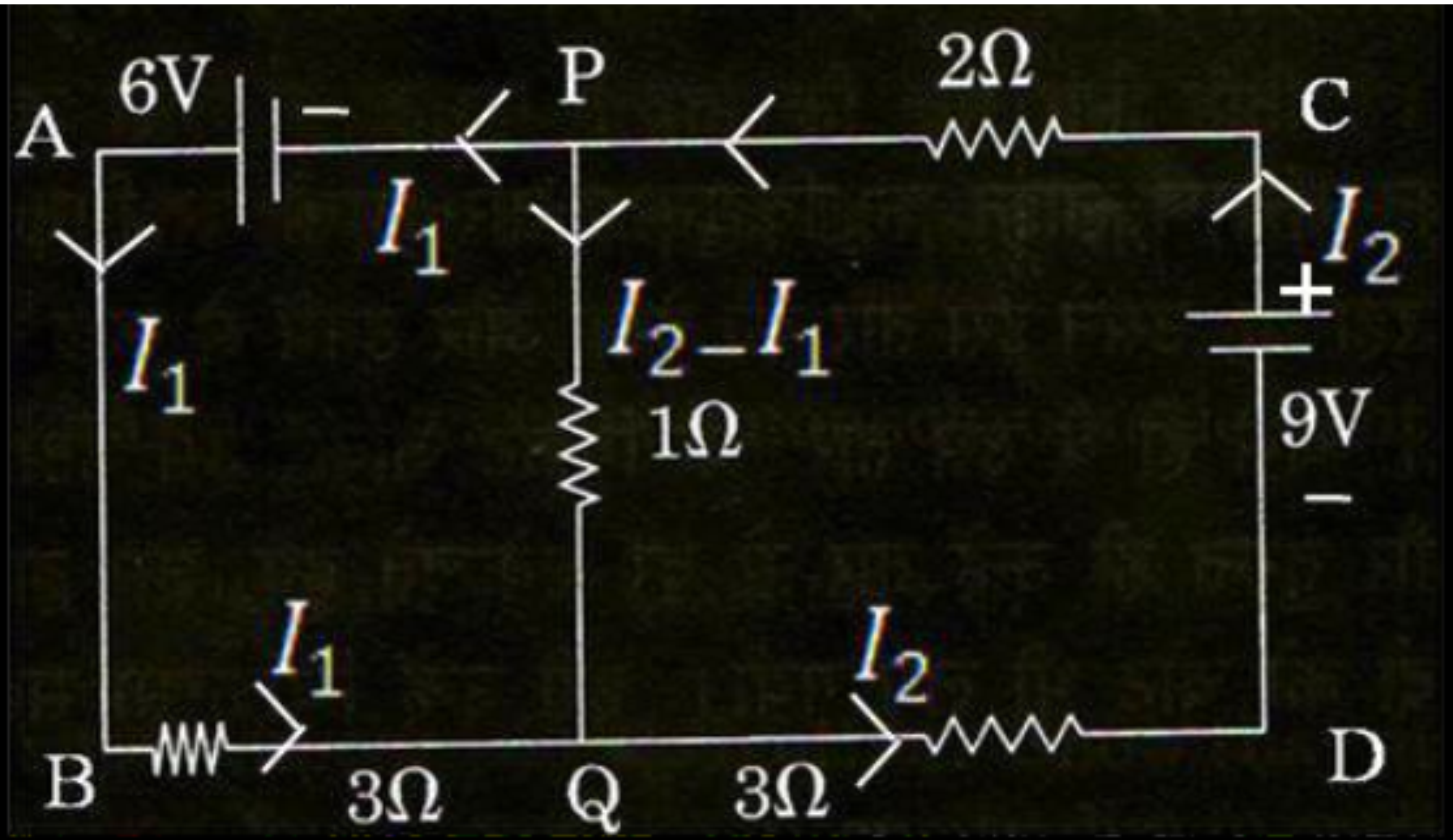
(ii) On the basis of electron drift, derive an expression for resistivity of a conductor in terms of number density of free electrons and relaxation time. On what factors does resistivity of a conductor depend ?

(iii) Why alloys like constantan and manganin are used for making standard resistors ?

i) Average velocity acquired by the electrons in the conductor in the presence of external electric field.	1
[Alternatively:	
$v_d = \frac{-eE\tau}{m}$ where τ is the relaxation time.]	
ii) $v_d = \frac{-eE\tau}{m}$	
We have $E = -\frac{V}{\ell}$, where V is potential difference across the length 'ℓ' of the conductor	
$v_d = \frac{eV\tau}{m\ell}$	½
Current flowing $I = neAv_d$	½
$I = neAv_d \frac{eV\tau}{m\ell} = \frac{ne^2AV\tau}{m\ell}$	½
$\frac{I}{V} = \frac{ne^2A\tau}{m\ell} = \frac{1}{R}$ -----------(i)	
Also, $R = \rho\frac{\ell}{A}$ -----------(ii)	
Comparing (i) and (ii)	
$\rho = \frac{m}{ne^2\tau}$	½
Resistivity of the material of a conductor depends on the relaxation time, i.e., temperature and the number density of electrons.	½+ ½
iii) Because constantan and manganin show very weak dependence of resistivity on temperature	1

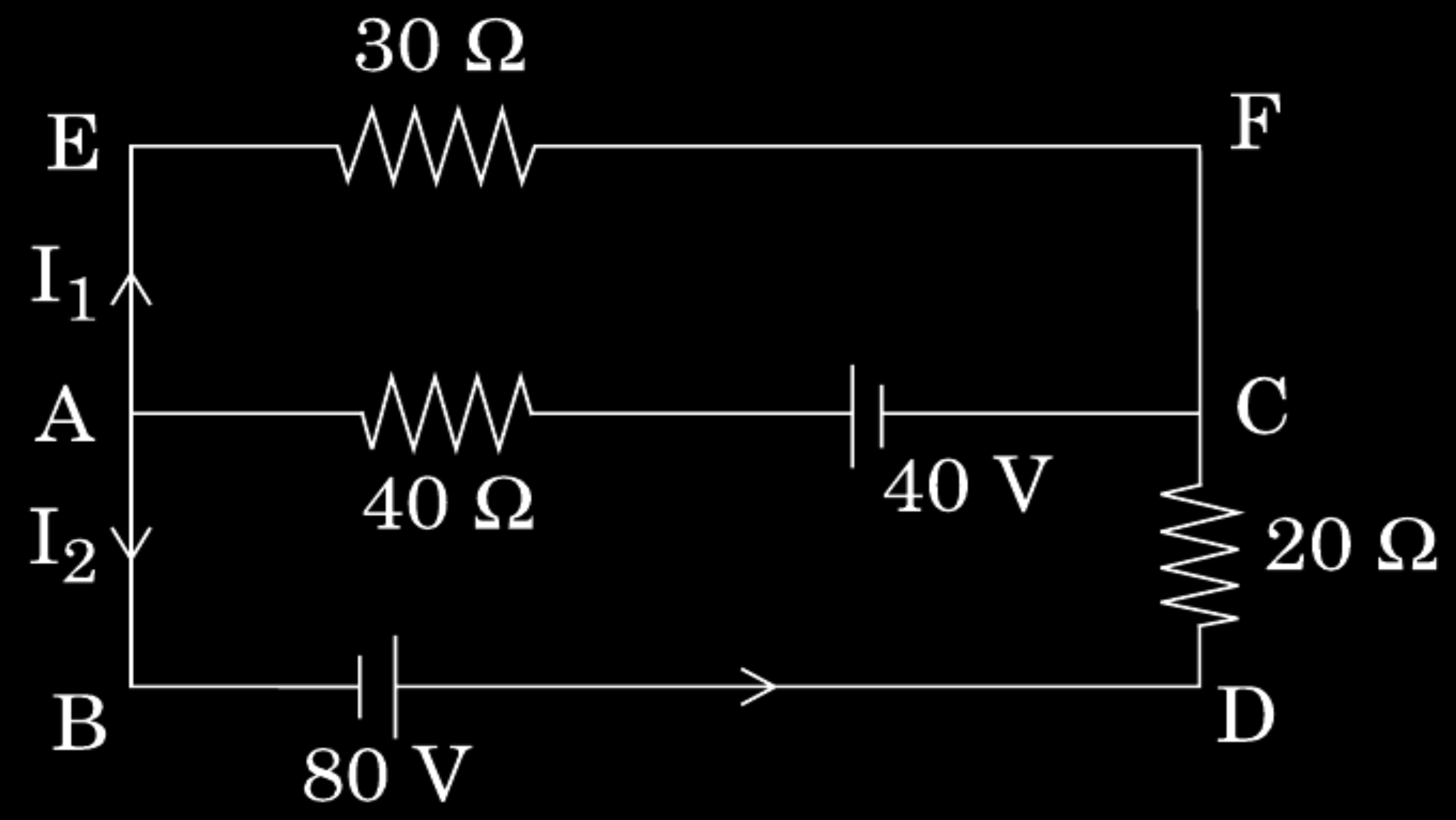
(i) Find the magnitude and direction of current in 1 Ω resistor in the given circuit.

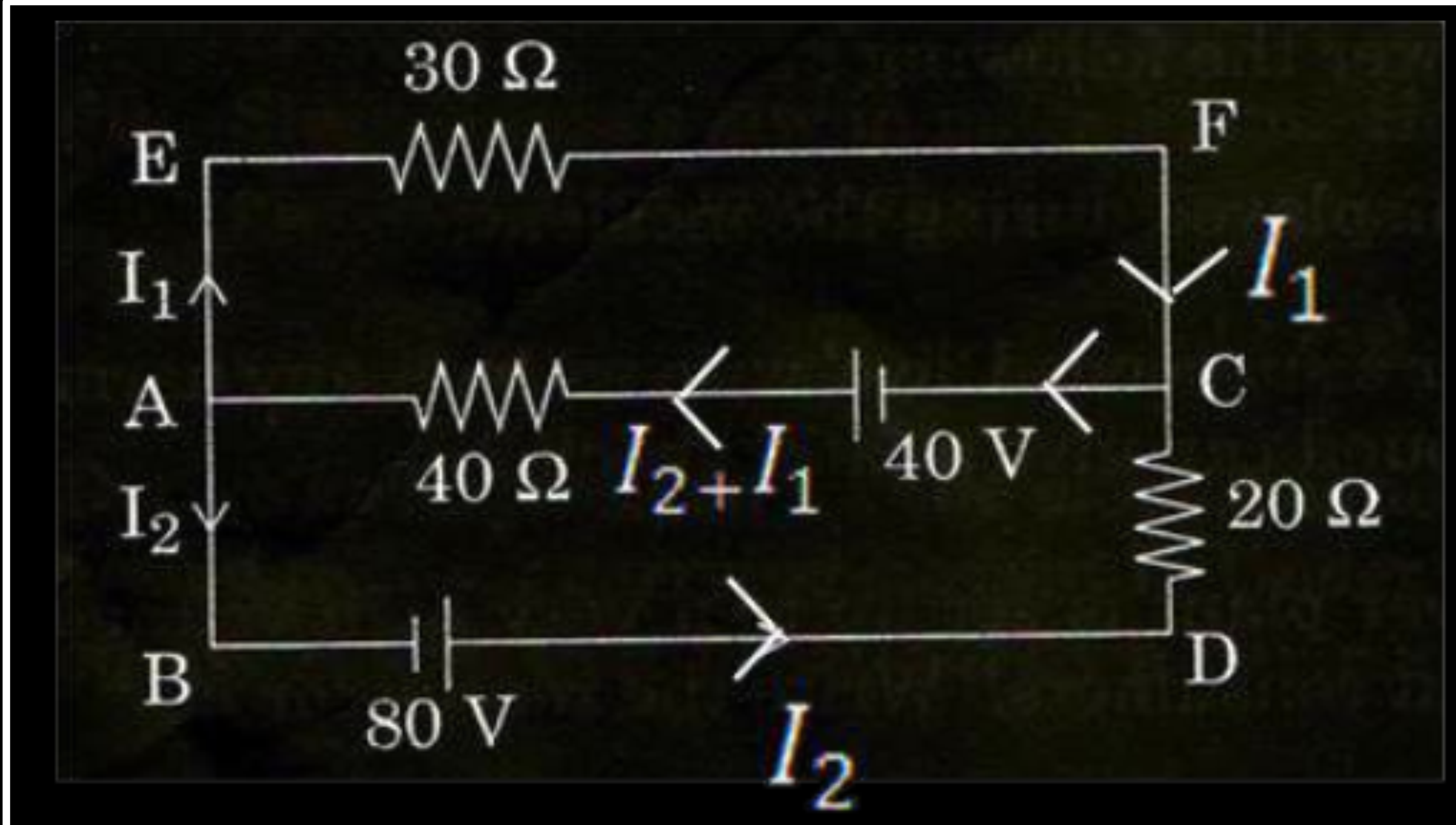




For the mesh APQBA	
$-6 - 1\,(I_2 - I_1) + 3\,I_1 = 0$	
Or $-I_2 + 4I_1 = 6$ ……………………………………(1)	1
For the mesh PCDQP	
$2I_2 - 9 + 3I_2 + 1\,(I_2 - I_1) = 0$	
Or $6I_2 - I_1 = 9$ ……………………………………(2)	1
Solving (1) and (2) , we get	
$$I_1 = \frac{45}{23}\,A$$	½
$$I_2 = \frac{42}{23}\,A$$	½
∴Current through the 1A resistor = $\frac{-3}{23}\,A$	½

(a) Use Kirchhoff 's rules, calculate the current in the arm AC of the given circuit.





For the mesh EFCAE

$$-30I_1 + 40 - 40\,(I_1 + I_2) = 0$$ 1

Or $-7I_1 - 4I_2 = -4$

Or $7I_1 + 4I_2 = 4$ ……………………………………(1)

For the mesh ACDBA

$$40\,(I_1 + I_2) - 40 + 20I_2 - 80 = 0$$ 1

Or $40I_1 + 60I_2 - 120 = 0$

Or $2I_1 + 3I_2 = 6$ ……………………………………(2)

Solving (1) and (2) , we get

$$I_1 = \frac{-12}{13}\ A$$

$$I_2 = \frac{34}{13}\ A$$

∴Current through arm AC = $I_1 + I_2$

$$= \frac{22}{13}\ A$$ 1

Define mobility of a charge carrier. What is its relation with relaxation time ?

When 5V potential difference is applied across a wire of length 0.1 m, the drift speed of electrons is 2.5×10^{-4} m/s. If the electron density in the wire is 8×10^{28} m^{-3}, calculate the resistivity of the material of wire.

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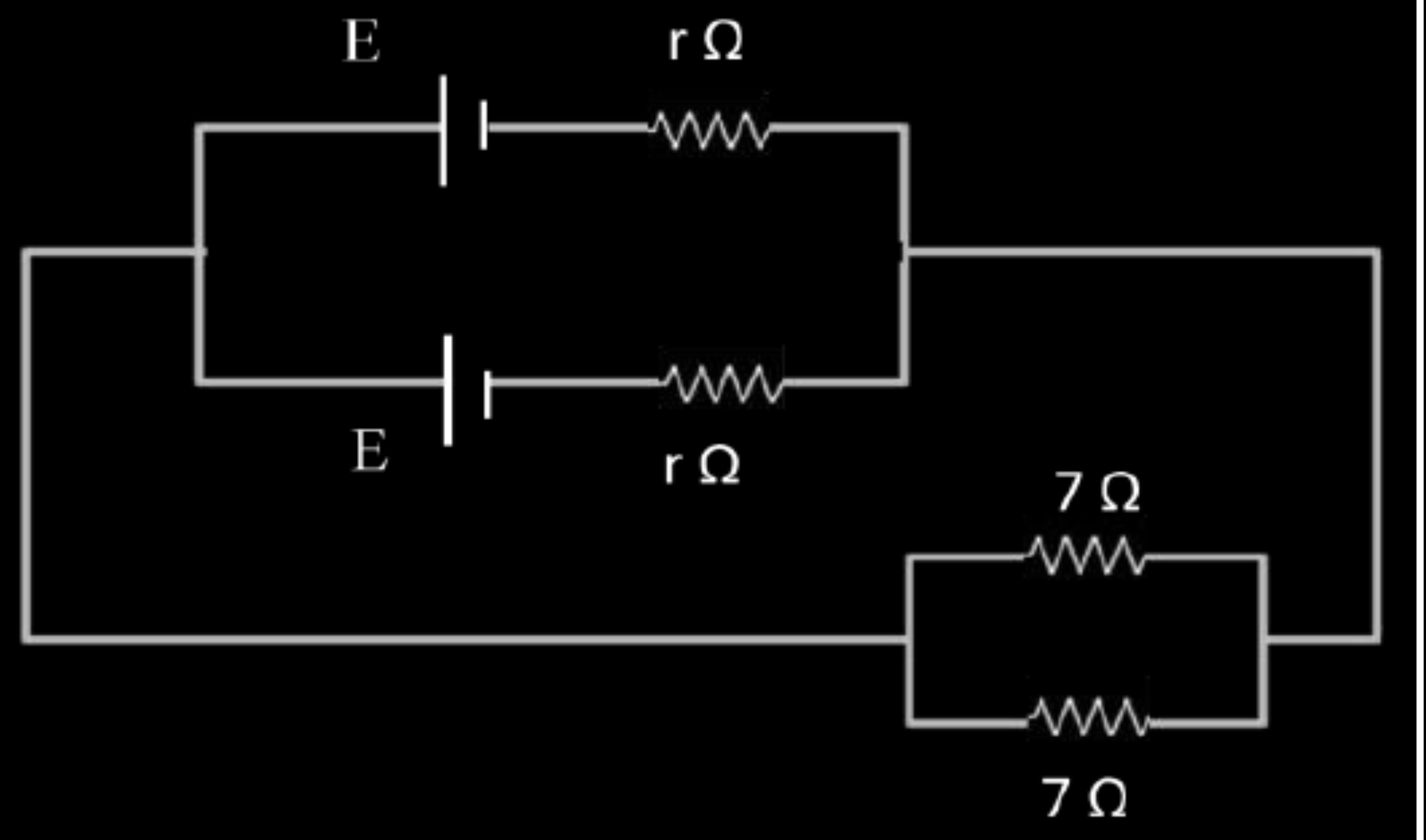
$R = \rho \frac{l}{A}$; $I = neAv_d$	½
$\therefore \rho = \frac{V}{nelv_d}$	½
Alternatively,	
$\left(\begin{array}{c} j = \sigma E = \frac{E}{\rho} \; or \; \frac{E}{j} = \rho \\ \therefore \rho = \frac{V}{lnev_d} \end{array} \right)$	
(Award this 1 mark even if the student writes the formula for ρ directly as such)	
$\therefore \rho = \frac{5}{0.1 \times 8 \times 10^{28} \times 1.6 \times 10^{-19} \times 2.5 \times 10^{-4}} \Omega - m$	½
$= 1.56 \times 10^{-5} \; \Omega - m$	
$\simeq 1.6 \times 10^{-5} \; \Omega - m$	½

Two identical cells of emf 1.5 V each joined in parallel supply energy to an external circuit consisting of two resistances of 7 Ω each joined in parallel. A very high resistance voltmeter reads the terminal voltage of cells to be 1.4 V. Calculate the internal resistance of each cell.

We have, for a single cell, $$r = \left(\frac{E}{V} - 1\right)R$$	1
$\therefore$ *For the parallel combination, as given in the question,* $\frac{r}{2} = \left(\frac{E}{V} - 1\right)\frac{R}{2}$	½
$\therefore r = \left(\frac{1.5}{1.4} - 1\right) \times 7\Omega$	½
$= \frac{0.1}{1.4} \times 7\ \Omega$	½
$=0.5\Omega$	½

(Alternatively,

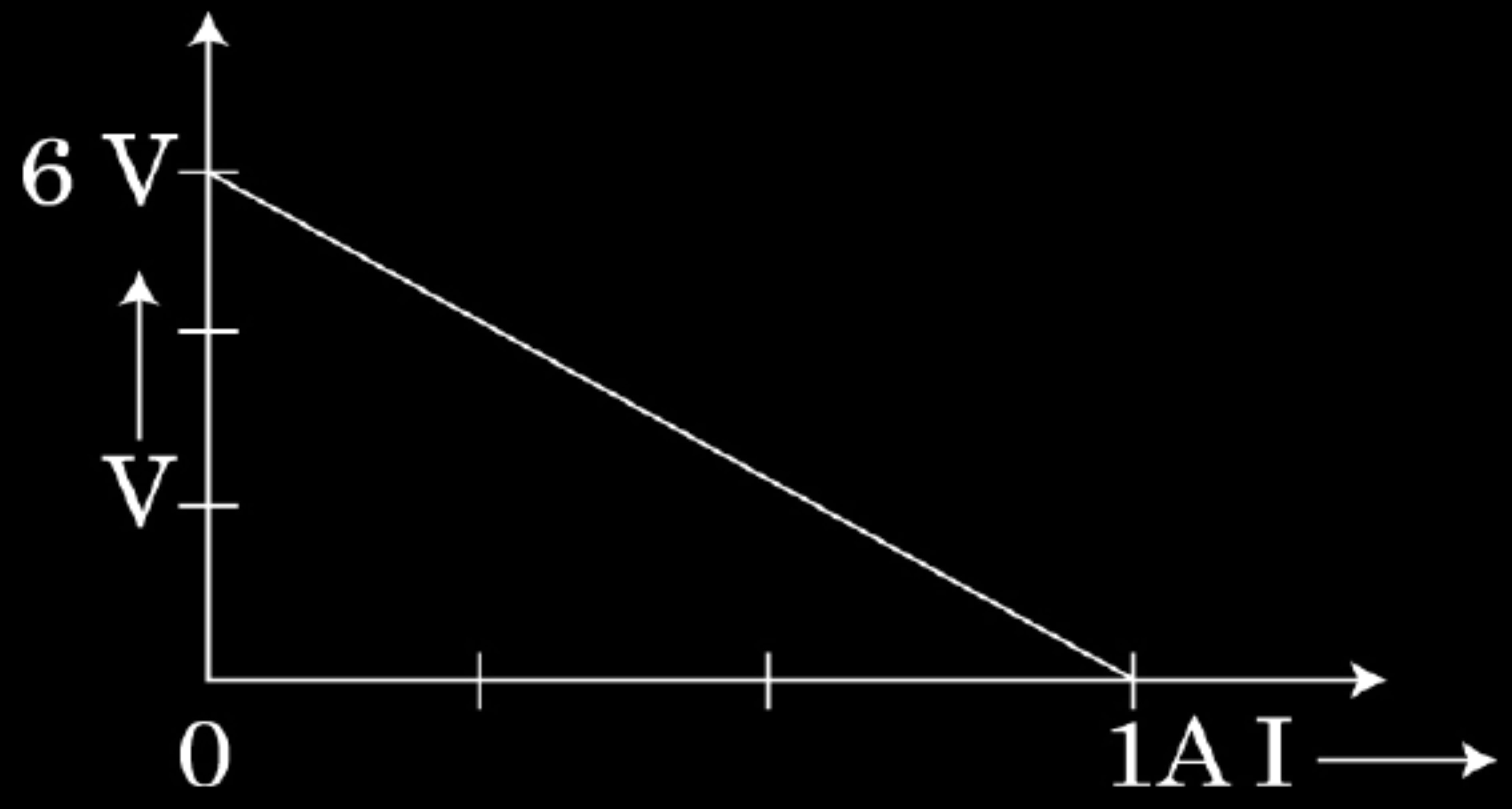
$I = \frac{V}{(R/2)}$	½	
And $E = V - I(r/2)$	½	
This gives $I = \frac{1.4}{7/2}\text{A} = 0.4\text{A}$	½	
$\therefore\ \frac{r}{2} = \frac{1.5-1.4}{0.4} = 0.25$	½	
$\therefore r = 0.5\Omega$)	½	
(Note: If the student just draws the circuit diagram of the setup but does not do any calculations, award 1 mark only.)		

Define the term conductivity of a conductor. On what factors does it depend ?

Define the term conductivity of a conductor. On what factors does it depend ?

Conductivity of a conductor is the current flowing per unit area per unit electric field applied. [Alternatively, conductivity $\sigma = \frac{J}{E}$]	½

The plot of the variation of potential difference across a combination of three identical cells in series, versus current is shown below. What is the emf and internal resistance of each cell ?



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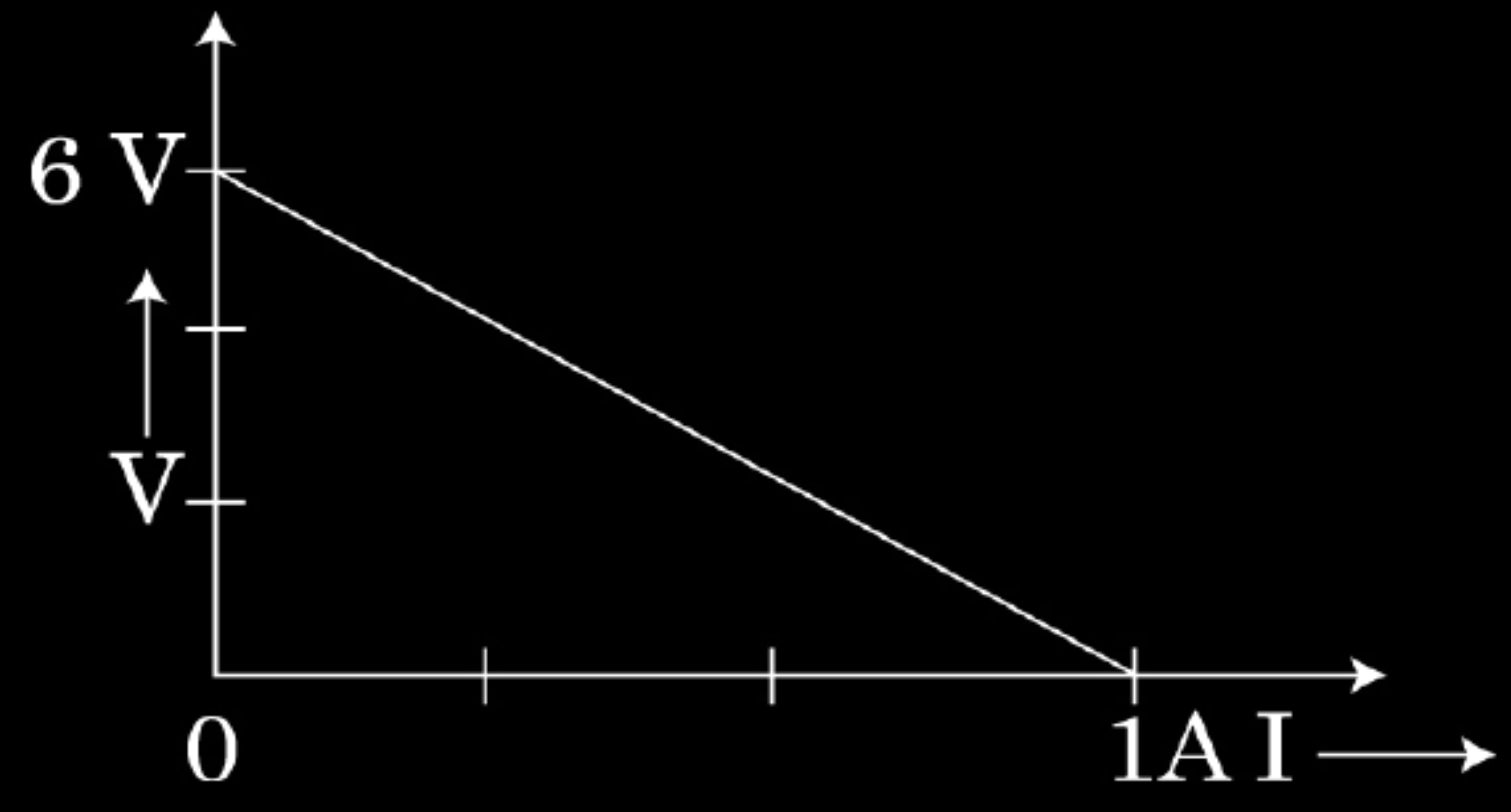
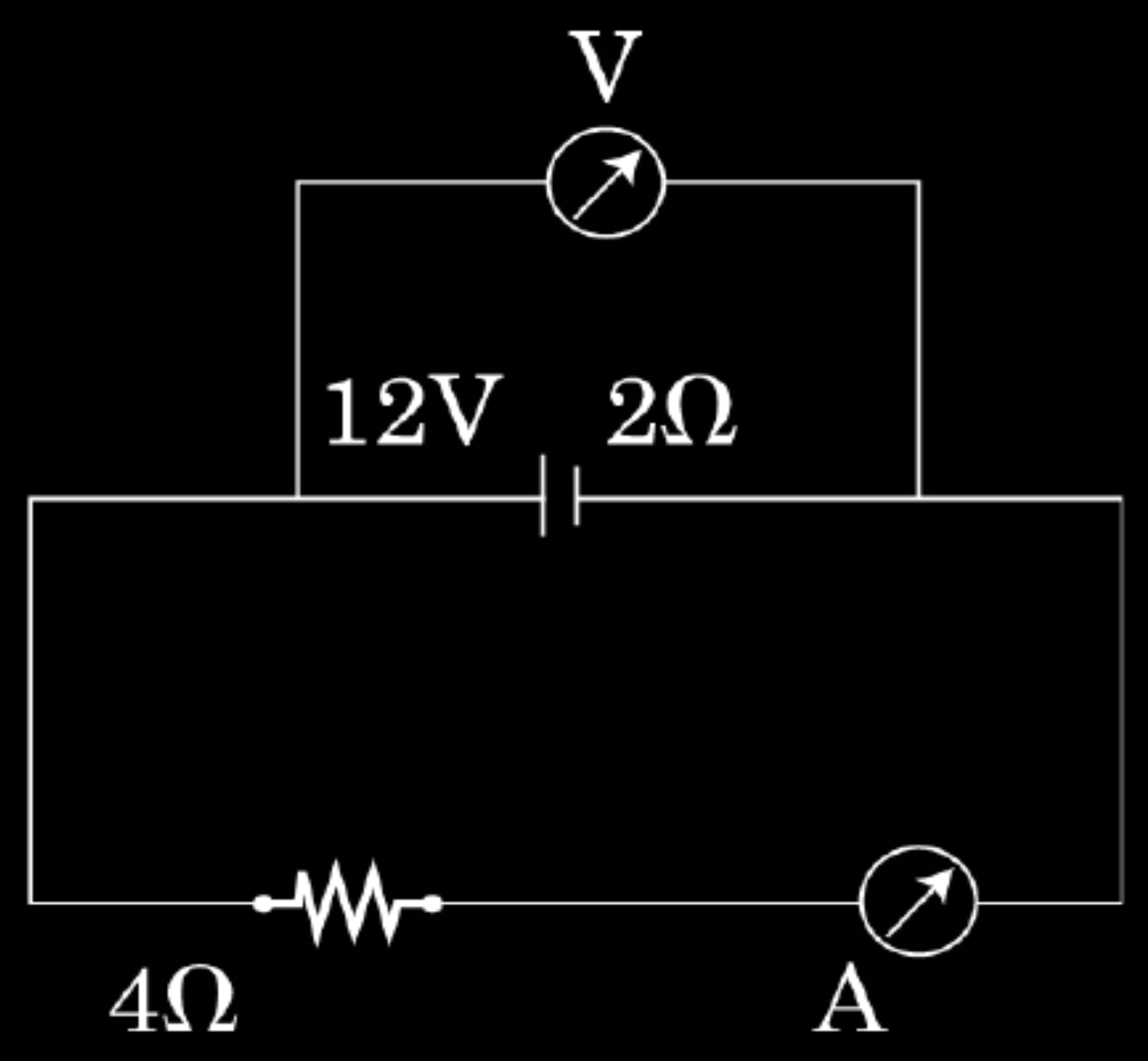


$E = 2V$	½
$r = 2\Omega$	½

A battery of emf 12V and internal resistance 2 Ω is connected to a 4 Ω resistor as shown in the figure.

(a) Show that a voltmeter when placed across the cell and across the resistor, in turn, gives the same reading.

(b) To record the voltage and the current in the circuit, why is voltmeter placed in parallel and ammeter in series in the circuit ?



a) Effective resistance of the circuit R_E= 6Ω	
$\therefore I = \frac{12A}{6} = 2A$	
Terminal potential difference across the cell, V=E-ir	½
Also p.d. across 4Ω resistor =4X2V= 8V	
Hence the volmeter gives the same reading in the two cases.	½
b) In series -current same	½
In parallel – potential same	½

(i) Derive an expression for drift velocity of free electrons.

(ii) How does drift velocity of electrons in a metallic conductor vary with increase in temperature ? Explain.

i. When a potential difference is applied across a conductor, an electric field is produced and free electrons are acted upon by an electric force **(= -Ee)**. Due to this, electrons accelerate and keep colliding with each other and acquire a constant (average) velocity v_d	½
$\therefore, F_e = -Ee$ $\therefore, F_e = \frac{-eV}{l}$	½
As a=$\frac{-F}{m}$ = $\frac{-eV}{m}$ as v = u+at u = 0 , t = τ (relaxation time)	½
v_d = -a τ $v_d = \frac{-eV}{lm}\tau$	½
ii. Decreases, as time of relaxation decreases.	½, ½